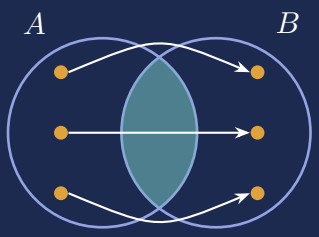
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DISCRETE MATHEMATICS AND GRAPH THEORY

Unit 1
Sets, Relations & Functions

Laws of sets · Cartesian products · Binary, partial-order and equivalence relations · Image, sum and product of functions · Bijective, inverse and composite functions · Finite, infinite, countable and uncountable sets · Cantor's diagonal argument · Power Set theorem · Schröder–Bernstein theorem

LECTURE NOTES · SOLVED PROBLEMS · EXERCISES WITH SOLUTIONS

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Syllabus hours	6 hrs (Unit 1.0)
Prepared for	B.Tech CSE students, GEC Munger
Exam pattern	University theory paper: 2, 5/7 and 10/15-mark questions

References: K. H. Rosen, *Discrete Mathematics and its Applications* (Ch. 2, 9); R. P. Grimaldi, *Discrete and Combinatorial Mathematics* (Ch. 3, 5, 7); S. S. Epp, *Discrete Mathematics with Applications*; C. L. Liu & D. P. Mohapatra, *Elements of Discrete Mathematics*; J. P. Tremblay & R. Manohar, *Discrete Mathematical Structures with Applications to Computer Science*; S. Lipschutz & M. Lipson, *Discrete Mathematics* (Schaum's Outlines).

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How to use these notes

Every topic follows the same pattern: **definition** (teal) → **theorems** (indigo) → **worked examples** → **pitfalls** (red) and **exam tips** (green). Sections 1–6 include exercise sets with full solutions (1A–1C, 2E, 3E–

3I, 4A–4B, 5A, 6A–6G); Section 11 has 13 fully solved university-pattern problems; Section 12 is a graded exercise set (Parts A–D) whose complete solutions are in Section 13. A one-page revision sheet closes the notes.

Notation: $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$; A^c = complement of A; $A \triangle B$ = symmetric difference; $|A|$ = cardinality; $\mathcal{P}(A)$ = power set; $a \mid b$ = "a divides b"; ■ = end of proof.

Syllabus item (Unit 1.0)	Section	Suggested hours
Operations and Laws of Sets	1	0.75
Cartesian Products	2	0.25
Binary Relation	3	0.75
Partial Ordering Relation	4	0.75
Equivalence Relation	5	0.5
Image of a Set; Sum and Product of Functions; Bijective, Inverse and Composite Functions (with floor/ceiling, projections, binary operations, Stirling numbers and the pigeonhole principle)	6	1.0
Size of a Set; Finite and Infinite Sets; Countable and Uncountable Sets	7	0.75
Cantor's Diagonal Argument	8	0.5
The Power Set Theorem	9	0.25
Schröder–Bernstein Theorem	10	0.5

1 Sets: Basics, Operations and Laws

Sets are the starting point of discrete mathematics: relations, functions, graphs and data structures are all built from them. This section covers sets, subsets, power sets, the set operations and the laws of set theory, with worked examples and exercises.

Notation used in this section

$\overline{A}$ = complement of A (also written A^c or A'); $\emptyset$ = null set (also written Φ, ϕ or $\{\ \}$); U = universal set; $A \triangle B$ = symmetric difference (also written $A + B$ or $A \oplus B$); $\mathcal{P}(A)$ = power set. $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{Z}$ = integers, $\mathbb{Z}^+$ = positive integers, $\mathbb{Q}$ = rationals, $\mathbb{R}$ = reals.

1.1 Sets and their description

Definition: Set, element

A **set** is a collection of objects that is **well defined**: given any object, we can decide without ambiguity whether it belongs to the collection. The objects are its **members** or **elements**. We write $a \in A$ ("a is an element of / belongs to A") and $x \notin A$ ("x is not an element of / does not belong to A").

- "All tall boys in a college" is *not* a set — "tall" is ambiguous. "All boys in the college taller than 170 cm" *is* a set: membership can be decided for every boy.
- If E is the set of all even integers, then $2 \in E$ and $3 \notin E$.
- A set is **finite** if it has finitely many elements (students of a college) and **infinite** otherwise ($\mathbb{N} = \{0, 1, 2, 3, \dots\}$).
- A set with exactly one element is a **singleton**, written $\{a\}$. Example: the set of integers between 2 and 8 that are squares of integers is $\{4\}$.

Two ways of describing a set. Let S be the set of all positive odd integers.

Method	Form	Read as
Tabulation (roster)	$S = \{1, 3, 5, 7, \dots\}$	"S is the set consisting of the elements $1, 3, 5, 7, \dots$"
Rule (set-builder)	$S = \{x \mid$ x is a positive odd integer$\}$	"S is the set of elements x such that x is a positive odd integer" (the bar $\mid$ means "such that")

In the tabulation method, **order does not matter and repetitions are ignored**: $\{1, 2, 3, 4\}$, $\{3, 1, 4, 2\}$ and $\{1, 2, 1, 4, 3, 4\}$ are the same set. (A collection in which repetitions *are* counted is called a *multiset*.)

Definition: Null set

The set with no elements is the **null set** or **empty set**, written $\emptyset$, Φ or $\{\ \}$. Examples: the set of positive integers less than 10 that are divisible by 11; the set of real numbers whose squares are negative.

1.2 Equal sets, subsets and Venn diagrams

Definition: Equal sets

$A = B$ when A and B have precisely the same elements.

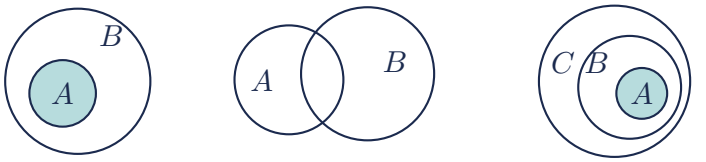
For example, $A = \{1, 2, 3, 4\}$ and $B = \{x \mid x \text{ is a positive integer with } x^2 < 20\}$ are equal, since $1^2, 2^2, 3^2, 4^2 < 20 \leq 5^2$.

Definition: Subset, proper subset

A is a **subset** of B (A is contained in B), written $A \subseteq B$ or $B \supseteq A$, if every element of A is also an element of B. If some element of A is not in B, we write $A \not\subseteq B$. A is a **proper subset** of B, written $A \subset B$ or $B \supset A$, if $A \subseteq B$ and B has at least one element not in A.

Example. $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $C = \{2, 3, 5, 6\}$. Every element of A is in B, so $A \subseteq B$; in fact $A \subset B$ because $4, 5 \in B - A$. Since $1 \in A$ but $1 \notin C$, $A \not\subseteq C$.

Venn diagrams (after John Venn, 1834–1923) picture sets as regions inside a rectangle representing U.



(a) $A \subset B$ (b) $A \not\subseteq B$ (c) $A \subseteq B \subseteq C$

Basic consequences of the definition of subset

1. Every set is a subset of itself: $A \subseteq A$.
2. $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. (This is how equality of sets is *proved*.)
3. The null set is a subset of every set: $\emptyset \subseteq A$.
4. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. (Figure (c) above.)
5. If $A = B$ and $B = C$, then $A = C$.

Statements 1, 2 and 5 are immediate; 3 and 4 are proved in the examples below. **The null set is unique:** if $\emptyset_1$ and $\emptyset_2$ were two null sets, then by (3) $\emptyset_1 \subseteq \emptyset_2$ and $\emptyset_2 \subseteq \emptyset_1$, so $\emptyset_1 = \emptyset_2$ by (2).

Definition: Universal set

In a given discussion, all sets considered are subsets of a fixed set U, the **universal set** (universe of discourse). In a study of integers, $U = \mathbb{Z}$; in a study of defective parts of a machine, U is the set of all its parts. The universal set depends on the context and is not unique.

Common pitfall

$\emptyset \neq \{\emptyset\}$: the first has 0 elements, the second has 1. Also distinguish $\in$ from $\subseteq$: for $A = \{1, \{2\}\}$, $\{2\} \in A$ but $\{2\} \not\subseteq A$ (since $2 \notin A$), while $\{1\} \subseteq A$ but $\{1\} \notin A$.

1.3 Power set

Definition: Power set

The **power set** $\mathcal{P}(A)$ is the set of all subsets of A, including $\emptyset$ and A itself.

A	$\mathcal{P}(A)$	$\lvert\mathcal{P}(A)\rvert$
$\{a, b\}$	$\{\emptyset, \{a\}, \{b\}, A\}$	$4 = 2^2$
$\{1, 2, 3\}$	$\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, A\}$	$8 = 2^3$
$\{1, 2, 3, 4\}$	$\{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, A\}$	$16 = 2^4$

Size of the power set

If a finite set A has n elements, then $\mathcal{P}(A)$ has 2^n elements.

Example 1.1 – The null set is a subset of every set

For a universal set U, let $A \subseteq U$. Prove that $\emptyset \subseteq A$. If $A \neq \emptyset$, prove that $\emptyset \subset A$.

Solution. Suppose $\emptyset \not\subseteq A$. Then $\emptyset$ contains an element (of U) that is not in A. This is impossible, because $\emptyset$ has no element at all. So the assumption is wrong and $\emptyset \subseteq A$. (This is an *indirect proof*.)
If $A \neq \emptyset$, then A has an element, which is not in $\emptyset$; together with $\emptyset \subseteq A$ this gives $\emptyset \subset A$. ■

Example 1.2 – Subset is transitive

For any sets A, B, C: if $A \subseteq B$ and $B \subseteq C$, prove that $A \subseteq C$.

Solution. Take any $x \in A$. Since $A \subseteq B$, $x \in B$. Since $B \subseteq C$, $x \in C$. Thus every element of A is in C, i.e. $A \subseteq C$. ■

Example 1.3 – Proof that $|\mathcal{P}(A)| = 2^n$

If a finite set A has n elements, prove that $\mathcal{P}(A)$ has 2^n elements.

Solution. A subset of A can have $0, 1, 2, \dots, n$ elements. There is one subset with no element ($\emptyset$) and one with all n elements (A). Each element gives a singleton subset: $n = \binom{n}{1}$ of them. Each combination of two elements gives a 2-element subset: $\binom{n}{2}$ of them; similarly there are $\binom{n}{r}$ subsets with r elements. Hence

$$\begin{aligned} |\mathcal{P}(A)| &= 1 + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} \\ &= \binom{n}{0}1^0 + \binom{n}{1}1^1 + \binom{n}{2}1^2 + \cdots + \binom{n}{n}1^n && \text{using } \tbinom{n}{0} = \tbinom{n}{n} = 1 \\ &= (1+1)^n = 2^n. && \text{binomial theorem} \end{aligned}$$

1.4 Operations on sets

Let A, B be subsets of a universal set U.

Definition: Union, intersection, disjoint sets

Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$, where "or" is *inclusive* (x in A, or in B, or in both). Clearly $A \subseteq A \cup B$ and $B \subseteq A \cup B$.
Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$. Clearly $A \cap B \subseteq A$ and $A \cap B \subseteq B$.
A and B are **disjoint** (mutually disjoint) if $A \cap B = \emptyset$.

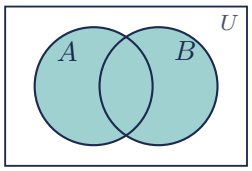
Examples. If $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{4, 5, 8, 9\}$, then $A \cup B = \{1, 2, \dots, 9\}$ and $A \cap B = \{4, 5\}$. The sets $\{1, 2, 3\}$ and $\{5, 7\}$ are disjoint.

Definition: Complement, relative complement, symmetric difference

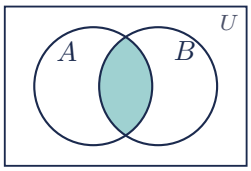
Complement of A (in U): $\overline{A} = \{x \mid x \in U \text{ and } x \notin A\}$.
Relative complement of A in B: $B - A = \{x \mid x \in B \text{ and } x \notin A\}$; $A - B$ is defined similarly. In general $B - A \neq A - B$.
Symmetric difference: $A \triangle B = (A \cup B) - (A \cap B) = \{x \mid x \in A \cup B \text{ and } x \notin A \cap B\}$ — the elements in exactly one of A, B.

Examples. If $U = \mathbb{Z}$ and E is the set of even integers, $\overline{E}$ is the set of odd integers. Always $\overline{U} = \emptyset$, $\overline{\emptyset} = U$,

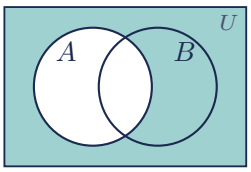
and A, $\overline{A}$ are disjoint. If $A = \{1,2,3,4\}$, $B = \{3,4,7,8\}$, then $B - A = \{7,8\}$ and $A - B = \{1,2\}$. If $A = \{1,2,3,4,5\}$, $B = \{4,5,7,8\}$, then $A \triangle B = \{1,2,3,7,8\}$.



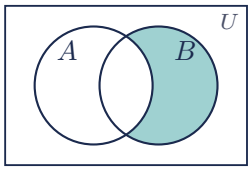
$A \cup B$



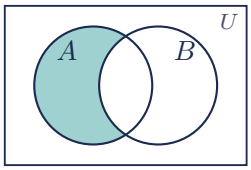
$A \cap B$



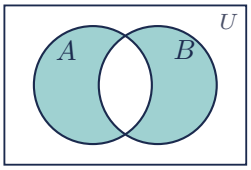
$\overline{A}$



$B - A$



$A - B$



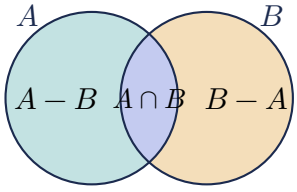
$A \triangle B$

Operation	Definition	Size (finite sets)
Union	$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$	$\lvert A\rvert + \lvert B\rvert - \lvert A \cap B\rvert$
Intersection	$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$	—
Difference	$A - B = \{x \mid x \in A \text{ and } x \notin B\} = A \cap \overline{B}$	$\lvert A\rvert - \lvert A \cap B\rvert$
Complement	$\overline{A} = \{x \in U \mid x \notin A\} = U - A$	$\lvert U\rvert - \lvert A\rvert$
Symmetric difference	$A \triangle B = (A - B) \cup (B - A)$	$\lvert A\rvert + \lvert B\rvert - 2\lvert A \cap B\rvert$

Results on relative complements

For any sets A, B (subsets of U):

1. $A - B$ and $B - A$ are disjoint.
2. If A and B are disjoint, then $A - B = A$ and $B - A = B$.
3. $U - A = \overline{A}$.
4. $A = (A \cup B) - (B - A)$ and $B = (A \cup B) - (A - B)$.
5. $A = (A \cap B) \cup (A - B)$ and $B = (A \cap B) \cup (B - A)$.



$A \cup B$ splits into three disjoint pieces: $A - B$, $A \cap B$, $B - A$. Results 4 and 5 can be read off this picture.

Example 1.4 – Computing with operations

$U = \{1, \ldots, 10\}$, $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7\}$, $C = \{2,4,6,8,10\}$. Find $A \cup B$, $A \cap C$, $A - B$, $B - A$, $A \triangle B$, $\overline{A}$, $\overline{A \cup B}$ and $A \cap (B \cup C)$.

Solution. $A \cup B = \{1, \ldots 7\}$; $A \cap C = \{2,4\}$; $A - B = \{1,2,3\}$; $B - A = \{6,7\}$; $A \triangle B = \{1,2,3,6,7\}$; $\overline{A} = \{6, \ldots, 10\}$; $\overline{A \cup B} = \{8,9,10\}$; $B \cup C = \{2,4,5,6,7,8,10\}$, so $A \cap (B \cup C) = \{2,4,5\}$.

Example 1.5 – Nine standard computations

If $U = \{1, 2, \dots, 9\}$, $A = \{1, 2, 4, 6, 8\}$ and $B = \{2, 4, 5, 9\}$, find (i) $\overline{A}$ (ii) $\overline{B}$ (iii) $\overline{A} \cup \overline{B}$ (iv) $\overline{A \cup B}$ (v) $\overline{A} \cap \overline{B}$ (vi) $\overline{A \cap B}$ (vii) $B - A$ (viii) $A - B$ (ix) $A \triangle B$.

Solution.

Part	Working	Result
(i)	$\overline{A} = U - A$	$\{3, 5, 7, 9\}$
(ii)	$\overline{B} = U - B$	$\{1, 3, 6, 7, 8\}$
(iii)	$\overline{A} \cup \overline{B}$	$\{1, 3, 5, 6, 7, 8, 9\}$
(iv)	$A \cup B = \{1, 2, 4, 5, 6, 8, 9\}$, so $\overline{A \cup B}$	$\{3, 7\}$
(v)	$\overline{A} \cap \overline{B}$	$\{3, 7\}$
(vi)	$A \cap B = \{2, 4\}$, so $\overline{A \cap B}$	$\{1, 3, 5, 6, 7, 8, 9\}$
(vii)	$B - A$	$\{5, 9\}$
(viii)	$A - B$	$\{1, 6, 8\}$
(ix)	$A \triangle B = (A \cup B) - (A \cap B)$	$\{1, 5, 6, 8, 9\}$

Notice (iv) = (v) and (vi) = (iii): these are De Morgan's laws in action (Section 1.5).

Example 1.6 – Finding A and B from $A - B$, $B - A$ and $A \cup B$

Determine A and B, given $A - B = \{1, 2, 4\}$, $B - A = \{7, 8\}$ and $A \cup B = \{1, 2, 4, 5, 7, 8, 9\}$.

Solution. Using result 4 (see the three-piece picture above):

$$A = (A \cup B) - (B - A) = \{1, 2, 4, 5, 9\}, \qquad B = (A \cup B) - (A - B) = \{5, 7, 8, 9\}.$$

Check: $A \cap B = \{5, 9\}$, and $A - B$, $B - A$ come out as given.

Example 1.7 – Finding A and B from $A - B$, $B - A$ and $A \cap B$

Determine A and B, given $A - B = \{1, 3, 7, 11\}$, $B - A = \{2, 6, 8\}$ and $A \cap B = \{4, 9\}$.

Solution. Using result 5:

$$A = (A \cap B) \cup (A - B) = \{1, 3, 4, 7, 9, 11\}, \qquad B = (A \cap B) \cup (B - A) = \{2, 4, 6, 8, 9\}.$$

Example 1.8 – Finding A and B from $A \cup B$, $A \cap B$ and $A - B$

Determine A and B, given $A \cup B = \{1, 2, 4, 5, 7, 8, 9, 10\}$, $A \cap B = \{2, 4, 7\}$ and $A - B = \{1, 8\}$.

Solution.

$$A = (A \cap B) \cup (A - B) = \{1, 2, 4, 7, 8\}, \qquad B = (A \cup B) - (A - B) = \{2, 4, 5, 7, 9, 10\}.$$

Example 1.9 – Power set of an intersection and of a union

For any two sets A, B, prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$. Give an example to show that $\mathcal{P}(A \cup B) \neq \mathcal{P}(A) \cup \mathcal{P}(B)$.

Solution. Take any $X \in \mathcal{P}(A \cap B)$. Then $X \subseteq A \cap B$, so $X \subseteq A$ and $X \subseteq B$; hence $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$, i.e. $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Thus $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$ (i)
Next take any $Y \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Then $Y \subseteq A$ and $Y \subseteq B$, so $Y \subseteq A \cap B$, i.e. $Y \in \mathcal{P}(A \cap B)$. Thus $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$ (ii)
(i) and (ii) give $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Counterexample for union: $A = \{1, 2\}, B = \{3, 4\}$, so $A \cup B = \{1, 2, 3, 4\}$. Then $\{1, 2, 3\} \subseteq A \cup B$, so $\{1, 2, 3\} \in \mathcal{P}(A \cup B)$; but $\{1, 2, 3\} \not\subseteq A$ and $\{1, 2, 3\} \not\subseteq B$, so $\{1, 2, 3\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$. Hence $\mathcal{P}(A \cup B) \neq \mathcal{P}(A) \cup \mathcal{P}(B)$. (Only $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$ holds in general.)

1.5 The laws of set theory

For subsets A, B, C of a universal set U:

No.	Law	Part (a)	Part (b) — the dual
1	Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
2	Associative	$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$
3	Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
4	Idempotent	$A \cup A = A$	$A \cap A = A$
5	Identity	$A \cup \emptyset = A$	$A \cap U = A$
6	Double complement	$\overline{\overline{A}} = A$	—
7	Inverse (complement)	$A \cup \overline{A} = U$	$A \cap \overline{A} = \emptyset$
8	De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$
9	Domination	$A \cup U = U$	$A \cap \emptyset = \emptyset$
10	Absorption	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$

These are exactly the laws of logic of Unit 3 with $\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, $\overline{\ } \leftrightarrow \neg$, $U \leftrightarrow T_0$, $\emptyset \leftrightarrow F_0$. Two further facts are used constantly: $A - B = A \cap \overline{B}$ and $A \triangle B = (A \cup B) \cap \overline{A \cap B}$.

1.6 Duality

Definition: Dual of a set expression

Let P be a set formed from sets (possibly including $\emptyset$ and U) using $\cup$ and $\cap$. The **dual** P^d is obtained by replacing every $\cup$ by $\cap$, every $\cap$ by $\cup$, U by $\emptyset$ and $\emptyset$ by U. Complements are left unchanged.

Example. If $P = \{(A \cup B) \cap \emptyset\} \cup \{(C \cap D) \cup U\}$, then $P^d = \{(A \cap B) \cup U\} \cap \{(C \cup D) \cap \emptyset\}$.

Principle of duality

If P and Q are set expressions with $P = Q$, then $P^d = Q^d$. (Proof omitted.) Every law in the table except double complement has two parts, each the dual of the other; so proving one part proves the other.

1.7 Three ways to prove a set identity

Method	How it works
Element (double inclusion)	Show $x \in$ LHS $\Rightarrow x \in$ RHS and conversely, or one chain of $\iff$
Laws of set theory	Transform one side into the other, naming a law at every step
Venn diagram / membership table	Label the 2^n regions (or rows); both sides must give the same regions (or the same final column)

1.8 Worked examples on the laws

Example 1.10 – Associative laws — element method

For any three sets A, B, C, prove (i) $A \cup (B \cup C) = (A \cup B) \cup C$ (ii) $A \cap (B \cap C) = (A \cap B) \cap C$.

Solution. (i) Let $D = B \cup C$ and $E = A \cup B$. Take any $x \in A \cup (B \cup C) = A \cup D$. Then

$$\begin{aligned} & x \in A \text{ or } x \in D \\ \text{i.e.}\quad & x \in A \text{ or } (x \in B \text{ or } x \in C) \\ \text{i.e.}\quad & (x \in A \text{ or } x \in B) \text{ or } x \in C \\ \text{i.e.}\quad & x \in E \text{ or } x \in C, \quad \text{i.e. } x \in E \cup C = (A \cup B) \cup C. \end{aligned}$$

Hence $A \cup (B \cup C) \subseteq (A \cup B) \cup C$. Reversing the steps gives $(A \cup B) \cup C \subseteq A \cup (B \cup C)$, so the two are equal.
(ii) Taking duals of both sides of (i) gives $A \cap (B \cap C) = (A \cap B) \cap C$ by the principle of duality. ■
Remark. Because of these laws we simply write $A \cup B \cup C$ and $A \cap B \cap C$.

Example 1.11 – Distributive laws — element method

For any three sets, prove (i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Solution. (i) Take any $x \in A \cap (B \cup C)$. Then $x \in A$ and ($x \in B$ or $x \in C$), i.e. ($x \in A$ and $x \in B$) or ($x \in A$ and $x \in C$), i.e. $x \in A \cap B$ or $x \in A \cap C$, i.e. $x \in (A \cap B) \cup (A \cap C)$. So $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.
Next take any $y \in (A \cap B) \cup (A \cap C)$. Then ($y \in A$ and $y \in B$) or ($y \in A$ and $y \in C$), i.e. $y \in A$ and ($y \in B$ or $y \in C$), i.e. $y \in A \cap (B \cup C)$. So $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. Hence (i).
(ii) follows from (i) by the principle of duality. ■

Example 1.12 – De Morgan's laws

For any two sets A and B, prove (i) $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (ii) $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Solution.

(i) $\overline{A} \cap \overline{B} = \{x \mid x \in \overline{A} \text{ and } x \in \overline{B}\} = \{x \mid x \notin A \text{ and } x \notin B\} = \{x \mid x \notin A \cup B\} = \overline{A \cup B}$.
(ii) $\overline{A} \cup \overline{B} = \{x \mid x \in \overline{A} \text{ or } x \in \overline{B}\} = \{x \mid x \notin A \text{ or } x \notin B\} = \{x \mid x \notin A \cap B\} = \overline{A \cap B}$.

Remark. Each law also follows from the other by the principle of duality. ■

Example 1.13 – Using De Morgan twice

For any three sets, prove $\overline{(A \cap B) \cup C} = (\overline{A} \cup \overline{B}) \cap \overline{C}$.

Solution. Let $D = A \cap B$. Then

$$\begin{aligned} \overline{(A \cap B) \cup C} = \overline{D \cup C} &= \overline{D} \cap \overline{C} && \textit{De Morgan} \\ &= \overline{(A \cap B)} \cap \overline{C} = (\overline{A} \cup \overline{B}) \cap \overline{C} && \textit{De Morgan} \quad ■ \end{aligned}$$

Example 1.14 – Five identities for the difference

For any two sets prove: (i) $A - B = A \cap \overline{B}$ (ii) $\overline{A - B} = \overline{A} \cup B$ (iii) $A - B = A - (A \cap B)$ (iv) $\overline{A - B} = \overline{A} \cup (A \cap B)$ (v) $A - (A - B) = A \cap B$.

Solution. (i) $A - B = \{x \in U \mid x \in A \text{ and } x \notin B\} = \{x \in U \mid x \in A \text{ and } x \in \overline{B}\} = A \cap \overline{B}$.

(ii) By (i), $\overline{A-B} = \overline{A \cap \overline{B}} = \overline{A} \cup \overline{\overline{B}} = \overline{A} \cup B$ (De Morgan, double complement).
(iii)

$$\begin{aligned} A-(A\cap B) &= A \cap \overline{(A\cap B)} && \textit{using (i)} \\ &= A \cap (\overline{A} \cup \overline{B}) && \textit{De Morgan} \\ &= (A\cap \overline{A}) \cup (A \cap \overline{B}) && \textit{Distributive} \\ &= \emptyset \cup (A \cap \overline{B}) = A \cap \overline{B} = A - B && \textit{Inverse, Identity, (i)} \end{aligned}$$

(iv) $\overline{A-B} = \overline{A-(A\cap B)}$ by (iii), $= \overline{A} \cup (A \cap B)$ by (ii) applied with $A \cap B$ in place of B.
(v)

$$\begin{aligned} A-(A-B) = A-(A\cap\overline{B}) &= A \cap \overline{(A\cap\overline{B})} && \textit{using (i) twice} \\ &= A \cap (\overline{A} \cup B) && \textit{De Morgan, double complement} \\ &= (A\cap\overline{A}) \cup (A\cap B) = \emptyset \cup (A\cap B) = A\cap B && \textit{Distributive, Inverse, Identity} \end{aligned}$$

Example 1.15 – Intersection and difference

For any sets A, B, C, prove (i) $A \cap (B-C) = (A\cap B) - C$ (ii) $(A-B)\cap(A-C) = A-(B\cup C)$.
Solution. (i) $A\cap(B-C) = A\cap(B\cap\overline{C}) = (A\cap B)\cap\overline{C} = (A\cap B)-C$ (difference law, associative).
(ii)

$$\begin{aligned} (A-B)\cap(A-C) = (A\cap\overline{B})\cap(A\cap\overline{C}) &= A\cap(\overline{B}\cap A)\cap\overline{C} && \textit{Associative, Commutative} \\ &= (A\cap A)\cap(\overline{B}\cap\overline{C}) = A\cap\overline{B\cup C} = A-(B\cup C) && \textit{Idempotent, De Morgan} \end{aligned}$$

Alternative (element method) for (ii): $x \in A-(B\cup C) \iff x\in A$ and $x\notin B$ and $x\notin C \iff (x\in A,\ x\notin B)$ and $(x\in A,\ x\notin C) \iff x\in(A-B)\cap(A-C)$. ■

Example 1.16 – Three expressions for the same set

For any three sets, prove that $(A-B)-C = A-(B\cup C) = (A-C)-(B-C)$.
Solution.

$$(A-B)-C = (A\cap\overline{B})\cap\overline{C} = A\cap(\overline{B}\cap\overline{C}) = A\cap\overline{(B\cup C)} = A-(B\cup C). \qquad \text{(i)}$$

Also,

$$\begin{aligned} (A-C)-(B-C) &= (A-C)\cap\overline{(B-C)} = (A\cap\overline{C})\cap\overline{(B\cap\overline{C})} \\ &= (A\cap\overline{C})\cap(\overline{B}\cup C) && \textit{De Morgan} \\ &= \{(A\cap\overline{C})\cap\overline{B}\}\cup\{(A\cap\overline{C})\cap C\} && \textit{Distributive} \\ &= \{A\cap(\overline{C}\cap\overline{B})\}\cup\{A\cap(\overline{C}\cap C)\} = \{A\cap\overline{(C\cup B)}\}\cup(A\cap\emptyset) \\ &= \{A\cap\overline{(B\cup C)}\}\cup\emptyset = A-(B\cup C). && \text{(ii)} \end{aligned}$$

(i) and (ii) together prove the result. ■

Example 1.17 – When union equals symmetric difference

If $S, T \subseteq U$, prove that S and T are disjoint if and only if $S\cup T = S\triangle T$.
Solution. By definition, $S\triangle T = (S\cup T)-(S\cap T) = (S\cup T)\cap\overline{(S\cap T)}$ (i)

Suppose S, T are disjoint, i.e. $S \cap T = \emptyset$. Then (i) gives $S \triangle T = (S \cup T) \cap \overline{\emptyset} = (S \cup T) \cap U = S \cup T$. Conversely, suppose $S \triangle T = S \cup T$. Then (i) becomes $S \cup T = (S \cup T) \cap \overline{(S \cap T)}$, so $S \cup T \subseteq \overline{S \cap T}$. Since $S \cap T \subseteq S \cup T$, every element of $S \cap T$ would lie in $\overline{S \cap T}$, which is impossible unless $S \cap T = \emptyset$. Hence S and T are disjoint. ■

Example 1.18 – Forms of the symmetric difference

Prove (i) $A \triangle B = (B \cap \overline{A}) \cup (A \cap \overline{B}) = (B - A) \cup (A - B)$ (ii) $\overline{A \triangle B} = \overline{A} \triangle B = A \triangle \overline{B}$.

Solution. (i)

$$\begin{aligned} A \triangle B &= (A \cup B) - (A \cap B) = (A \cup B) \cap \overline{(A \cap B)} \\ &= (A \cup B) \cap (\overline{A} \cup \overline{B}) && \textit{De Morgan} \\ &= \{(A \cup B) \cap \overline{A}\} \cup \{(A \cup B) \cap \overline{B}\} && \textit{Distributive} \\ &= \{(A \cap \overline{A}) \cup (B \cap \overline{A})\} \cup \{(A \cap \overline{B}) \cup (B \cap \overline{B})\} && \textit{Distributive} \\ &= \{\emptyset \cup (B \cap \overline{A})\} \cup \{(A \cap \overline{B}) \cup \emptyset\} \\ &= (B \cap \overline{A}) \cup (A \cap \overline{B}) = (B - A) \cup (A - B). \end{aligned}$$

(ii) Using (i) with $\overline{A}$ in place of A:

$$\begin{aligned} \overline{A} \triangle B &= (B \cap \overline{\overline{A}}) \cup (\overline{A} \cap \overline{B}) = (A \cap B) \cup (\overline{A} \cap \overline{B}) \\ &= \overline{\overline{(A \cap B)} \cap \overline{(\overline{A} \cap \overline{B})}} && \textit{De Morgan, double complement} \\ &= \overline{(\overline{A} \cup \overline{B}) \cap (A \cup B)} = \overline{A \triangle B}. && \textit{De Morgan; definition} \end{aligned}$$

Similarly $A \triangle \overline{B} = (\overline{B} \cap \overline{A}) \cup (A \cap B)$, the same set, so $A \triangle \overline{B} = \overline{A \triangle B}$. Thus $\overline{A} \triangle B = A \triangle \overline{B} = \overline{A \triangle B}$. ■

Example 1.19 – Simplification using the laws

Using the laws of set theory, simplify (i) $A \cap (B - A)$ (ii) $(A - B) \cup (A \cap B)$ (iii) $\overline{A} \cup \overline{B} \cup (A \cap B \cap \overline{C})$ (iv) $\overline{\overline{(A \cup B) \cap C} \cup \overline{B}}$.

Solution. (i) $A \cap (B - A) = A \cap (B \cap \overline{A}) = A \cap (\overline{A} \cap B) = (A \cap \overline{A}) \cap B = \emptyset \cap B = \emptyset$.

(ii) $(A - B) \cup (A \cap B) = (A \cap \overline{B}) \cup (A \cap B) = A \cap (\overline{B} \cup B) = A \cap U = A$.

(iii)

$$\begin{aligned} (\overline{A} \cup \overline{B}) \cup (A \cap B \cap \overline{C}) &= \overline{(A \cap B)} \cup \{(A \cap B) \cap \overline{C}\} && \textit{De Morgan} \\ &= \{\overline{(A \cap B)} \cup (A \cap B)\} \cap \{\overline{(A \cap B)} \cup \overline{C}\} && \textit{Distributive} \\ &= U \cap (\overline{A} \cup \overline{B} \cup \overline{C}) = \overline{A} \cup \overline{B} \cup \overline{C} && \textit{Inverse, De Morgan, Identity} \end{aligned}$$

(iv)

$$\begin{aligned} \overline{\overline{(A \cup B) \cap C} \cup \overline{B}} &= \overline{\overline{(A \cup B) \cap C}} \cap \overline{\overline{B}} && \textit{De Morgan} \\ &= \{(A \cup B) \cap C\} \cap B && \textit{Double complement} \\ &= (A \cup B) \cap (B \cap C) && \textit{Associative, Commutative} \\ &= \{(A \cup B) \cap B\} \cap C = B \cap C && \textit{Associative, Absorption} \end{aligned}$$

Example 1.20 – A four-set identity

For any sets A, B, C, D, prove using the laws of set theory that $(A\cap B)\cup(A\cap B\cap\overline{C}\cap D)\cup(\overline{A}\cap B) = B$.

Solution. First, $(A \cap B) \cup (A \cap B \cap \overline{C} \cap D) = (A \cap B) \cup \{(A \cap B) \cap (\overline{C} \cap D)\} = A \cap B$ by an absorption law. Therefore

$$\begin{aligned}(A \cap B) \cup (A \cap B \cap \overline{C} \cap D) \cup (\overline{A} \cap B) &= (A \cap B) \cup (\overline{A} \cap B) \\ &= (B \cap A) \cup (B \cap \overline{A}) && \textit{Commutative} \\ &= B \cap (A \cup \overline{A}) = B \cap U = B. && \textit{Distributive, Inverse, Identity}\end{aligned}$$

Example 1.21 – Membership table

Prove $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$ using a membership table (1 = "belongs", 0 = "does not belong").

Solution.

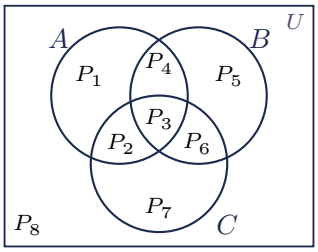
A	B	$A-B$	$B-A$	$(A-B)\cup(B-A)$	$A\cup B$	$A\cap B$	$(A\cup B)-(A\cap B)$
1	1	0	0	**0**	1	1	**0**
1	0	1	0	**1**	1	0	**1**
0	1	0	1	**1**	1	0	**1**
0	0	0	0	**0**	0	0	**0**

The two bold columns are identical, so the identity holds. (This is part (i) of the symmetric-difference example above, proved a third way.)

Example 1.22 – Venn diagram proof with three sets

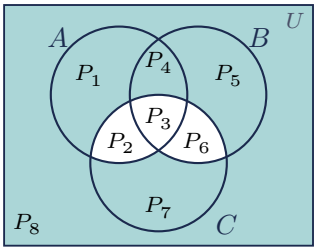
Using a Venn diagram, prove that for any three sets $\overline{(A \cup B) \cap C} = (\overline{A} \cap \overline{B}) \cup \overline{C}$.

Solution. Label the eight regions of the diagram $P_1, \dots, P_8$:

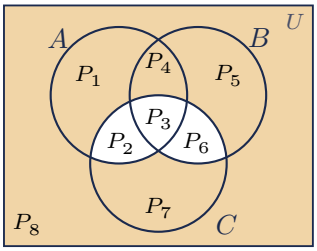


Regions $P_1, \dots, P_8$ for A, B, C

Then $A = P_1 \cup P_2 \cup P_3 \cup P_4$, $B = P_3 \cup P_4 \cup P_5 \cup P_6$, $C = P_2 \cup P_3 \cup P_6 \cup P_7$.
(i) LHS: $(A\cup B)\cap C = P_2\cup P_3\cup P_6$, so $\overline{(A \cup B) \cap C} = U-(P_2\cup P_3\cup P_6) = P_1\cup P_4\cup P_5\cup P_7\cup P_8$.
(ii) RHS: $\overline{A} = P_5\cup P_6\cup P_7\cup P_8$, $\overline{B} = P_1\cup P_2\cup P_7\cup P_8$, $\overline{C} = P_1\cup P_4\cup P_5\cup P_8$. So $\overline{A}\cap\overline{B} = P_7\cup P_8$ and $(\overline{A} \cap \overline{B}) \cup \overline{C} = P_1 \cup P_4 \cup P_5 \cup P_7 \cup P_8$.
(i) and (ii) are the same regions, so the identity holds.



$\overline{(A \cup B) \cap C}$ shaded

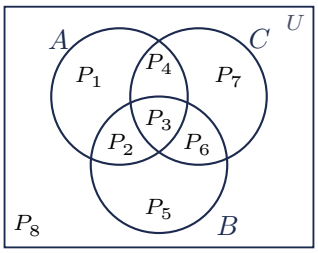


$(\overline{A} \cap \overline{B}) \cup \overline{C}$ shaded

Example 1.23 – Symmetric difference is associative — Venn proof

Using a Venn diagram, prove that $A \triangle (B \triangle C) = (A \triangle B) \triangle C$.

Solution. Label the regions as below (here B is the lower circle):



Regions $P_1, \ldots, P_8$

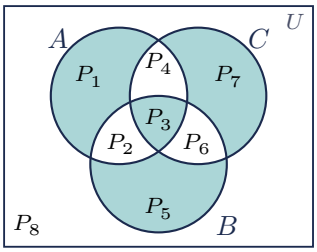
$A = P_1 \cup P_2 \cup P_3 \cup P_4$, $B = P_2 \cup P_3 \cup P_5 \cup P_6$, $C = P_3 \cup P_4 \cup P_6 \cup P_7$.
$B \cup C = P_2 \cup P_3 \cup P_4 \cup P_5 \cup P_6 \cup P_7$ and $B \cap C = P_3 \cup P_6$, so $B \triangle C = P_2 \cup P_4 \cup P_5 \cup P_7$.
Then $A \cup (B \triangle C) = P_1 \cup P_2 \cup P_3 \cup P_4 \cup P_5 \cup P_7$ and $A \cap (B \triangle C) = P_2 \cup P_4$, so

$$A \triangle (B \triangle C) = P_1 \cup P_3 \cup P_5 \cup P_7 \tag{i}$$

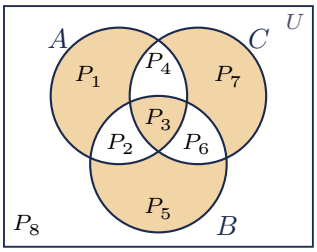
On the other hand, $A \cup B = P_1 \cup \cdots \cup P_6$ and $A \cap B = P_2 \cup P_3$, so $A \triangle B = P_1 \cup P_4 \cup P_5 \cup P_6$. Then $(A \triangle B) \cup C = P_1 \cup P_3 \cup P_4 \cup P_5 \cup P_6 \cup P_7$ and $(A \triangle B) \cap C = P_4 \cup P_6$, so

$$(A \triangle B) \triangle C = P_1 \cup P_3 \cup P_5 \cup P_7 \tag{ii}$$

(i) and (ii) show $A \triangle (B \triangle C) = (A \triangle B) \triangle C$: both consist of the points lying in an *odd* number of the three sets.



$A \triangle (B \triangle C)$ shaded



$(A \triangle B) \triangle C$ shaded

Example 1.24 – Writing dual statements

Write the dual of each statement: (1) $A = A \cap (A \cup B)$ (2) $A = (A \cup B) \cap (A \cup \emptyset)$ (3) $A \cup B = (A \cap B) \cup (A \cap \overline{B}) \cup (\overline{A} \cap B)$ (4) $U = (A \cap B) \cup (A \cap \overline{B}) \cup (\overline{A} \cap B) \cup (\overline{A} \cap \overline{B})$.

Solution.

(1) $A = A \cup (A \cap B)$

(2) $A = (A \cap B) \cup (A \cap U)$

(3) $A \cap B = (A \cup B) \cap (A \cup \overline{B}) \cap (\overline{A} \cup B)$

(4) $\emptyset = (A \cup B) \cap (A \cup \overline{B}) \cap (\overline{A} \cup B) \cap (\overline{A} \cup \overline{B})$

Each original statement is true, so by the principle of duality each dual is true as well.

Example 1.25 – A subset criterion

Show that $A \subseteq B \iff A \cup B = B \iff A \cap B = A \iff A \cap \overline{B} = \emptyset$.

Solution. If $A \subseteq B$, every element of $A \cup B$ is already in B, so $A \cup B = B$; also every element of A is in $A \cap B$, so $A \cap B = A$. If $A \cap B = A$, then no element of A lies outside B, i.e. $A \cap \overline{B} = \emptyset$. Finally, if $A \cap \overline{B} = \emptyset$, every $x \in A$ must lie in B, so $A \subseteq B$. ■

Example 1.26 – The dual of $A \subseteq B$

For two sets A and B, show that the dual of the statement $A \subseteq B$ is the statement $B \subseteq A$.

Solution. By the subset criterion, $A \subseteq B$ is equivalent to $A \cup B = B$. The dual of $A \cup B = B$ is $A \cap B = B$, which is equivalent to $B \subseteq A$. Hence the dual of $A \subseteq B$ is $B \subseteq A$. ■

Examination tip

For a 10-mark "prove the identity" question, use the laws method and name every law; add a Venn diagram or membership table only as a check (unless the question asks for a Venn proof). Never "prove" an identity with one numerical example — that only verifies it. A numerical example *is*, however, the right way to show that a statement is *false*.

1.9 Exercises on sets

Attempt these before reading Section 1.10, which has complete solutions.

Exercise Set 1A — Sets, subsets and power sets

1A.1 Which of the following sets are equal? (1) $A = \{x \mid x$ is a letter in the word *follow*$\}$ (2) $B = \{x \mid x$ is a letter in the word *wolf*$\}$ (3) $C = \{x \mid x$ is a letter in the word *flow*$\}$.

1A.2 Which of the following sets are equal? $A = \{1, 2, 3, 4\}$, $B = \{2, 1, 4, 3\}$, $C = \{1, 2, 4, 1, 3, 2, 4\}$, $D = \{x \mid x$ is a positive integer less than $5\}$, $S = \{x \mid x$ is an integer less than or equal to $4\}$.

1A.3 Which of the following sets are different? $\emptyset$, $\{\emptyset\}$, $\{0\}$.

1A.4 Let $A = \{1, \{1\}, 2, \{2\}\}$. Which of the following are true? (i) $1 \in A$ (ii) $1 \subset A$ (iii) $\{1\} \subset A$ (iv) $\{1, 2\} \in A$ (v) $\{\{1\}, \{2\}\} \subset A$.

1A.5 Describe by the tabulation method: (i) $A = \{1 + (-1)^n \mid n$ is a positive integer$\}$ (ii) $B = \{n + 1/n \mid n \in \{1, 2, 3, 5, 7\}\}$.

1A.6 Which of the following sets are equal to the null set? (1) $\{x \mid x$ is an integer and $x^2 + 4 = 6\}$ (2) $\{x \mid x$ is an integer and $3x + 5 = 9\}$ (3) $\{x \mid x$ is real and $x^3 = -1\}$ (4) $\{x \mid x$ is real and $x = x + 1\}$ (5) $\{x \mid x$ is real and $x + 8 = 8\}$.

1A.7 Let A, B, C be sets. Prove: (1) If $A \subset B$ and $B \subseteq C$, then $A \subset C$. (2) If $A \subseteq B$ and $B \subset C$, then $A \subset C$. (3) If $A \subset B$ and $B \subset C$, then $A \subset C$.

1A.8 Find all subsets of (i) $A = \{0, \emptyset\}$ (ii) $A = \{0, \{1, 2\}\}$.

1A.9 Find the elements of $\mathcal{P}(A)$ for (1) $A = \{3, 7\}$ (2) $A = \{2, 3, 7\}$ (3) $A = \{a, b, c, d\}$.

Exercise Set 1B — Operations on sets

1B.1 Let $U = \{1, \dots, 9\}$, $A = \{1, 2, 3, 7\}$, $B = \{4, 5, 6, 7\}$, $C = \{1, 3, 6\}$, $D = \{6, 8, 9\}$. Compute (i) $A \cap B$ (ii) $B \cap \overline{C}$ (iii) $\overline{A} \cup \overline{C}$ (iv) $\overline{A} \cap \overline{D}$ (v) $\overline{A \cup C}$ (vi) $B \cup \overline{D}$ (vii) $A \cup B \cup C$ (viii) $A \cap B \cap C$ (ix) $A \cap (B \cup C)$ (x) $(A \cup B) \cap C$ (xi) $A \triangle B$ (xii) $\overline{A} \triangle \overline{C}$.

1B.2 Let $U = \{1, \dots, 9\}$, $A = \{1, 2, 4, 6, 8\}$, $B = \{2, 4, 5, 9\}$, $C = \{x \mid x \text{ is a positive integer and } x^2 \leq 16\}$. Compute (i) $A - B$ (ii) $B - A$ (iii) $(A \cup B) - C$ (iv) $A \cup (B - C)$ (v) $(A - B) - C$ (vi) $(A \cup B) - (B \cap C)$.

1B.3 Through examples, prove that the following statements are false: (1) If $A \cap C = B \cap C$, then $A = B$. (2) If $A \cup C = B \cup C$, then $A = B$. (3) If $A - C = B - C$, then $A = B$.

1B.4 If A, B, C, D are sets with $A \subseteq B$ and $C \subseteq D$, prove (1) $A \cup C \subseteq B \cup D$ (2) $A \cap C \subseteq B \cap D$.

Exercise Set 1C — Laws of set theory

1C.1 For any sets A and B, prove: (1) $A \triangle B = B \triangle A$ (2) $A \triangle \overline{A} = U$ (3) $A \triangle U = \overline{A}$ (4) $A \triangle \emptyset = A$ (5) $\overline{(A \cup B) \cap C} = (\overline{A} \cap \overline{B}) \cup \overline{C}$ (6) $A = A \cup (A \cap B) = A \cap (A \cup B) = (A - B) \cup (A \cap B) = (A \cap B) \cup (A \cap \overline{B}) = (A \cup B) \cap (A \cup \overline{B})$ (7) $A \cup B = (A \cap B) \cup (A \cap \overline{B}) \cup (\overline{A} \cap B) = (A - B) \cup (A \cap B) \cup (B - A)$.

1C.2 For any four sets A, B, C, D, prove: (i) $(A \cap B) \subseteq \{(A \cap C) \cup (B \cap \overline{C})\}$ (ii) $\{(A \cup C) \cap (B \cup \overline{C})\} \subseteq (A \cup B)$ (iii) $A - (B \cap C) = (A - B) \cup (A - C)$ (iv) $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$ (v) $(A \cap B) \cup [B \cap \{(C \cap D) \cup (C \cap \overline{D})\}] = B \cap (A \cup C)$.

1C.3 Let A, B, C be any three sets. Prove: (i) If $A = B \cup C$ and $B \cap C = \emptyset$, then $B = A - C$ and $C = A - B$. (ii) If $B = A - C$ and $C \subseteq A$, then $B \cap C = \emptyset$ and $A = B \cup C$. (iii) If $A \cap C = B \cap C$ and $A - C = B - C$, then $A = B$.

1C.4 Using Venn diagrams, prove that for any sets A, B, C: (i) $\overline{(A \cap B) \cup C} = (\overline{A} \cup \overline{B}) \cap \overline{C}$
(ii) $\overline{(A \cup B) \cap C} \cup \overline{B} = \overline{B \cap C}$.

1.10 Solutions to Section 1 exercises

Set 1A

1A.1 $A = \{f, o, l, w\}$ (repeated letters counted once), $B = \{w, o, l, f\}$, $C = \{f, l, o, w\}$. **All three are equal.**

1A.2 A, B, C and D all equal $\{1, 2, 3, 4\}$ (order and repetition do not matter). $S = \{\dots, -2, -1, 0, 1, 2, 3, 4\}$ is infinite and contains 0, so S differs. Hence $\mathbf{A = B = C = D}$, S is different.

1A.3 All three are different. $\emptyset$ has no element; $\{\emptyset\}$ has one element, namely $\emptyset$; $\{0\}$ has one element, the number 0, and $0 \neq \emptyset$.

1A.4 Only **(i), (iii) and (v)** are true.

Part	Verdict	Reason
(i) $1 \in A$	True	1 is listed as an element
(ii) $1 \subset A$	False	1 is an element, not a set of elements of A; $\subset$ relates sets
(iii) $\{1\} \subset A$	True	$1 \in A$, and A has other elements (note $\{1\} \in A$ is also true)
(iv) $\{1, 2\} \in A$	False	$\{1, 2\}$ is not one of the four listed elements (though $\{1, 2\} \subset A$)
(v) $\{\{1\}, \{2\}\} \subset A$	True	both $\{1\}$ and $\{2\}$ are elements of A

1A.5 (i) For odd n, $1 + (-1)^n = 0$; for even n, $1 + (-1)^n = 2$. So $A = \{0, 2\}$.
(ii) $n = 1, 2, 3, 5, 7$ give $2, \frac{5}{2}, \frac{10}{3}, \frac{26}{5}, \frac{50}{7}$. So $B = \{2, \frac{5}{2}, \frac{10}{3}, \frac{26}{5}, \frac{50}{7}\}$.

1A.6 Sets **(1), (2) and (4)** are null. (1) $x^2 = 2$ has no integer solution: $\emptyset$. (2) $3x = 4$ gives $x = \frac{4}{3}$, not an integer: $\emptyset$. (3) $x^3 = -1$ gives $x = -1$: the set is $\{-1\}$. (4) $x = x + 1$ gives $0 = 1$, impossible: $\emptyset$. (5) $x + 8 = 8$ gives $x = 0$: the set is $\{0\}$.

1A.7 (1) $A \subset B \subseteq C$ gives $A \subseteq C$ by transitivity. Since $A \subset B$, there is $b \in B$ with $b \notin A$; as $B \subseteq C$, $b \in C$. So C has an element not in A, and $A \subset C$.
(2) $A \subseteq B \subseteq C$ gives $A \subseteq C$. Since $B \subset C$, there is $c \in C$ with $c \notin B$; then $c \notin A$ (as $A \subseteq B$). So $A \subset C$.
(3) $B \subset C$ implies $B \subseteq C$, so (3) follows from (1). ■

1A.8 (i) $A = \{0, \emptyset\}$ has 2 elements, so 4 subsets: $\emptyset,\ \{0\},\ \{\emptyset\},\ A$.
(ii) $A = \{0, \{1, 2\}\}$ has 2 elements (0 and the set $\{1, 2\}$), so 4 subsets: $\emptyset,\ \{0\},\ \{\{1, 2\}\},\ A$.

1A.9 (1) $\mathcal{P}(\{3, 7\}) = \{\emptyset, \{3\}, \{7\}, A\}$ — 4 elements.
(2) $\mathcal{P}(\{2, 3, 7\}) = \{\emptyset, \{2\}, \{3\}, \{7\}, \{2, 3\}, \{2, 7\}, \{3, 7\}, A\}$ — 8 elements.
(3) $\mathcal{P}(\{a, b, c, d\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, A\}$ — 16 elements.

Set 1B

1B.1 First, $\overline{A} = \{4, 5, 6, 8, 9\}$, $\overline{C} = \{2, 4, 5, 7, 8, 9\}$, $\overline{D} = \{1, 2, 3, 4, 5, 7\}$.

Part	Answer	Part	Answer
(i)	$A \cap B = \{7\}$	(ii)	$B \cap \overline{C} = \{4, 5, 7\}$
(iii)	$\overline{A} \cup \overline{C} = \{2, 4, 5, 6, 7, 8, 9\}$	(iv)	$\overline{A} \cap \overline{D} = \{4, 5\}$
(v)	$A \cup C = \{1, 2, 3, 6, 7\}$; $\overline{A \cup C} = \{4, 5, 8, 9\}$	(vi)	$B \cup \overline{D} = \{1, 2, 3, 4, 5, 6, 7\}$
(vii)	$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7\}$	(viii)	$A \cap B \cap C = \{7\} \cap C = \emptyset$
(ix)	$A \cap (B \cup C) = \{1, 3, 7\}$	(x)	$(A \cup B) \cap C = \{1, 3, 6\}$
(xi)	$A \triangle B = \{1, 2, 3, 4, 5, 6\}$	(xii)	$\overline{A} \triangle \overline{C} = \{2, 6, 7\}$

Note on (xii). $\overline{A} \cup \overline{C} = \{2, 4, 5, 6, 7, 8, 9\}$ and $\overline{A} \cap \overline{C} = \{4, 5, 8, 9\}$, so $\overline{A} \triangle \overline{C} = \{2, 6, 7\}$. As a check, $\overline{A} \triangle \overline{C} = A \triangle C = \{2, 7\} \cup \{6\}$ (by the symmetric-difference example, applied twice).

1B.2 $C = \{1, 2, 3, 4\}$ (since $4^2 = 16$, $5^2 = 25$). $A \cup B = \{1, 2, 4, 5, 6, 8, 9\}$, $B \cap C = \{2, 4\}$.
(i) $A - B = \{1, 6, 8\}$ (ii) $B - A = \{5, 9\}$ (iii) $(A \cup B) - C = \{5, 6, 8, 9\}$ (iv) $B - C = \{5, 9\}$, so $A \cup (B - C) = \{1, 2, 4, 5, 6, 8, 9\}$ (v) $(A - B) - C = \{6, 8\}$ (vi) $(A \cup B) - (B \cap C) = \{1, 5, 6, 8, 9\}$.

1B.3 One counterexample is enough to show that a statement is false.

Statement	Counterexample	Check
(1)	$U = \{1, 2, 3\}$, $A = \{1\}$, $B = \{2\}$, $C = \{3\}$	$A \cap C = \emptyset = B \cap C$, but $A \neq B$
(2)	$U = \{1, 2\}$, $A = \{1\}$, $B = \{2\}$, $C = U$	$A \cup C = U = B \cup C$, but $A \neq B$
(3)	$U = \{1, 2, 3\}$, $A = \{1, 2\}$, $B = \{1\}$, $C = \{2\}$	$A - C = \{1\} = B - C$, but $A \neq B$

(Compare 1C.3(iii): if *both* $A \cap C = B \cap C$ *and* $A - C = B - C$, then $A = B$.)

1B.4 (1) Let $x \in A \cup C$. If $x \in A$, then $x \in B$ (as $A \subseteq B$); if $x \in C$, then $x \in D$ (as $C \subseteq D$). Either way $x \in B \cup D$. So $A \cup C \subseteq B \cup D$.
(2) Let $x \in A \cap C$. Then $x \in A \subseteq B$ and $x \in C \subseteq D$, so $x \in B \cap D$. So $A \cap C \subseteq B \cap D$. ■

Set 1C

1C.1
(1) $A \triangle B = (A \cup B) - (A \cap B) = (B \cup A) - (B \cap A) = B \triangle A$ (commutative laws).
(2) $A \triangle \overline{A} = (A \cup \overline{A}) - (A \cap \overline{A}) = U - \emptyset = U$ (inverse laws).
(3) $A \triangle U = (A \cup U) - (A \cap U) = U - A = \overline{A}$ (domination, identity).
(4) $A \triangle \emptyset = (A \cup \emptyset) - (A \cap \emptyset) = A - \emptyset = A$ (identity, domination).
(5) $\overline{(A \cup B) \cap C} = \overline{A \cup B} \cup \overline{C} = (\overline{A} \cap \overline{B}) \cup \overline{C}$ (De Morgan twice). (A Venn proof is the worked example in

Section 1.8.)
(6) $A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$ are the absorption laws. $(A - B) \cup (A \cap B) = A$ is result 5 on relative complements (or the simplification example (ii)). Since $A - B = A \cap \overline{B}$, $(A \cap B) \cup (A \cap \overline{B}) = (A - B) \cup (A \cap B) = A$. Finally $(A \cup B) \cap (A \cup \overline{B}) = A \cup (B \cap \overline{B}) = A \cup \emptyset = A$ (distributive, inverse, identity).
(7)

$$\begin{aligned}(A \cap B) \cup (A \cap \overline{B}) \cup (\overline{A} \cap B) &= \{A \cap (B \cup \overline{B})\} \cup (\overline{A} \cap B) && \textit{Distributive}\\ &= A \cup (\overline{A} \cap B) && \textit{Inverse, Identity}\\ &= (A \cup \overline{A}) \cap (A \cup B) && \textit{Distributive}\\ &= U \cap (A \cup B) = A \cup B. && \textit{Inverse, Identity}\end{aligned}$$

Since $A \cap \overline{B} = A - B$ and $\overline{A} \cap B = B - A$, this also equals $(A - B) \cup (A \cap B) \cup (B - A)$. ■

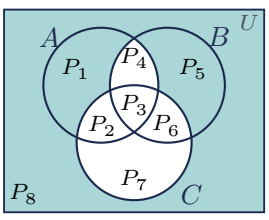
1C.2
(i) $A \cap B = (A \cap B) \cap U = (A \cap B) \cap (C \cup \overline{C}) = (A \cap B \cap C) \cup (A \cap B \cap \overline{C})$. Now $A \cap B \cap C \subseteq A \cap C$ and $A \cap B \cap \overline{C} \subseteq B \cap \overline{C}$, so $A \cap B \subseteq (A \cap C) \cup (B \cap \overline{C})$.
(ii) $(A \cup C) \cap (B \cup \overline{C}) = (A \cap B) \cup (A \cap \overline{C}) \cup (C \cap B) \cup (C \cap \overline{C}) = (A \cap B) \cup (A \cap \overline{C}) \cup (B \cap C)$ (distributive, inverse, identity). Each of the three pieces lies inside A or inside B, so the union lies in $A \cup B$.
(iii) $A - (B \cap C) = A \cap \overline{B \cap C} = A \cap (\overline{B} \cup \overline{C}) = (A \cap \overline{B}) \cup (A \cap \overline{C}) = (A - B) \cup (A - C)$ (difference law, De Morgan, distributive).
(iv) Use $X \triangle Y = (X \cap \overline{Y}) \cup (Y \cap \overline{X})$ (symmetric-difference example, part (i)). With $X = A \cap B, Y = A \cap C$:

$$(A \cap B) \cap \overline{(A \cap C)} = (A \cap B) \cap (\overline{A} \cup \overline{C}) = (A \cap B \cap \overline{A}) \cup (A \cap B \cap \overline{C}) = A \cap (B \cap \overline{C}),$$
$$(A \cap C) \cap \overline{(A \cap B)} = A \cap (C \cap \overline{B}) \quad \text{(similarly)}.$$

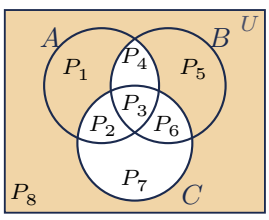
Hence $(A \cap B) \triangle (A \cap C) = A \cap \{(B \cap \overline{C}) \cup (C \cap \overline{B})\} = A \cap (B \triangle C)$ (distributive law).
(v) $(C \cap D) \cup (C \cap \overline{D}) = C \cap (D \cup \overline{D}) = C \cap U = C$. So the left side is $(A \cap B) \cup (B \cap C) = (B \cap A) \cup (B \cap C) = B \cap (A \cup C)$ (commutative, distributive). ■

1C.3
(i) Since $B \cap C = \emptyset$, $B - C = B$ (result 2). Then $A - C = (B \cup C) \cap \overline{C} = (B \cap \overline{C}) \cup (C \cap \overline{C}) = (B - C) \cup \emptyset = B$. Interchanging the roles of B and C gives $A - B = C$.
(ii) $B \cap C = (A \cap \overline{C}) \cap C = A \cap (\overline{C} \cap C) = A \cap \emptyset = \emptyset$. Also $B \cup C = (A \cap \overline{C}) \cup C = (A \cup C) \cap (\overline{C} \cup C) = (A \cup C) \cap U = A \cup C = A$, because $C \subseteq A$.
(iii) By result 5, $A = (A \cap C) \cup (A - C)$ and $B = (B \cap C) \cup (B - C)$. The two right-hand sides are equal by hypothesis, so $A = B$. ■

1C.4 Use the region labels of the Venn-diagram example in Section 1.8: $A = P_1 \cup P_2 \cup P_3 \cup P_4$, $B = P_3 \cup P_4 \cup P_5 \cup P_6$, $C = P_2 \cup P_3 \cup P_6 \cup P_7$, P_8 outside all three.
(i) $A \cap B = P_3 \cup P_4$; $(A \cap B) \cup C = P_2 \cup P_3 \cup P_4 \cup P_6 \cup P_7$; so $\overline{(A \cap B) \cup C} = P_1 \cup P_5 \cup P_8$.
$\overline{A} \cup \overline{B} = (P_5 \cup P_6 \cup P_7 \cup P_8) \cup (P_1 \cup P_2 \cup P_7 \cup P_8) = P_1 \cup P_2 \cup P_5 \cup P_6 \cup P_7 \cup P_8$; intersecting with $\overline{C} = P_1 \cup P_4 \cup P_5 \cup P_8$ gives $P_1 \cup P_5 \cup P_8$. Same regions, so the identity holds.

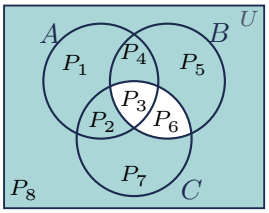


$\overline{(A \cap B) \cup C}$

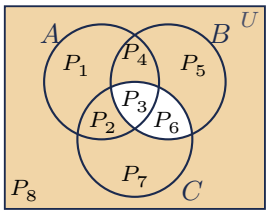


$(\overline{A} \cup \overline{B}) \cap \overline{C}$

(ii) $(A \cup B) \cap C = P_2 \cup P_3 \cup P_6$, so $\overline{(A \cup B) \cap C} = P_1 \cup P_4 \cup P_5 \cup P_7 \cup P_8$; adding $\overline{B} = P_1 \cup P_2 \cup P_7 \cup P_8$ gives $P_1 \cup P_2 \cup P_4 \cup P_5 \cup P_7 \cup P_8$.
$B \cap C = P_3 \cup P_6$, so $\overline{B \cap C} = P_1 \cup P_2 \cup P_4 \cup P_5 \cup P_7 \cup P_8$. Same regions, so the identity holds.



$(A \cup B) \cap C \cup \overline{B}$



$\overline{B \cap C}$

2 Cartesian Products

Relations and functions are subsets of a Cartesian product, so this short section is the foundation of Sections 3–6.

2.1 Ordered pairs and the product of two sets

Definition: Cartesian product

For sets A and B, the set of all **ordered pairs** (a, b) with $a \in A$ and $b \in B$ is the **Cartesian product** (cross product, product set) of A and B, in this order:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Two ordered pairs are equal, $(a, b) = (c, d)$, if and only if $a = c$ and $b = d$.

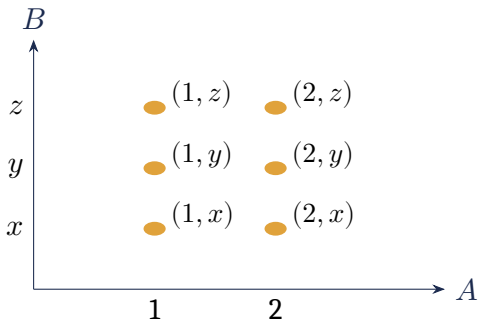
Since $(a, b) \neq (b, a)$ in general, $B \times A = \{(b, a) \mid b \in B,\ a \in A\}$ is in general different from $A \times B$. (Contrast with sets: $\{1, 2\} = \{2, 1\}$, but $(1, 2) \neq (2, 1)$.)

Example. If $A = \{1, 0, -1\}$ and $B = \{2, 3\}$, then

$$A \times B = \{(1, 2), (1, 3), (0, 2), (0, 3), (-1, 2), (-1, 3)\},$$
$$B \times A = \{(2, 1), (2, 0), (2, -1), (3, 1), (3, 0), (3, -1)\} \neq A \times B.$$

The product of a set with itself, $A \times A = \{(a, b) \mid a, b \in A\}$, is written A^2. For $A = \{1, 0, -1\}$,

$$A^2 = \{(1, 1), (1, 0), (1, -1), (0, 1), (0, 0), (0, -1), (-1, 1), (-1, 0), (-1, -1)\}.$$



$A \times B$ for $A = \{1, 2\}$, $B = \{x, y, z\}$ drawn as a grid: $2 \times 3 = 6$ ordered pairs.

2.2 Size of a product; k-fold products

If $|A| = m$ and $|B| = n$, the first entry of (a, b) can be chosen in m ways and, for each choice, the second in n ways. Hence:

Size of a Cartesian product

If A and B are finite sets, then $|A \times B| = |A|\,|B|$. Consequently $|B \times A| = |A \times B|$ and $|A \times A| = |A|^2$.

For example, if $|A| = 5$ and $|B| = 8$, then $A \times B$ and $B \times A$ have $5 \times 8 = 40$ elements each, $A \times A$ has 25 and $B \times B$ has 64.

Definition: k-fold product

For non-empty sets $A_1, \dots, A_k$, the **k-fold product** is the set of ordered **k-tuples**

$$A_1 \times A_2 \times \cdots \times A_k = \{(a_1, a_2, \dots, a_k) \mid a_i \in A_i,\ i = 1, \dots, k\},$$

with $(a_1, \dots, a_k) = (b_1, \dots, b_k)$ iff $a_i = b_i$ for every i. For finite sets, $|A_1 \times \cdots \times A_k| = |A_1| \cdot |A_2| \cdots |A_k|$.

Example. $A = \{1, 0\}$, $B = \{2, -2\}$, $C = \{0, -1\}$:

$$A \times B \times C = \{(1,2,0), (1,2,-1), (1,-2,0), (1,-2,-1), (0,2,0), (0,2,-1), (0,-2,0), (0,-2,-1)\},$$

which has $2 \cdot 2 \cdot 2 = 8$ elements.

Algebraic properties of the product

For any sets A, B, C, D:

1. $A \times \emptyset = \emptyset \times A = \emptyset$.
2. $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and $(A \cup B) \times C = (A \times C) \cup (B \times C)$.
3. $A \times (B \cap C) = (A \times B) \cap (A \times C)$ and $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
4. $A \times (B - C) = (A \times B) - (A \times C)$.
5. For non-empty sets, $A \times B \subseteq C \times D$ iff $A \subseteq C$ and $B \subseteq D$.
6. $A \times B = B \times A$ iff $A = \emptyset$, or $B = \emptyset$, or $A = B$.
7. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$, but only $(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$.

Items 1–5 are proved in the worked examples below; item 6 is Exercise 2E.7.

2.3 Worked examples

Example 2.1 – Equality of ordered pairs

Find x and y in each case: (i) $(2x,\ x + y) = (6, 1)$ (ii) $(y - 2,\ 2x + 1) = (x - 1,\ y + 2)$.

Solution. (i) $(2x, x + y) = (6, 1)$ iff $2x = 6$ and $x + y = 1$. These give $x = 3$ and $y = 1 - x = -2$.
(ii) The pairs are equal iff $y - 2 = x - 1$ and $2x + 1 = y + 2$, i.e. $x - y + 1 = 0$ and $2x - y - 1 = 0$. Subtracting, $x - 2 = 0$, so $x = 2$ and $y = 3$.

Example 2.2 – Eleven standard computations

Let $A = \{1, 3, 5\}$, $B = \{2, 3\}$, $C = \{4, 6\}$. Write down (1) $A \times B$ (2) $B \times A$ (3) $B \times C$ (4) $A \times C$ (5) $(A \cup B) \times C$ (6) $A \cup (B \times C)$ (7) $(A \times B) \cup C$ (8) $A \cap (B \times C)$ (9) $(A \times B) \cup (B \times C)$ (10) $(A \times B) \cap (B \times A)$ (11) $(A \times B) \cap (B \times C)$.

Solution. Using the definition of the product:

$$
\begin{aligned}
&(1)\ A \times B = \{(1,2),(1,3),(3,2),(3,3),(5,2),(5,3)\}\\
&(2)\ B \times A = \{(2,1),(2,3),(2,5),(3,1),(3,3),(3,5)\}\\
&(3)\ B \times C = \{(2,4),(2,6),(3,4),(3,6)\}\\
&(4)\ A \times C = \{(1,4),(1,6),(3,4),(3,6),(5,4),(5,6)\}\\
&(5)\ (A \cup B) \times C = \{1,2,3,5\} \times \{4,6\}\\
&\qquad = \{(1,4),(1,6),(2,4),(2,6),(3,4),(3,6),(5,4),(5,6)\}\\
&(6)\ A \cup (B \times C) = \{1,3,5,(2,4),(2,6),(3,4),(3,6)\}\\
&(7)\ (A \times B) \cup C = \{(1,2),(1,3),(3,2),(3,3),(5,2),(5,3),4,6\}\\
&(8)\ A \cap (B \times C) = \emptyset \quad \text{(numbers and ordered pairs have nothing in common)}\\
&(9)\ (A \times B) \cup (B \times C) = \{(1,2),(1,3),(3,2),(3,3),(5,2),(5,3),\\
&\qquad (2,4),(2,6),(3,4),(3,6)\}\\
&(10)\ (A \times B) \cap (B \times A) = \{(3,3)\}\\
&(11)\ (A \times B) \cap (B \times C) = \emptyset
\end{aligned}
$$

Common pitfall

In (6)–(8) a set of numbers is combined with a set of ordered pairs. Such unions are *mixed* sets (allowed, but unusual), and such intersections are always empty, because a number is never an ordered pair.

Example 2.3 – Subsets of $\mathbb{Z} \times \mathbb{Z}$ given by equations

Suppose $A, B, C \subseteq \mathbb{Z} \times \mathbb{Z}$ with $A = \{(x,y) \mid y = 5x - 1\}$, $B = \{(x,y) \mid y = 6x\}$, $C = \{(x,y) \mid 3x - y = -7\}$. Find (i) $A \cap B$ (ii) $B \cap C$ (iii) $\overline{A} \cup \overline{C}$ (iv) $\overline{B} \cup \overline{C}$.

Solution. (i) $(x,y) \in A \cap B \iff y = 5x-1$ and $y = 6x \iff 5x - 1 = 6x \iff x = -1,\ y = -6$. Thus $A \cap B = \{(-1,-6)\}$.
(ii) $(x,y) \in B \cap C \iff y = 6x$ and $3x - y = -7 \iff 6x = 3x + 7 \iff x = 7/3$, which is not an integer. Thus $B \cap C = \emptyset$.
(iii) By De Morgan, $\overline{A} \cup \overline{C} = \overline{A \cap C}$, so $\overline{\overline{A} \cup \overline{C}} = A \cap C$. Now $y = 5x - 1$ and $y = 3x + 7$ give $x = 4$, $y = 19$. Thus the set is $\{(4,19)\}$.
(iv) $\overline{B} \cup \overline{C} = \overline{B \cap C} = \overline{\emptyset} = \mathbb{Z} \times \mathbb{Z}$, the universal set.

Example 2.4 – Product with the null set

For any set A, prove that $A \times \emptyset = \emptyset \times A = \emptyset$.

Solution. Suppose $A \times \emptyset \neq \emptyset$. Then it has an element (a,b) with $a \in A$ and $b \in \emptyset$. But $b \in \emptyset$ means $\emptyset$ is not empty — a contradiction. Hence $A \times \emptyset = \emptyset$; similarly $\emptyset \times A = \emptyset$. ■

Example 2.5 – When is $A \times B \subseteq C \times D$?

Let A, B, C, D be non-empty sets. Prove that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$. What happens if one of the sets is empty?

Solution. Suppose $A \times B \subseteq C \times D$. Take any $a \in A$ and any $b \in B$ (possible because $A, B \neq \emptyset$). Then $(a,b) \in A \times B \subseteq C \times D$, so $a \in C$ and $b \in D$. Hence $A \subseteq C$ and $B \subseteq D$.
Conversely, suppose $A \subseteq C$ and $B \subseteq D$. For any $(x,y) \in A \times B$ we have $x \in A \subseteq C$ and $y \in B \subseteq D$, so $(x,y) \in C \times D$. Hence $A \times B \subseteq C \times D$. ■
If a set is empty the first half fails. Take $A = \emptyset$, $B = \{1,2\}$, $C = \{2,4,5\}$, $D = \{1,3\}$. Then $A \times B = \emptyset \subseteq C \times D$, but $B \not\subseteq D$.

Example 2.6 – Distributive laws for the product

For non-empty sets A, B, C, prove (1) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (2) $(A \cup B) \times C = (A \times C) \cup (B \times C)$ (3) $A \times (B \cap C) = (A \times B) \cap (A \times C)$ (4) $(A \cap B) \times C = (A \times C) \cap (B \times C)$ (5) $A \times (B - C) = (A \times B) - (A \times C)$.

Solution. For any ordered pair (x, y):

$$\begin{aligned}
\text{(1) } (x,y) \in A \times (B \cup C) &\iff x \in A \text{ and } (y \in B \text{ or } y \in C) \\
&\iff (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C) \\
&\iff (x,y) \in (A \times B) \cup (A \times C). \\
\text{(2) } (x,y) \in (A \cup B) \times C &\iff (x \in A \text{ or } x \in B) \text{ and } y \in C \\
&\iff (x \in A,\ y \in C) \text{ or } (x \in B,\ y \in C) \\
&\iff (x,y) \in (A \times C) \cup (B \times C). \\
\text{(3) } (x,y) \in A \times (B \cap C) &\iff x \in A,\ y \in B \text{ and } y \in C \\
&\iff (x \in A,\ y \in B) \text{ and } (x \in A,\ y \in C) \\
&\iff (x,y) \in (A \times B) \cap (A \times C). \\
\text{(4) } (x,y) \in (A \cap B) \times C &\iff x \in A,\ x \in B \text{ and } y \in C \\
&\iff (x \in A,\ y \in C) \text{ and } (x \in B,\ y \in C) \\
&\iff (x,y) \in (A \times C) \cap (B \times C). \\
\text{(5) } (x,y) \in A \times (B - C) &\iff x \in A,\ y \in B \text{ and } y \notin C \\
&\iff (x,y) \in A \times B \text{ and } (x,y) \notin A \times C \\
&\iff (x,y) \in (A \times B) - (A \times C).
\end{aligned}$$

In (5), since $x \in A$ already, $(x, y) \notin A \times C$ exactly when $y \notin C$. ■

Example 2.7 – Union does not distribute both ways

Show by an example that $(A \times B) \cup (C \times D) \neq (A \cup C) \times (B \cup D)$ in general.

Solution. $A = B = \{1\}, C = D = \{2\}$: the left side is $\{(1,1), (2,2)\}$ (2 elements); the right side is $\{1,2\} \times \{1,2\}$ (4 elements). The pairs $(1,2)$ and $(2,1)$ are missing on the left.

2.4 Exercises on Cartesian products

2E.1 Find x and y in each case: (1) $(2x-3,\ 3y+1) = (5, 7)$ (2) $(x+2,\ 4) = (5,\ 2x+y)$ (3) $(x, y) = (x^2, y^2)$ (4) $(x, y) = (y^2, x^2)$.

2E.2 Given $A = \{a, b\}$ and $B = \{1, 2, 3\}$, find $A \times B$, $B \times A$, $A \times A$ and $B \times B$.

2E.3 Given $A = \{1, 2\}$, $B = \{a, b, c\}$ and $C = \{3, 4\}$, find $A \times B \times C$ and $B \times C \times A$.

2E.4 Let $A = \{1, 2, 3, 4\}, B = \{2, 5\}, C = \{3, 4, 7\}$. Write down $A \times B$, $B \times A$, $A \cup (B \times C)$, $(A \cup B) \times C$, $(A \times C) \cup (B \times C)$.

2E.5 If $A = \{2, 3\}$, $B = \{-1, 2\}$ and $C = \{a, b\}$, verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

2E.6 If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$, verify the same two identities.

2E.7 For which sets A, B do we have $A \times B = B \times A$?

2.5 Solutions to Section 2 exercises

2E.1 (1) $2x-3=5$ and $3y+1=7$: $x=4$, $y=2$.
(2) $x+2=5$ gives $x=3$; then $4=2x+y=6+y$ gives $y=-2$.
(3) $x=x^2$ and $y=y^2$: $x\in\{0,1\}$ and $y\in\{0,1\}$, so $(x,y)\in\{(0,0),(0,1),(1,0),(1,1)\}$.
(4) $x=y^2$ and $y=x^2$ give $x=x^4$, i.e. $x(x^3-1)=0$. Over the reals, $x=0$ or $x=1$, giving $(x,y)=(0,0)$ or $(1,1)$. (In general: $x=y=0$, or $x=y^2$ with $x^3=1$.)

2E.2

$$\begin{aligned}
A\times B &= \{(a,1),(a,2),(a,3),(b,1),(b,2),(b,3)\},\\
B\times A &= \{(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)\},\\
A\times A &= \{(a,a),(a,b),(b,a),(b,b)\},\\
B\times B &= \{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)\}.
\end{aligned}$$

2E.3 Each product has $2\cdot 3\cdot 2=12$ triples:

$$\begin{aligned}
A\times B\times C &= \{(1,a,3),(1,a,4),(1,b,3),(1,b,4),(1,c,3),(1,c,4),\\
&\qquad (2,a,3),(2,a,4),(2,b,3),(2,b,4),(2,c,3),(2,c,4)\},\\
B\times C\times A &= \{(a,3,1),(a,3,2),(a,4,1),(a,4,2),(b,3,1),(b,3,2),\\
&\qquad (b,4,1),(b,4,2),(c,3,1),(c,3,2),(c,4,1),(c,4,2)\}.
\end{aligned}$$

2E.4

$$\begin{aligned}
A\times B &= \{(1,2),(2,2),(3,2),(4,2),(1,5),(2,5),(3,5),(4,5)\},\\
B\times A &= \{(2,1),(2,2),(2,3),(2,4),(5,1),(5,2),(5,3),(5,4)\},\\
A\cup(B\times C) &= \{1,2,3,4,(2,3),(2,4),(2,7),(5,3),(5,4),(5,7)\},\\
(A\cup B)\times C &= \{(1,3),(2,3),(3,3),(4,3),(5,3),(1,4),(2,4),(3,4),\\
&\qquad (4,4),(5,4),(1,7),(2,7),(3,7),(4,7),(5,7)\}\\
&= (A\times C)\cup(B\times C).
\end{aligned}$$

2E.5 $B\cup C=\{-1,2,a,b\}$ and $B\cap C=\emptyset$.
$A\times(B\cup C)=\{(2,-1),(2,2),(2,a),(2,b),(3,-1),(3,2),(3,a),(3,b)\}$; $A\times B=\{(2,-1),(2,2),(3,-1),(3,2)\}$ and $A\times C=\{(2,a),(2,b),(3,a),(3,b)\}$, whose union is the same 8 pairs. ✓
$A\times(B\cap C)=A\times\emptyset=\emptyset$, and $(A\times B)\cap(A\times C)=\emptyset$ because the second entries come from the disjoint sets B and C. ✓

2E.6 $B\cup C=\{2,3,4,5\}$ and $B\cap C=\{3,4\}$.
$A\times(B\cup C)$ has $3\times 4=12$ pairs (a,b) with $a\in\{1,2,3\}$, $b\in\{2,3,4,5\}$. $A\times B$ supplies those with $b\in\{2,3,4\}$, $A\times C$ those with $b\in\{3,4,5\}$; together they give every $b\in\{2,3,4,5\}$. ✓
$A\times(B\cap C)=\{(1,3),(1,4),(2,3),(2,4),(3,3),(3,4)\}$, and the pairs common to $A\times B$ and $A\times C$ are exactly those with $b\in\{3,4\}$ — the same six. ✓

2E.7 $A\times B=B\times A$ **iff** $A=\emptyset$, or $B=\emptyset$, or $A=B$.
If one set is empty, both products are $\emptyset$; if $A=B$ the products coincide. Conversely, let $A,B\neq\emptyset$ and $A\times B=B\times A$. For $a\in A$, pick any $b\in B$; then $(a,b)\in B\times A$, so $a\in B$. Hence $A\subseteq B$, and similarly $B\subseteq A$, so $A=B$. ■

3 Binary Relations

A relation records which elements of one set are linked to which elements of another; formally it is just a set of ordered pairs.

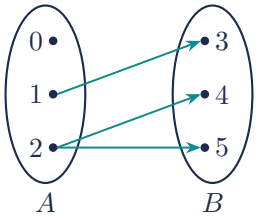
3.1 Definition and arrow diagrams

Definition: Binary relation

Let A and B be sets. Any subset R of $A \times B$ is called a **binary relation** (or just a **relation**) from A to B. Conversely, every set of ordered pairs (a, b) with $a \in A$, $b \in B$ is a relation from A to B. If $(a, b) \in R$ we say "a is related to b by R" and write $a\,R\,b$ (the statements $(a, b) \in R$ and aRb mean the same). A relation from A to A, i.e. a subset of $A \times A$, is a **binary relation on** A.

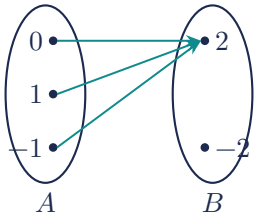
- **Domain** of $R = \{a \mid (a, b) \in R \text{ for some } b\}$; **range** $= \{b \mid (a, b) \in R \text{ for some } a\}$.
- **Inverse relation** $R^{-1} = \{(b, a) \mid (a, b) \in R\}$; **complement** $\overline{R} = (A \times B) - R$.

Example (arrow diagram). $A = \{0, 1, 2\}$, $B = \{3, 4, 5\}$, $R = \{(1, 3), (2, 4), (2, 5)\}$. Since $R \subseteq A \times B$, R is a relation from A to B: $1R3$, $2R4$, $2R5$. Draw A and B as ovals and an arrow $a \to b$ for each pair; this is the **arrow diagram**. Here 0 is related to nothing and 2 to two elements.

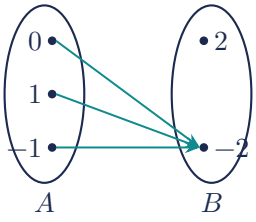


$R = \{(1, 3), (2, 4), (2, 5)\}$

Example ("less than" and "greater than"). $A = \{0, 1, -1\}$, $B = \{2, -2\}$, $R_1 = \{(0, 2), (1, 2), (-1, 2)\}$ and $R_2 = \{(0, -2), (1, -2), (-1, -2)\}$. R_1 consists of the pairs $(a, b) \in A \times B$ with $a < b$, so aR_1b reads "a is less than b"; R_2 consists of the pairs with $a > b$.



(a) R_1: $a < b$



(b) R_2: $a > b$

Example (a relation on a set). On $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\} \subseteq A \times A$ is a relation on A. Here $(x, y) \in R$ iff x divides y, so R stands for "divides".

3.2 Counting relations

Number of relations

If $|A| = m$ and $|B| = n$, there are 2^{mn} relations from A to B, and 2^{m^2} binary relations on A.

Reason. A relation from A to B is a subset of $A \times B$, so the relations are exactly the elements of $\mathcal{P}(A \times B)$. As $|A \times B| = mn$, there are 2^{mn} of them. For instance, if $|A| = 3$ and $|B| = 2$ there are $2^6 = 64$ relations from A to B.

3.3 Worked examples

Example 3.1 – Finding $|A|$ from the number of relations

Let A and B be finite sets with $|B| = 3$. If there are 4096 relations from A to B, what is $|A|$?

Solution. With $|A| = m$ and $|B| = n = 3$, there are $2^{mn} = 2^{3m}$ relations. So $2^{3m} = 4096 = 2^{12}$, i.e. $3m = 12$. (Equivalently $3m \log_e 2 = \log_e 4096$, $m = \frac{\log_e 4096}{3 \log_e 2} = 4$.) Thus $|A| = 4$.

Example 3.2 – Six counting questions

Let $A = \{1, 2, 3\}$ and $B = \{2, 4, 5\}$. Determine (1) $|A \times B|$ (2) the number of relations from A to B (3) the number of binary relations on A (4) the number of relations from A to B that contain $(1, 2)$ and $(1, 5)$ (5) the number of relations from A to B that contain exactly five ordered pairs (6) the number of binary relations on A that contain at least seven ordered pairs.

Solution. $|A| = m = 3$, $|B| = n = 3$.

(1) $|A \times B| = mn = 9$.

(2) Relations from A to B: $2^{mn} = 2^9 = 512$.

(3) Binary relations on A: $2^{m \cdot m} = 2^{m^2} = 2^9 = 512$.

(4) Let $R_1 = \{(1, 2), (1, 5)\}$. Every relation containing these two pairs has the form $R_1 \cup R_2$, where R_2 is any subset of $\overline{R_1} = (A \times B) - R_1$. Since $|\overline{R_1}| = 9 - 2 = 7$, there are $2^7 = 128$ such relations.

(5) Choose 5 of the 9 ordered pairs: $\binom{9}{5} = 126$.

(6) On A there are 9 ordered pairs; choose 7, 8 or 9 of them: $\binom{9}{7} + \binom{9}{8} + \binom{9}{9} = 36 + 9 + 1 = 46$.

Example 3.3 – A relation given by a rule

Let $A = \{1, 2, 3, 4, 6\}$ and $(a, b) \in R$ iff a is a multiple of b. Write R as a set of ordered pairs.

Solution. $R = \{(a, b) \mid a, b \in A,\ a \text{ is a multiple of } b\}$

$$= \{(1,1), (2,1), (2,2), (3,1), (3,3), (4,1), (4,2), (4,4), (6,1), (6,2), (6,3), (6,6)\}.$$

Example 3.4 – A recursively defined relation

A relation R on $\mathbb{Z}$ is defined recursively by (i) $(0, 0) \in R$, and (ii) if $(s, t) \in R$, then $(s + 1,\ t + 7) \in R$. Find R as a set of ordered pairs.

Solution. By (i), $(0, 0) \in R$. Applying (ii): $(0 + 1, 0 + 7) = (1, 7) \in R$; again: $(2, 14) \in R$; repeating, $(3, 21), (4, 28), \cdots \in R$. Thus

$$R = \{(0,0), (1,7), (2,14), (3,21), (4,28), \ldots\} = \{(m, n) \mid m \in \mathbb{N},\ n = 7m\}.$$

Note that the recursion only moves forward from $(0, 0)$, so m runs over $0, 1, 2, \ldots$; pairs such as $(-1, -7)$ are *not* produced.

3.4 Exercises on relations

3E.1 Consider the sets $A = \{0, 1, 2\}$ and $B = \{8, 9\}$. Indicate whether the following sets of ordered pairs are relations from A to B: $R_1 = \{(0, 8), (1, 8), (2, 9)\}$, $R_2 = \{(1, 8), (1, 9), (2, 2)\}$, $R_3 = \{(1, 9), (0, 8), (8, 0)\}$.

3E.2 Indicate whether the following sets of ordered pairs are relations on $A = \{4, 5, 7, 12\}$: $R_1 = \{(4, 4),$

$(5,12),(7,7)\}$, $R_2=\{(5,4),(4,12),(12,12)\}$, $R_3=\{(4,4),(5,5),(12,2)\}$.

3E.3 Let $A=\{1,2,3,4\}$ and R be the relation on A defined by $(a,b)\in R$ iff $a\le b$. Write down R as a set of ordered pairs.

3E.4 Let A be the set of all positive integers and R the relation on A defined by aRb iff $a=b^k$ for some positive integer k. Which of the following belong to R? (a) $(3,9)$ (b) $(9,3)$ (c) $(2,5)$ (d) $(5,2)$ (e) $(6,6)$ (f) $(32,2)$.

3E.5 A binary relation R on $\mathbb{N}$ is defined by $R=\{(m,n)\in\mathbb{N}\times\mathbb{N}\mid n=5m+2\}$. Give a recursive definition of R. Hence decide whether $(7,22)\in R$.

3.5 Solutions to Section 3 exercises

3E.1 $A\times B=\{(0,8),(0,9),(1,8),(1,9),(2,8),(2,9)\}$. $R_1\subseteq A\times B$, so R_1 **is a relation** from A to B. R_2 contains $(2,2)$ with $2\notin B$, and R_3 contains $(8,0)$ with $8\notin A$; so R_2 **and** R_3 **are not** relations from A to B.

3E.2 R_1 **and** R_2 **are relations on** A (all entries lie in A). R_3 **is not**, because $(12,2)$ has $2\notin A$.

3E.3 $R=\{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)\}$ (10 pairs).

3E.4 (a) $3=9^k$: impossible, so $(3,9)\notin R$. (b) $9=3^2$: $(9,3)\in R$. (c) $2=5^k$: no. (d) $5=2^k$: no. (e) $6=6^1$: $(6,6)\in R$. (f) $32=2^5$: $(32,2)\in R$. **Answer:** $(9,3)$, $(6,6)$, $(32,2)$ belong to R; the others do not.

3E.5 Recursive definition: (i) $(0,2)\in R$; (ii) if $(m,n)\in R$, then $(m+1,\ n+5)\in R$. (Increasing m by 1 increases $5m+2$ by 5.)
Starting from $(0,2)$: $(1,7)$, $(2,12)$, $(3,17)$, $(4,22)$, $(5,27)$, $(6,32)$, $(7,37),\dots$. Hence $(4,22)\in R$ but $(7,22)\notin R$, since $5\cdot7+2=37\neq22$.

3.6 Zero–one matrices and directed graphs

A relation between finite sets can be stored and pictured in two standard ways: as a table of 0s and 1s (useful for computation) and as a diagram of dots and arrows (useful for seeing properties at a glance).

Definition: Matrix of a relation

Let $A=\{a_1,a_2,\dots,a_m\}$ and $B=\{b_1,b_2,\dots,b_n\}$, and let R be a relation from A to B. Put

$$m_{ij}=\begin{cases}1 & \text{if } (a_i,b_j)\in R\\ 0 & \text{if } (a_i,b_j)\notin R,\end{cases}\qquad 1\le i\le m,\ 1\le j\le n.$$

The $m\times n$ matrix $M_R=[m_{ij}]$ (also written $M(R)$) is the **matrix of the relation** R, or the **relation matrix**. Since it contains only 0s and 1s it is also called the **zero–one matrix** of R. *Rows correspond to the elements of A and columns to the elements of B*, in a fixed order. When $B=A$, M_R is a square $n\times n$ matrix with $m_{ij}=1$ iff $(a_i,a_j)\in R$.

The zero–one matrix is the standard data structure for storing a relation in a computer: membership of a pair is just a table look-up.

Example 3.5 – Reading off a matrix

(i) $A=\{0,1,2\}$, $B=\{p,q\}$ and $R=\{(0,p),(1,q),(2,p)\}$. Here $m_{11}=1$ as $(0,p)\in R$; $m_{12}=0$ as $(0,q)\notin R$; $m_{21}=0$, $m_{22}=1$, $m_{31}=1$, $m_{32}=0$. So

$$M_R=\begin{bmatrix}1&0\\0&1\\1&0\end{bmatrix}.$$

(ii) $A=\{1,2,3,4\}$ and $R=\{(1,2),(1,3),(2,4),(3,2)\}$ on A. Here $m_{ij}=1$ exactly for $(i,j)=$

$(1,2),(1,3),(2,4),(3,2)$, so

$$M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Examination tip

For any relation R from a finite set A to a finite set B: (1) M_R is the zero matrix if and only if $R = \emptyset$; (2) every entry of M_R is 1 if and only if $R = A \times B$.

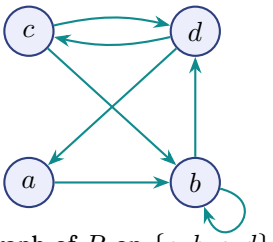
Definition: Digraph of a relation

Let R be a relation on a finite set A. Draw a small circle (a **vertex** or **node**) for each element of A, and draw an arrow (a directed **edge**) from vertex x to vertex y if and only if $(x, y) \in R$. The resulting picture is the **directed graph** or **digraph** of R.

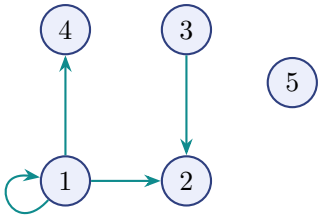
- The vertex an edge leaves is its **origin** (**source**); the vertex where it ends is its **terminus**.
- A vertex that is neither the source nor the terminus of any edge is an **isolated vertex**.
- An edge whose source and terminus coincide is a **loop**; a loop at a means $(a, a) \in R$.
- The **in-degree** of a vertex is the number of edges ending at it; the **out-degree** is the number of edges leaving it. (A loop counts once in each.)

Example 3.6 – Digraph, loops and degrees

(i) $A = \{a, b, c, d\}$ and $R = \{(a,b),(b,b),(b,d),(c,b),(c,d),(d,a),(d,c)\}$. Since $(b,b) \in R$ there is a loop at b. Edges $c \to d$ and $d \to c$ both appear, so they are drawn as a pair of arrows.
(ii) $A = \{1,2,3,4,5\}$ and $R = \{(1,1),(1,2),(1,4),(3,2)\}$. There is a loop at 1, and vertex 5 is isolated.



Digraph of R on $\{a, b, c, d\}$



5 is an isolated vertex

(i) Vertex	a	b	c	d	(ii) Vertex	1	2	3	4	5
In-degree	1	3	1	2	In-degree	1	2	0	1	0
Out-degree	1	2	2	2	Out-degree	3	0	1	0	0

As a check, in each digraph the in-degrees and the out-degrees both add up to $|R|$ (7 in (i), 4 in (ii)), because every edge has exactly one source and one terminus.

Example 3.7 – Matrix of a relation from A to B

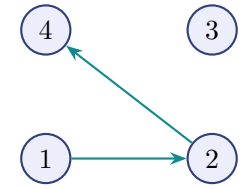
Let $A = \{1,2\}$, $B = \{p,q,r,s\}$ and $R = \{(1,q),(1,r),(2,p),(2,q),(2,s)\}$. Write down M_R.

Solution. Checking all eight pairs of $A \times B$: $(1,p) \notin R$, $(1,q) \in R$, $(1,r) \in R$, $(1,s) \notin R$; $(2,p) \in R$, $(2,q) \in R$, $(2,r) \notin R$, $(2,s) \in R$. Hence $M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$.

Example 3.8 – $y = 2x$

Let $A = \{1, 2, 3, 4\}$ and $x\,R\,y$ iff $y = 2x$. (a) Write R as a set of ordered pairs. (b) Draw its digraph. (c) Find the in-degrees and out-degrees.

Solution. (a) $(x, y) \in R$ iff $y = 2x$, so $R = \{(1, 2), (2, 4)\}$. (b) See the figure; 3 is an isolated vertex.



$y = 2x$: 3 is isolated

(c)

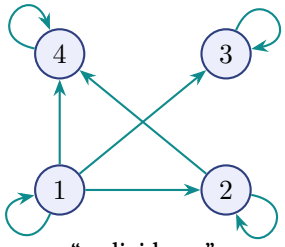
Vertex	1	2	4
In-degree	0	1	1
Out-degree	1	1	0

(vertex 3 has in- and out-degree 0).

Example 3.9 – "x divides y"

Let $A = \{1, 2, 3, 4\}$ and $x\,R\,y$ iff $x \mid y$. Write R, draw its digraph and find the degrees.

Solution. $1 \mid 1,\ 1 \mid 2,\ 1 \mid 3,\ 1 \mid 4,\ 2 \mid 2,\ 2 \mid 4,\ 3 \mid 3,\ 4 \mid 4$, so $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.



"x divides y"

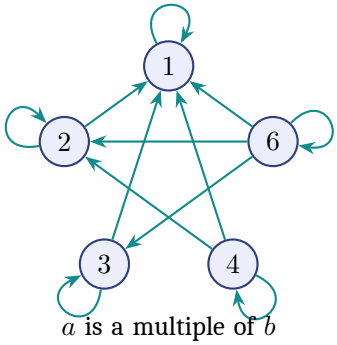
Vertex	1	2	3	4
In-degree	1	2	2	3
Out-degree	4	2	1	1

Example 3.10 – "a is a multiple of b"

Let $A = \{1, 2, 3, 4, 6\}$ and $a\,R\,b$ iff a is a multiple of b. Represent R as a matrix and draw its digraph.

Solution. $R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 3), (4, 1), (4, 2), (4, 4), (6, 1), (6, 2), (6, 3), (6, 6)\}$. Taking the elements in the order $1, 2, 3, 4, 6$:

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$



a is a multiple of b

(Row a has a 1 in column b exactly when b divides a; e.g. row 6 is $1\ 1\ 1\ 0\ 1$ because 1, 2, 3, 6 divide 6.)

Example 3.11 – From a matrix to a relation

Determine the relation R from A to B whose matrix is

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Solution. M_R is 4×3, so $|A| = 4$ and $|B| = 3$; write $A = \{a_1, a_2, a_3, a_4\}$, $B = \{b_1, b_2, b_3\}$. Reading the 1s row by row: $R = \{(a_1, b_1), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_3, b_3), (a_4, b_1)\}$.

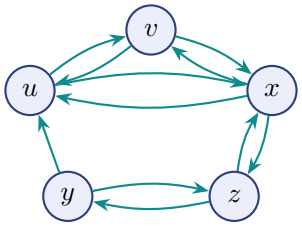
Example 3.12 – Matrix to digraph

Let $A = \{u, v, x, y, z\}$ and let R be the relation on A with matrix (rows and columns in the order u, v, x, y, z)

$$M(R) = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

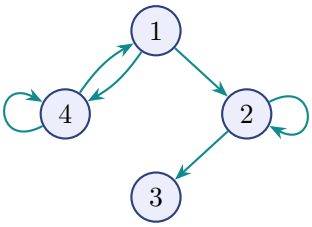
Write R and draw its digraph.

Solution. Reading the rows: $R = \{(u, v), (u, x), (v, u), (v, x), (x, u), (x, v), (x, z), (y, u), (y, z), (z, x), (z, y)\}$.



Example 3.13 – Digraph to relation and matrix

Find the relation represented by the digraph below and write its matrix.

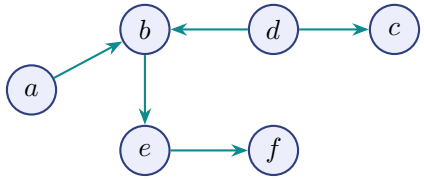


Solution. The digraph has four vertices, so R is a relation on $A = \{1, 2, 3, 4\}$: $R = \{(1, 2), (1, 4), (2, 2), (2, 3), (4, 1), (4, 4)\}$ and

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

Example 3.14 – A six-element digraph

For $A = \{a, b, c, d, e, f\}$ the digraph below represents a relation R on A. Determine R and its matrix.



Solution. $R = \{(a,b), (b,e), (d,b), (d,c), (e,f)\}$, and with the order a, b, c, d, e, f:

$$M(R) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

(Rows c and f are zero: no edge leaves c or f.)

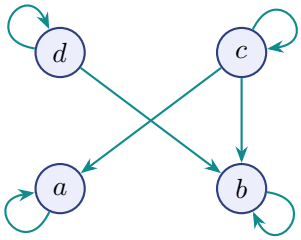
Example 3.15 – Degrees from a matrix

Let $A = \{a, b, c, d\}$ and let R have the matrix

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Construct the digraph and list the in- and out-degrees.

Solution. $R = \{(a,a), (b,b), (c,a), (c,b), (c,c), (d,b), (d,d)\}$.



Vertex	a	b	c	d
In-degree	2	3	1	1
Out-degree	1	1	3	2

Shortcut: the out-degree of a vertex is the number of 1s in its *row* of M_R; the in-degree is the number of 1s in its *column*.

3.7 Exercise Set 3F — matrices and digraphs

3F.1 Let $A = \{a, b, c\}$, $B = \{0, 1\}$ and $R = \{(a, 0), (b, 0), (c, 1)\}$ be a relation from A to B. Write down the matrix of this relation.

3F.2 Let $A = \{1, 2, 3, 4\}$ and $(a, b) \in R$ iff $a \leq b$. Write R as a set of ordered pairs and write down its matrix.

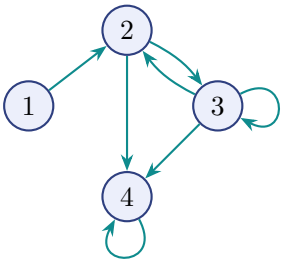
3F.3 Determine the relation R from a set A to a set B represented by $M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$.

3F.4 Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1)\}$. Draw the digraph of R.

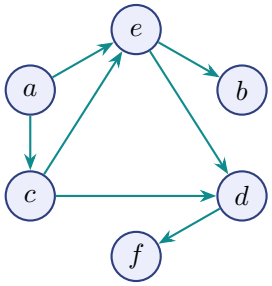
3F.5 Let $A = \{a, b, c, d, e, f\}$ and $R = \{(a, b), (a, d), (b, c), (b, e), (c, f), (d, e), (e, c), (e, f), (f, d)\}$. Draw the digraph of R.

3F.6 If $R = \{(x, y) \mid x > y\}$ is a relation on $A = \{1, 2, 3, 4\}$, write down the matrix and the digraph of R.

3F.7 Find the relation determined by each digraph below and write down its matrix.



(a)



(b)

3F.8 Let $A = \{2, 4, 5, 7\}$ and let R be the relation on A with matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Construct the digraph of R.

3F.9 Find the relation R on $A = \{a, b, c, d, e\}$ and draw its digraph, given

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

3.8 Solutions to Exercise Set 3F

3F.1 – Matrix of a relation from A to B

Rows a, b, c; columns $0, 1$. $(a, 0), (b, 0), (c, 1) \in R$ and the other three pairs are not.

Conclusion. $M_R = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

3F.2 – The relation $\leq$ on $\{1, 2, 3, 4\}$

Pairs. For each a list every $b \geq a$: $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$ (10 pairs).

Conclusion. $M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ — an upper-triangular matrix of 1s (row a has 1s from column a onwards).

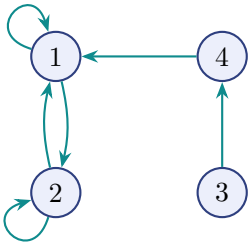
3F.3 – Relation from a 3×4 matrix

M_R is 3×4, so take $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3, b_4\}$. Row 1 has a 1 in column 1; row 2 in column 2; row 3 in columns 2 and 3.

Conclusion. $R = \{(a_1, b_1), (a_2, b_2), (a_3, b_2), (a_3, b_3)\}$.

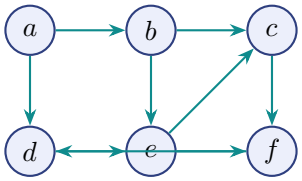
3F.4 – Digraph of a relation on $\{1, 2, 3, 4\}$

Loops at 1 and 2; a two-way pair between 1 and 2; arrows $3 \to 4$ and $4 \to 1$.



3F.5 – Digraph on six vertices

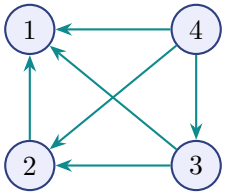
Draw one arrow for each of the nine pairs; there are no loops and no two-way pairs.



3F.6 – The relation $x > y$ on $\{1, 2, 3, 4\}$

Pairs. $R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$.

Matrix. $M_R = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ (1s strictly below the diagonal).



3F.7 – Relations from digraphs

(a) $R = \{(1,2), (2,3), (2,4), (3,2), (3,3), (3,4), (4,4)\}$ on $\{1, 2, 3, 4\}$, with

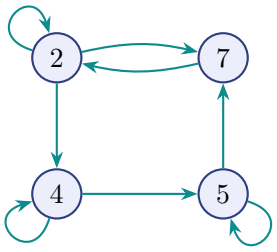
$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(b) $R = \{(a,c),(a,e),(c,d),(c,e),(d,f),(e,b),(e,d)\}$ on $\{a,b,c,d,e,f\}$, with (order $a,\dots,f$)

$$M_R = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

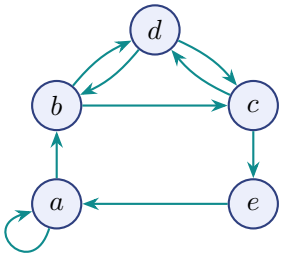
3F.8 – Digraph from a matrix on $\{2,4,5,7\}$

Rows and columns are in the order $2,4,5,7$, so $R = \{(2,2),(2,4),(2,7),(4,4),(4,5),(5,5),(5,7),(7,2)\}$: loops at 2, 4 and 5, and a two-way pair between 2 and 7.



3F.9 – Relation and digraph from a 5×5 matrix

Reading the rows: $R = \{(a,a),(a,b),(b,c),(b,d),(c,d),(c,e),(d,b),(d,c),(e,a)\}$.



3.9 Operations on relations

A relation from A to B is a subset of $A \times B$, so the set operations give new relations from old ones.

Definition: Union, intersection, complement, converse

Let R, R_1, R_2 be relations from A to B.

- **Union:** $(a,b) \in R_1 \cup R_2$ iff $(a,b) \in R_1$ or $(a,b) \in R_2$.
- **Intersection:** $(a,b) \in R_1 \cap R_2$ iff $(a,b) \in R_1$ and $(a,b) \in R_2$.
- **Complement:** $(a,b) \in \overline{R}$ iff $(a,b) \notin R$; that is, $\overline{R} = (A \times B) - R$, the complement of R in the universal set $A \times B$.
- **Converse:** the relation R^c from B to A with $(a,b) \in R^c$ iff $(b,a) \in R$. (Some authors call R^c the *inverse* of R and write R^{-1}.)

Matrix forms

$M(R_1 \cup R_2)$ is the entry-wise OR of $M(R_1)$ and $M(R_2)$; $M(R_1 \cap R_2)$ is the entry-wise AND; $M(\overline{R})$ is obtained by interchanging 0 and 1; and $M(R^c) = (M_R)^T$, the transpose. Also $(R^c)^c = R$.

Example 3.16 – All six operations

Let $A = \{a, b, c\}$, $B = \{1, 2, 3\}$, $R = \{(a, 1), (b, 1), (c, 2), (c, 3)\}$ and $S = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$. Find $\overline{R}$, $\overline{S}$, $R \cup S$, $R \cap S$, R^c and S^c.

Solution. $A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3), (c, 1), (c, 2), (c, 3)\}$. Hence

$$\overline{R} = (A \times B) - R = \{(a, 2), (a, 3), (b, 2), (b, 3), (c, 1)\},$$
$$\overline{S} = (A \times B) - S = \{(a, 3), (b, 3), (c, 1), (c, 2), (c, 3)\},$$
$$R \cup S = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 2), (c, 3)\}, \qquad R \cap S = \{(a, 1), (b, 1)\},$$
$$R^c = \{(1, a), (1, b), (2, c), (3, c)\}, \qquad S^c = \{(1, a), (1, b), (2, a), (2, b)\}.$$

Example 3.17 – Operations through matrices

Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and let R, S from A to B have matrices

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \qquad M_S = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Determine $\overline{R}$, $R \cup S$, $R \cap S$ and S^c, and their matrices.

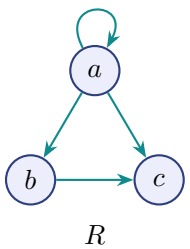
Solution. $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 2), (3, 3)\}$ and $S = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (3, 2), (3, 4)\}$. Therefore $\overline{R} = \{(1, 2), (1, 4), (2, 1), (2, 2), (2, 3), (3, 4)\}$; $R \cup S = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$; $R \cap S = \{(1, 1), (1, 3), (2, 4), (3, 2)\}$; $S^c = \{(1, 1), (2, 1), (2, 3), (3, 1), (4, 1), (4, 2), (4, 3)\}$.

$$M(\overline{R}) = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad M(R \cup S) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix},$$

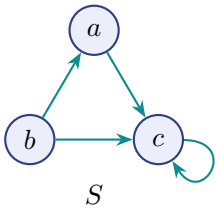
$$M(R \cap S) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad M(S^c) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} = (M_S)^T.$$

Example 3.18 – Operations on digraphs

The digraphs of two relations R and S on $A = \{a, b, c\}$ are given below. Draw the digraphs of $\overline{R}$, $R \cup S$, $R \cap S$ and R^c.



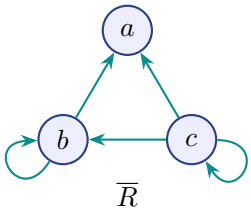
R



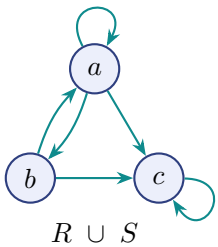
S

Solution. From the digraphs, $R = \{(a, a), (a, b), (a, c), (b, c)\}$ and $S = \{(a, c), (b, a), (b, c), (c, c)\}$.

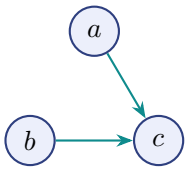
Hence $\overline{R} = \{(b,a),(b,b),(c,a),(c,b),(c,c)\}$, $R \cup S = \{(a,a),(a,b),(a,c),(b,a),(b,c),(c,c)\}$, $R \cap S = \{(a,c),(b,c)\}$ and $R^c = \{(a,a),(b,a),(c,a),(c,b)\}$ (every arrow of R reversed).



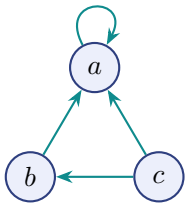
$\overline{R}$



$R \cup S$



$R \cap S$



R^c

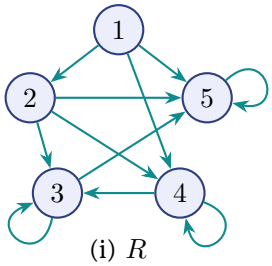
3.10 Exercise Set 3G — operations on relations

3G.1 Let $A = \{1,2,3\}$, $R = \{(1,1),(1,2),(2,3),(3,1)\}$ and $S = \{(2,1),(3,1),(3,2),(3,3)\}$. Compute $\overline{R}$, $R \cap S$, $R \cup S$ and S^c.

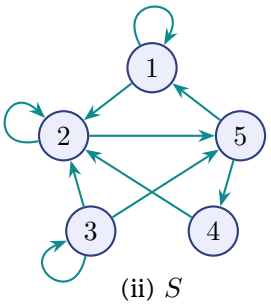
3G.2 Let $A = \{a,b,c,d\}$, $R = \{(a,a),(a,c),(b,c),(c,a),(d,b),(d,d)\}$ and $S = \{(a,b),(b,c),(c,a),(c,b),(d,c)\}$. Compute $M(R \cap S)$, $M(R \cup S)$, $M(R^c)$ and $M(\overline{S})$.

3G.3 Let $A = \{1,2,3\}$ and let R, S be relations on A with $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Find $\overline{R}$, R^c, $R \cap S$ and $R \cup S$.

3G.4 Let $A = \{1,2,3,4,5\}$ and let R and S be the relations on A whose digraphs are given below. Find $\overline{R}$, R^c and $R \cap S$.



(i) R



(ii) S

3G.5 Let R and S be relations from A to B. Prove: (1) if $R \subseteq S$ then $R^c \subseteq S^c$; (2) $(R \cap S)^c = R^c \cap S^c$; (3) $(R \cup S)^c = R^c \cup S^c$.

3G.6 Prove that a relation R from A to B is an invertible function from A to B if and only if R^c is an invertible function from B to A.

3.11 Solutions to Exercise Set 3G

3G.1 – Complement, intersection, union, converse

$A \times A$ has 9 pairs.
Complement. $\overline{R} = (A \times A) - R = \{(1,3),(2,1),(2,2),(3,2),(3,3)\}$.
Intersection and union. $R \cap S = \{(3,1)\}$; $R \cup S = \{(1,1),(1,2),(2,1),(2,3),(3,1),(3,2),(3,3)\}$.
Converse. Reverse each pair of S: $S^c = \{(1,2),(1,3),(2,3),(3,3)\}$.

3G.2 – Matrices of combined relations

With the order a, b, c, d:

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \qquad M_S = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

AND / OR.

$$M(R \cap S) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad M(R \cup S) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}.$$

Transpose / swap.

$$M(R^c) = (M_R)^T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad M(\overline{S}) = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

(the 0s and 1s of M_S interchanged).

3G.3 – Operations from matrices

$R = \{(1,1),(1,3),(2,2),(2,3)\}$, $S = \{(1,2),(1,3),(2,1),(2,2),(3,2)\}$.
Conclusion.

$$M(\overline{R}) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \qquad M(R^c) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix},$$

$$M(R \cap S) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad M(R \cup S) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

3G.4 – Operations from digraphs

Read the digraphs. $R = \{(1,2),(1,4),(1,5),(2,3),(2,4),(2,5),(3,3),(3,5),(4,3),(4,4),(5,5)\}$ (11 pairs) and $S = \{(1,1),(1,2),(2,2),(2,5),(3,2),(3,3),(3,5),(4,2),(5,1),(5,4)\}$.
Complement. $A \times A$ has 25 pairs, so $\overline{R}$ has $25 - 11 = 14$ pairs: $\overline{R} = \{(1,1),(1,3),(2,1),(2,2),(3,1),(3,2),(3,4),(4,1),(4,2),(4,5),(5,1),(5,2),(5,3),(5,4)\}$.
Converse. $R^c = \{(2,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,4),(5,1),(5,2),(5,3),(5,5)\}$.
Intersection. $R \cap S = \{(1,2),(2,5),(3,3),(3,5)\}$.

3G.5 – Converse of subsets, intersections and unions

(1) Let $(b,a) \in R^c$. Then $(a,b) \in R \subseteq S$, so $(b,a) \in S^c$. Hence $R^c \subseteq S^c$.
(2) $(b,a) \in (R \cap S)^c \iff (a,b) \in R \cap S \iff (a,b) \in R$ and $(a,b) \in S \iff (b,a) \in R^c$ and $(b,a) \in S^c \iff (b,a) \in R^c \cap S^c$.
(3) The same chain with "or" in place of "and" gives $(R \cup S)^c = R^c \cup S^c$. ■

3G.6 – R is invertible iff R^c is

Suppose R is an invertible function $A \to B$, i.e. a bijection. Then each $b \in B$ is the image of exactly one $a \in A$; so for each b there is exactly one a with $(b,a) \in R^c$, which means R^c is a function $B \to A$. It is one-to-one (two different b's cannot share an a, as R is a function) and onto (every a has an image under R). So R^c is invertible; in fact $R^c = R^{-1}$.
Conversely, if R^c is an invertible function $B \to A$, apply the first part to R^c: then $(R^c)^c = R$ is an invertible function $A \to B$. ■

3.12 Composition of relations

Definition: Composite (product) relation

Let R be a relation from A to B and S a relation from B to C. The **composition** (or **product**) of R and S is the relation from A to C

$$R \circ S = \{(a,c) \mid a \in A,\ c \in C,\ \text{and there exists } b \in B \text{ with } (a,b) \in R \text{ and } (b,c) \in S\}.$$

So a is related to c by $R \circ S$ when we can reach c from a in two stages — first to some b by R, then from b to c by S. Read $R \circ S$ as "R followed by S". Note $R \subseteq A \times B$, $S \subseteq B \times C$ and $R \circ S \subseteq A \times C$.

Common pitfall

R and S cannot be interchanged: in general $S \circ R \neq R \circ S$, and $S \circ R$ may not even be defined when A, B, C are different sets. Note: some books (e.g. Rosen) write this relation as $S \circ R$. In an examination, state which order you use.

Powers. If R is a relation on A, then $R \circ R$ is again a relation on A, denoted R^2; similarly $R^3 = R \circ R^2$, and in general $R^1 = R$ and $R^n = R \circ R^{n-1}$ for $n \geq 2$.

Matrix of a composite

Let R be a relation from $A = \{a_1, \ldots, a_m\}$ to $B = \{b_1, \ldots, b_n\}$ and S a relation from B to $C = \{c_1, \ldots, c_p\}$. Then

$$M(R) \cdot M(S) = M(R \circ S),$$

where the matrix product is computed with the Boolean rule $1 + 1 = 1$.

Proof. Let $M(R) = [r_{ij}]$, $M(S) = [s_{ij}]$, $M(R \circ S) = [p_{ij}]$ and $M(R) \cdot M(S) = [q_{ij}]$, so $q_{ij} = \sum_{k=1}^{n} r_{ik} s_{kj}$. Now $q_{ij} = 1$ iff $r_{ik} = 1$ and $s_{kj} = 1$ for some k, i.e. iff $(a_i, b_k) \in R$ and $(b_k, c_j) \in S$ for some b_k, i.e. iff $(a_i, c_j) \in R \circ S$, i.e. iff $p_{ij} = 1$. Hence $q_{ij} = p_{ij}$ for all i, j. ■

Corollary. If R is a relation on a finite set A, then $M(R^2) = [M(R)]^2$. **Generalisation.** $[M(R)]^n = M(R^n)$ for every positive integer n (by induction, with the corollary as the basis step).

Composition is associative

If R is a relation from A to B, S from B to C and T from C to D, then $R \circ (S \circ T) = (R \circ S) \circ T$.

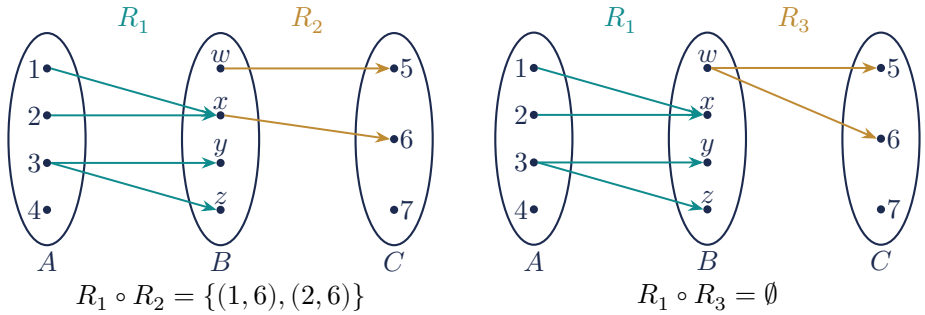
Proof. Both sides are relations from A to D. Let $(a,d) \in R \circ (S \circ T)$. Then there is $b \in B$ with $(a,b) \in R$

and $(b,d) \in S \circ T$; the latter gives $c \in C$ with $(b,c) \in S$ and $(c,d) \in T$. Now $(a,b) \in R$, $(b,c) \in S$ give $(a,c) \in R \circ S$, and with $(c,d) \in T$ we get $(a,d) \in (R \circ S) \circ T$. So $R \circ (S \circ T) \subseteq (R \circ S) \circ T$. The reverse inclusion is proved in the same way. ■

Example 3.19 – Composition with arrow diagrams

Let $A = \{1,2,3,4\}$, $B = \{w,x,y,z\}$, $C = \{5,6,7\}$, $R_1 = \{(1,x),(2,x),(3,y),(3,z)\}$ from A to B, and $R_2 = \{(w,5),(x,6)\}$, $R_3 = \{(w,5),(w,6)\}$ from B to C. Find $R_1 \circ R_2$ and $R_1 \circ R_3$.

Solution. $(1,x) \in R_1$ and $(x,6) \in R_2$, so $(1,6) \in R_1 \circ R_2$; $(2,x) \in R_1$ and $(x,6) \in R_2$, so $(2,6) \in R_1 \circ R_2$. For no other $(a,b) \in R_1$ is there a $(b,c) \in R_2$. Hence $R_1 \circ R_2 = \{(1,6),(2,6)\}$. Also, no $(a,b) \in R_1$ has $(b,c) \in R_3$ (the only element of B used by R_3 is w, and no arrow of R_1 reaches w), so $R_1 \circ R_3 = \emptyset$.



$R_1 \circ R_2 = \{(1,6),(2,6)\}$ $\qquad$ $R_1 \circ R_3 = \emptyset$

Example 3.20 – Checking $M(R_1 \circ R_2) = M(R_1) \cdot M(R_2)$

For R_1, R_2 of the previous example, keeping the orders of A, B, C as listed,

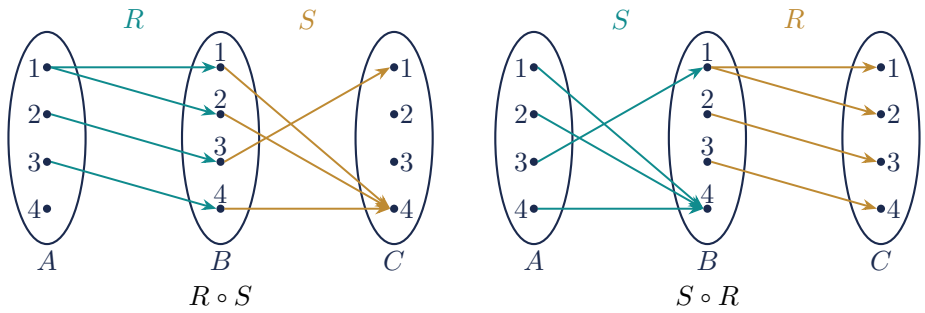
$$M(R_1) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad M(R_2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad M(R_1 \circ R_2) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Multiplying with $1 + 1 = 1$: row 1 of $M(R_1)$ is $(0,1,0,0)$, which picks out row 2 of $M(R_2)$, namely $(0,1,0)$; the same for row 2; row 3 is $(0,0,1,1)$, which adds rows 3 and 4 of $M(R_2)$, both zero; row 4 is zero. So $M(R_1) \cdot M(R_2) = M(R_1 \circ R_2)$.

Example 3.21 – $R \circ S$, $S \circ R$, R^2, S^2

Let $A = \{1,2,3,4\}$, $R = \{(1,1),(1,2),(2,3),(3,4)\}$ and $S = \{(3,1),(4,4),(2,4),(1,4)\}$. Determine $R \circ S$, $S \circ R$, R^2 and S^2.

Solution. Follow two-step paths. $R \circ S$: $1 \xrightarrow{R} 1 \xrightarrow{S} 4$, $1 \xrightarrow{R} 2 \xrightarrow{S} 4$, $2 \xrightarrow{R} 3 \xrightarrow{S} 1$, $3 \xrightarrow{R} 4 \xrightarrow{S} 4$; so $R \circ S = \{(1,4),(2,1),(3,4)\}$. $S \circ R$: only $3 \xrightarrow{S} 1 \xrightarrow{R} 1, 2$ works, so $S \circ R = \{(3,1),(3,2)\}$. Similarly $R^2 = R \circ R = \{(1,1),(1,2),(1,3),(2,4)\}$ and $S^2 = S \circ S = \{(1,4),(2,4),(3,4),(4,4)\}$. Note $R \circ S \neq S \circ R$.



Example 3.22 – Composites and their matrices

Let $A = \{1,2,3,4\}$, $R = \{(1,2),(1,3),(2,4),(4,4)\}$ and $S = \{(1,1),(1,2),(1,3),(1,4),(2,3),(2,4)\}$. Find $R \circ S$, $S \circ R$, R^2, S^2 and their matrices.

Solution. $R \circ S = \{(1,3),(1,4)\}$, $S \circ R = \{(1,2),(1,3),(1,4),(2,4)\}$, $R^2 = \{(1,4),(2,4),(4,4)\}$, $S^2 = \{(1,1),(1,2),(1,3),(1,4)\}$.

$$M(R \circ S) = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad M(S \circ R) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$M(R^2) = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad M(S^2) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Common pitfall

Note that $(4,4) \in R^2$: since $(4,4) \in R$, the path $4 \xrightarrow{R} 4 \xrightarrow{R} 4$ puts $(4,4)$ in R^2. A loop in R always survives into R^2.

Example 3.23 – Composites from matrices

Let $A = \{a,b,c\}$ and $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $M_S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$. Find $R \circ S$, $S \circ R$, $R \circ R$, $S \circ S$ and their matrices.

Solution. $R = \{(a,a),(a,c),(b,a),(b,b),(b,c),(c,b)\}$ and $S = \{(a,a),(b,b),(b,c),(c,a),(c,c)\}$. Hence $R \circ S = \{(a,a),(a,c),(b,a),(b,b),(b,c),(c,b),(c,c)\}$; $S \circ R = \{(a,a),(a,c),(b,a),(b,b),(b,c),(c,a),(c,b),(c,c)\}$; $R^2 = \{(a,a),(a,b),(a,c),(b,a),(b,b),(b,c),(c,a),(c,b),(c,c)\}$ (all of $A \times A$); $S^2 = \{(a,a),(b,a),(b,b),(b,c),(c,a),(c,c)\}$.

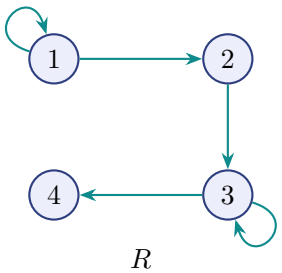
$$M(R \circ S) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad M(S \circ R) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix},$$

$$M(R \circ R) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad M(S \circ S) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

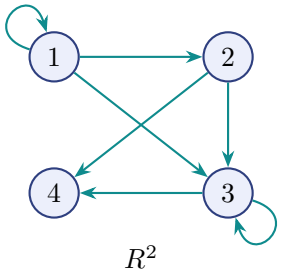
Example 3.24 – Digraph of R^2

Let $R = \{(1,1),(1,2),(2,3),(3,3),(3,4)\}$ on $A = \{1,2,3,4\}$. Draw the digraph of R, obtain R^2 and draw its digraph.

Solution. Every two-step path gives a pair of R^2: $R^2 = \{(1,1),(1,2),(1,3),(2,3),(2,4),(3,3),(3,4)\}$.



R



R^2

Example 3.25 – R^2 through $[M(R)]^2$

Let $R = \{(1,2),(1,3),(2,4),(3,2)\}$ on $A = \{1,2,3,4\}$. Write $M(R)$, compute $[M(R)]^2$ and hence obtain R^2.

Solution.

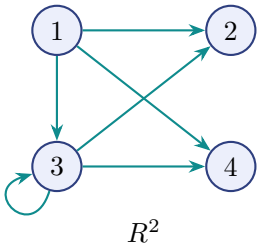
$$M(R) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad [M(R)]^2 = M(R) \cdot M(R) = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since $[M(R)]^2 = M(R^2)$, $R^2 = \{(1,2),(1,4),(3,4)\}$.

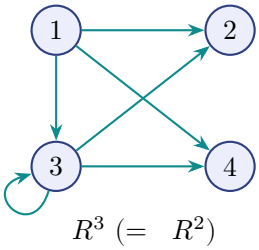
Example 3.26 – R^2 and R^3

If $A = \{1,2,3,4\}$ and $R = \{(1,2),(1,3),(2,4),(3,2),(3,3),(3,4)\}$, find R^2 and R^3 and draw their digraphs.

Solution. $R^2 = R \circ R = \{(1,2),(1,3),(1,4),(3,2),(3,3),(3,4)\}$ and $R^3 = R^2 \circ R = R^2$ (the same six pairs).



R^2



$R^3 \ (= \ R^2)$

Example 3.27 – Verifying $M(R^2) = [M(R)]^2$ and $M(R^3) = [M(R)]^3$

For the relation of Example 3.26, keeping the order $1, 2, 3, 4$:

$$M(R) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad M(R^2) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M(R^3).$$

Multiplying with $1+1=1$, $[M(R)]^2 = M(R)\cdot M(R) = \begin{bmatrix}0&1&1&1\\0&0&0&0\\0&1&1&1\\0&0&0&0\end{bmatrix} = M(R^2)$, and $[M(R)]^3 = M(R)\cdot[M(R)]^2 = \begin{bmatrix}0&1&1&1\\0&0&0&0\\0&1&1&1\\0&0&0&0\end{bmatrix} = M(R^3)$.

Example 3.28 – Cancellation fails

With $A=\{1,2,3,4\}$ and $R=\{(1,1),(1,2),(2,3),(3,3),(3,4),(4,4)\}$, find two relations $S\neq T$ on A with $R\circ S = R\circ T = \{(1,1),(1,2),(1,4)\}$.

Solution. Many pairs work; one is $S=\{(1,1),(1,4),(2,2)\}$ and $T=\{(1,1),(1,2),(2,4)\}$. Check: $R\circ S$: $1 \xrightarrow{R} 1 \xrightarrow{S} 1,4$ and $1 \xrightarrow{R} 2 \xrightarrow{S} 2$ give $\{(1,1),(1,4),(1,2)\}$, and nothing starting at $2,3,4$ continues in S. $R\circ T$: $1 \xrightarrow{R} 1 \xrightarrow{T} 1,2$ and $1 \xrightarrow{R} 2 \xrightarrow{T} 4$ give the same set. So $R\circ S = R\circ T$ does *not* imply $S=T$.

Example 3.29 – Triple composites

Let $R=\{(1,2),(3,4),(2,2)\}$ and $S=\{(4,2),(2,5),(3,1),(1,3)\}$ on $A=\{1,2,3,4,5\}$. Find $R\circ(R\circ S)$, $R\circ(S\circ R)$, $S\circ(R\circ S)$ and $S\circ(S\circ R)$.

Solution. First $R\circ S=\{(1,5),(3,2),(2,5)\}$ and $S\circ R=\{(4,2),(3,2),(1,4)\}$. Therefore $R\circ(R\circ S)=\{(1,5),(2,5)\}$, $R\circ(S\circ R)=\{(3,2)\}$, $S\circ(R\circ S)=\{(4,5),(3,5),(1,2)\}$, $S\circ(S\circ R)=\{(3,4),(1,2)\}$.

3.13 Exercise Set 3H — composition of relations

3H.1 Let $A=\{1,2,3,4\}$, $R=\{(1,1),(1,3),(3,2),(3,4),(4,2)\}$ and $S=\{(2,1),(3,3),(3,4),(4,1)\}$. Find $R\circ S$, $S\circ R$, $R\circ R$, $S\circ S$.

3H.2 For $A=\{1,2,3,4\}$ let $R=\{(1,1),(1,2),(2,3),(3,3),(3,4)\}$. Find R^2, R^3, R^4 and draw their digraphs.

3H.3 Let $A=\{1,2,3,4\}$ and

$$M_R=\begin{bmatrix}1&1&1&0\\0&0&0&1\\0&1&0&0\\0&0&0&0\end{bmatrix},\qquad M_S=\begin{bmatrix}0&0&1&1\\0&0&1&0\\1&0&0&0\\1&0&0&0\end{bmatrix}.$$

Find the matrices of $S\circ R$ and $R\circ S$.

3H.4 If $R=\{(a,a),(a,b),(b,c),(c,d),(c,e),(d,e)\}$ is a relation on $A=\{a,b,c,d,e\}$, find R^2 and verify that $M(R^2)=[M(R)]^2$.

3H.5 For the relations R and S of worked Examples 3.22 and 3.23, verify that $M(R\circ S)=M(R)\cdot M(S)$, $M(S\circ R)=M(S)\cdot M(R)$, $M(R^2)=[M(R)]^2$ and $M(S^2)=[M(S)]^2$.

3H.6 If R and S are relations on a set A, prove that $(R\cap S)^2\subseteq R^2\cap S^2$.

3H.7 Let R and S be relations from A to B and T a relation from B to C. Prove that if $R\subseteq S$ then $R\circ T\subseteq S\circ T$.

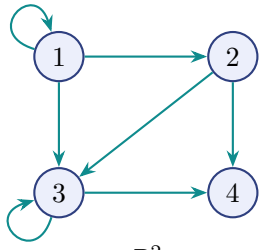
3.14 Solutions to Exercise Set 3H

3H.1 – Four composites

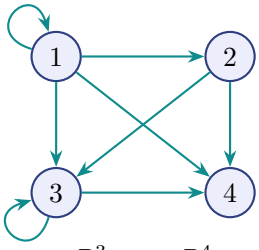
$R \circ S$. $1 \to 3 \to 3, 4$; $3 \to 2 \to 1$; $3 \to 4 \to 1$; $4 \to 2 \to 1$ (first arrow in R, second in S): $R \circ S = \{(1,3), (1,4), (3,1), (4,1)\}$.
$S \circ R$. $2 \to 1 \to 1, 3$; $3 \to 3 \to 2, 4$; $3 \to 4 \to 2$; $4 \to 1 \to 1, 3$: $S \circ R = \{(2,1), (2,3), (3,2), (3,4), (4,1), (4,3)\}$.
$R \circ R$. $1 \to 1 \to 1, 3$; $1 \to 3 \to 2, 4$; $3 \to 4 \to 2$: $R \circ R = \{(1,1), (1,2), (1,3), (1,4), (3,2)\}$.
$S \circ S$. $3 \to 3 \to 3, 4$; $3 \to 4 \to 1$: $S \circ S = \{(3,1), (3,3), (3,4)\}$.

3H.2 – Powers R^2, R^3, R^4

R^2. $1 \to 1 \to 1, 2$; $1 \to 2 \to 3$; $2 \to 3 \to 3, 4$; $3 \to 3 \to 3, 4$: $R^2 = \{(1,1), (1,2), (1,3), (2,3), (2,4), (3,3), (3,4)\}$.
$R^3 = R \circ R^2$. $1 \to 1$ then R^2 from 1 gives $1, 2, 3$; $1 \to 2$ then R^2 from 2 gives $3, 4$; $2 \to 3$ and $3 \to 3$ give $3, 4$: $R^3 = \{(1,1), (1,2), (1,3), (1,4), (2,3), (2,4), (3,3), (3,4)\}$.
$R^4 = R \circ R^3$. The same computation gives $R^4 = R^3$ (and then $R^n = R^3$ for all $n \geq 3$).



R^2



$R^3 = R^4$

Remark. Do not forget the loops: since $(1,1), (3,3) \in R$, the paths $1 \to 1 \to \cdots \to 1$ and $3 \to 3 \to \cdots \to 3$ keep these loops in every power.

3H.3 – Composites through matrices

$R = \{(1,1), (1,2), (1,3), (2,4), (3,2)\}$, $S = \{(1,3), (1,4), (2,3), (3,1), (4,1)\}$.
$R \circ S$.

$$M(R \circ S) = M_R \cdot M_S = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

i.e. $R \circ S = \{(1,1), (1,3), (1,4), (2,1), (3,3)\}$.
$S \circ R$.

$$M(S \circ R) = M_S \cdot M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix},$$

i.e. $S \circ R = \{(1,2), (2,2), (3,1), (3,2), (3,3), (4,1), (4,2), (4,3)\}$.

3H.4 – R^2 on five elements

Two-step paths: $a \to a \to a, b$; $a \to b \to c$; $b \to c \to d, e$; $c \to d \to e$. So $R^2 = \{(a,a), (a,b), (a,c),$

$(b,d),(b,e),(c,e)\}$. With the order $a,\dots,e$:

$$M(R)=\begin{bmatrix}1&1&0&0&0\\0&0&1&0&0\\0&0&0&1&1\\0&0&0&0&1\\0&0&0&0&0\end{bmatrix},\qquad [M(R)]^2=\begin{bmatrix}1&1&1&0&0\\0&0&0&1&1\\0&0&0&0&1\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}=M(R^2).$$

3H.5 – Verifying the matrix rule

Example 3.22. With $M_R=\begin{bmatrix}0&1&1&0\\0&0&0&1\\0&0&0&0\\0&0&0&1\end{bmatrix}$ and $M_S=\begin{bmatrix}1&1&1&1\\0&0&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix}$, Boolean multiplication gives

$$M_R\cdot M_S=\begin{bmatrix}0&0&1&1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}=M(R\circ S),\qquad M_S\cdot M_R=\begin{bmatrix}0&1&1&1\\0&0&0&1\\0&0&0&0\\0&0&0&0\end{bmatrix}=M(S\circ R),$$

$$[M_R]^2=\begin{bmatrix}0&0&0&1\\0&0&0&1\\0&0&0&0\\0&0&0&1\end{bmatrix}=M(R^2),\qquad [M_S]^2=\begin{bmatrix}1&1&1&1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}=M(S^2).$$

Example 3.23. Similarly

$$M_R\cdot M_S=\begin{bmatrix}1&0&1\\1&1&1\\0&1&1\end{bmatrix},\qquad M_S\cdot M_R=\begin{bmatrix}1&0&1\\1&1&1\\1&1&1\end{bmatrix},$$

$$[M_R]^2=\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix},\qquad [M_S]^2=\begin{bmatrix}1&0&0\\1&1&1\\1&0&1\end{bmatrix},$$

exactly the matrices found there. Each product is computed with the rule $1+1=1$.

3H.6 – $(R\cap S)^2\subseteq R^2\cap S^2$

Let $(a,c)\in(R\cap S)^2$. Then there is b with $(a,b)\in R\cap S$ and $(b,c)\in R\cap S$. Since $(a,b),(b,c)\in R$, $(a,c)\in R^2$; since $(a,b),(b,c)\in S$, $(a,c)\in S^2$. So $(a,c)\in R^2\cap S^2$. ■

Remark. Equality need not hold: on $\{1,2,3\}$ take $R=\{(1,2),(2,3)\}$ and $S=\{(1,3),(3,3)\}$. Then $R^2\cap S^2=\{(1,3)\}$ but $R\cap S=\emptyset$, so $(R\cap S)^2=\emptyset$.

3H.7 – Composition preserves inclusion

Let $(a,c)\in R\circ T$. Then $(a,b)\in R$ and $(b,c)\in T$ for some $b\in B$. As $R\subseteq S$, $(a,b)\in S$, so $(a,c)\in S\circ T$. Hence $R\circ T\subseteq S\circ T$. ■

3.15 Properties of relations on a set

Throughout, R is a relation on a set A (a subset of $A\times A$).

Reflexive relation

Definition: Reflexive

R is **reflexive** if $(a, a) \in R$ for all $a \in A$, i.e. every element is related to itself. R is *not* reflexive if there is some $a \in A$ with $(a, a) \notin R$.

"$\leq$" on $\mathbb{R}$ is reflexive ($a \leq a$ for every real a); "$<$" and "$>$" are not. On $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (2, 2), (3, 3)\}$ is not reflexive, because $4 \in A$ but $(4, 4) \notin R$.

(1) The matrix of a reflexive relation has 1s on its main diagonal.

(2) In the digraph of a reflexive relation there is a cycle of length 1 (a loop) at every vertex.

(3) On any set A the relation $\Delta_A = \{(a, a) \mid a \in A\}$ — the **equality relation** or **diagonal** of $A \times A$ — is reflexive, and it is a subset of every reflexive relation on A. Its matrix is the identity matrix.

Irreflexive relation

Definition: Irreflexive

R is **irreflexive** if $(a, a) \notin R$ for every $a \in A$: no element is related to itself.

"$<$" and "$>$" on $\mathbb{R}$ are irreflexive. **Irreflexive is not the same as non-reflexive:** $R = \{(1, 1), (1, 2)\}$ on $A = \{1, 2, 3\}$ is not reflexive (as $(2, 2), (3, 3) \notin R$) and not irreflexive (as $(1, 1) \in R$). The matrix of an irreflexive relation has 0s on its main diagonal, and its digraph has no loops.

Symmetric relation

Definition: Symmetric

R is **symmetric** if $(b, a) \in R$ whenever $(a, b) \in R$. It is not symmetric if there are a, b with $(a, b) \in R$ but $(b, a) \notin R$.

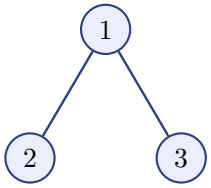
On $A = \{1, 2, 3\}$, $R_1 = \{(1, 1), (1, 2), (2, 1)\}$ is symmetric, while $R_2 = \{(1, 2), (2, 1), (1, 3)\}$ is not, because $(1, 3) \in R_2$ but $(3, 1) \notin R_2$.

Common pitfall

Do not confuse **asymmetric** with *not symmetric*. R is asymmetric if $(a, b) \in R \Rightarrow (b, a) \notin R$ for all a, b (so it has no loops and no two-way pairs). R is merely *not symmetric* if at least one pair $(a, b) \in R$ has $(b, a) \notin R$.

The matrix $M_R = [m_{ij}]$ of a symmetric relation satisfies $m_{ij} = 1 \Rightarrow m_{ji} = 1$ and $m_{ij} = 0 \Rightarrow m_{ji} = 0$; that is, M_R is a *symmetric matrix* ($M_R = M_R^T$). In the digraph of a symmetric relation every edge between two vertices comes with the reverse edge, so the arrows are dropped: the picture is called the **graph** of the relation, an edge joining a and b is written $\{a, b\}$ (a *bidirected* edge), and a, b are called **adjacent vertices**.

For example, $R = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$ on $A = \{1, 2, 3\}$ is symmetric. In its graph, 1 and 2 are adjacent and 1 and 3 are adjacent, but 2 and 3 are not.



Graph of R

Antisymmetric relation

Definition: Antisymmetric

R is **antisymmetric** if $(a,b) \in R$ and $(b,a) \in R$ together imply $a = b$. It is not antisymmetric if there are $a \neq b$ with both $(a,b) \in R$ and $(b,a) \in R$.

"$\leq$" on $\mathbb{R}$ is antisymmetric ($a \leq b$ and $b \leq a$ give $a = b$). "Not symmetric" and "antisymmetric" are different ideas; a relation can be both symmetric and antisymmetric, or neither. On $A = \{1,2,3\}$: $R_1 = \{(1,1), (2,2)\}$ is both symmetric and antisymmetric, and $R_2 = \{(1,2),(2,1),(2,3)\}$ is neither.

(1) If $M_R = [m_{ij}]$ is the matrix of an antisymmetric relation, then for $i \neq j$, either $m_{ij} = 0$ or $m_{ji} = 0$.

(2) In the digraph of an antisymmetric relation there is no bidirectional edge between two different vertices.

Transitive relation

Definition: Transitive

R is **transitive** if $(a,b) \in R$ and $(b,c) \in R$ imply $(a,c) \in R$. It is not transitive if there are a, b, c with $(a,b), (b,c) \in R$ but $(a,c) \notin R$.

"$\leq$" and "$\geq$" on $\mathbb{R}$ are transitive. On $A = \{1,2,3\}$, $R_1 = \{(1,1),(1,2),(2,3),(1,3)\}$ is transitive, but $R_2 = \{(1,2),(2,3),(1,3),(3,1)\}$ is not: $(1,3),(3,1) \in R_2$ while $(1,1) \notin R_2$.

Common pitfall

For example, $\{(1,1),(1,2),(2,3),(1,3),(3,1),(3,2)\}$ looks transitive but is *not*: $(2,3)$ and $(3,1)$ are in it but $(2,1)$ is not. When checking transitivity, test every pair $(a,b),(b,c)$ — including those that pass through a loop or return to the start.

Matrix test. R is transitive if and only if its matrix has the property: if $m_{ik} = 1$ and $m_{kj} = 1$, then $m_{ij} = 1$.

Property	Definition (for all $a, b, c \in A$)	Matrix / digraph test
Reflexive	$(a,a) \in R$	all diagonal entries 1; loop at every vertex
Irreflexive	$(a,a) \notin R$	all diagonal entries 0; no loops
Symmetric	$aRb \Rightarrow bRa$	$M_R = M_R^T$; every arrow is two-way
Antisymmetric	$aRb \wedge bRa \Rightarrow a = b$	$m_{ij} = 1,\ i \neq j \Rightarrow m_{ji} = 0$; no two-way arrows
Asymmetric	$aRb \Rightarrow \neg(bRa)$	antisymmetric *and* irreflexive
Transitive	$aRb \wedge bRc \Rightarrow aRc$	$M_R^2 \leq M_R$; every 2-step path has a shortcut

Example 3.30 – Seven relations on $\{1,2,3\}$

Let $A = \{1,2,3\}$. Determine the nature of $R_1 = \{(1,2),(2,1),(1,3),(3,1)\}$, $R_2 = \{(1,1),(2,2),(3,3),(2,3)\}$, $R_3 = \{(1,1),(2,2),(3,3)\}$, $R_4 = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}$, $R_5 = \{(1,1),(2,3),(3,3)\}$, $R_6 = \{(2,3),(3,4),(2,4)\}$, $R_7 = \{(1,3),(3,2)\}$.

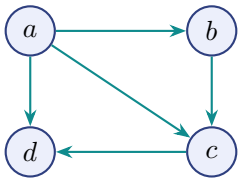
Solution. Checking each property directly (R_6 contains the element 4, so take $A = \{1,2,3,4\}$ for it):

	Ref	Irref	Sym	Antisym	Trans	Reason for a failure
R_1	×	✓	✓	×	×	$(1,2),(2,1)\in R_1$, $(1,1)\notin R_1$
R_2	✓	×	×	✓	✓	$(2,3)\in R_2$, $(3,2)\notin R_2$
R_3	✓	×	✓	✓	✓	— (it is Δ_A)
R_4	✓	×	✓	×	✓	$(2,3),(3,2)\in R_4$, $2\neq 3$
R_5	×	×	×	✓	✓	$(2,2)\notin R_5$; $(1,1)\in R_5$
R_6	×	✓	×	✓	✓	no loops; $(2,3)\in R_6$, $(3,2)\notin R_6$
R_7	×	✓	×	✓	×	$(1,3),(3,2)\in R_7$, $(1,2)\notin R_7$

In words: R_1 is symmetric and irreflexive, but neither reflexive nor transitive; R_2 is reflexive and transitive but not symmetric; R_3 and R_4 are reflexive and symmetric; R_5 is neither reflexive nor symmetric; R_6 is transitive and irreflexive but not symmetric; R_7 is irreflexive but neither transitive nor symmetric.

Example 3.31 – Nature of three relations on $\{1,2,3,4\}$

Let $A=\{1,2,3,4\}$, $R_1=\{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4),(4,3),(4,4)\}$, $R_2=\{(1,2),(1,3),(3,1),(1,1),(3,3),(3,2),(1,4),(4,2),(3,4)\}$, and let R_3 be the relation on $\{a,b,c,d\}$ given by the digraph below. Determine their nature.



R_3

Solution. R_1 is reflexive, symmetric and transitive (it relates elements within the blocks $\{1,2\}$ and $\{3,4\}$). R_2 is transitive: every two-step path has its shortcut, e.g. $(3,1),(1,2)$ with $(3,2)$; $(1,3),(3,4)$ with $(1,4)$; $(1,4),(4,2)$ with $(1,2)$. (R_2 is not reflexive, as $(2,2)\notin R_2$; not symmetric; and not antisymmetric, as $(1,3),(3,1)\in R_2$.) From the digraph, $R_3=\{(a,b),(a,c),(a,d),(b,c),(c,d)\}$: it has no loops and no two-way arrows, so R_3 is irreflexive and antisymmetric. (It is not transitive: $b\to c\to d$ but there is no $b\to d$.)

Example 3.32 – Nature from matrices

Find the nature of the relations represented by the matrices

$$\text{(a)}\ \begin{bmatrix}1&1&1&0\\1&0&0&0\\1&0&1&1\\0&0&1&1\end{bmatrix}\qquad \text{(b)}\ \begin{bmatrix}1&0&1&0\\0&1&0&1\\1&0&1&0\\0&1&0&1\end{bmatrix}\qquad \text{(c)}\ \begin{bmatrix}0&0&1&1\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$$

Solution. (a) The matrix is symmetric ($a_{ij}=a_{ji}$), so the relation is symmetric. (b) The matrix has 1s on the main diagonal and is symmetric, so the relation is reflexive and symmetric. (c) The matrix is not symmetric ($a_{13}=1$, $a_{31}=0$), so the relation is not symmetric; the 1s in positions $(1,4)$ and $(4,1)$ show that it is not antisymmetric either.

Example 3.33 – Transitivity from a matrix

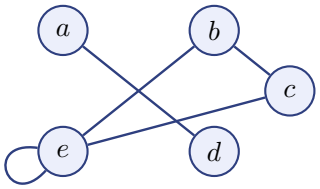
Show that the relation R on $A = \{a, b, c\}$ with $M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ is transitive.

Solution. $R = \{(a,a), (a,b), (a,c), (b,c), (c,c)\}$. The two-step paths are $a \to a \to a, b, c$; $a \to b \to c$; $a \to c \to c$; $b \to c \to c$; $c \to c \to c$. Each end-pair $(a,a), (a,b), (a,c), (b,c), (c,c)$ is already in R, so R is transitive.

Example 3.34 – Graph of a symmetric relation

Let $A = \{a, b, c, d, e\}$ and $R = \{(a,d), (d,a), (c,b), (b,c), (c,e), (e,c), (b,e), (e,b), (e,e)\}$, a symmetric relation on A. Draw the graph of R.

Solution. Join each two-way pair by a single undirected edge and draw the loop at e:



Graph of R

Example 3.35 – The divisibility relation

On $\mathbb{Z}^+$ let $a\,R\,b$ iff a divides b (written $a \mid b$; R is called the **divisibility relation**). Prove that R is reflexive, transitive and antisymmetric, but not symmetric.

Solution. *Reflexive:* $a \mid a$ for all $a \in \mathbb{Z}^+$. *Not symmetric:* $3 \mid 6$ but $6 \nmid 3$. *Antisymmetric:* $a \mid b$ and $b \mid a$ give $b = ka$, $a = lb$ with $k, l \in \mathbb{Z}^+$, so $kl = 1$, $k = l = 1$ and $a = b$. *Transitive:* $a \mid b$ and $b \mid c$ give $b = ka$, $c = lb = (kl)a$, so $a \mid c$.

Example 3.36 – The subset relation

Let S be a universal set and define R on $\mathcal{P}(S)$ by $(A, B) \in R$ iff $A \subseteq B$ (the **subset relation**). Prove that R is reflexive, antisymmetric and transitive, but not symmetric.

Solution. $A \subseteq A$ for every $A \in \mathcal{P}(S)$ (reflexive). $A \subseteq B$ and $B \subseteq A$ imply $A = B$ (antisymmetric). $A \subseteq B$ and $B \subseteq C$ imply $A \subseteq C$ (transitive). But $A \subseteq B$ need not give $B \subseteq A$; e.g. (for $S \neq \emptyset$) $\emptyset \subseteq S$ while $S \not\subseteq \emptyset$. So R is not symmetric.

Example 3.37 – Properties passed to $\overline{R}$ and R^c

Let R be a relation on A. Prove: (1) R is reflexive iff $\overline{R}$ is irreflexive; (2) if R is reflexive, so is R^c; (3) if R is symmetric, so are R^c and $\overline{R}$; (4) if R is transitive, so is R^c.

Solution. (1) R reflexive $\iff$ $(a,a) \in R$ for all a $\iff$ $(a,a) \notin \overline{R}$ for all a $\iff$ $\overline{R}$ irreflexive.
(2) $(a,a) \in R$ for all a gives $(a,a) \in R^c$ for all a.
(3) If $(a,b) \in R^c$ then $(b,a) \in R$, so $(a,b) \in R$ by symmetry, i.e. $(b,a) \in R^c$. If $(a,b) \in \overline{R}$ then $(a,b) \notin R$, so $(b,a) \notin R$ (otherwise symmetry would put (a,b) in R), i.e. $(b,a) \in \overline{R}$.
(4) If $(a,b), (b,c) \in R^c$ then $(b,a), (c,b) \in R$, so $(c,a) \in R$ by transitivity, i.e. $(a,c) \in R^c$. ■

Example 3.38 – Intersections and unions

Let R, S be relations on A. Prove: (1) if R, S are reflexive, so are $R \cap S$ and $R \cup S$; (2) if R, S are symmetric, so are $R \cap S$ and $R \cup S$; (3) if R, S are antisymmetric, so is $R \cap S$; (4) if R, S are transitive, so is $R \cap S$.

Solution. (1) (a, a) lies in both R and S, hence in $R \cap S$ and in $R \cup S$.
(2) If $(a, b) \in R \cap S$, then $(b, a) \in R$ and $(b, a) \in S$, so $(b, a) \in R \cap S$. If $(x, y) \in R \cup S$, then $(y, x) \in R$ or $(y, x) \in S$, so $(y, x) \in R \cup S$.
(3) If $(a, b), (b, a) \in R \cap S$, they are both in R, so $a = b$ by antisymmetry of R.
(4) If $(a, b), (b, c) \in R \cap S$, transitivity of R and of S gives $(a, c) \in R$ and $(a, c) \in S$, so $(a, c) \in R \cap S$. ■

Remark. $R \cup S$ need not be antisymmetric or transitive. On $A = \{1, 2, 3\}$ take $R = \{(1, 1), (1, 2), (2, 3), (1, 3)\}$ and $S = \{(2, 2), (2, 3), (3, 1), (2, 1)\}$. Both are antisymmetric and transitive, but $R \cup S = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1)\}$ contains $(1, 2), (2, 1)$ (not antisymmetric) and $(3, 1), (1, 2)$ without $(3, 2)$ (not transitive).

Example 3.39 – Two of three properties exclude the third

Let R be a non-empty relation on A. Prove that if R has any two of the properties *irreflexive*, *symmetric*, *transitive*, it cannot have the third.

Solution. Take any $(a, b) \in R$ (possible since $R \neq \emptyset$).
Irreflexive and symmetric ⇒ not transitive: by symmetry $(b, a) \in R$; if R were transitive, $(a, b), (b, a)$ would give $(a, a) \in R$, contradicting irreflexivity.
Irreflexive and transitive ⇒ not symmetric: if R were symmetric, $(b, a) \in R$ and transitivity would give $(a, a) \in R$ — impossible.
Symmetric and transitive ⇒ not irreflexive: $(b, a) \in R$ by symmetry, then $(a, a) \in R$ by transitivity. ■

Example 3.40 – Counting relations with given properties

If a finite set A has n elements, prove that there are (1) 2^{n^2-n} reflexive relations, (2) 2^{n^2-n} irreflexive relations, (3) $2^{(n^2+n)/2}$ symmetric relations, (4) $2^{(n^2-n)/2}$ relations that are both reflexive and symmetric, and (5) $2^n \cdot 3^{(n^2-n)/2}$ antisymmetric relations on A.

Solution. Let $A = \{a_1, \dots, a_n\}$, $\Delta = \{(a_i, a_i)\}$ ($|\Delta| = n$) and $\overline{\Delta} = (A \times A) - \Delta = \{(a_i, a_j) \mid i \neq j\}$, so $|\overline{\Delta}| = n^2 - n$.
(1) Δ is contained in every reflexive relation, so each reflexive relation is $\Delta \cup \Delta_1$ with $\Delta_1 \subseteq \overline{\Delta}$ arbitrary: 2^{n^2-n} choices.
(2) An irreflexive relation contains no element of Δ, so it is just a subset of $\overline{\Delta}$: 2^{n^2-n}.
(3) A symmetric relation is $\Delta_2 \cup \Delta_3$, where $\Delta_2 \subseteq \Delta$ (2^n choices) and $\Delta_3 \subseteq \overline{\Delta}$ is a union of complete pairs $\{(a_i, a_j), (a_j, a_i)\}$; there are $n(n-1)/2$ such pairs, so $2^{n(n-1)/2}$ choices. Total $2^n \cdot 2^{n(n-1)/2} = 2^{n(n+1)/2}$.
(4) Reflexive and symmetric: Δ is forced, Δ_3 is free: $2^{(n^2-n)/2}$.
(5) Antisymmetric: $\Delta_2 \subseteq \Delta$ is free (2^n), and for each of the $(n^2 - n)/2$ pairs $\{(a_i, a_j), (a_j, a_i)\}$ there are three choices — take (a_i, a_j) only, (a_j, a_i) only, or neither. Total $2^n \cdot 3^{(n^2-n)/2}$. ■

Type of relation	Number	$n = 3$
All relations	2^{n^2}	512
Reflexive / Irreflexive	2^{n^2-n}	64
Symmetric	$2^{n(n+1)/2}$	64
Reflexive and symmetric	$2^{n(n-1)/2}$	8
Antisymmetric	$2^n \cdot 3^{n(n-1)/2}$	216
Asymmetric	$3^{n(n-1)/2}$	27

(The $n = 3$ column was confirmed by checking all 512 relations on a 3-element set.)

Example 3.41 – Matrix criteria

For a relation R on $A = \{a_1, \dots, a_n\}$ with matrix M, prove: (1) R is reflexive iff $I_n \le M$; (2) R is transitive iff $M^2 \le M$; (3) R is antisymmetric iff $M \cap M^T \le I_n$. (Here I_n is the identity matrix; for 0–1 matrices, $A \le B$ means $a_{ij} \le b_{ij}$ for all i, j, and $A \cap B$ is the 0–1 matrix with $c_{ij} = 1$ iff $a_{ij} = b_{ij} = 1$.)

Solution. (1) $I_n = [\delta_{ij}]$ has 1s only on the diagonal. R reflexive $\iff m_{ii} = 1$ for all $i \iff \delta_{ij} \le m_{ij}$ for all $i, j \iff I_n \le M$.

(2) Let $M^2 = [p_{ij}]$, $p_{ij} = \sum_k m_{ik} m_{kj}$ (with $1 + 1 = 1$). Suppose $M^2 \le M$ and $(a_r, a_s), (a_s, a_t) \in R$. Then $m_{rs} m_{st} = 1$, so $p_{rt} = 1 \le m_{rt}$, i.e. $(a_r, a_t) \in R$: R is transitive. Conversely, if R is transitive and $p_{ij} = 1$, then $m_{ik} = m_{kj} = 1$ for some k, i.e. $(a_i, a_k), (a_k, a_j) \in R$, so $(a_i, a_j) \in R$ and $m_{ij} = 1$. If $p_{ij} = 0$, then $p_{ij} \le m_{ij}$ trivially. So $M^2 \le M$.

(3) The (i, j) entry of $M \cap M^T$ is 1 iff $m_{ij} = m_{ji} = 1$, i.e. iff both (a_i, a_j) and (a_j, a_i) are in R. R is antisymmetric iff this never happens for $i \ne j$, i.e. iff all off-diagonal entries of $M \cap M^T$ are 0, i.e. iff $M \cap M^T \le I_n$. ■

3.16 Exercise Set 3I — properties of relations

3I.1 Consider $R_1 = \{(1,1), (1,2), (1,3), (3,3)\}$, $R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$ and $R_3 = \{(1,1), (1,2), (2,2), (2,3)\}$ on $A = \{1, 2, 3\}$. Which of these are (i) reflexive, (ii) symmetric, (iii) transitive, (iv) antisymmetric?

3I.2 Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,2), (1,3), (4,2)\}$. Is R transitive?

3I.3 Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,2), (2,2), (3,4), (4,1)\}$. Verify that R is (i) not symmetric, (ii) not asymmetric, (iii) antisymmetric.

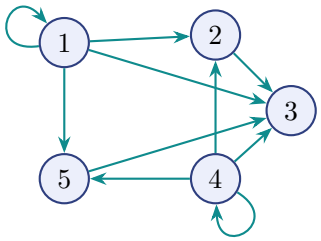
3I.4 On $A = \{1, 2, 3, 4\}$ let $R = \{(1,1), (2,2), (2,3), (3,2), (4,2), (4,4)\}$ and $S = \{(1,3), (1,1), (3,1), (1,2), (3,3), (4,4)\}$. (a) Verify that R is not (i) reflexive, (ii) symmetric, (iii) transitive, (iv) antisymmetric. (b) Determine whether S is reflexive, irreflexive, symmetric, asymmetric, antisymmetric or transitive.

3I.5 Let $A = \{1, 2, 3\}$. Verify: (a) $R_1 = \{(1,1), (2,2)\}$ is both symmetric and antisymmetric; (b) $R_2 = \{(1,2), (2,1), (2,3)\}$ is neither symmetric nor antisymmetric; (c) $R_3 = \{(1,1), (1,2), (3,2), (2,3), (3,3)\}$ is neither reflexive nor irreflexive.

3I.6 Let $A = \{1, 2, 3\}$, $R_1 = \{(1,1), (1,2), (2,2), (3,3), (3,1)\}$, $R_2 = \{(1,1), (1,2), (2,1), (2,3), (3,2)\}$, $R_3 = \{(1,1), (1,2), (2,3), (1,3)\}$. Verify that (i) R_1 is reflexive and antisymmetric but neither symmetric nor transitive; (ii) R_2 is symmetric but not reflexive, antisymmetric or transitive; (iii) R_3 is transitive and antisymmetric but not reflexive or symmetric.

3I.7 For $A = \{1, 2, 3, 4\}$ give an example of a relation that is (i) reflexive and symmetric but not transitive; (ii) reflexive and transitive but not symmetric; (iii) symmetric and transitive but not reflexive.

3I.8 Verify that the relation represented by the following digraph is antisymmetric and transitive.



3I.9 Verify that the relation with matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

is irreflexive and symmetric.

3I.10 Let $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1, 2), (2, 1), (1, 5), (5, 1), (2, 2), (3, 4), (4, 3), (3, 5), (5, 3)\}$, a symmetric relation on A. Draw the graph of R.

3I.11 Let U be a universal set and V a fixed subset of U. On $\mathcal{P}(U)$ define $(A, B) \in R$ iff $A \cap V = B \cap V$. Prove that R is reflexive, symmetric and transitive.

3I.12 On $\mathbb{Z}$ define $a\, R\, b$ iff $ab \geq 0$. Prove that R is reflexive and symmetric but not transitive.

3I.13 On $\mathbb{Z}$ define $a\, R_1\, b$ iff $a + b$ is odd, and $a\, R_2\, b$ iff $a - b$ is even. Prove that (i) R_1 is symmetric, (ii) R_2 is symmetric, reflexive and transitive.

3I.14 Let A be the set of all non-zero rational numbers, and for $a, b \in A$ let $a\, R\, b$ iff a/b is an integer. Prove that R is reflexive and transitive, but neither symmetric nor antisymmetric.

3I.15 On $\mathbb{Z} \times \mathbb{Z}$ define $(a, b)\, R\, (c, d)$ iff $a \leq c$. Prove that R is reflexive and transitive.

3I.16 Prove: (1) on $\mathbb{Z}$, $a\, R\, b$ iff $a \leq b+1$ is reflexive; (2) on $\mathbb{Z}$, $a\, R\, b$ iff $|a-b| = 2$ is irreflexive and symmetric; (3) on $\mathbb{Z}^+$, $a\, R\, b$ iff $a \mid b$ or $b \mid a$ is reflexive and symmetric but not transitive; (4) on $\mathbb{Z}^+$, $a\, R\, b$ iff a equals some positive integral power of b is reflexive, antisymmetric and transitive.

3I.17 Prove that if a relation on a set is irreflexive and transitive, then it is antisymmetric.

3I.18 Let R be a relation on A. Prove: (1) if R is symmetric then R^2 is symmetric; (2) if R is symmetric then R^n is symmetric for $n \geq 2$; (3) if R is non-empty, symmetric and transitive, then R is not irreflexive.

3I.19 Let R be a relation on A. Prove: (1) R is symmetric iff $R = R^c$; (2) R is antisymmetric iff $R \cap R^c \subseteq \Delta_A$; (3) R is asymmetric iff $R \cap R^c = \emptyset$.

3I.20 Let $R_1 \subseteq R_2$ be relations on A. Prove or disprove: (i) if R_1 is reflexive, so is R_2; (ii) if R_1 is symmetric, so is R_2; (iii) if R_1 is antisymmetric, so is R_2; (iv) if R_1 is transitive, so is R_2.

3.17 Solutions to Exercise Set 3I

3I.1 – Three relations on $\{1,2,3\}$

	Reflexive	Symmetric	Transitive	Antisymmetric
R_1	× ($(2,2) \notin R_1$)	× ($(1,2)$, not $(2,1)$)	✓	✓
R_2	✓	✓	✓	× ($(1,2),(2,1)$)
R_3	× ($(3,3) \notin R_3$)	×	× ($(1,2),(2,3)$, not $(1,3)$)	✓

Conclusion. R_1 is transitive and antisymmetric; R_2 is reflexive, symmetric and transitive; R_3 is antisymmetric only.

3I.2 – Is $\{(1,2),(1,3),(4,2)\}$ transitive?

For transitivity we need pairs $(a,b),(b,c)$ in R. The second coordinates are 2 and 3, and no pair of R starts with 2 or 3. So there is nothing to check.

Conclusion. **Yes** — R is (vacuously) transitive.

3I.3 – Not symmetric, not asymmetric, antisymmetric

(i) $(1,2) \in R$ but $(2,1) \notin R$: not symmetric. (ii) $(2,2) \in R$, so "$(a,b) \in R \Rightarrow (b,a) \notin R$" fails for $a = b = 2$: not asymmetric. (iii) The off-diagonal pairs $(1,2),(3,4),(4,1)$ have no reverses in R: antisymmetric.

3I.4 – Checking R and S

(a) R: not reflexive ($(3,3) \notin R$); not symmetric ($(4,2) \in R, (2,4) \notin R$); not transitive ($(4,2),(2,3) \in R, (4,3) \notin R$); not antisymmetric ($(2,3),(3,2) \in R$).
(b) S: not reflexive ($(2,2) \notin S$); not irreflexive ($(1,1) \in S$); not symmetric ($(1,2) \in S, (2,1) \notin S$); not asymmetric and not antisymmetric ($(1,3),(3,1) \in S$); not transitive ($(3,1),(1,2) \in S, (3,2) \notin S$).

Conclusion. S has **none** of the six properties.

3I.5 – Both, neither, neither

(a) R_1 has only loops: every pair's reverse is itself (symmetric) and no two distinct elements are related both ways (antisymmetric).
(b) $(2,3) \in R_2$ but $(3,2) \notin R_2$: not symmetric; $(1,2),(2,1) \in R_2$ with $1 \neq 2$: not antisymmetric.
(c) $(2,2) \notin R_3$: not reflexive; $(1,1) \in R_3$: not irreflexive.

3I.6 – Three verifications

(i) R_1: all loops present (reflexive); no two-way pair (antisymmetric); $(1,2) \in R_1, (2,1) \notin R_1$ (not symmetric); $(3,1),(1,2) \in R_1, (3,2) \notin R_1$ (not transitive).
(ii) R_2: each pair has its reverse (symmetric); $(3,3) \notin R_2$ (not reflexive); $(1,2),(2,1)$ (not antisymmetric); $(2,1),(1,2) \in R_2, (2,2) \notin R_2$ (not transitive).
(iii) R_3: the only two-step paths $1 \to 1 \to 1, 2, 3$ and $1 \to 2 \to 3$ end in pairs of R_3 (transitive); no two-way pairs (antisymmetric); $(2,2) \notin R_3$ (not reflexive); $(1,2) \in R_3, (2,1) \notin R_3$ (not symmetric).

3I.7 – Examples on $\{1,2,3,4\}$

(i) $\{(1,1),(2,2),(3,3),(4,4),(1,2),(2,1),(2,3),(3,2)\}$: reflexive and symmetric; not transitive since $(1,2),(2,3)$ are in but $(1,3)$ is not.
(ii) $\{(1,1),(2,2),(3,3),(4,4),(1,2)\}$: reflexive and transitive; not symmetric.

(iii) $\{(1,1),(2,2),(1,2),(2,1)\}$: symmetric and transitive; not reflexive since $(3,3) \notin R$.

3I.8 – A digraph that is antisymmetric and transitive

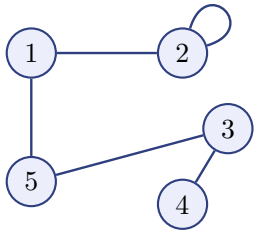
$R = \{(1,1),(1,2),(1,3),(1,5),(2,3),(4,2),(4,3),(4,4),(4,5),(5,3)\}$.
Antisymmetric. No two distinct vertices are joined by arrows in both directions.
Transitive. The two-step paths between different vertices are $1 \to 2 \to 3$, $1 \to 5 \to 3$, $4 \to 2 \to 3$, $4 \to 5 \to 3$; the shortcuts $1 \to 3$ and $4 \to 3$ are present. Paths through a loop ($1 \to 1 \to x$, $4 \to 4 \to x$) end at pairs already in R.

3I.9 – Irreflexive and symmetric from a matrix

All diagonal entries are 0, so the relation is irreflexive. The matrix equals its transpose ($m_{12} = m_{21} = 1$, $m_{14} = m_{41} = 1$, $m_{23} = m_{32} = 1$, $m_{24} = m_{42} = 1$, all other off-diagonal entries 0), so the relation is symmetric.

3I.10 – Graph of a symmetric relation

The undirected edges are $\{1,2\},\{1,5\},\{3,4\},\{3,5\}$, with a loop at 2.



Graph of R

3I.11 – $A \cap V = B \cap V$

R is defined by an equation $f(A) = f(B)$ with $f(X) = X \cap V$. Equality is reflexive, symmetric and transitive, so R is too: $A \cap V = A \cap V$; $A \cap V = B \cap V \Rightarrow B \cap V = A \cap V$; and $A \cap V = B \cap V$, $B \cap V = C \cap V$ give $A \cap V = C \cap V$. ■

3I.12 – $ab \geq 0$ on $\mathbb{Z}$

Reflexive: $a \cdot a = a^2 \geq 0$. Symmetric: $ab = ba$. Not transitive: $(-1) \cdot 0 \geq 0$ and $0 \cdot 1 \geq 0$, so $-1\,R\,0$ and $0\,R\,1$, but $(-1) \cdot 1 = -1 < 0$. (The integer 0 is related to everything, which breaks transitivity.)

3I.13 – $a + b$ odd; $a - b$ even

(i) $a + b = b + a$, so $a\,R_1\,b \Rightarrow b\,R_1\,a$. (Note that R_1 is irreflexive, since $a + a = 2a$ is even.)
(ii) $a - a = 0$ is even (reflexive); $b - a = -(a - b)$ (symmetric); $a - c = (a - b) + (b - c)$ is a sum of even numbers (transitive). ■

3I.14 – $a/b \in \mathbb{Z}$ on non-zero rationals

Reflexive: $a/a = 1$. Transitive: $a/b = m$ and $b/c = n$ with $m, n \in \mathbb{Z}$ give $a/c = mn \in \mathbb{Z}$. Not symmetric: $2/1 = 2 \in \mathbb{Z}$ but $1/2 \notin \mathbb{Z}$. Not antisymmetric: $2/(-2) = -1$ and $(-2)/2 = -1$ are integers, yet $2 \neq -2$.

3I.15 – $(a,b)\,R\,(c,d)$ iff $a \le c$

Reflexive: $a \le a$. Transitive: $(a,b)\,R\,(c,d)$ and $(c,d)\,R\,(e,f)$ mean $a \le c \le e$, so $(a,b)\,R\,(e,f)$. (It is not antisymmetric: $(1,2)\,R\,(1,3)$ and $(1,3)\,R\,(1,2)$.)

3I.16 – Four relations

(1) $a \le a+1$: reflexive. (It is not symmetric — $0 \le 5+1$ but $5 \not\le 0+1$ — and not transitive — $3 \le 2+1, 2 \le 1+1$, but $3 \not\le 1+1$.)
(2) $|a-a| = 0 \neq 2$: irreflexive; $|b-a| = |a-b|$: symmetric.
(3) $a \mid a$: reflexive; the condition "$a \mid b$ or $b \mid a$" is symmetric in a, b. Not transitive: $2\,R\,6$ ($2 \mid 6$) and $6\,R\,3$ ($3 \mid 6$) but $2 \nmid 3$ and $3 \nmid 2$.
(4) $a = a^1$: reflexive. If $a = b^k$ and $b = a^l$ then $a = a^{kl}$; for $a \ge 2$ this forces $kl = 1$, so $k = 1$ and $a = b$; for $a = 1$, $b = 1^l = 1 = a$. So R is antisymmetric. If $a = b^k$ and $b = c^l$ then $a = c^{kl}$, so R is transitive.

3I.17 – Irreflexive + transitive $\Rightarrow$ antisymmetric

Suppose $(a,b), (b,a) \in R$ with $a \neq b$. Transitivity gives $(a,a) \in R$, contradicting irreflexivity. Hence no such pair exists, i.e. R is antisymmetric (in fact asymmetric). ■

3I.18 – Symmetry of powers

(1) Let $(a,c) \in R^2$: $(a,b), (b,c) \in R$ for some b. By symmetry $(c,b), (b,a) \in R$, so $(c,a) \in R^2$.
(2) Induction: if R^{n-1} is symmetric and $(a,c) \in R^n = R \circ R^{n-1}$, then $(a,b) \in R$ and $(b,c) \in R^{n-1}$ for some b, so $(c,b) \in R^{n-1}$ and $(b,a) \in R$. A path $c \to b \to a$ of the form "R^{n-1} then R" lies in $R^{n-1} \circ R = R^n$ (powers of one relation commute, by associativity). So $(c,a) \in R^n$.
(3) This is part of the "two of three" example: take $(a,b) \in R$; symmetry gives (b,a), and transitivity gives $(a,a) \in R$. ■

3I.19 – Characterisations through R^c

(1) If R is symmetric, $(a,b) \in R \iff (b,a) \in R \iff (a,b) \in R^c$, so $R = R^c$. Conversely, if $R = R^c$ and $(a,b) \in R$, then $(a,b) \in R^c$, i.e. $(b,a) \in R$.
(2) $(a,b) \in R \cap R^c$ iff both $(a,b), (b,a) \in R$. R is antisymmetric iff this forces $a = b$, i.e. iff $R \cap R^c \subseteq \Delta_A$.
(3) R is asymmetric iff no pair (a,b) (with $a = b$ allowed) has both $(a,b), (b,a) \in R$, i.e. iff $R \cap R^c = \emptyset$.
■

3I.20 – Properties inherited by a larger relation

(i) **True**: $\Delta_A \subseteq R_1 \subseteq R_2$.
(ii) **False**: on $\{1,2\}$, $R_1 = \emptyset$ is symmetric, $R_2 = \{(1,2)\}$ is not.
(iii) **False**: $R_1 = \{(1,2)\}$ is antisymmetric, $R_2 = \{(1,2),(2,1)\}$ is not.
(iv) **False**: $R_1 = \{(1,2)\}$ is transitive, $R_2 = \{(1,2),(2,3)\}$ on $\{1,2,3\}$ is not.

4 Partial Ordering Relation

Definition: Partial order, poset

A relation R on a set A is a **partial ordering relation** (a **partial order**) on A if R is (i) reflexive, (ii) antisymmetric and (iii) transitive. A set A with a partial order R is a **partially ordered set** or **poset**, written (A, R). A partial order is often written $\preceq$.

The most familiar partial order is "less than or equal to" on $\mathbb{Z}$: it is reflexive, antisymmetric and transitive, so $(\mathbb{Z}, \leq)$ is a poset; so is $(\mathbb{Z}, \geq)$. The divisibility relation on $\mathbb{Z}^+$ and the subset relation on $\mathcal{P}(S)$ are partial orders (Section 3, properties examples), so $(\mathbb{Z}^+, |)$ and $(\mathcal{P}(S), \subseteq)$ are posets. "$<$" and "$>$" are *not* partial orders on $\mathbb{Z}$, because they are not reflexive. Congruence modulo n (Section 5) is not a partial order either: it is not antisymmetric, since $2 \equiv 8 \pmod 3$ and $8 \equiv 2 \pmod 3$ but $2 \neq 8$.

Relation	Poset?	Reason
$(\mathbb{Z}, \leq)$	Yes, total	any two integers are comparable
$(\mathbb{Z}^+, \mid)$	Yes, partial	2 and 3 are incomparable
$(\mathcal{P}(S), \subseteq)$	Yes, partial	$\{a\}$, $\{b\}$ incomparable
$(\mathbb{Z}, <)$	No	not reflexive
$(\mathbb{Z} \setminus \{0\}, \mid)$	No	$2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$ (not antisymmetric)

4.1 Total orders

Definition: Total order, chain

A partial order R on A is a **total order** (or **linear order**) if for all $x, y \in A$, either $x \, R \, y$ or $y \, R \, x$ — every two elements are **comparable**. Then (A, R) is a **totally ordered set** (also called a **linearly ordered set** or a **chain**).

$(\mathbb{R}, \leq)$ is totally ordered, since for any reals $x \leq y$ or $y \leq x$. Divisibility is a total order on $\{1, 2, 4, 8\}$, but only a partial order on $\{1, 2, 4, 6, 8\}$: neither $4 \mid 6$ nor $6 \mid 4$. The subset relation on $\mathcal{P}(S)$ is not a total order in general: for $S = \{1, 2, 3\}$, $S_1 = \{1, 2\}$ and $S_2 = \{1, 3\}$, neither $S_1 \subseteq S_2$ nor $S_2 \subseteq S_1$. Thus *every total order is a partial order, but not every partial order is a total order.*

4.2 Hasse diagrams

Since a partial order is reflexive, its digraph has a loop at every vertex; since it is transitive, an edge $a \to b$ and an edge $b \to c$ force an edge $a \to c$. These edges carry no information, so the digraph of a finite poset is simplified as follows.

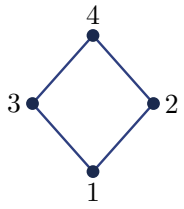
1. Delete every loop (reflexivity is understood by convention).
2. Delete every edge implied by transitivity: keep $a \to b$ only if b **covers** a, i.e. $a \prec b$ and there is no c with $a \prec c \prec b$.
3. Draw the vertices as dots and place each element *above* the elements it covers, so that every edge points upward; then remove the arrowheads.

The resulting picture is the **poset diagram** or **Hasse diagram** of the partial order. $a \preceq b$ exactly when one can climb from a to b along edges.

Example 4.1 – A partial order and its Hasse diagram

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), (4,4)\}$. Verify that R is a partial order and draw its Hasse diagram.

Solution. R contains (a, a) for every a (reflexive); it contains no pair (a, b) together with (b, a) for $a \neq b$ (antisymmetric); and every two-step path $1 \to 2 \to 4$, $1 \to 3 \to 4$ has its shortcut $(1, 4)$ (transitive). So R is a partial order. The covering pairs are $(1, 2), (1, 3), (2, 4), (3, 4)$, giving a diamond.

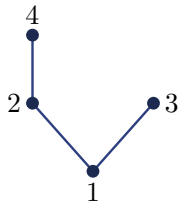


Example 1

Example 4.2 – Divisibility on $\{1, 2, 3, 4\}$

Let $A = \{1, 2, 3, 4\}$ and $x \, R \, y$ iff $x \mid y$. Prove that (A, R) is a poset and draw its Hasse diagram.

Solution. $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$. It is reflexive; whenever $(a, b) \in R$ with $a \neq b$, $(b, a) \notin R$ (antisymmetric); and whenever $(a, b), (b, c) \in R$, $(a, c) \in R$ (transitive). The covering pairs are $(1, 2), (1, 3), (2, 4)$.

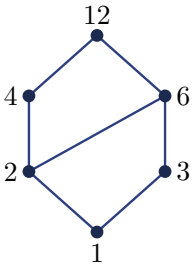


$(\{1, 2, 3, 4\}, |)$

Example 4.3 – Divisibility on $\{1, 2, 3, 4, 6, 12\}$

Let $A = \{1, 2, 3, 4, 6, 12\}$ and $a \, R \, b$ iff $a \mid b$. Prove that R is a partial order and draw its Hasse diagram.

Solution. $R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,12), (2,2), (2,4), (2,6), (2,12), (3,3), (3,6), (3,12), (4,4), (4,12), (6,6), (6,12), (12,12)\}$. As $a \mid a$, R is reflexive; $a \mid b$ and $b \mid c$ give $a \mid c$ (transitive); $a \mid b$ and $b \mid a$ give $a = b$ (antisymmetric). The covering pairs are $(1, 2), (1, 3), (2, 4), (2, 6), (3, 6), (4, 12), (6, 12)$.



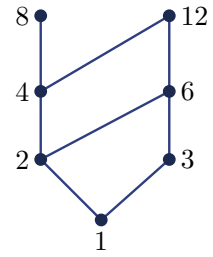
D_{12}

Example 4.4 – Divisibility on $\{1, 2, 3, 4, 6, 8, 12\}$

Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ and $x \, R \, y$ iff $x \mid y$. Draw the Hasse diagram of R.

Solution. $R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,8), (1,12), (2,2), (2,4), (2,6), (2,8), (2,12),$

$(3,3),(3,6),(3,12),(4,4),(4,8),(4,12),(6,6),(6,12),(8,8),(12,12)\}$ (22 pairs). The covering pairs are $(1,2),(1,3),(2,4),(2,6),(3,6),(4,8),(4,12),(6,12)$.

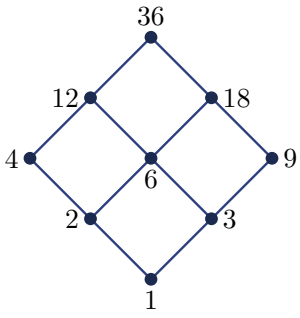


$\{1,2,3,4,6,8,12\}$

Example 4.5 – The divisors of 36

Draw the Hasse diagram of the positive divisors of 36 under divisibility. (For a positive integer n, D_n denotes the set of all positive divisors of n.)

Solution. $D_{36} = \{1,2,3,4,6,9,12,18,36\}$. Under divisibility: 1 is related to every element; 2 to $2,4,6,12,18,36$; 3 to $3,6,9,12,18,36$; 4 to $4,12,36$; 6 to $6,12,18,36$; 9 to $9,18,36$; 12 to $12,36$; 18 to $18,36$; 36 to 36. Keeping only covering pairs gives a 3×3 grid.



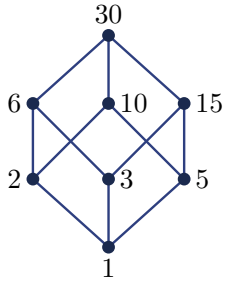
D_{36}

Example 4.6 – $\mathbb{Z}^+$ is not totally ordered by divisibility

For $(\mathbb{Z}^+,|)$ to be totally ordered we would need $a \mid b$ or $b \mid a$ for every $a,b \in \mathbb{Z}^+$. But for $a=2, b=3$, neither $2 \mid 3$ nor $3 \mid 2$. So $\mathbb{Z}^+$ is not totally ordered by divisibility.

Example 4.7 – Total or not?

Consider the partial order of divisibility on (i) $A=\{1,2,3,5,6,10,15,30\}$, (ii) $A=\{2,4,8,16,32\}$. Draw the Hasse diagrams and decide whether each poset is totally ordered.



(i) D_{30}: not total



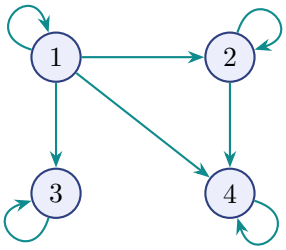
(ii) a chain: total

Solution. The diagrams are shown. In (ii) the elements form a single vertical chain, so any two are comparable: the poset is **totally ordered**. In (i), 2 and 3 (for example) are incomparable, so it is **not** totally

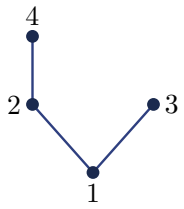
ordered.

Example 4.8 – Hasse diagram from a digraph

A partial order R on $A = \{1, 2, 3, 4\}$ is represented by the digraph below. Draw the Hasse diagram of R.

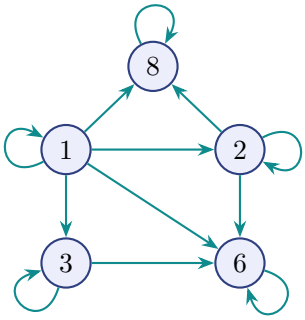


Solution. $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (1,4), (2,4)\}$. Removing the loops and the transitive edge $1 \to 4$ (implied by $1 \to 2 \to 4$) leaves the covering pairs $(1,2), (1,3), (2,4)$.

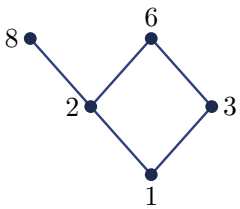


Example 4.9 – Poset from a digraph

The digraph of a relation on $A = \{1, 2, 3, 6, 8\}$ is shown below. Verify that (A, R) is a poset and draw its Hasse diagram.



Solution. $R = \{(1,1), (1,2), (1,3), (1,6), (1,8), (2,2), (2,6), (2,8), (3,3), (3,6), (6,6), (8,8)\}$ — exactly divisibility on A. It is reflexive, transitive and antisymmetric, so (A, R) is a poset. The covering pairs are $(1,2), (1,3), (2,6), (2,8), (3,6)$.

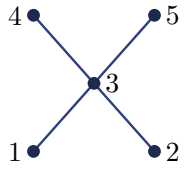


Example 4.10 – Hasse diagram from a matrix

Draw the Hasse diagram of the relation R on $A = \{1, 2, 3, 4, 5\}$ whose matrix is

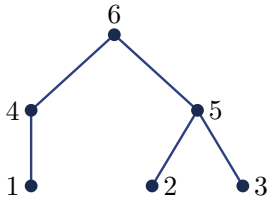
$$M_R = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Solution. Reading the rows, $R = \{(1,1), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,4), (5,5)\}$. Removing the loops and the pairs implied by transitivity ($(1,4), (1,5), (2,4), (2,5)$, which come from $1, 2 \prec 3 \prec 4, 5$) leaves the covering pairs $(1,3), (2,3), (3,4), (3,5)$.



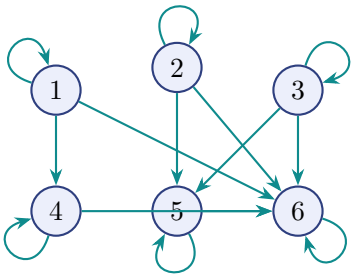
Example 4.11 – From a Hasse diagram to the relation and digraph

The Hasse diagram of a partial order R on $A = \{1, 2, 3, 4, 5, 6\}$ is shown below. Write R as a subset of $A \times A$ and construct its digraph.



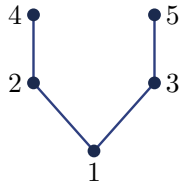
Hasse diagram

Solution. From the diagram, $1\,R\,4$, $4\,R\,6$, $2\,R\,5$, $3\,R\,5$, $5\,R\,6$. By the conventions of Hasse diagrams we must add the loops $a\,R\,a$ and the transitive pairs $1\,R\,6$, $2\,R\,6$, $3\,R\,6$. Therefore $R = \{(1,1), (1,4), (1,6), (2,2), (2,5), (2,6), (3,3), (3,5), (3,6), (4,4), (4,6), (5,5), (5,6), (6,6)\}$.



Example 4.12 – Matrix from a Hasse diagram

Determine the matrix of the partial order whose Hasse diagram is given below.



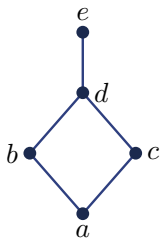
Solution. The order is on $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,2), (2,4), (3,3), (3,5), (4,4), (5,5)\}$. Consequently

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(This matrix can be written directly from the diagram: row a has a 1 in column b exactly when b is a itself or can be reached by climbing from a.)

Example 4.13 – Relation matrix and digraph from a Hasse diagram

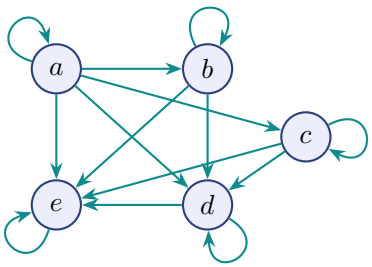
For $A = \{a, b, c, d, e\}$ the Hasse diagram of a poset (A, R) is shown. (a) Determine the relation matrix of R. (b) Construct the digraph of R.



Solution. $R = \{(a,a), (a,b), (a,c), (a,d), (a,e), (b,b), (b,d), (b,e), (c,c), (c,d), (c,e), (d,d), (d,e), (e,e)\}$.
(a)

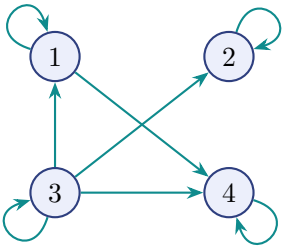
$$M(R) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(b) The digraph:

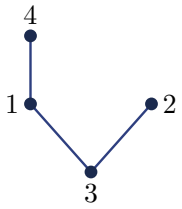


Example 4.14 – Extending a partial order to a total order

The digraph of a relation R on $A = \{1, 2, 3, 4\}$ is shown below. (a) Verify that (A, R) is a poset and draw its Hasse diagram. (b) How many more directed edges are needed in the digraph of R to extend R to a total order?



Solution. From the digraph, $R = \{(1,1), (1,4), (2,2), (3,3), (3,1), (3,2), (3,4), (4,4)\}$.
(a) R is reflexive, transitive and antisymmetric, so it is a partial order. Its covering pairs are $(3,1), (3,2), (1,4)$:



(b) **Topological sorting** produces a total order containing R: repeatedly pick a maximal element of what remains, label it $v_1, v_2, \ldots$ in turn and delete it; then list the elements in the reverse order. Here $v_1 = 4$; then $v_2 = 1$ (or 2), $v_3 = 2$ (or 1), and $v_4 = 3$. So a total order T containing R is

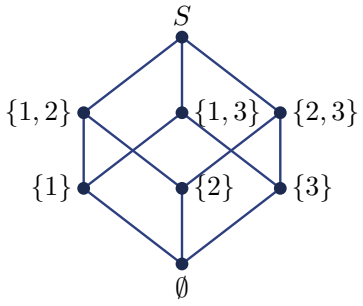
$$3 < 2 < 1 < 4 \qquad \text{or} \qquad 3 < 1 < 2 < 4.$$

For the first, $T = \{(3,3), (3,2), (3,1), (3,4), (2,2), (2,1), (2,4), (1,1), (1,4), (4,4)\}$. Comparing with R, the new pairs are $(2,1)$ and $(2,4)$; for the second order they are $(1,2)$ and $(2,4)$. Either way, **two** more directed edges are needed.

Example 4.15 – The power set of a 3-element set

Let $S = \{1, 2, 3\}$ and on $\mathcal{P}(S)$ define $X \, R \, Y$ iff $X \subseteq Y$. Show that R is a partial order on $\mathcal{P}(S)$ and draw its Hasse diagram.

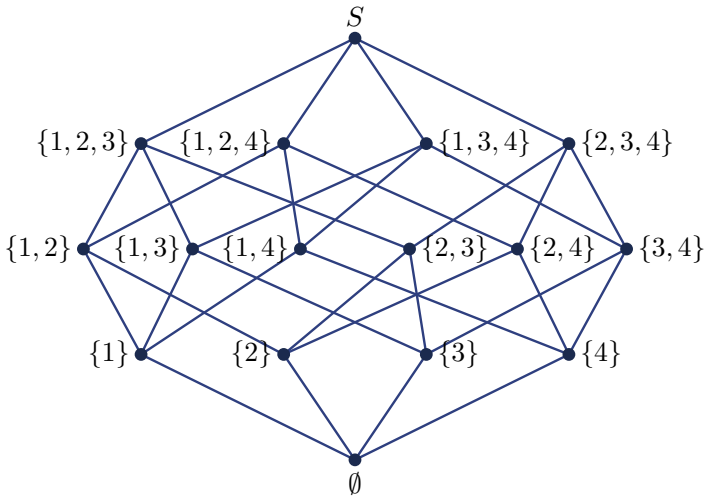
Solution. The subset relation is reflexive, antisymmetric and transitive on $\mathcal{P}(S)$ for any set S (Section 3), so it is a partial order. For $S = \{1, 2, 3\}$ the subsets are $\emptyset$, $S_1 = \{1\}$, $S_2 = \{2\}$, $S_3 = \{3\}$, $S_4 = \{1, 2\}$, $S_5 = \{2, 3\}$, $S_6 = \{3, 1\}$ and S. We have $\emptyset \subseteq S_r$ $(r = 1, \ldots, 6)$, $\emptyset \subseteq S$, $S_r \subseteq S$, $S_r \subseteq S_r$, and $S_1 \subseteq S_4, S_6$; $S_2 \subseteq S_4, S_5$; $S_3 \subseteq S_5, S_6$. The Hasse diagram is a cube:



Example 4.16 – The power set of a 4-element set

Draw the Hasse diagram of $(\mathcal{P}(S), \subseteq)$ for $S = \{1, 2, 3, 4\}$.

Solution. $|\mathcal{P}(S)| = 2^4 = 16$: $\emptyset$; four singletons; six 2-element subsets; four 3-element subsets; and S. Each k-element subset is covered by the $(k+1)$-element subsets containing it (e.g. $\{1\} \subseteq \{1,2\}, \{1,3\}, \{1,4\}$; $\{1,2\} \subseteq \{1,2,3\}, \{1,2,4\}$). There are 32 covering pairs and the diagram is a 4-dimensional cube.



Example 4.17 – Product partial order

Let (A, R_1) and (B, R_2) be posets. On $A \times B$ define $(a, b)\, R\, (x, y)$ iff $a\, R_1\, x$ and $b\, R_2\, y$. Prove that R is a partial order.

Solution. *Reflexive:* for $(a, b) \in A \times B$, $a\, R_1\, a$ and $b\, R_2\, b$, so $(a, b)\, R\, (a, b)$.
Antisymmetric: if $(a, b)\, R\, (a', b')$ and $(a', b')\, R\, (a, b)$, then $a\, R_1\, a'$, $a'\, R_1\, a$ and $b\, R_2\, b'$, $b'\, R_2\, b$, so $a = a'$ and $b = b'$.
Transitive: $(a, b)\, R\, (a', b')$ and $(a', b')\, R\, (a'', b'')$ give $a\, R_1\, a''$ and $b\, R_2\, b''$, so $(a, b)\, R\, (a'', b'')$.
Hence $(A \times B, R)$ is a poset; R is called the **product partial order**. ■

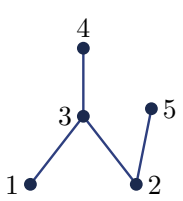
4.3 Exercise Set 4A — partial orders and Hasse diagrams

4A.1 In each case decide whether R is a partial order on $\mathbb{Z}$: (1) $a\, R\, b$ iff $a = 3b$; (2) $a\, R\, b$ iff a is a multiple of b^2; (3) $a\, R\, b$ iff a equals some positive integral power of b.

4A.2 Let $A = \{1, 2, 3, 4, 5\}$ and $x\, R\, y$ iff $x \le y$. Verify that R is a partial order on A. If $B = \{1, 2, 4\}$, verify that $(B \times B) \cap R$ is a partial order on B.

4A.3 Give an example of a relation which is both a partial order and an equivalence relation on a set.

4A.4 Let $A = \{1, 2, 3, 4, 5\}$. Determine the partial orders represented by the Hasse diagrams below.

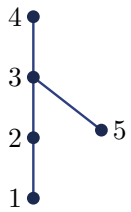


(a)



(b)

4A.5 Determine the matrix of the partial order whose Hasse diagram is given below.



4A.6 Write down the matrix of the partial order considered in worked Example 4.4.

4A.7 Let $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,1), (2,2), (3,3), (1,3), (3,4), (3,5), (1,4), (4,4), (1,5), (2,3), (2,4), (2,5), (5,5)\}$. Draw the Hasse diagram of R.

4A.8 Draw the Hasse diagram of the partial order on $A = \{1, 2, 3, 4, 5\}$ whose matrix is

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

4A.9 The digraph of a relation R on $A = \{1, 2, 3, 4\}$ is the one in worked Example 4.14. Verify that (A, R) is a poset and draw its Hasse diagram.

4A.10 Draw the Hasse diagram of the positive divisors of 45.

4A.11 Draw the Hasse diagram of divisibility on A when
(a) $A = \{3, 6, 12, 36, 72\}$; (b) $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$; (c) $A = \{2, 3, 6, 12, 24, 36\}$;
(d) $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$; (e) $A = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$.

4A.12 Show that divisibility is not a total order on $A = \{1, 2, 3, 6\}$ but is a total order on $B = \{1, 3, 6, 12, 24\}$.

4A.13 On $\mathbb{Z}$ define $a \, R \, b$ if $a - b$ is a non-negative even integer. Verify that R is a partial order but not a total order.

4A.14 Let $A = \{2, 3, 6, 9, 12, 18, 24\}$. On $A \times A$ define $(a, b) \, R \, (a', b')$ iff $a \mid a'$ and $b \leq b'$. Prove that R is a partial order.

4A.15 If R is a partial order on a set A and $B \subseteq A$, prove that $R \cap (B \times B)$ is a partial order on B.

4.4 Solutions to Exercise Set 4A

4A.1 – Partial orders on $\mathbb{Z}$?

(1) **No**: not reflexive, since $1 \neq 3 \cdot 1$. (2) **No**: not reflexive, since 2 is not a multiple of $2^2 = 4$.
(3) **Yes**. Reflexive: $a = a^1$. Transitive: $a = b^k$, $b = c^l$ give $a = c^{kl}$. Antisymmetric: if $a = b^k$ and $b = a^l$, then $a = a^{kl}$; for $|a| \geq 2$ this forces $kl = 1$, so $a = b$; and for $a \in \{0, 1, -1\}$ we get $b = a^l \in \{0, 1, -1\}$ with $a = b^k$, which again forces $b = a$.

4A.2 – $\leq$ on A and its restriction to B

$\leq$ is reflexive ($x \leq x$), antisymmetric ($x \leq y \leq x \Rightarrow x = y$) and transitive on A, so it is a partial order. $(B \times B) \cap R = \{(1,1), (1,2), (1,4), (2,2), (2,4), (4,4)\}$ is just $\leq$ on B, and has the same three properties (see also 4A.15).

4A.3 – Both a partial order and an equivalence relation

On any set A the equality relation $\Delta_A = \{(a, a) \mid a \in A\}$ is reflexive, symmetric, antisymmetric and transitive, so it is both. (In fact it is the only such relation: symmetric and antisymmetric together allow only loops, and reflexive forces all of them.)

4A.4 – Relations from Hasse diagrams

(a) Covering pairs $1 \prec 3$, $2 \prec 3$, $3 \prec 4$, $2 \prec 5$; adding loops and the transitive pairs $(1,4), (2,4)$: $R = \{(1,1), (1,3), (1,4), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (4,4), (5,5)\}$.
(b) The chain $1 \prec 2 \prec 3 \prec 4 \prec 5$: $R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,4), (4,5), (5,5)\}$ (15 pairs, the relation $\leq$).

4A.5 – Matrix from a Hasse diagram

The diagram shows the chain $1 \prec 2 \prec 3 \prec 4$ with $5 \prec 3$. So $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4), (5,3), (5,4), (5,5)\}$ and

Conclusion.

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

4A.6 – Matrix of divisibility on $\{1, 2, 3, 4, 6, 8, 12\}$

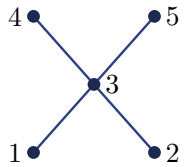
Order the elements $1, 2, 3, 4, 6, 8, 12$; row a has a 1 in column b iff $a \mid b$:

Conclusion.

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

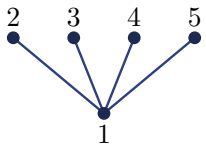
4A.7 – Hasse diagram of a given order

The covering pairs are $(1,3), (2,3), (3,4), (3,5)$; the pairs $(1,4), (1,5), (2,4), (2,5)$ follow by transitivity.



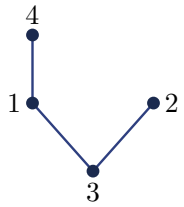
4A.8 – Hasse diagram from a matrix

Row 1 is all 1s and the other rows have only their diagonal 1: so $1 \prec 2, 3, 4, 5$ and nothing else.



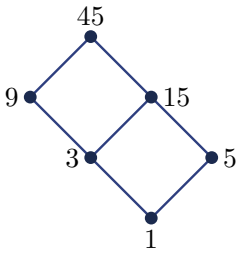
4A.9 – Poset from a digraph

As in worked Example 4.14, $R = \{(1,1),(1,4),(2,2),(3,3),(3,1),(3,2),(3,4),(4,4)\}$ is reflexive, antisymmetric and transitive. Hasse diagram:



4A.10 – Divisors of 45

$D_{45} = \{1, 3, 5, 9, 15, 45\}$; covering pairs $(1,3),(1,5),(3,9),(3,15),(5,15),(9,45),(15,45)$.

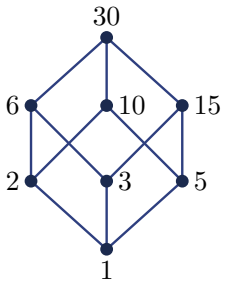


D_{45}

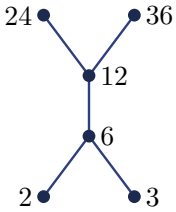
4A.11 – Five divisibility posets



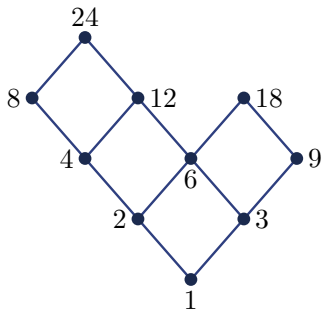
(a)



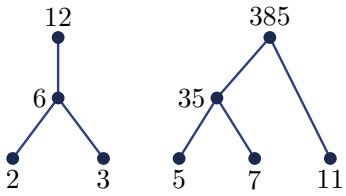
(b)



(c)



(d)



(e)

4A.12 – Total or not

On $A = \{1, 2, 3, 6\}$: neither $2 \mid 3$ nor $3 \mid 2$, so divisibility is **not** a total order. On $B = \{1, 3, 6, 12, 24\}$: $1 \mid 3 \mid 6 \mid 12 \mid 24$, so any two elements are comparable — a **total** order (the Hasse diagram is a single chain).

4A.13 – $a - b$ a non-negative even integer

Reflexive: $a - a = 0$. Antisymmetric: if $a - b \geq 0$ and $b - a \geq 0$ then $a = b$. Transitive: if $a - b = 2k \geq 0$ and $b - c = 2l \geq 0$ then $a - c = 2(k + l) \geq 0$. So R is a partial order. It is not total: $1 - 0 = 1$ is odd and $0 - 1 < 0$, so 0 and 1 are incomparable.

4A.14 – A product-type order on $A \times A$

This is the product partial order of $(A, |)$ and $(A, \leq)$, so it is a partial order by the product example. Directly: $a \mid a, b \leq b$ (reflexive); $a \mid a' \mid a$ and $b \leq b' \leq b$ give $a = a', b = b'$ (antisymmetric, as $A \subseteq \mathbb{Z}^+$); $a \mid a' \mid a''$ and $b \leq b' \leq b''$ give $a \mid a'', b \leq b''$ (transitive).

4A.15 – Restriction of a partial order

Let $R_B = R \cap (B \times B)$. For $b \in B$, $(b, b) \in R$ and $(b, b) \in B \times B$ (reflexive). If $(x, y), (y, x) \in R_B$ they lie in R, so $x = y$ (antisymmetric). If $(x, y), (y, z) \in R_B$ then $(x, z) \in R$ and $x, z \in B$, so $(x, z) \in R_B$ (transitive). ■

4.5 Extremal elements in posets

Let (A, R) be a poset.

Definition: Maximal, minimal, greatest, least

(1) $a \in A$ is a **maximal element** if there is no $x \neq a$ in A with $a \, R \, x$. In the Hasse diagram, no edge goes upward from a.

(2) $a \in A$ is a **minimal element** if there is no $x \neq a$ in A with $x \, R \, a$. In the Hasse diagram, no edge comes up into a from below.

(3) $a \in A$ is a **greatest element** if $x \, R \, a$ for all $x \in A$.

(4) $a \in A$ is a **least element** if $a \, R \, x$ for all $x \in A$.

In poset (i) below, 5 and 6 are maximal elements and 1 is a minimal element; 1 is also the least element, and there is no greatest element. In poset (ii), 5 is both maximal and the greatest element, and 1 is both minimal and the least element.

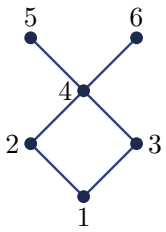


Figure (i)

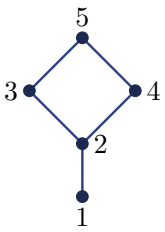


Figure (ii)

Definition: Bounds of a subset

Let $B \subseteq A$.

(5) $a \in A$ is an **upper bound** of B if $x \, R \, a$ for all $x \in B$.

(6) $a \in A$ is a **lower bound** of B if $a \, R \, x$ for all $x \in B$.

(7) $a \in A$ is the **least upper bound** (LUB, **supremum**, Sup) of B if (i) a is an upper bound of B and (ii) $a \, R \, a'$ for every upper bound a' of B.

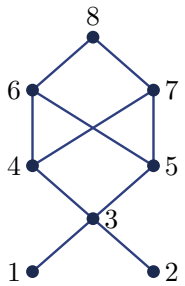
(8) $a \in A$ is the **greatest lower bound** (GLB, **infimum**, Inf) of B if (i) a is a lower bound of B and (ii) $a' \, R \, a$ for every lower bound a' of B.

Remark. Maximal, minimal, greatest and least elements are defined with reference to the whole set A; upper and lower bounds, LUB and GLB are defined with reference to a specified subset B of A — and need not belong to B.

Term	Meaning	In the Hasse diagram
Maximal m	no x with $m \prec x$	nothing above m
Minimal m	no x with $x \prec m$	nothing below m
Greatest (top)	$x \preceq g$ for every x	unique maximal element, above all
Least (bottom)	$\ell \preceq x$ for every x	unique minimal element, below all
Upper bound of B	$s \preceq u$ for all $s \in B$	reachable upward from each s
LUB (supremum)	least of all upper bounds of B	lowest common point above B
Lower bound / GLB	dual notions	highest common point below B

Example 4.18 – Bounds of two subsets

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ with the partial order shown, and consider $B_1 = \{1, 2\}$ and $B_2 = \{3, 4, 5\}$.



(A, R)

Solution. (1) $1 \, R \, 3$ and $2 \, R \, 3$, so 3 is an upper bound of B_1; for the same reason $4, 5, 6, 7, 8$ are upper bounds of B_1.
(2) The upper bound 3 satisfies $3 \, R \, x$ for every upper bound x of B_1, so $\text{LUB}(B_1) = 3$.
(3) No $x \in A$ has $x \, R \, 1$ and $x \, R \, 2$ (apart from themselves, 1 and 2 have nothing below them, and they

are incomparable). So B_1 has no lower bounds, and hence no GLB.
(4) For each $x \in B_2$, $x\,R\,6$; so 6 is an upper bound of B_2, and so are 7 and 8.
(5) Although 6 is "lowest" among the upper bounds, 6 is not related to the upper bound 7. So B_2 has **no** least upper bound.
(6) $1\,R\,x$ for each $x \in B_2$, so 1 is a lower bound of B_2; so are 2 and 3.
(7) Among the lower bounds 1, 2, 3 of B_2, $1\,R\,3$ and $2\,R\,3$ and $3\,R\,3$, so $\text{GLB}(B_2) = 3$.

Existence of maximal and minimal elements

If (A, R) is a poset and A is finite and non-empty, then A has at least one maximal element and at least one minimal element.

Proof. Take $a \in A$. If a is not maximal, there is $x_1 \neq a$ with $a\,R\,x_1$. If x_1 is not maximal, there is $x_2 \neq x_1$ with $x_1\,R\,x_2$. Continuing, we obtain a chain $a\,R\,x_1$, $x_1\,R\,x_2$, $x_2\,R\,x_3, \ldots$ of distinct elements (a repetition would contradict antisymmetry and transitivity). Since A is finite the chain must stop at some x_k, which is then maximal. A similar argument gives a minimal element. ■

Uniqueness of greatest and least elements

Every poset has at most one greatest element and at most one least element.

Proof. Let a and b be greatest elements. Since b is greatest, $a\,R\,b$; since a is greatest, $b\,R\,a$. By antisymmetry $a = b$. The argument for least elements is similar. ■

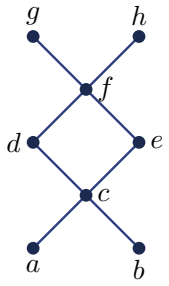
Uniqueness of LUB and GLB

If (A, R) is a poset and $B \subseteq A$, then B has at most one LUB and at most one GLB.

Proof. Let a and b both be LUBs of B. Both are upper bounds. As a is a LUB and b an upper bound, $a\,R\,b$; similarly $b\,R\,a$. By antisymmetry $a = b$. Similarly for GLBs. ■

Example 4.19 – Bounds read from a Hasse diagram

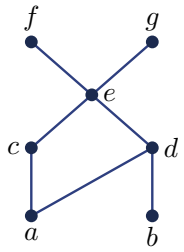
Consider the poset below. If $B = \{c, d, e\}$, find (i) all upper bounds of B, (ii) all lower bounds of B, (iii) the LUB of B, (iv) the GLB of B.



Solution. (i) c, d, e are all related to f, g, h; so f, g, h are the upper bounds of B.
(ii) a, b, c are related to all of c, d, e; so a, b, c are the lower bounds of B.
(iii) The upper bound f is related to the other upper bounds g and h: $\text{LUB}(B) = f$.
(iv) The lower bounds a and b are related to the lower bound c: $\text{GLB}(B) = c$.

Example 4.20 – LUB and GLB by upward paths

Find the LUB and GLB of $B = \{c, d, e\}$ in the poset below.



Solution. Following all upward paths from c, d and e, the first element common to all of them is e: $\text{LUB}(B) = e$. Following all downward paths, the only element below all of c, d, e is a (b lies below d but not below c): $\text{GLB}(B) = a$.

Example 4.21 – Bounds in $(\mathbb{R}, \leq)$

Let $\mathbb{R}$ be ordered by $\leq$ and $B = (1, 2) = \{x \in \mathbb{R} \mid 1 < x < 2\}$. Find (i) all upper bounds, (ii) all lower bounds, (iii) the LUB and (iv) the GLB of B.

Solution. (i) Every real number ≥ 2 is an upper bound of B. (ii) Every real number ≤ 1 is a lower bound. (iii) 2 is the LUB. (iv) 1 is the GLB.
Remark. The LUB and GLB of a set need not belong to the set: here $1, 2 \notin B$.

Example 4.22 – Bounds in D_{12}

In $(D_{12}, |)$: least element 1, greatest element 12. Upper bounds of $\{2, 3\}$ are 6, 12, so $\text{LUB}\{2, 3\} = 6$; lower bounds of $\{4, 6\}$ are 1, 2, so $\text{GLB}\{4, 6\} = 2$. In general, in $(D_n, |)$, LUB $=$ lcm and GLB $=$ gcd.

Example 4.23 – Divisibility on a set without 1

Is $\{2, 3, 4\}$ under divisibility a poset with a least element?

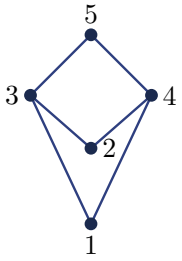
Solution. It is a poset (divisibility is always reflexive, antisymmetric and transitive on positive integers). Minimal elements are 2 and 3, maximal elements are 3 and 4; since there are two minimal elements, there is **no least element**. Note that 3 is both maximal and minimal (it is comparable with nothing else).

4.6 Lattices

Definition: Lattice

A poset (A, R) is a **lattice** if for all $x, y \in A$ the elements $\text{LUB}\{x, y\}$ and $\text{GLB}\{x, y\}$ exist in A.

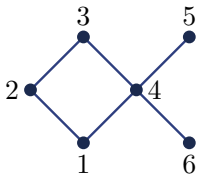
- $(\mathbb{N}, \leq)$ is a lattice: $\text{LUB}\{x, y\} = \max\{x, y\}$ and $\text{GLB}\{x, y\} = \min\{x, y\}$, both in $\mathbb{N}$.
- $(\mathbb{Z}^+, |)$ is a lattice: $\text{LUB}\{a, b\} = \text{lcm}(a, b)$ and $\text{GLB}\{a, b\} = \gcd(a, b)$, both in $\mathbb{Z}^+$.
- The poset below is **not** a lattice: the lower bounds of $\{3, 4\}$ are 1 and 2, which are incomparable, so $\text{GLB}\{3, 4\}$ does not exist.



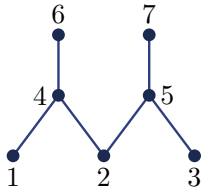
not a lattice

4.7 Exercise Set 4B — extremal elements and lattices

4B.1 For the posets shown, find all maximal and all minimal elements.

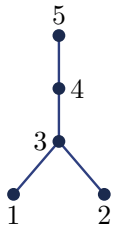


(a)

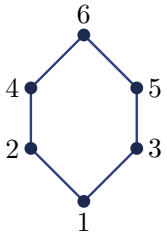


(b)

4B.2 For the posets shown, find the greatest and least elements, if they exist.



(a)

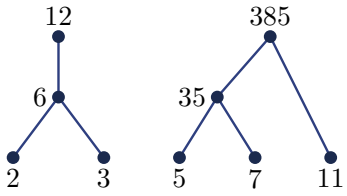


(b)

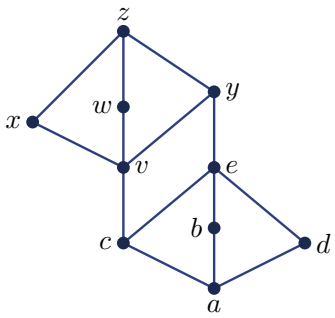
4B.3 For the poset $(\mathbb{R}, \leq)$ verify: (i) if $B = [0, 1]$ then 0 is the GLB and 1 is the LUB of B; (ii) if $B = (0, 1]$ then 0 is the GLB and 1 is the LUB; (iii) if $B = \{q \in \mathbb{Q} \mid q^2 < 2\}$ then $\sqrt{2}$ is the LUB and $-\sqrt{2}$ is the GLB. Identify the maximal and minimal elements of B in each case.

4B.4 In the poset $(\mathcal{P}(S), \subseteq)$ with $S = \{1, 2, 3, 4\}$, find the LUB and GLB of: (i) $B = \{\{1\}, \{2\}\}$; (ii) $B = \{\{1\}, \{2\}, \{1, 2\}\}$; (iii) $B = \{\{1\}, \{2\}, \{3\}, \{1, 2\}\}$; (iv) $B = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$; (v) $B = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$.

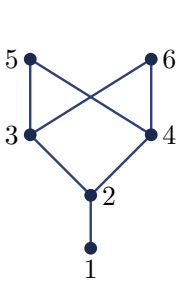
4B.5 The poset (A, R) on $A = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$ is represented by the Hasse diagram below. Find, if they exist:
LUB$\{2, 3\}$, LUB$\{2, 12\}$, LUB$\{3, 6\}$, LUB$\{5, 7\}$, LUB$\{7, 11\}$, LUB$\{11, 35\}$;
GLB$\{2, 3\}$, GLB$\{3, 6\}$, GLB$\{2, 12\}$, GLB$\{6, 12\}$, GLB$\{35, 385\}$.



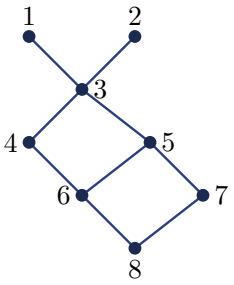
4B.6 For the poset represented by the Hasse diagram below, find (i) GLB$\{b, c\}$, (ii) GLB$\{b, w\}$, (iii) GLB$\{e, x\}$, (iv) LUB$\{c, b\}$, (v) LUB$\{d, x\}$, (vi) LUB$\{c, e\}$, (vii) LUB$\{a, v\}$.



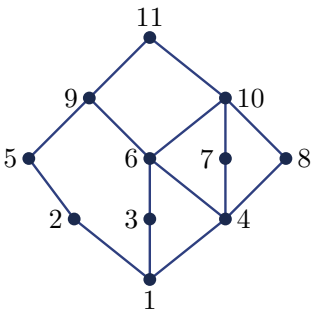
4B.7 For the posets shown, find (i) all upper bounds, (ii) all lower bounds and (iii) the LUB and GLB of B, where $B = \{3, 4, 5\}$ in case (a), $B = \{2, 3, 4\}$ in case (b) and $B = \{6, 7, 10\}$ in case (c).



(a)



(b)

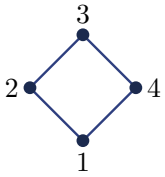


(c)

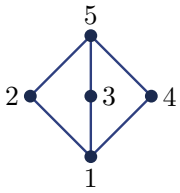
4B.8 Let $A = \{0, 1, 2\}$. On $A \times A$ define $(a, b)\, R\, (c, d)$ iff (i) $a < c$, or (ii) $a = c$ and $b \leq d$. Verify that R is a partial order, and determine the maximal/minimal and greatest/least elements.

4B.9 Let $S = \{1, 2, 3\}$ and let R be the subset relation. Show that the poset $(\mathcal{P}(S), \subseteq)$ is a lattice.

4B.10 Show that the posets represented by the following Hasse diagrams are lattices.



(a)



(b)

4B.11 Show that the poset D_{36} (Hasse diagram in Section 4.2) is a lattice.

4B.12 Show that the poset considered in Exercise 4B.5 is not a lattice.

4.8 Solutions to Exercise Set 4B

4B.1 – Maximal and minimal elements

(a) Nothing lies above 3 or 5, and nothing lies below 1 or 6. **Maximal: 3, 5; minimal: 1, 6.**
(b) Nothing lies above 6 or 7, and nothing lies below 1, 2 or 3. **Maximal: 6, 7; minimal: 1, 2, 3.**

4B.2 – Greatest and least elements

(a) 5 lies above every element: **greatest 5**. The two minimal elements 1 and 2 are incomparable, so there is **no least element**.
(b) **Greatest 6, least 1**: every element lies on an upward path from 1 to 6.

4B.3 – Bounds in $(\mathbb{R}, \leq)$

(i) $B = [0, 1]$: the lower bounds are all $x \leq 0$, the greatest of which is 0; the upper bounds are all $x \geq 1$, the least of which is 1. Here $0, 1 \in B$: 1 is the maximal (greatest) and 0 the minimal (least) element of B.
(ii) $B = (0, 1]$: the same bounds, so GLB $= 0$ and LUB $= 1$. Now $1 \in B$ is maximal, but $0 \notin B$ and B has **no minimal element** (for any $x \in B$, $x/2 \in B$ is smaller).
(iii) $B = \{q \in \mathbb{Q} \mid q^2 < 2\} = \{q \in \mathbb{Q} \mid -\sqrt{2} < q < \sqrt{2}\}$. Every real $\geq \sqrt{2}$ is an upper bound, and rationals approach $\sqrt{2}$ arbitrarily closely, so LUB $= \sqrt{2}$; similarly GLB $= -\sqrt{2}$. Neither belongs to B, and B has **no maximal and no minimal element**.

4B.4 – LUB and GLB in $(\mathcal{P}(S), \subseteq)$

In $(\mathcal{P}(S), \subseteq)$ the LUB of a family is the union of its members and the GLB is their intersection.
(i) LUB $= \{1, 2\}$, GLB $= \emptyset$. (ii) LUB $= \{1, 2\}$, GLB $= \emptyset$. (iii) LUB $= \{1, 2, 3\}$, GLB $= \emptyset$.
(iv) LUB $= \{1, 2\}$, GLB $= \emptyset$. (v) LUB $= \{1, 2, 3\}$, GLB $= \{1\}$.

4B.5 – Bounds in a divisibility poset

LUB$\{2, 3\} = 6$, LUB$\{2, 12\} = 12$, LUB$\{3, 6\} = 6$, LUB$\{5, 7\} = 35$, LUB$\{7, 11\} = 385$, LUB$\{11, 35\} = 385$;
GLB$\{3, 6\} = 3$, GLB$\{2, 12\} = 2$, GLB$\{6, 12\} = 6$, GLB$\{35, 385\} = 35$; GLB$\{2, 3\}$ **does not exist** (nothing in A lies below both 2 and 3, since $1 \notin A$).

4B.6 – GLBs and LUBs by paths

(i) GLB$\{b, c\} = a$ (ii) GLB$\{b, w\} = a$ (iii) GLB$\{e, x\} = c$ (below e: b, c, d, a; below x: v, c, a; common: c, a)
(iv) LUB$\{c, b\} = e$ (v) LUB$\{d, x\} = z$ (vi) LUB$\{c, e\} = e$ (as $c \prec e$) (vii) LUB$\{a, v\} = v$ (as $a \prec v$).

4B.7 – Upper and lower bounds of subsets

(a) $B = \{3, 4, 5\}$: (i) upper bounds: **5** only (6 is not above 5); (ii) lower bounds: **1, 2**; (iii) LUB $= 5$, GLB $= 2$.
(b) $B = \{2, 3, 4\}$: since $4 \prec 3 \prec 2$, B is a chain. (i) upper bounds: **2**; (ii) lower bounds: **4, 6, 8**; (iii) LUB $= 2$, GLB $= 4$.
(c) $B = \{6, 7, 10\}$: (i) upper bounds: **10, 11**; (ii) lower bounds: **1, 4**; (iii) LUB $= 10$, GLB $= 4$.

4B.8 – Lexicographic order on $\{0,1,2\}^2$

Reflexive: $a = a$ and $b \leq b$. Antisymmetric: if $(a,b)\,R\,(c,d)$ and $(c,d)\,R\,(a,b)$, then $a < c$ and $c < a$ is impossible, so $a = c$, and then $b \leq d \leq b$ gives $b = d$. Transitive: if $(a,b)\,R\,(c,d)\,R\,(e,f)$ then either $a < e$, or $a = c = e$ and $b \leq d \leq f$. So R is a partial order — in fact a total order: $(0,0) < (0,1) < (0,2) < (1,0) < \cdots < (2,2)$.

Conclusion. $(0,0)$ is the minimal and least element; $(2,2)$ is the maximal and greatest element.

4B.9 – $(\mathcal{P}(\{1,2,3\}), \subseteq)$ is a lattice

For any $X, Y \subseteq S$, $X \cup Y$ is an upper bound of $\{X,Y\}$ and is contained in every upper bound, so $\text{LUB}\{X,Y\} = X \cup Y \in \mathcal{P}(S)$; similarly $\text{GLB}\{X,Y\} = X \cap Y \in \mathcal{P}(S)$. Hence it is a lattice (the Hasse diagram is the cube of Section 4.2).

4B.10 – Two lattices

(a) For comparable $x \preceq y$, LUB $= y$ and GLB $= x$. The only incomparable pair is $\{2,4\}$, with LUB 3 and GLB 1.
(b) For comparable pairs the same remark applies. Any two of $2, 3, 4$ are incomparable, with LUB 5 and GLB 1. So every pair has a LUB and a GLB in both posets: both are lattices.

4B.11 – D_{36} is a lattice

For $a, b \in D_{36}$, $\text{lcm}(a,b)$ and $\gcd(a,b)$ divide 36, so they lie in D_{36}, and they are the LUB and GLB of $\{a,b\}$ under divisibility. Hence D_{36} is a lattice. (The Hasse diagram of D_{36} is the 3×3 grid drawn in Section 4.2.)

4B.12 – The poset of 4B.5 is not a lattice

$\text{GLB}\{2,3\}$ does not exist (4B.5); also, e.g., $\text{LUB}\{12, 385\}$ does not exist, since no element of A lies above both. So the poset is not a lattice.

5 Equivalence Relation

Definition: Equivalence relation

A relation R on a set A is an **equivalence relation** on A if (i) R is reflexive, (ii) R is symmetric and (iii) R is transitive on A.

A trivial example is "is equal to" on $\mathbb{R}$. "Is less than" on $\mathbb{R}$ is not an equivalence relation (it is neither reflexive nor symmetric).

Examination tip

To prove a relation is an equivalence relation, first rewrite the condition in the form "$f(a) = f(b)$" (same remainder, same parity, same difference, same ratio). Then all three properties follow from those of $=$, and the classes are the sets on which f is constant.

Example 5.1 – Verifying an equivalence relation

Let $A = \{1,2,3,4\}$ and $R = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,3),(3,3),(4,4)\}$. Verify that R is an equivalence relation.

Solution. *Reflexive:* $(1,1),(2,2),(3,3),(4,4) \in R$. *Symmetric:* the only off-diagonal pairs are $(1,2)$,

$(2,1)$ and $(3,4),(4,3)$, which come in reverse pairs. *Transitive:* $(1,2),(2,1)$ give $(1,1) \in R$; $(2,1),(1,2)$ give $(2,2) \in R$; $(3,4),(4,3)$ give $(3,3)$; $(4,3),(3,4)$ give $(4,4)$; paths through a loop end in pairs already in R. Hence R is an equivalence relation.

Example 5.2 – A relation that fails

Let $A = \{1,2,3,4\}$ and $R = \{(1,1),(1,2),(2,1),(2,2),(3,1),(3,3),(1,3),(4,1),(4,4)\}$. Is R an equivalence relation?

Solution. $(a,a) \in R$ for every $a \in A$, so R is reflexive. But $(4,1) \in R$ and $(1,4) \notin R$, so R is not symmetric. Hence R is **not** an equivalence relation (there is no need to check transitivity).

Example 5.3 – Difference a multiple of 5

Let $A = \{1,2,\ldots,12\}$ and $(x,y) \in R$ iff $x-y$ is a multiple of 5. Verify that R is an equivalence relation.

Solution. For any $x \in A$, $x-x = 0 = 5\cdot 0$, so $(x,x) \in R$ (reflexive). If $(x,y) \in R$, then $x-y = 5k$ for some integer k, so $y-x = 5(-k)$ and $(y,x) \in R$ (symmetric). If $(x,y),(y,z) \in R$, then $x-y = 5k_1$ and $y-z = 5k_2$, so $x-z = (x-y)+(y-z) = 5(k_1+k_2)$ and $(x,z) \in R$ (transitive). Thus R is an equivalence relation.

Example 5.4 – Overlapping sets

Let $A = A_1 \cup A_2 \cup A_3$ with $A_1 = \{1,2\}$, $A_2 = \{2,3,4\}$, $A_3 = \{5\}$, and define $x\,R\,y$ iff x and y lie in the same set A_i. Is R an equivalence relation?

Solution. $x\,R\,x$ for every x, since x and x lie in the same A_i (reflexive). If $x,y \in A_i$ then $y,x \in A_i$ (symmetric). But $(1,2) \in R$ (both in A_1) and $(2,3) \in R$ (both in A_2), while $(1,3) \notin R$ (1 and 3 are in no common A_i). So R is **not** transitive and not an equivalence relation. (The trouble is that A_1 and A_2 overlap — compare with partitions below.)

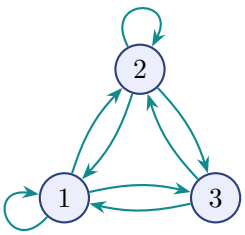
Example 5.5 – From a matrix

A relation R on $A = \{a,b,c\}$ has matrix $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Is R an equivalence relation?

Solution. $R = \{(a,a),(a,c),(b,b),(c,c)\}$. Since $(a,c) \in R$ but $(c,a) \notin R$, R is not symmetric, so it is **not** an equivalence relation. (Equivalently, M_R is not a symmetric matrix.)

Example 5.6 – From a digraph

The digraph of a relation R on $A = \{1,2,3\}$ is given below. Is R an equivalence relation?



$(3,3) \notin R$

Solution. $R = \{(1,1),(2,2),(1,2),(2,1),(1,3),(3,1),(2,3),(3,2)\}$. It is symmetric (every arrow is two-way), but it is not reflexive, since $(3,3) \notin R$. So R is **not** an equivalence relation.

Common pitfall

Do not call this R transitive: $(3,1),(1,3) \in R$ but $(3,3) \notin R$. In fact a symmetric, transitive relation in which 3 is related to anything must contain $(3,3)$.

Example 5.7 – Equal ratios

Let S be the set of all non-zero integers and $A = S \times S$. On A define $(a,b)\,R\,(c,d)$ iff $ad = bc$. Show that R is an equivalence relation.

Solution. *Reflexive:* $ab = ba$, so $(a,b)\,R\,(a,b)$. *Symmetric:* if $ad = bc$ then $cb = da$, i.e. $(c,d)\,R\,(a,b)$. *Transitive:* if $ad = bc$ and $cf = de$, multiply: $adcf = bcde$; since $c,d \neq 0$ we may cancel cd to get $af = be$, i.e. $(a,b)\,R\,(e,f)$. (When $b,d \neq 0$, $ad = bc$ just says $a/b = c/d$: this is how fractions are built from pairs of integers.)

Example 5.8 – Congruence modulo n

For a fixed integer $n > 1$, a is **congruent to** b **modulo** n, written $a \equiv b \pmod{n}$, if $a - b$ is a multiple of n, i.e. $a - b = kn$ for some $k \in \mathbb{Z}$. Prove that congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Solution. Write $a\,R\,b$ for $a \equiv b \pmod{n}$. *Reflexive:* $a - a = 0$ is a multiple of n. *Symmetric:* $a\,R\,b \Rightarrow a - b = kn \Rightarrow b - a = (-k)n \Rightarrow b\,R\,a$. *Transitive:* $a\,R\,b$ and $b\,R\,c \Rightarrow a-b$ and $b-c$ are multiples of $n \Rightarrow (a-b)+(b-c) = a-c$ is a multiple of $n \Rightarrow a\,R\,c$. There are exactly n classes $[0],[1],\ldots,[n-1]$ (the possible remainders).

Example 5.9 – New equivalence relations from old

Prove: (1) if R is an equivalence relation on A, so is R^c; (2) if R and S are equivalence relations on A, so is $R \cap S$; (3) if R and S are equivalence relations on A, $R \cup S$ need not be one.

Solution. (1) R is reflexive, symmetric and transitive, hence so is R^c (properties example in Section 3); in fact $R^c = R$ since R is symmetric.
(2) $R \cap S$ is reflexive, symmetric and transitive by the intersection example in Section 3.
(3) $R \cup S$ is reflexive and symmetric, but need not be transitive. On $A = \{1,2,3\}$ let $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ and $S = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}$, both equivalence relations. Then $(1,2),(2,3) \in R \cup S$ but $(1,3) \notin R \cup S$.

Example 5.10 – Classes on a finite set

On $A = \{1,2,3,4,5\}$, $x\,R\,y$ iff $x + y$ is even. Show R is an equivalence relation, find its classes and $|R|$.

Solution. $x + x = 2x$ is even (reflexive); $x + y = y + x$ (symmetric); if $x + y$ and $y + z$ are even then $x + z = (x + y) + (y + z) - 2y$ is even (transitive). $x\,R\,y$ exactly when x, y have the same parity, so the classes are $[1] = \{1,3,5\}$ and $[2] = \{2,4\}$. Each block of size k contributes k^2 pairs: $|R| = 3^2 + 2^2 = 13$.

5.1 Equivalence classes

Definition: Equivalence class

Let R be an equivalence relation on A and $a \in A$. The set of all elements of A related to a is the **equivalence class** of a with respect to R, denoted $R(a)$, $[a]$ or $\bar{a}$:

$$\bar{a} = [a] = R(a) = \{x \in A \mid (x,a) \in R\}.$$

- For $R = \{(1,1),(1,3),(2,2),(3,1),(3,3)\}$ on $A = \{1,2,3\}$, the elements x with $(x,1) \in R$ are 1 and 3, so $[1] = \{1,3\}$. Similarly $[2] = \{2\}$ and $[3] = \{1,3\}$.
- On $\mathbb{Z}$ let $x\,R\,y$ iff $x - y$ is even. Then $[a] = \{x \in \mathbb{Z} \mid x - a \text{ is even}\} = \{a + 2k \mid k \in \mathbb{Z}\}$. Thus $[0] = \{0, \pm 2, \pm 4, \pm 6, \dots\}$, $[1] = \{\dots, -5, -3, -1, 1, 3, 5, \dots\}$, $[2] = \{\dots, -4, -2, 0, 2, 4, \dots\} = [0]$, and so on.

Fundamental properties of equivalence classes

Let R be an equivalence relation on A and $a, b \in A$.

(1) $a \in [a]$.

(2) $a\,R\,b$ if and only if $[a] = [b]$.

(3) If $[a] \cap [b] \neq \emptyset$, then $[a] = [b]$.

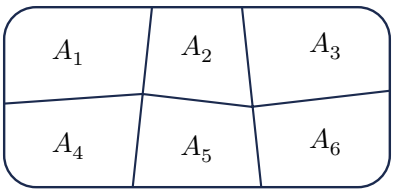
Proof. (1) R is reflexive, so $a\,R\,a$, i.e. $a \in [a]$.
(2) Suppose $a\,R\,b$ and let $x \in [a]$. Then $x\,R\,a$ and $a\,R\,b$, so $x\,R\,b$ by transitivity, i.e. $x \in [b]$. Thus $[a] \subseteq [b]$; similarly (using symmetry, $b\,R\,a$) $[b] \subseteq [a]$, so $[a] = [b]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$, i.e. $a\,R\,b$.
(3) If $x \in [a] \cap [b]$, then $x\,R\,a$ and $x\,R\,b$; by symmetry $a\,R\,x$, and by transitivity $a\,R\,b$. Hence $[a] = [b]$ by (2). ■

Note. The contrapositive of (3) reads: if $[a] \neq [b]$, then $[a] \cap [b] = \emptyset$. So *two equivalence classes are either disjoint or identical*: if they have one element in common, they have all elements in common.
Corollary. For $a, b \in A$, if $b \in [a]$ then $[b] = [a]$. (Since $b \in [b]$, $b \in [a]$ gives $[a] \cap [b] \neq \emptyset$.)

5.2 Partitions and the fundamental theorem

Definition: Partition

Let A be a non-empty set. A collection $P = \{A_1, A_2, \dots, A_k\}$ of *non-empty* subsets of A is a **partition** of A if (1) $A = A_1 \cup A_2 \cup \cdots \cup A_k$, and (2) the A_i are pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$. The sets $A_1, \dots, A_k$ are the **blocks** or **cells** of the partition. (The representation $A = A_1 \cup \cdots \cup A_k$ is also referred to as a partition of A.)



A partition of A into six blocks (cells)

For example, let $A = \{1,2,3,4,5,6,7,8\}$ with subsets $A_1 = \{1,3,5,7\}$, $A_2 = \{2,4\}$, $A_3 = \{6,8\}$. Then $A = A_1 \cup A_2 \cup A_3$ and any two of A_1, A_2, A_3 are disjoint, so $P = \{A_1, A_2, A_3\}$ is a partition of A with blocks A_1, A_2, A_3. If $A_4 = \{1,3,5\}$, then $P_1 = \{A_2, A_3, A_4\}$ is *not* a partition: the blocks are disjoint but their union is not A. If $A_5 = \{5,6,8\}$, then $P_2 = \{A_1, A_2, A_5\}$ is not a partition either: the union is A, but A_1 and A_5 are not disjoint.

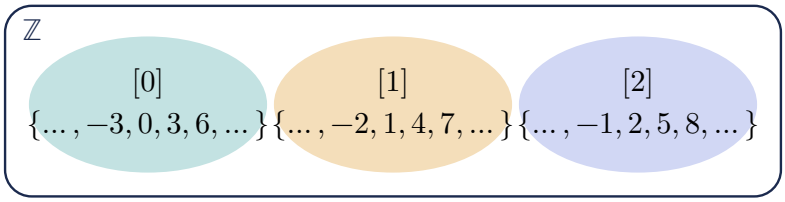
Fundamental theorem on equivalence relations

Let A be a non-empty set.

(1) Any equivalence relation R on A induces a partition of A, whose blocks are the distinct equivalence classes.

(2) Any partition of A gives rise to an equivalence relation R on A: $a\,R\,b$ iff a and b lie in the same block.

Proof. (1) Let $P = \{[a] \mid a \in A\}$, the set of all *distinct* classes. Every $a \in A$ lies in a class of P, namely $[a]$, so A is the union of the classes; and two distinct classes are disjoint by the note above. So P is a partition of A. (2) Let $P = \{A_1, \dots, A_k\}$ be a partition and define $a \, R \, b$ iff $a, b \in A_i$ for some i. Any a lies in some A_i, so $a \, R \, a$ (reflexive). If $a, b \in A_i$ then $b, a \in A_i$ (symmetric). If $a, b \in A_i$ and $b, c \in A_j$, then $b \in A_i \cap A_j$; since blocks are disjoint this forces $A_i = A_j$, so $a, c \in A_i$ and $a \, R \, c$ (transitive). ■



Congruence modulo 3 partitions $\mathbb{Z}$ into three classes.

Example 5.11 – Partition induced by R

For the equivalence relation $R = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,3),(3,3),(4,4)\}$ on $A = \{1,2,3,4\}$, determine the induced partition.

Solution. $[1] = \{1,2\}$, $[2] = \{1,2\}$, $[3] = \{3,4\}$, $[4] = \{3,4\}$. Only $[1]$ and $[3]$ are distinct, so the partition is $P = \{[1],[3]\} = \{\{1,2\},\{3,4\}\}$, and $A = [1] \cup [3] = \{1,2\} \cup \{3,4\}$.

Example 5.12 – Partition of a five-element set

Let $A = \{1,2,3,4,5\}$ and $R = \{(1,1),(2,2),(2,3),(3,2),(3,3),(4,4),(4,5),(5,4),(5,5)\}$, an equivalence relation on A. Find the partition of A induced by R.

Solution. $[1] = \{1\}$, $[2] = \{2,3\} = [3]$, $[4] = \{4,5\} = [5]$. The distinct classes are $[1], [2], [4]$, so $P = \{\{1\},\{2,3\},\{4,5\}\}$ and $A = \{1\} \cup \{2,3\} \cup \{4,5\}$.

Example 5.13 – Partition for the "multiple of 5" relation

For the set A and relation R of Example 5.3, find the partition of A induced by R.

Solution. Listing the multiples of 5 present in A, the classes are $[1] = [6] = [11] = \{1,6,11\}$, $[2] = [7] = [12] = \{2,7,12\}$, $[3] = [8] = \{3,8\}$, $[4] = [9] = \{4,9\}$, $[5] = [10] = \{5,10\}$. These five classes are distinct, so

$$A = \{1,6,11\} \cup \{2,7,12\} \cup \{3,8\} \cup \{4,9\} \cup \{5,10\}.$$

Example 5.14 – Relation induced by a partition

Let $A = \{a,b,c,d,e\}$ and consider the partition $P = \{\{a,b\},\{c,d\},\{e\}\}$ of A. Find the equivalence relation inducing this partition.

Solution. Since a, b belong to the same block, $a \, R \, a$, $a \, R \, b$, $b \, R \, a$, $b \, R \, b$; similarly $c \, R \, c$, $c \, R \, d$, $d \, R \, c$, $d \, R \, d$; and $e \, R \, e$ for the block $\{e\}$. So

$$R = \{(a,a),(a,b),(b,a),(b,b),(c,c),(c,d),(d,c),(d,d),(e,e)\}.$$

Example 5.15 – Another relation from a partition

Let $A = \{1,2,3,4,5,6,7\}$ and let R be the equivalence relation on A that induces the partition $A = \{1,2\} \cup \{3\} \cup \{4,5,7\} \cup \{6\}$. Find R.

Solution. Pair every element with every element of its own block. A block of size k gives k^2 pairs:

$$R = \{(1,1),(1,2),(2,1),(2,2),\ (3,3),\ (6,6),\\ (4,4),(4,5),(4,7),(5,4),(5,5),(5,7),(7,4),(7,5),(7,7)\},$$

with $|R| = 4 + 1 + 9 + 1 = 15$.

Example 5.16 – Pairs with equal sums

Let $A = \{1, 2, 3, 4, 5\}$ and on $A \times A$ define $(a, b)\, R\, (c, d)$ iff $a + b = c + d$. Verify that R is an equivalence relation, list its classes and give the partition of $A \times A$ induced by R.

Solution. R has the form "$f(a, b) = f(c, d)$" with $f(a, b) = a + b$, so it is reflexive, symmetric and transitive. The possible sums are $2, 3, \ldots, 10$, one class for each:

Sum	Class	Elements
2	$[(1,1)]$	$\{(1,1)\}$
3	$[(1,2)]$	$\{(1,2),(2,1)\}$
4	$[(1,3)]$	$\{(1,3),(2,2),(3,1)\}$
5	$[(1,4)]$	$\{(1,4),(2,3),(3,2),(4,1)\}$
6	$[(1,5)]$	$\{(1,5),(2,4),(3,3),(4,2),(5,1)\}$
7	$[(2,5)]$	$\{(2,5),(3,4),(4,3),(5,2)\}$
8	$[(3,5)]$	$\{(3,5),(4,4),(5,3)\}$
9	$[(4,5)]$	$\{(4,5),(5,4)\}$
10	$[(5,5)]$	$\{(5,5)\}$

These nine classes are the only distinct classes, so the induced partition is

$$A \times A = [(1,1)] \cup [(1,2)] \cup [(1,3)] \cup [(1,4)] \cup [(1,5)] \cup [(2,5)] \cup [(3,5)] \cup [(4,5)] \cup [(5,5)].$$

Example 5.17 – Symmetric part of a preorder

Let R be an arbitrary reflexive and transitive relation on A. On A define $a\, S\, b$ iff $a\, R\, b$ and $b\, R\, a$. Prove that S is an equivalence relation.

Solution. *Reflexive:* $a\, R\, a$ for all a, so $a\, S\, a$. *Symmetric:* $a\, S\, b$ means $a\, R\, b$ and $b\, R\, a$, which is the same as $b\, R\, a$ and $a\, R\, b$, i.e. $b\, S\, a$. *Transitive:* $a\, S\, b$ and $b\, S\, c$ give $a\, R\, b$, $b\, R\, a$, $b\, R\, c$, $c\, R\, b$; by transitivity of R, $a\, R\, c$ and $c\, R\, a$, so $a\, S\, c$. ■

Example 5.18 – Counting equivalence relations

Find the number of equivalence relations that can be defined on a finite set A with $|A| = 6$.

Solution. By the fundamental theorem, equivalence relations on A correspond one-to-one to partitions of A. A partition of a 6-element set has at most 6 cells. Treating the elements of A as objects and the cells as identical containers, the number of partitions with exactly k cells is the Stirling number of the second kind $S(6, k)$ (Section 6), where $S(m, n) = \frac{1}{n!}\sum_{k=0}^{n}(-1)^k\binom{n}{k}(n-k)^m$:

$$S(6,1) = 1, \qquad S(6,2) = \tfrac{1}{2}(2^6 - 2) = 31, \qquad S(6,3) = \tfrac{1}{6}(3^6 - 3\cdot 2^6 + 3\cdot 1^6) = 90,$$
$$S(6,4) = \tfrac{1}{24}(4^6 - 4\cdot 3^6 + 6\cdot 2^6 - 4\cdot 1^6) = 65,$$
$$S(6,5) = \tfrac{1}{120}(5^6 - 5\cdot 4^6 + 10\cdot 3^6 - 10\cdot 2^6 + 5\cdot 1^6) = 15, \qquad S(6,6) = 1.$$

So the number of partitions of A is $p(6) = 1+31+90+65+15+1 = \mathbf{203}$, and 203 equivalence relations can be defined on A. (These totals $1, 2, 5, 15, 52, 203, \ldots$ for $|A| = 1, 2, 3, \ldots$ are the *Bell numbers*.)

5.3 Exercise Set 5A — equivalence relations and partitions

5A.1 Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3)\}$. Verify that R is an equivalence relation.

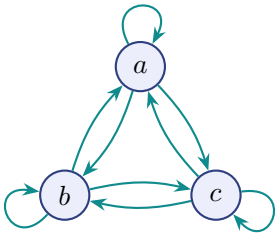
5A.2 Let $A = \{1, 2, 3, 4\}$, $R_1 = \{(1, 1), (2, 1), (2, 2), (3, 3), (4, 4), (4, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 3), (1, 3), (4, 1), (4, 4)\}$. Verify that R_1 and R_2 are not equivalence relations.

5A.3 The matrix of a relation R on $A = \{1, 2, 3\}$ is

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Show that R is an equivalence relation.

5A.4 A relation R on $\{a, b, c\}$ is represented by the digraph below. Show that R is an equivalence relation.



$R = A \times A$

5A.5 On $\mathbb{Z}$ let $a\, R\, b$ iff $a \leq b$. Is R an equivalence relation?

5A.6 On $\mathbb{Z}$ define $(a, b) \in R$ iff $a^2 - b^2$ is an even integer. Show that R is an equivalence relation.

5A.7 On $\mathbb{Z}$ define $x\, R\, y$ iff $4 \mid (x - y)$. Verify that R is an equivalence relation and find the equivalence class of any integer i.

5A.8 Let $\mathbb{N}$ be the set of natural numbers. On $\mathbb{N} \times \mathbb{N}$ define $(a, b)\, R\, (c, d)$ iff $a + d = b + c$. Show that R is an equivalence relation and find the equivalence class of $(2, 5)$.

5A.9 If $A = \{1, 2, \ldots, 10\}$, prove that in each case the given subsets form a partition of A: (a) $A_1 = \{1, 2, 3, 4, 5\}$, $A_2 = \{6, 7, 8, 9, 10\}$; (b) $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 6, 7, 9\}$, $A_3 = \{5, 8, 10\}$; (c) $A_i = \{i, i + 5\}$, $i = 1, 2, 3, 4, 5$.

5A.10 Let $A = \{a, b, c, d, e, f, g, h\}$, $A_1 = \{d, e\}$, $A_2 = \{a, c, d\}$, $A_3 = \{f, h\}$, $A_4 = \{b, g\}$. Determine whether $\{A_1, A_2, A_3\}$ is a partition of A.

5A.11 Let $A = \{1, 2, \ldots, 10\}$, $A_1 = \{1, 2, 3, 4\}$, $A_2 = \{5, 6, 7\}$, $A_3 = \{4, 5, 7, 9\}$, $A_4 = \{4, 8, 10\}$, $A_5 = \{8, 9, 10\}$, $A_6 = \{1, 2, 3, 6, 8, 10\}$. Which of the following are partitions of A? (a) $\{A_1, A_2, A_5\}$ (b) $\{A_1, A_3, A_5\}$ (c) $\{A_3, A_6\}$ (d) $\{A_2, A_3, A_4\}$.

5A.12 If $A_i = [i, i + 1)$ for $i \in \mathbb{Z}$, prove that $\{A_i\}$ is a partition of $\mathbb{R}$.

5A.13 On $A = \{1, 2, 3, 4\}$ verify that $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$ is an equivalence relation, and find the corresponding partition.

5A.14 If $A = \{1, 2, 3, 4, 5\}$ and R is the equivalence relation on A that induces the partition $A = \{1, 2\} \cup \{3, 4\} \cup \{5\}$, find R.

5A.15 For $A = \{1, 2, 3, 4, 5, 6\}$ consider the partition $P = \{A_1, A_2\}$ with $A_1 = \{1, 3, 5\}$, $A_2 = \{2, 4, 6\}$. Determine the corresponding equivalence relation.

5A.16 On the set of non-zero integers define $a\, R\, b$ iff $ab > 0$. Show that R is an equivalence relation and that the corresponding partition consists of exactly two infinite sets.

5A.17 Consider the set $A = \{(-4, -20), (-3, -9), (-2, -4), (-1, -11), (-1, -3), (1, 2), (1, 5), (2, 10), (2, 14), (3, 6), (4, 8), (4, 12)\}$, and define $(a, b)\, R\, (c, d)$ iff $ad = bc$. (i) Verify that R is an equivalence relation on A. (ii) Find $[(2, 14)]$, $[(-3, -9)]$ and $[(4, 8)]$. (iii) Find the partition of A induced by R.

5A.18 Let $A = \{1, 2, \dots, 20\}$ and let R be the equivalence relation $a\,R\,b$ iff $a - b$ is divisible by 5. Find the partition of A induced by R.

5A.19 If R and S are equivalence relations on A, prove or disprove that $R \circ S$ is an equivalence relation on A.

5A.20 If R is an equivalence relation on A with $|A| = n$, prove that (i) $n \le |R| \le n^2$ and (ii) $|R| - n$ is even.

5A.21 Given that a relation R on A is symmetric and transitive, prove that R is an equivalence relation if and only if for every $a \in A$ there exists $b \in A$ with $(a, b) \in R$.

5A.22 Let R be a reflexive relation on A. Show that R is an equivalence relation if and only if $(a, b), (a, c) \in R$ imply $(b, c) \in R$.

5A.23 Find the number of equivalence relations that can be defined on a finite set A with $|A| = 5$.

5A.24 Let $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{x, y, z\}$ and let $f : A \to B$ be the onto function $f = \{(1, x), (2, z), (3, x), (4, y), (5, z), (6, y), (7, x)\}$. On A define $a\,R\,b$ iff $f(a) = f(b)$. Prove that R is an equivalence relation and find the partition of A induced by R.

5.4 Solutions to Exercise Set 5A

5A.1 – The diagonal relation

$R = \Delta_A$: it contains every (a, a) (reflexive); its pairs are their own reverses (symmetric); and every two-step path $a \to a \to a$ ends in $(a, a) \in R$ (transitive). Its classes are $\{1\}, \{2\}, \{3\}$.

5A.2 – Two failures

R_1 is not symmetric: $(2, 1) \in R_1$ but $(1, 2) \notin R_1$. R_2 is not symmetric: $(4, 1) \in R_2$ but $(1, 4) \notin R_2$. So neither is an equivalence relation.

5A.3 – Equivalence from a matrix

$R = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3)\}$. Diagonal all 1s (reflexive); $M_R = M_R^T$ (symmetric); the paths $2 \to 3 \to 2$ and $3 \to 2 \to 3$ end in $(2, 2), (3, 3) \in R$ (transitive). Classes: $\{1\}$ and $\{2, 3\}$.

5A.4 – Equivalence from a digraph

Every vertex has a loop and every pair of distinct vertices is joined in both directions, so $R = A \times A$, which is reflexive, symmetric and transitive. There is a single class $\{a, b, c\}$.

5A.5 – $\le$ on $\mathbb{Z}$

No: $1 \le 2$ but $2 \not\le 1$, so $\le$ is not symmetric.

5A.6 – $a^2 - b^2$ even

$a^2 - b^2$ is even iff a^2 and b^2 have the same parity iff a and b have the same parity. So R is "same parity", of the form $f(a) = f(b)$, and hence reflexive, symmetric and transitive. Directly: $a^2 - a^2 = 0$; $b^2 - a^2 = -(a^2 - b^2)$; $a^2 - c^2 = (a^2 - b^2) + (b^2 - c^2)$. Classes: the even and the odd integers.

5A.7 – Congruence modulo 4

This is congruence modulo 4 (Example 5.8 with $n = 4$), so it is an equivalence relation.

Conclusion. $[i] = \{x \in \mathbb{Z} \mid 4 \mid (x - i)\} = \{4k + i \mid k \in \mathbb{Z}\}$. There are four distinct classes $[0], [1], [2], [3]$.

5A.8 – $a+d=b+c$ on $\mathbb{N}\times\mathbb{N}$

$a+d=b+c \iff b-a=d-c$: pairs are related when they have the same difference "second − first". Being of the form $f(a,b)=f(c,d)$, R is an equivalence relation.

Conclusion. $[(2,5)]=\{(a,b)\mid b-a=3\}=\{(1,4),(2,5),(3,6),(4,7),(5,8),(6,9),\dots\}$.

5A.9 – Partitions of $\{1,\dots,10\}$

(a) $A_1\cup A_2=A$, $A_1\cap A_2=\emptyset$, both non-empty: a partition.
(b) $A_1\cup A_2\cup A_3=A$ and the blocks are pairwise disjoint, so it is a partition.
(c) $A_1=\{1,6\}$, $A_2=\{2,7\}$, $A_3=\{3,8\}$, $A_4=\{4,9\}$, $A_5=\{5,10\}$: non-empty, pairwise disjoint, union A — a partition.

5A.10 – Is $\{A_1,A_2,A_3\}$ a partition?

No. $A_1\cap A_2=\{d\}\neq\emptyset$, and $b,g\notin A_1\cup A_2\cup A_3$. (Even adding A_4 does not help, because of the overlap at d.)

5A.11 – Which are partitions?

(a) $\{1,2,3,4\},\{5,6,7\},\{8,9,10\}$: disjoint, union A — **yes**. (b) $A_1\cap A_3=\{4\}$ — **no**.
(c) $\{4,5,7,9\}$ and $\{1,2,3,6,8,10\}$: disjoint, union A — **yes**. (d) $A_2\cap A_3=\{5,7\}$ — **no**.

5A.12 – $\{[i,i+1)\}$ partitions $\mathbb{R}$

Each A_i is non-empty ($i\in A_i$). Every real x lies in $A_{\lfloor x\rfloor}$, since $\lfloor x\rfloor\le x<\lfloor x\rfloor+1$; so $\bigcup A_i=\mathbb{R}$. If $x\in A_i\cap A_j$, then $i=\lfloor x\rfloor=j$; so distinct blocks are disjoint. ■

5A.13 – Partition from a relation

R is Δ_A together with the two-way pair $(1,2),(2,1)$; it is reflexive, symmetric, and transitive (the paths $1\to2\to1$, $2\to1\to2$ end at loops).

Conclusion. Classes $[1]=[2]=\{1,2\}$, $[3]=\{3\}$, $[4]=\{4\}$: the partition is $\{\{1,2\},\{3\},\{4\}\}$.

5A.14 – Relation from a partition

Conclusion. $R=\{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4),(4,3),(4,4),(5,5)\}$ ($4+4+1=9$ pairs).

5A.15 – Odd and even blocks

Each block of size 3 gives $3^2=9$ pairs:

Conclusion. $R=\{(1,1),(1,3),(1,5),(2,2),(2,4),(2,6),(3,1),(3,3),(3,5),(4,2),(4,4),(4,6),(5,1),(5,3),(5,5),(6,2),(6,4),(6,6)\}$ (18 pairs).

5A.16 – $ab>0$ on non-zero integers

$ab>0$ iff a and b have the same sign. Reflexive: $a^2>0$ for $a\neq0$. Symmetric: $ab=ba$. Transitive: $ab>0$ and $bc>0$ give $ab^2c>0$, and since $b^2>0$, $ac>0$.

Conclusion. The classes are $[1]=\{1,2,3,\dots\}$ and $[-1]=\{-1,-2,-3,\dots\}$: exactly two infinite sets.

5A.17 – Equal ratios on a finite set

(i) As in the "equal ratios" example, $ad = bc$ means $b/a = d/c$ (all entries non-zero), so R is an equivalence relation.
(ii) Group by the ratio b/a: $[(2,14)] = \{(2,14)\}$ (ratio 7); $[(-3,-9)] = \{(-3,-9),(-1,-3),(4,12)\}$ (ratio 3); $[(4,8)] = \{(-2,-4),(1,2),(3,6),(4,8)\}$ (ratio 2).
(iii) The remaining classes are $[(-4,-20)] = \{(-4,-20),(1,5),(2,10)\}$ (ratio 5) and $[(-1,-11)] = \{(-1,-11)\}$ (ratio 11). So

$$A = [(-4,-20)] \cup [(-3,-9)] \cup [(-2,-4)] \cup [(-1,-11)] \cup [(2,14)].$$

5A.18 – Remainders modulo 5 in $\{1,\dots,20\}$

Conclusion. $A = \{1,6,11,16\} \cup \{2,7,12,17\} \cup \{3,8,13,18\} \cup \{4,9,14,19\} \cup \{5,10,15,20\}$.

5A.19 – Is $R \circ S$ an equivalence relation?

Not necessarily. On $A = \{1,2,3\}$ take $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ and $S = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}$. Then $R \circ S = \{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,2),(3,3)\}$, which contains $(1,3)$ but not $(3,1)$: it is not symmetric.

Remark. $R \circ S$ is always reflexive. It is an equivalence relation exactly when $R \circ S = S \circ R$.

5A.20 – Size of an equivalence relation

(i) R is reflexive, so $\Delta_A \subseteq R \subseteq A \times A$, giving $n \le |R| \le n^2$.
(ii) $R - \Delta_A$ consists of pairs (a,b) with $a \neq b$. By symmetry these come in couples $\{(a,b),(b,a)\}$ of two distinct pairs, so $|R - \Delta_A| = |R| - n$ is even. (Alternatively, if the classes have sizes $k_1,\dots,k_r$, then $|R| - n = \sum(k_i^2 - k_i) = \sum k_i(k_i - 1)$, a sum of even numbers.) ■

5A.21 – When symmetric + transitive is enough

If R is an equivalence relation, take $b = a$: $(a,a) \in R$. Conversely, suppose every a has some b with $(a,b) \in R$. By symmetry $(b,a) \in R$, and by transitivity $(a,b),(b,a)$ give $(a,a) \in R$. So R is reflexive, hence an equivalence relation. ■

5A.22 – A one-line criterion

($\Rightarrow$) If $(a,b),(a,c) \in R$, symmetry gives $(b,a) \in R$, and transitivity with (a,c) gives $(b,c) \in R$.
($\Leftarrow$) Assume R is reflexive and (P): $(a,b),(a,c) \in R \Rightarrow (b,c) \in R$. *Symmetric:* if $(a,b) \in R$, then using (a,b) and (a,a) in (P) gives $(b,a) \in R$. *Transitive:* if $(a,b),(b,c) \in R$, then $(b,a) \in R$ by symmetry, and (P) applied to $(b,a),(b,c)$ gives $(a,c) \in R$. ■

5A.23 – Equivalence relations on a 5-element set

The number equals the number of partitions: $p(5) = \sum_{k=1}^{5} S(5,k) = S(5,1) + S(5,2) + S(5,3) + S(5,4) + S(5,5) = 1 + 15 + 25 + 10 + 1 = \mathbf{52}$.
(For example $S(5,2) = \frac{1}{2}(2^5 - 2) = 15$ and $S(5,3) = \frac{1}{6}(3^5 - 3 \cdot 2^5 + 3) = 25$.)

5A.24 – The relation $f(a) = f(b)$

R has the form $f(a) = f(b)$, so it is reflexive, symmetric and transitive. Its classes are the preimages of the elements of B: $f^{-1}(x) = \{1,3,7\}$, $f^{-1}(y) = \{4,6\}$, $f^{-1}(z) = \{2,5\}$.

Conclusion. $A = [1] \cup [4] \cup [2] = \{1, 3, 7\} \cup \{4, 6\} \cup \{2, 5\}$.

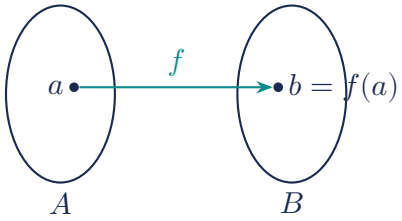
6 Functions

A function is a special relation in which every element of the first set is related to exactly one element of the second. This section covers functions in full: images and pre-images, restrictions and extensions, the types of function, counting results (including Stirling numbers), floor and ceiling, projections, binary operations, the pigeonhole principle, composition and inverses.

6.1 Definition, image and pre-image

Definition: Function

Let A and B be non-empty sets. A **function** (or **mapping**) f from A to B is a relation from A to B such that for **each** $a \in A$ there is a **unique** $b \in B$ with $(a, b) \in f$. We write $b = f(a)$; b is the **image** (or **value**) of a, and a is a **preimage** (or **argument**) of b, under f. A function from A to B is denoted $f : A \to B$; A is its **domain**, B its **codomain**, and the set $f(A)$ of all images is its **range**.

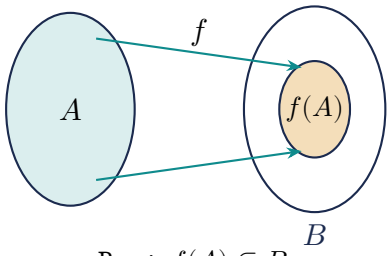


Every function is a relation, but a relation need not be a function: under a relation an element of A may be related to two different elements of B (or to none); under a function it is related to exactly one.

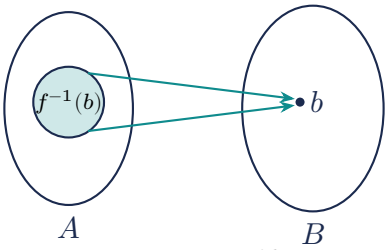
Immediate consequences of the definition

For a function $f : A \to B$:

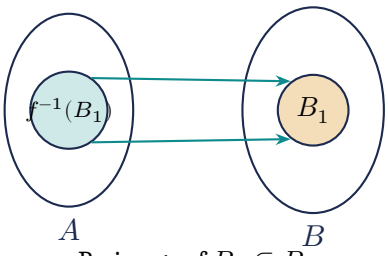
1. Every $a \in A$ belongs to some pair $(a, b) \in f$; and if $(a, b_1) \in f$ and $(a, b_2) \in f$, then $b_1 = b_2$. Every element of A has an image, and only one.
2. An element $b \in B$ *need not* have a preimage in A.
3. If b has a preimage, the preimage *need not* be unique: two different elements of A can have the same image.
4. The statements $(a, b) \in f$, afb and $b = f(a)$ are equivalent.
5. If $g : A \to B$ is also a function, then $f = g$ iff $f(a) = g(a)$ for every $a \in A$.
6. The range is $f(A) = \{f(x) \mid x \in A\} \subseteq B$.
7. For $A_1 \subseteq A$, the **image of** A_1 is $f(A_1) = \{f(x) \mid x \in A_1\}$, and $f(A_1) \subseteq f(A)$.
8. For $b \in B$, the **preimage set of** b is $f^{-1}(b) = \{x \in A \mid f(x) = b\} \subseteq A$.
9. For $B_1 \subseteq B$, the **preimage of** B_1 is $f^{-1}(B_1) = \{x \in A \mid f(x) \in B_1\} \subseteq A$.



Range $f(A) \subseteq B$



Preimage set of b



Preimage of $B_1 \subseteq B$

Common pitfall

The notation $f^{-1}(b)$ and $f^{-1}(B_1)$ stands for a *set* of preimages and is defined for *every* function. It does not mean that f has an inverse function; that is a separate idea (see *Invertible functions* below).

Laws for images and pre-images

For $f : A \to B$, $S_1, S_2 \subseteq A$ and $C, D \subseteq B$:

1. $f(S_1 \cup S_2) = f(S_1) \cup f(S_2)$, but only $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$ (equality if f is one-to-one).
2. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$ and $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.
3. $f^{-1}(\overline{C}) = \overline{f^{-1}(C)}$; $\quad S_1 \subseteq f^{-1}(f(S_1))$; $\quad f(f^{-1}(C)) \subseteq C$.

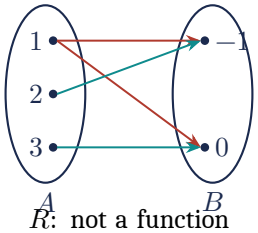
Part 1 is proved under *Properties of functions* below, and part 2 in the worked examples that follow.

6.2 Worked examples on functions, images and pre-images

Example 6.1 – A relation that is not a function

Let $A = \{1, 2, 3\}$, $B = \{-1, 0\}$ and $R = \{(1, -1), (1, 0), (2, -1), (3, 0)\}$, a relation from A to B. Is R a function from A to B?

Solution. From the arrow diagram, the element 1 of A is related to two different elements, -1 and 0, of B. Therefore R is **not** a function.

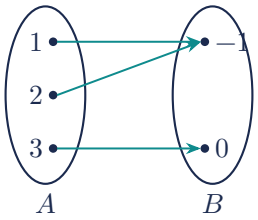


R: not a function
(1 has two images)

Example 6.2 – A relation that is a function

With A and B as above, let $S = \{(1, -1), (2, -1), (3, 0)\}$. Is S a function?

Solution. Each element of A is related to a unique element of B: 1 only to -1, 2 only to -1, 3 only to 0. So S **is** a function from A to B, and its range is $\{-1, 0\} = B$.



S: a function

Example 6.3 – Three relations on a set

Let $A = \{1, 2, 3, 4\}$. Determine whether the following relations on A are functions: (1) $f = \{(2, 3), (1, 4), (2, 1), (3, 2), (4, 4)\}$ (2) $g = \{(3, 1), (4, 2), (1, 1)\}$ (3) $h = \{(2, 1), (3, 4), (1, 4), (2, 1), (4, 4)\}$.

Solution. (1) $(2, 3) \in f$ and $(2, 1) \in f$: the element 2 is related to two different elements, so f is **not** a function.
(2) The element 2 is not related to any element under g, so g is **not** a function.
(3) Every element of A is related to a unique element of A, so h **is** a function from A to A, with range $h(A) = \{1, 4\}$. (The pair $(2, 1)$ is listed twice; in a set, repetition has no significance.)

Example 6.4 – Finding a range

Let $A = \{0, \pm 1, \pm 2, 3\}$ and $f : A \to \mathbb{R}$, $f(x) = x^3 - 2x^2 + 3x + 1$. Find the range of f.

Solution. $f(0) = 1$, $f(1) = 3$, $f(-1) = -5$, $f(2) = 7$, $f(-2) = -21$, $f(3) = 19$. Hence $f(A) = \{1, 3, -5, 7, -21, 19\}$.

Example 6.5 – Images of subsets

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + 1$. Find the images of (i) $A_1 = \{2, 3\}$ (ii) $A_2 = \{-2, 0, 3\}$ (iii) $A_3 = (0, 1)$ (iv) $A_4 = [-6, 3]$.

Solution. (i) $f(2) = 5$, $f(3) = 10$: $f(A_1) = \{5, 10\}$.
(ii) $f(-2) = 5$, $f(0) = 1$, $f(3) = 10$: $f(A_2) = \{5, 1, 10\}$.
(iii) $f(A_3) = \{x^2 + 1 \mid 0 < x < 1\}$. As x runs over $(0, 1)$, x^2 runs over $(0, 1)$, so $f(A_3) = (1, 2)$.
(iv) $f(A_4) = \{x^2 + 1 \mid -6 \le x \le 3\}$. On $[-6, 3]$, x^2 takes every value from 0 (at $x = 0$) to 36 (at $x = -6$), so $f(A_4) = [1, 37]$.

Example 6.6 – Preimages for a finite function

Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{6, 7, 8, 9, 10\}$ and $f = \{(1, 7), (2, 7), (3, 8), (4, 6), (5, 9), (6, 9)\}$. Find $f^{-1}(6)$ and $f^{-1}(9)$. If $B_1 = \{7, 8\}$ and $B_2 = \{8, 9, 10\}$, find $f^{-1}(B_1)$ and $f^{-1}(B_2)$.

Solution. $f^{-1}(6) = \{x \in A \mid f(x) = 6\} = \{4\}$ and $f^{-1}(9) = \{5, 6\}$.
$f(x) \in B_1$ when $f(x) = 7$ ($x = 1, 2$) or $f(x) = 8$ ($x = 3$): $f^{-1}(B_1) = \{1, 2, 3\}$. Similarly $f^{-1}(B_2) = \{3, 5, 6\}$ (nothing maps to 10).

Example 6.7 – Preimages for a piecewise function

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3x - 5$ for $x > 0$ and $f(x) = -3x + 1$ for $x \le 0$. (i) Determine $f(0)$, $f(-1)$, $f(5/3)$, $f(-5/3)$. (ii) Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(-3)$, $f^{-1}(-6)$. (iii) What are $f^{-1}([-5, 5])$ and $f^{-1}([-6, 5])$?

Solution. (i) $f(0) = -3(0) + 1 = 1$; $f(-1) = 4$; $f(5/3) = 3(5/3) - 5 = 0$; $f(-5/3) = -3(-5/3) + 1 = 6$.
(ii) Solve $f(x) = c$ separately on each piece and keep only the solutions lying in that piece:

c	$3x-5=c$, need $x>0$	$-3x+1=c$, need $x\le 0$	$f^{-1}(c)$
0	$x=5/3$ ✓	$x=1/3$ ×	$\{5/3\}$
1	$x=2$ ✓	$x=0$ ✓	$\{2,0\}$
-1	$x=4/3$ ✓	$x=2/3$ ×	$\{4/3\}$
3	$x=8/3$ ✓	$x=-2/3$ ✓	$\{8/3,-2/3\}$
-3	$x=2/3$ ✓	$x=4/3$ ×	$\{2/3\}$
-6	$x=-1/3$ ×	$x=7/3$ ×	$\emptyset$

(iii) For $x>0$: $-5\le 3x-5\le 5 \iff 0\le 3x\le 10$, i.e. $0<x\le 10/3$. For $x\le 0$: $-5\le -3x+1\le 5 \iff -6\le -3x\le 4 \iff -4/3\le x\le 2$, i.e. $-4/3\le x\le 0$. Hence

$$f^{-1}([-5,5])=[-4/3,0]\cup(0,10/3]=[-4/3,10/3].$$

For $[-6,5]$: when $x>0$, $3x-5>-5\ge -6$ automatically, and when $x\le 0$, $-3x+1\ge 1$; so the lower bound changes nothing and $f^{-1}([-6,5])=[-4/3,10/3]$ as well.

Example 6.8 – Pre-images of unions, intersections and complements

Let $f:A\to B$ and C,D be non-empty subsets of B. Prove (1) $f^{-1}(C\cup D)=f^{-1}(C)\cup f^{-1}(D)$ (2) $f^{-1}(C\cap D)=f^{-1}(C)\cap f^{-1}(D)$ (3) $f^{-1}(\overline{C})=\overline{f^{-1}(C)}$.

Solution. For any $x\in A$:

$$\begin{aligned}(1)\ x\in f^{-1}(C\cup D) &\iff f(x)\in C\cup D \iff f(x)\in C \text{ or } f(x)\in D\\ &\iff x\in f^{-1}(C) \text{ or } x\in f^{-1}(D) \iff x\in f^{-1}(C)\cup f^{-1}(D).\\ (2)\ x\in f^{-1}(C\cap D) &\iff f(x)\in C \text{ and } f(x)\in D \iff x\in f^{-1}(C)\cap f^{-1}(D).\\ (3)\ x\in f^{-1}(\overline{C}) &\iff f(x)\in \overline{C} \iff f(x)\notin C \iff x\notin f^{-1}(C) \iff x\in \overline{f^{-1}(C)}.\end{aligned}$$

(Here $\overline{C}=B-C$ and $\overline{f^{-1}(C)}=A-f^{-1}(C)$.) ■

Example 6.9 – Image and pre-image on $\mathbb{Z}$

$f:\mathbb{Z}\to\mathbb{Z}$, $f(x)=x^2$. Find $f(\{-2,-1,0,1,3\})$ and $f^{-1}(\{1,4,5\})$.

Solution. $f(S)=\{4,1,0,1,9\}=\{0,1,4,9\}$ (repeated values listed once). $f^{-1}(\{1,4,5\})=\{x\mid x^2\in\{1,4,5\}\}=\{-2,-1,1,2\}$; no integer squares to 5.

6.3 Restriction and extension of a function

Definition: Restriction, extension

Let $\emptyset\neq A_1\subseteq A$. If $f:A\to B$ and $f_1:A_1\to B$ satisfy $f_1(x)=f(x)$ for all $x\in A_1$, then f_1 is the **restriction of** f **to** A_1 (from A), and f is an **extension of** f_1 **to** A (from A_1). A restriction is unique; extensions usually are not.

Example 6.10 – A restriction

Let $A=\{1,2,3,4,5\}$ and $f:A\to\mathbb{R}$, $f=\{(1,10),(2,13),(3,16),(4,19),(5,22)\}$. If $g:\mathbb{Q}\to\mathbb{R}$ is $g(x)=3x+7$, show that f is the restriction of g to A (from $\mathbb{Q}$).

Solution. $A\subset\mathbb{Q}$, and $g(1)=10=f(1)$, $g(2)=13=f(2)$, $g(3)=16=f(3)$, $g(4)=19=f(4)$, $g(5)=22=f(5)$. Thus $f(x)=g(x)$ for all $x\in A$, so f is the restriction of g to A.

Example 6.11 – An extension

Let $A = \{1, 2, 3, 4, 5\}$ and $f = \{(1, 10), (2, 15), (3, 20), (4, 25), (5, 30)\}$. If $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 5x+5$, show that g is an extension of f to $\mathbb{R}$ (from A).

Solution. $A \subset \mathbb{R}$ and $g(1) = 10$, $g(2) = 15$, $g(3) = 20$, $g(4) = 25$, $g(5) = 30$, i.e. $g(x) = f(x)$ for all $x \in A$. So g extends f.

Example 6.12 – Counting extensions

Let $A = \{1, 2, 3, 4, 5\}$, $B = \{w, x, y, z\}$, $A_1 = \{2, 3, 5\}$ and $g : A_1 \to B$. In how many ways can g be extended to a function $f : A \to B$?

Solution. An extension has the form $f = \{(1, b), (2, b_1), (3, b_2), (4, b'), (5, b_3)\}$, where $b_1 = g(2)$, $b_2 = g(3)$, $b_3 = g(5)$ are fixed by g, and the images of 1 and 4 may each be any of the 4 elements of B. Hence there are $4 \times 4 = 4^2 = 16$ extensions.

Example 6.13 – Counting all functions

(a) If $|A| = m$ and $|B| = n$, how many functions are there from A to B? (b) If there are 2187 functions from A to B and $|B| = 3$, find $|A|$.

Solution. (a) Let $A = \{a_1, \dots, a_m\}$, $B = \{b_1, \dots, b_n\}$. A function is $f = \{(a_1, x_1), (a_2, x_2), \dots, (a_m, x_m)\}$ where each x_i is one of the n elements of B, chosen independently. So there are $n \times n \times \cdots \times n = n^m = |B|^{|A|}$ functions.
(b) $3^m = 2187 = 3^7$, so $m = 7$; i.e. $|A| = 7$.

6.4 Sum and product of functions

Definition: Sum and product

For $f, g : A \to \mathbb{R}$: $(f+g)(x) = f(x)+g(x)$ and $(fg)(x) = f(x)\,g(x)$ for every $x \in A$. Both are again functions $A \to \mathbb{R}$; they are defined point by point.

Example 6.14 – Sum and product — formula and table

(a) $f(x) = x^2$, $g(x) = 3x - 1$ on $\mathbb{R}$. (b) On $A = \{1, 2, 3\}$, $f = \{(1, 2), (2, 5), (3, 1)\}$, $g = \{(1, 4), (2, -1), (3, 3)\}$.

Solution. (a) $(f + g)(x) = x^2 + 3x - 1$; $(fg)(x) = 3x^3 - x^2$; at $x = 2$: 9 and 20.
(b) $f + g = \{(1, 6), (2, 4), (3, 4)\}$; $fg = \{(1, 8), (2, -5), (3, 3)\}$.

Common pitfall

fg (product, multiply values) is not $f \circ g$ (composition, substitute). For $f(x) = x^2$, $g(x) = 3x - 1$: $(fg)(x) = x^2(3x - 1)$ but $(f \circ g)(x) = (3x - 1)^2$.

6.5 Exercise Set 6A — functions, images, pre-images

6A.1 Consider the following relations on $A = \{1, 2, 3\}$: $f = \{(1, 3), (2, 3), (3, 1)\}$, $g = \{(1, 2), (3, 1)\}$, $h = \{(1, 3), (2, 1), (1, 2), (3, 1)\}$. Which of these are functions?

6A.2 Let $X = \{1, 2, 3, 4\}$ and $Y = \{a, b, c\}$. Determine whether each relation R from X to Y is a function; if so, give its range. (i) $R = \{(1, a), (2, b), (3, a), (4, b)\}$ (ii) $R = \{(1, a), (2, b), (1, b), (3, a), (4, b)\}$ (iii) $R = \{(1, c), (2, b), (3, a)\}$ (iv) $R = \{(1, a), (2, a), (3, a), (4, a)\}$.

6A.3 Determine whether the following relations are functions; if so, find the range. (i) $\{(x, y) \mid x, y \in$

$\mathbb{Z},\ y = 3x + 1\}$ (ii) $\{(x, y) \mid x, y \in \mathbb{Z},\ y = x^2 + 3\}$ (iii) $\{(x, y) \mid x, y \in \mathbb{R},\ y^2 = x\}$ (iv) $\{(x, y) \mid x, y \in \mathbb{Q},\ x^2 + y^2 = 1\}$.

6A.4 If R is a relation from A to B with $|A| = 5$, $|B| = 6$ and $|R| = 6$, prove that R is not a function.

6A.5 If $A = \{0, \pm 1, \pm 2\}$ and $f : A \to \mathbb{R}$, $f(x) = x^2 - x + 1$, find the range of f.

6A.6 If $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, what is the range of f? What is $f(\mathbb{Z})$? What is $f([-2, 1])$?

6A.7 For $A = \{1, 2, 3, 4, 5\}$, $B = \{w, x, y, z\}$ and $f = \{(1, w), (2, x), (3, x), (4, y), (5, y)\}$, find the images of $A_1 = \{1\}$, $A_2 = \{2, 3\}$, $A_3 = \{1, 2, 3\}$ and $A_4 = \{2, 3, 4, 5\}$.

6A.8 For $A = \{2, 3, 4, 5, 6\}$, let $f : A \to \mathbb{R}$ be $f = \{(2, 5), (3, 10), (4, 17), (5, 26), (6, 37)\}$. If $g : \mathbb{Q} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ are $g(x) = x^2 + 1$ and $h(x) = x^2 + 1$, verify that (i) f is a restriction of both g and h; (ii) g is an extension of f and a restriction of h; (iii) h is an extension of both f and g.

6A.9 Let $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5\}$ and $f = \{(1, 3), (2, 3), (3, 4), (4, 5), (5, 4)\}$. Find $f^{-1}(3)$, $f^{-1}(4)$, $f^{-1}(B_1)$, $f^{-1}(B_2)$ where $B_1 = \{3, 4\}$, $B_2 = \{4, 5\}$.

6A.10 If $f : \mathbb{Z} \to \mathbb{R}$, $f(x) = x^2 + 5$, find $f^{-1}(\{6\})$, $f^{-1}([6, 7])$, $f^{-1}([6, 10])$, $f^{-1}([-4, 5))$, $f^{-1}([-4, 5])$ and $f^{-1}([5, \infty))$.

6A.11 If $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + 5$, find $f^{-1}([6, 7])$, $f^{-1}([6, 10])$ and $f^{-1}([5, \infty))$.

6A.12 Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x + 7$ for $x \le 0$; $f(x) = -2x + 5$ for $0 < x < 3$; $f(x) = x - 1$ for $x \ge 3$. Find $f^{-1}(-10)$, $f^{-1}(0)$, $f^{-1}(4)$, $f^{-1}(6)$, $f^{-1}(7)$, $f^{-1}(8)$. Also determine $f^{-1}([-5, -1])$, $f^{-1}([-5, 0])$, $f^{-1}([-2, 4])$, $f^{-1}((5, 10))$ and $f^{-1}([11, 17])$.

6A.13 If $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8, 9, 10, 11, 12\}$, how many functions $f : A \to B$ satisfy $f^{-1}(\{6, 7, 8\}) = \{1, 2\}$?

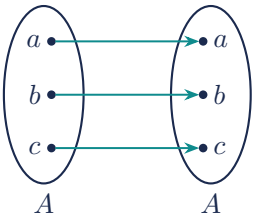
6A.14 If A is a finite set with $|A| = n$, how many functions are there from A to A?

6.6 Types of functions

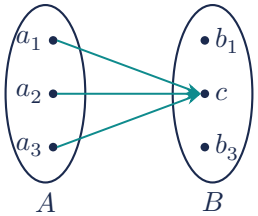
Definition: Identity, constant, onto, one-to-one, bijection

Let $f : A \to B$.

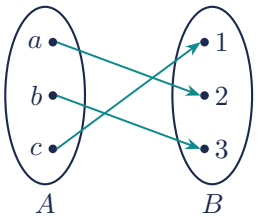
- $f : A \to A$ with $f(a) = a$ for every $a \in A$ is the **identity function** (identity mapping) on A, written I_A or 1_A. Here $f(A) = A$.
- f is a **constant function** if $f(a) = c$ for every $a \in A$, where $c \in B$ is fixed; then $f(A) = \{c\}$.
- f is **onto** (**surjective**) if for every $b \in B$ there is an $a \in A$ with $f(a) = b$, i.e. every element of B has a preimage; equivalently, the range of f equals B. We then say f maps A *onto* B.
- f is **one-to-one** (**one-one**, **1-1**, **injective**) if different elements of A have different images: $a_1 \ne a_2 \Rightarrow f(a_1) \ne f(a_2)$; equivalently (contrapositive) $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$. Then every element of $f(A)$ has a *unique* preimage.
- f is a **one-to-one correspondence** (**bijection**) if it is both one-to-one and onto: every element of A has a unique image and every element of B has a unique preimage. A bijection from a set A to itself is a **permutation** of A.



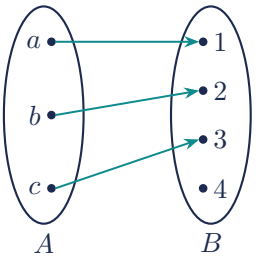
Identity I_A



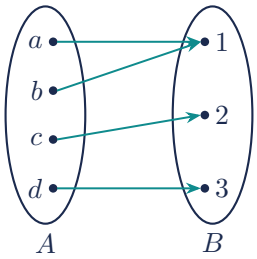
Constant: $f(A) = \{c\}$



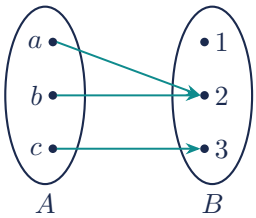
One-to-one correspondence



One-to-one, not onto



Onto, not one-to-one



Neither

Function	One-to-one?	Onto?	Reason
$f:\mathbb{R}\to\mathbb{R},\ x^2$	No	No	$f(-1)=f(1)$; -1 has no preimage
$f:\mathbb{R}\to[0,\infty),\ x^2$	No	Yes	codomain now equals range
$f:[0,\infty)\to[0,\infty),\ x^2$	Yes	Yes	restricting the domain removes $-x$
$f:\mathbb{R}\to\mathbb{R},\ x^3$	Yes	Yes	strictly increasing, unbounded both ways
$f:\mathbb{Z}\to\mathbb{Z},\ 2x+1$	Yes	No	even numbers are never reached
$f:\mathbb{R}\to\mathbb{Z},\ \lfloor x\rfloor$	No	Yes	$\lfloor 0.2\rfloor=\lfloor 0.7\rfloor=0$
$f:\mathbb{R}\to\mathbb{R},\ e^x$	Yes	No	range is $(0,\infty)$

Examination tip

To prove **one-to-one**: assume $f(x_1)=f(x_2)$ and deduce $x_1=x_2$. To disprove it, exhibit $x_1\neq x_2$ with $f(x_1)=f(x_2)$. To prove **onto**: take an arbitrary y in the *codomain*, solve $f(x)=y$, and check that the solution lies in the *domain*. To disprove it, exhibit one y with no preimage.

6.7 Worked examples on types of functions

Example 6.15 – Identity, constant, neither, permutation

Find the nature of the following functions on $A=\{1,2,3\}$: (1) $f=\{(1,1),(2,2),(3,3)\}$ (2) $g=\{(1,2),(2,2),(3,2)\}$ (3) $h=\{(1,2),(2,2),(3,1)\}$ (4) $p=\{(1,2),(2,3),(3,1)\}$.

Solution. (1) $(a,a)\in f$ for every a, i.e. $f(a)=a$: f is the **identity** function on A.
(2) $g(1)=g(2)=g(3)=2$: g is a **constant** function.
(3) h is neither identity nor constant. Its range is $\{2,1\}\subset A$; 3 has no preimage, so h is **not onto**; 1 and 2 have the same image 2, so h is **not one-to-one**.
(4) Every element has a unique image and every element has a unique preimage: p is **one-to-one and onto**, a one-to-one correspondence; being a bijection of A to itself, it is a **permutation** of A.

Example 6.16 – One-to-one but not onto

Let $A=\{1,2,3\}$, $B=\{1,2,3,4,5\}$. Determine whether the following functions from A to B are (a) one-to-one (b) onto: (i) $f=\{(1,1),(2,3),(3,4)\}$ (ii) $g=\{(1,1),(2,2),(3,3)\}$.

Solution. (i) No two elements of A have the same image, so f is **one-to-one**. The element 5 (also 2) of B has no preimage, so f is **not onto**.
(ii) g is also **one-to-one** (images 1, 2, 3 are distinct) and **not onto** (4 and 5 have no preimage). In fact no function from a 3-element set onto a 5-element set exists.

Common pitfall

Example 6.17 – One-to-one versus not one-to-one

$f, g : \mathbb{R} \to \mathbb{R}$ are $f(x) = 3x + 7$ and $g(x) = x(x^3 - 1)$. Verify that f is one-to-one but g is not.

Solution. If $f(x_1) = f(x_2)$, then $3x_1 + 7 = 3x_2 + 7$, so $x_1 = x_2$: f is one-to-one. But $g(0) = 0 = g(1)$ with $0 \neq 1$, so g is not one-to-one.

Example 6.18 – A function on $\mathbb{Z} \times \mathbb{Z}$

The function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is $f(x, y) = 2x + 3y$. Verify that f is onto but not one-to-one. If $A = \{(0, n) \mid n \in \mathbb{Z}^+\}$, find $f(A)$.

Solution. *Onto:* for any $n \in \mathbb{Z}$, $n = 4n - 3n = 2(2n) + 3(-n) = f(2n, -n)$, so $(2n, -n)$ is a preimage of n.
Not one-to-one: $f(0, 2) = 6 = f(3, 0)$, but $(0, 2) \neq (3, 0)$.
For $(0, n) \in A$, $f(0, n) = 2 \cdot 0 + 3n = 3n$, so $f(A) = \{3n \mid n \in \mathbb{Z}^+\} = \{3, 6, 9, 12, \dots\}$.

Example 6.19 – A bijection on $\mathbb{Z}$

Let $f : \mathbb{Z} \to \mathbb{Z}$, $f(a) = a + 1$. Is f one-to-one, onto, both or neither?

Solution. If $a_1 \neq a_2$ then $a_1 + 1 \neq a_2 + 1$: **one-to-one**. For any $b \in \mathbb{Z}$, $f(b - 1) = b$: **onto**. So f is a one-to-one correspondence.

Example 6.20 – Onto but not one-to-one

Let $A = \mathbb{R}$ and $B = \{x \mid x \text{ is real and } x \geq 0\}$. Is $f : A \to B$, $f(a) = a^2$, onto? one-to-one?

Solution. Take any $b \in B$. Then $b \geq 0$, so its square roots $\pm\sqrt{b}$ are real, i.e. $\pm\sqrt{b} \in A$, and $f(\sqrt{b}) = f(-\sqrt{b}) = b$. Every $b \in B$ has a preimage, so f is **onto**. A $b > 0$ has the two preimages $\pm\sqrt{b}$, so f is **not one-to-one**.

Example 6.21 – Proving a function is bijective

Show $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$, is bijective and find f^{-1}.

Solution. One-to-one: $2x_1 + 3 = 2x_2 + 3 \Rightarrow x_1 = x_2$. Onto: for any $y \in \mathbb{R}$, $x = \frac{y-3}{2} \in \mathbb{R}$ gives $f(x) = y$. So f is bijective and $f^{-1}(y) = \frac{y-3}{2}$. Check: $f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = x$.

6.8 Exercise Set 6B — types of functions

6B.1 Let $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2\}$, $D = \{d_1, d_2, d_3, d_4\}$ and let $f_1 : A \to B$, $f_2 : A \to D$, $f_3 : B \to C$, $f_4 : D \to B$ be: (i) $f_1 = \{(a_1, b_2), (a_2, b_3), (a_3, b_1)\}$ (ii) $f_2 = \{(a_1, d_2), (a_2, d_1), (a_3, d_4)\}$ (iii) $f_3 = \{(b_1, c_2), (b_2, c_2), (b_3, c_1)\}$ (iv) $f_4 = \{(d_1, b_1), (d_2, b_2), (d_3, b_1), (d_4, b_2)\}$. Find the nature of each function.

6B.2 Prove: (i) If $A = \{1, 2, 3, 4\}$, $B = \{w, x, y, z\}$, $f = \{(1, w), (2, x), (3, y), (4, z)\}$, then f is both one-to-one and onto. (ii) If $A = \{1, 2, 3, 4, 5\}$, $B = \{w, x, y, z\}$, $f = \{(1, w), (2, w), (3, x), (4, y), (5, z)\}$, then f is onto but not one-to-one. (iii) If $A = \{1, 2, 3, 4\}$, $B = \{v, w, x, y, z\}$, $f = \{(1, v), (2, x), (3, z), (4, y)\}$ and $g = \{(1, v), (2, v), (3, w), (4, x)\}$, then f is one-to-one but not onto, and g is neither one-to-one nor onto.

6B.3 Determine whether $f : A \to B$ is one-to-one or onto (or both or neither): (i) $A = B = \{1, 2, 3, 4\}$, $f = \{(1, 1), (2, 3), (3, 4), (4, 2)\}$ (ii) $A = \{a, b, c\}$, $B = \{1, 2, 3, 4\}$, $f = \{(a, 1), (b, 1), (c, 3)\}$ (iii) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $f = \{(1, 1), (2, 3), (3, 4)\}$ (iv) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $f = \{(1, 1), (2, 3), (3, 3)\}$ (v) $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $f = \{(1, a), (2, a), (3, d), (4, c)\}$ (vi) $A = \mathbb{R}$, $B = \{x \mid x \geq 0\}$, $f(a) = |a|$ (vii) $A = \mathbb{R} \times \mathbb{R}$, $B = \mathbb{R}$, $f((a, b)) = ab$ (viii) $A = B = \mathbb{R} \times \mathbb{R}$, $f((a, b)) = (a + b,\ a - b)$.

6B.4 For each $f : \mathbb{R} \to \mathbb{R}$ determine whether f is one-to-one or onto; if f is not onto, find its range: (i) $2x-3$ (ii) x^3 (iii) x^2 (iv) x^2+x (v) e^{x^2} (vi) $\sin x$.

6B.5 $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is $f((x,y)) = 4x+5y$. Prove that f is not one-to-one but is onto. Find $f(A)$ where $A = \{(n,0) \mid n \in \mathbb{Z}^+\}$.

6B.6 Let $f : A \to B$ where $A = X \cup Y$ with $X \cap Y = \emptyset$; let f_1 be the restriction of f to X and f_2 the restriction of f to Y. If f_1 and f_2 are one-to-one, does it follow that f is one-to-one?

6.9 Properties of functions

Images of subsets

Let $f : X \to Y$ and let A, B be non-empty subsets of X. Then (1) if $A \subseteq B$, then $f(A) \subseteq f(B)$; (2) $f(A \cup B) = f(A) \cup f(B)$; (3) $f(A \cap B) \subseteq f(A) \cap f(B)$, with equality if f is one-to-one.

Proof. (1) $y \in f(A) \Rightarrow y = f(x)$ for some $x \in A \subseteq B \Rightarrow y \in f(B)$.
(2) $y \in f(A \cup B) \Rightarrow y = f(x)$ with $x \in A$ or $x \in B \Rightarrow y \in f(A)$ or $y \in f(B)$; so $f(A \cup B) \subseteq f(A) \cup f(B)$. Since $A, B \subseteq A \cup B$, (1) gives $f(A), f(B) \subseteq f(A \cup B)$, so $f(A) \cup f(B) \subseteq f(A \cup B)$. Hence equality.
(3) $y \in f(A \cap B) \Rightarrow y = f(x)$ with $x \in A$ and $x \in B \Rightarrow y \in f(A) \cap f(B)$. Conversely, if f is one-to-one and $y \in f(A) \cap f(B)$, then $y = f(x_1) = f(x_2)$ with $x_1 \in A$, $x_2 \in B$; one-to-one gives $x_1 = x_2 \in A \cap B$, so $y \in f(A \cap B)$. ■
Remark. Without one-to-one, equality in (3) can fail. For $f(x) = x^2$ on $\mathbb{R}$, $A = [-1, 0]$, $B = [0, 1]$: $A \cap B = \{0\}$, so $f(A \cap B) = \{0\}$; but $f(A) = f(B) = [0, 1]$, so $f(A) \cap f(B) = [0, 1]$.

Functions between finite sets

Let A, B be finite and $f : A \to B$. Then (1) if f is one-to-one, $|A| \leq |B|$; (2) if f is onto, $|B| \leq |A|$; (3) if f is a one-to-one correspondence, $|A| = |B|$; (4) if $|A| > |B|$, then at least two different elements of A have the same image under f.

Proof. Let $A = \{a_1, \dots, a_n\}$, $B = \{b_1, \dots, b_m\}$. (1) If f is one-to-one, $f(a_1), \dots, f(a_n)$ are n different elements of B, so $m \geq n$. (2) If f is onto, each b_j has a preimage; different b's have different preimages (a function sends each a to one b), so A has at least m elements: $m \leq n$. (3) follows from (1) and (2). (4) is the contrapositive of (1). ■
Part (4) says: if n objects (pigeons) are assigned to m places (pigeonholes) and $n > m$, some place receives two objects — the **pigeonhole principle** (studied below).

One-to-one $\iff$ onto, for equal finite sizes

If A and B are finite with $|A| = |B|$ and $f : A \to B$, then f is one-to-one if and only if f is onto.

Proof. Let $A = \{a_1, \dots, a_n\}$ and $|B| = n$. If f is one-to-one, the n images $f(a_1), \dots, f(a_n)$ are distinct, so $|f(A)| = n = |B|$; as $f(A) \subseteq B$, $f(A) = B$, i.e. f is onto. Conversely, if f is onto, $B = \{f(a_1), \dots, f(a_n)\}$; since $|B| = n$, these n images must all be different, so f is one-to-one. ■
Corollary. For finite A, B with $|A| = |B|$, $f : A \to B$ is a bijection iff it is one-to-one, iff it is onto. So the four statements "f is one-to-one", "onto", "a bijection", "a one-to-one correspondence" are equivalent; to prove one of them it suffices to prove any one.

6.10 Counting one-to-one and onto functions; Stirling numbers

Counting functions between finite sets

Let $|A| = m$ and $|B| = n$.

1. Number of functions $A \to B$: n^m.
2. Number of one-to-one functions $A \to B$: 0 if $m > n$; and, if $m \le n$,
$$P(n, m) = n(n-1)(n-2)\cdots(n-m+1) = \frac{n!}{(n-m)!}.$$
3. Number of onto functions $A \to B$ ($m \ge n$; there are none if $m < n$):
$$p(m, n) = \sum_{k=0}^{n} (-1)^k \binom{n}{n-k} (n-k)^m.$$
4. Number of bijections $A \to B$: $n!$ if $m = n$, and 0 otherwise.

Proof of 2. In $f = \{(a_1, x), (a_2, x), \ldots, (a_m, x)\}$, the image of a_1 can be chosen in n ways; as f is one-to-one, that of a_2 in $n-1$ ways, of a_3 in $n-2$ ways, ..., of a_m in $n-(m-1)$ ways. Multiplying gives $n!/(n-m)!$. (Formula 3 is proved by inclusion–exclusion, Unit 2.)

Definition: Stirling numbers of the second kind

For $m \ge n$, the number $S(m, n) = \dfrac{p(m, n)}{n!} = \dfrac{1}{n!} \displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{n-k} (n-k)^m$ is the **Stirling number of the second kind**. It counts the ways of distributing m *distinct* objects into n *identical* containers with **no container empty**.

- $S(m, 1) = 1$ and $S(m, m) = 1$ for all $m \ge 1$.
- The number of ways to distribute m distinct objects into n identical containers, **empty containers allowed**, is $p(m) = \displaystyle\sum_{i=1}^{n} S(m, i)$ for $m \ge n$.
- Recurrence: $S(m+1, n) = S(m, n-1) + n\, S(m, n)$ for $1 \le n \le m$ (proved below).

$S(m,n)$	$n=1$	2	3	4	5	6
$m=1$	1					
2	1	1				
3	1	3	1			
4	1	7	6	1		
5	1	15	25	10	1	
6	1	31	90	65	15	1
7	1	63	301	350	140	21
8	1	127	966	1701	1050	266

6.11 Worked examples on properties and counting

Example 6.22 – Counting one-to-one functions

(a) Let $|A| = m$ and $|B| = n$. How many one-to-one functions are there from A to B? (b) If there are 60 one-to-one functions from A to B and $|A| = 3$, what is $|B|$?

Solution. (a) None if $m > n$; $\dfrac{n!}{(n-m)!} = P(n,m)$ if $m \le n$ (see the proof above).

(b) $\dfrac{n!}{(n-3)!} = n(n-1)(n-2) = 60 = 5 \cdot 4 \cdot 3$, so $n = 5$: $|B| = 5$.

Example 6.23 – Onto functions from a 7-set to a 4-set

Let $A = \{1,2,3,4,5,6,7\}$ and $B = \{w,x,y,z\}$. Find the number of onto functions from A to B.

Solution. $m = 7, n = 4$:

$$p(7,4) = \sum_{k=0}^{4} (-1)^k \binom{4}{4-k} (4-k)^7 = \binom{4}{4}4^7 - \binom{4}{3}3^7 + \binom{4}{2}2^7 - \binom{4}{1}1^7 + 0$$
$$= 16384 - 4 \times 2187 + 6 \times 128 - 4 = 16384 - 8748 + 768 - 4 = 8400.$$

Example 6.24 – All, one-to-one and onto functions both ways

Let $A = \{1,2,3,4\}$ and $B = \{1,2,3,4,5,6\}$. (a) How many functions are there from A to B; how many are one-to-one; how many onto? (b) The same questions from B to A.

Solution. $m = |A| = 4, n = |B| = 6$.

(a) Functions: $6^4 = 1296$. One-to-one: $\dfrac{6!}{2!} = 360$. Onto: none, since $4 < 6$.

(b) Functions: $4^6 = 4096$. One-to-one: none, since $6 > 4$. Onto: $p(6,4) = 4^6 - 4 \times 3^6 + 6 \times 2^6 - 4 = 4096 - 2916 + 384 - 4 = 1560$.

Example 6.25 – Stirling numbers from $p(m,n)$

Given $p(6,4) = 1560$ and $p(7,4) = 8400$, evaluate $S(6,4)$ and $S(7,4)$.

Solution. $S(6,4) = \dfrac{p(6,4)}{4!} = \dfrac{1560}{24} = 65$ and $S(7,4) = \dfrac{8400}{24} = 350$.

Example 6.26 – Evaluating $S(5,4)$ and $S(8,6)$

Evaluate $S(5,4)$ and $S(8,6)$ from the formula.

Solution.

$$S(5,4) = \frac{1}{4!}\left\{4^5 - \binom{4}{3}3^5 + \binom{4}{2}2^5 - \binom{4}{1}1^5\right\} = \frac{1024 - 972 + 192 - 4}{24} = \frac{240}{24} = 10,$$

$$S(8,6) = \frac{1}{6!}\left\{6^8 - \binom{6}{5}5^8 + \binom{6}{4}4^8 - \binom{6}{3}3^8 + \binom{6}{2}2^8 - \binom{6}{1}1^8\right\}$$
$$= \frac{1679616 - 6 \times 390625 + 15 \times 65536 - 20 \times 6561 + 15 \times 256 - 6}{720} = \frac{191520}{720} = 266.$$

Example 6.27 – Programmers and executives

There are six programmers who can assist eight executives. In how many ways can the executives be assisted so that each programmer assists at least one executive?

Solution. Let A be the set of 8 executives and B the set of 6 programmers. Assigning a programmer to each executive so that every programmer is used is an onto function $A \to B$. The number is $p(8,6) = 6!\,S(8,6) = 720 \times 266 = 191520$.

Example 6.28 – Distinct objects, identical containers, empty allowed

Find the number of ways of distributing four distinct objects among three identical containers, with some container(s) possibly empty.

Solution. $m = 4$, $n = 3$: $p(4) = S(4,1) + S(4,2) + S(4,3)$, where

$$S(4,1) = 1, \quad S(4,2) = \tfrac{1}{2}\{2^4 - 2 \times 1^4\} = 7, \quad S(4,3) = \tfrac{1}{6}\{3^4 - 3 \times 2^4 + 3 \times 1^4\} = 6.$$

So the required number is $1 + 7 + 6 = 14$.

Example 6.29 – The Stirling recurrence

If m, n are positive integers with $1 \le n \le m$, prove that $S(m+1, n) = S(m, n-1) + n\,S(m, n)$.

Solution. Let $A = \{a_1, \dots, a_m, a_{m+1}\}$; count the distributions of A among n identical containers, none empty, by what happens to a_{m+1}.
Case 1: a_{m+1} is alone in its container. Then $a_1, \dots, a_m$ fill the other $n-1$ containers with none empty: $S(m, n-1)$ ways.
Case 2: a_{m+1} shares its container. Then $a_1, \dots, a_m$ already fill all n containers, none empty, in $S(m,n)$ ways, and a_{m+1} can be added to any one of the n (now distinguishable) containers: $n\,S(m,n)$ ways.
Adding the two cases, $S(m+1,n) = S(m,n-1) + nS(m,n)$. ■

Example 6.30 – Using the recurrence once

Evaluate $S(8,7)$, given $S(7,6) = 21$.

Solution. $S(8,7) = S(7,6) + 7\,S(7,7) = 21 + 7 \times 1 = 28$ (as $S(7,7) = 1$).

Example 6.31 – Using the recurrence twice

Given $S(8,4) = 1701$, $S(8,5) = 1050$ and $S(8,6) = 266$, evaluate $S(10,6)$.

Solution.

$$\begin{aligned} S(10,6) &= S(9,5) + 6S(9,6) = \{S(8,4) + 5S(8,5)\} + 6\{S(8,5) + 6S(8,6)\} \\ &= S(8,4) + 11S(8,5) + 36S(8,6) \\ &= 1701 + 11 \times 1050 + 36 \times 266 = 1701 + 11550 + 9576 = 22827. \end{aligned}$$

6.12 Exercise Set 6C — counting functions

6C.1 If $A = \{2,6,9\}$ and $B = \{p,q,r,s,t\}$, find the number of one-to-one functions from A to B.

6C.2 If $A = \{1,2,3,4,5\}$ and there are 6720 one-to-one functions $f : A \to B$, what is $|B|$?

6C.3 If $A = \{w,x,y,z\}$ and $B = \{1,2,3\}$, how many onto functions are there from A to B?

6C.4 If A and B are finite sets with $|A| = 5$ and $|B| = 3$, find the number of onto functions from A to B.

6C.5 Let $A = \{1,b,c,d,e\}$ and $B = \{1,2,3,4,5,6,7,8\}$. (i) Determine the number of functions from A to

B; how many of these are one-to-one, and how many onto? (ii) The same questions for functions from B to A.

6C.6 If a set A has n elements, how many one-to-one correspondences are there from A to A?

6C.7 If $A = \{1, 2, 3, \ldots, n\}$ for some fixed $n \in \mathbb{Z}^+$, find how many bijective functions $f : A \to A$ satisfy $f(1) \neq 1$.

6C.8 Verify that $S(m, 1) = 1$ and $S(m, m) = 1$ for all $m \geq 1$.

6C.9 Prove that $S(5, 3) = 25$, $S(7, 2) = 63$ and $S(8, 5) = 1050$.

6C.10 Verify that $\displaystyle\sum_{k=0}^{6} \binom{6}{k} (k!)\, S(8, k) = 6^8$.

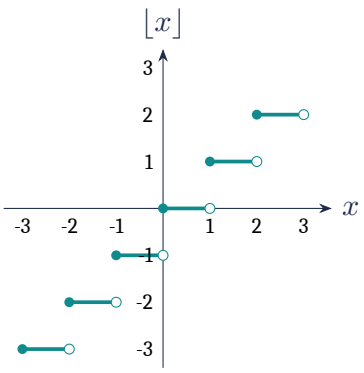
6.13 Some particular functions

Floor and ceiling functions

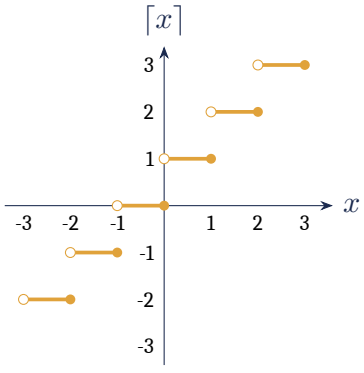
Definition: Floor, ceiling

For a real number x, $\lfloor x \rfloor$ (the **floor** of x) is the greatest integer less than or equal to x, and $\lceil x \rceil$ (the **ceiling** of x) is the least integer greater than or equal to x. The function $\mathbb{R} \to \mathbb{Z}$, $x \mapsto \lfloor x \rfloor$ is the **floor function** (greatest integer function); $x \mapsto \lceil x \rceil$ is the **ceiling function** (least integer function).

- $\lfloor x \rfloor = \lceil x \rceil = x$ if $x \in \mathbb{Z}$. If $x \in \mathbb{R} - \mathbb{Z}$, $\lfloor x \rfloor$ is the integer immediately to the left of x on the real line and $\lceil x \rceil$ the integer immediately to the right, so $\lceil x \rceil = \lfloor x \rfloor + 1$.
- Floor: $\lfloor 2 \rfloor = 2$, $\lfloor 2.1 \rfloor = 2$, $\lfloor \sqrt{2} \rfloor = 1$, $\lfloor -4 \rfloor = -4$, $\lfloor -6.3 \rfloor = -7$, $\lfloor 0 \rfloor = 0$.
- Ceiling: $\lceil 2 \rceil = 2$, $\lceil 2.1 \rceil = 3$, $\lceil \sqrt{2} \rceil = 2$, $\lceil -4 \rceil = -4$, $\lceil -6.3 \rceil = -6$, $\lceil 0 \rceil = 0$.
- Useful facts: $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$; $\lceil -x \rceil = -\lfloor x \rfloor$; $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ for $n \in \mathbb{Z}$.



Floor: closed dot = value taken



Ceiling

Example 6.32 – Fourteen evaluations

Evaluate: (i) $\lceil 3.6 \rceil + \lceil 4.2 \rceil$ (ii) $\lceil 3.6 + 4.2 \rceil$ (iii) $\lceil 3.6 \rceil - \lceil 4.2 \rceil$ (iv) $\lceil 3.6 - 4.2 \rceil$ (v) $\lfloor 3.6 \rfloor + \lfloor 4.2 \rfloor$ (vi) $\lfloor 3.6 + 4.2 \rfloor$ (vii) $\lfloor 3.6 \rfloor - \lfloor 4.2 \rfloor$ (viii) $\lfloor 3.6 - 4.2 \rfloor$ (ix) $\lfloor 3.6 \rfloor + \lceil 4.2 \rceil$ (x) $\lfloor 3.6 \rfloor - \lceil 4.2 \rceil$ (xi) $\lceil 2.8 \rceil \times \lceil 2.9 \rceil$ (xii) $\lceil 2.8 \times 2.9 \rceil$ (xiii) $\lfloor 2.8 \rfloor \times \lceil 2.9 \rceil$ (xiv) $\lfloor 2.8 \times 2.9 \rfloor$.

Solution.

Part	Value	Part	Value
(i)	$4+5=9$	(ii)	$\lceil 7.8\rceil = 8$
(iii)	$4-5=-1$	(iv)	$\lceil -0.6\rceil = 0$
(v)	$3+4=7$	(vi)	$\lfloor 7.8\rfloor = 7$
(vii)	$3-4=-1$	(viii)	$\lfloor -0.6\rfloor = -1$
(ix)	$3+5=8$	(x)	$3-5=-2$
(xi)	$3\times 3=9$	(xii)	$\lceil 8.12\rceil = 9$
(xiii)	$2\times 3=6$	(xiv)	$\lfloor 8.12\rfloor = 8$

Comparing (i)/(ii), (v)/(vi) and (xi)/(xii): the floor and ceiling of a sum or product are *not* in general the sum or product of the floors or ceilings.

Example 6.33 – Four floor/ceiling facts

Prove: (1) $\lfloor 2x\rfloor = \lfloor x\rfloor + \lfloor x+\frac{1}{2}\rfloor$ for any real x. (2) $\lfloor x\rfloor = \lceil x\rceil - 1$ for any $x \in \mathbb{R}-\mathbb{Z}$. (3) $\lfloor x\rfloor = \lceil x\rceil$ is not true for every $x \in \mathbb{R}$. (4) $-\lceil x\rceil = \lceil -x\rceil$ is not true for every $x \in \mathbb{R}$.

Solution. Write $N = \lfloor x\rfloor$, so $N \le x < N+1$.
(1) If $x \ge N+\frac{1}{2}$, then $2N+1 \le 2x < 2N+2$ and $N+1 \le x+\frac{1}{2} < N+\frac{3}{2}$, so $\lfloor 2x\rfloor = 2N+1 = N+(N+1) = \lfloor x\rfloor + \lfloor x+\frac{1}{2}\rfloor$. If $x < N+\frac{1}{2}$, then $2N \le 2x < 2N+1$ and $N \le x+\frac{1}{2} < N+1$, so $\lfloor 2x\rfloor = 2N = N+N = \lfloor x\rfloor + \lfloor x+\frac{1}{2}\rfloor$.
(2) If $x \notin \mathbb{Z}$, then $N < x < N+1$, so $\lfloor x\rfloor = N$ and $\lceil x\rceil = N+1$; hence $\lfloor x\rfloor = \lceil x\rceil - 1$.
(3) Take $x = 2.6$: $\lfloor 2.6\rfloor = 2$ but $\lceil 2.6\rceil = 3$. (The equality *does* hold when x is an integer, so "not true for every x" is the right statement.)
(4) Take $x = 1.7$: $-\lceil 1.7\rceil = -2$ but $\lceil -1.7\rceil = -1$.

Common pitfall

Note: $\lceil -x\rceil = -\lfloor x\rfloor$ *does* hold for every real x (e.g. $-\lfloor 1.7\rfloor = -1 = \lceil -1.7\rceil$). The statement that fails is $-\lceil x\rceil = \lceil -x\rceil$ (equivalently $-\lfloor x\rfloor = \lfloor -x\rfloor$).

Example 6.34 – Two identities for $a \ge 1$

If a is a positive real number with $a \ge 1$, prove (i) $\lfloor \lceil a\rceil / a\rfloor = 1$ (ii) $\lceil \lfloor a\rfloor / a\rceil = 1$.

Solution. (i) If a is a positive integer, $\lceil a\rceil = a$ and $\lfloor a/a\rfloor = \lfloor 1\rfloor = 1$. Otherwise write $a = n+c$ with $n \in \mathbb{Z}^+$ and $0 < c < 1$. Then

$$\frac{\lceil a\rceil}{a} = \frac{n+1}{n+c} = 1 + \frac{1-c}{n+c} = 1+d, \qquad 0 < d = \frac{1-c}{n+c} < 1,$$

so $\lfloor \lceil a\rceil / a\rfloor = \lfloor 1+d\rfloor = 1$.

(ii) If $a \in \mathbb{Z}^+$, $\lceil a/a\rceil = 1$. Otherwise, with $a = n+c$ as above, $\dfrac{\lfloor a\rfloor}{a} = \dfrac{n}{n+c} = 1 - \dfrac{c}{n+c}$, where $0 < \dfrac{c}{n+c} < 1$; hence $\left\lceil 1 - \dfrac{c}{n+c}\right\rceil = 1$. ■

Projections

Definition: Projection

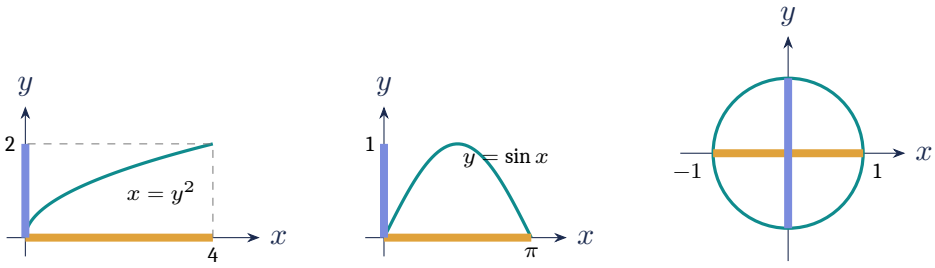
For sets A, B and $D \subseteq A \times B$, the function $\pi_A : D \to A$, $\pi_A(a,b) = a$, is the **projection of D on A**, and $\pi_B : D \to B$, $\pi_B(a,b) = b$, is the **projection of D on B**. If $D = A \times B$, both projections are onto.

Example. $A = \{a,b,c\}$, $B = \{1,2,3,4\}$, $D = \{(a,1),(a,2),(b,1),(b,2),(c,3)\}$. Then $\pi_A(a,1) = \pi_A(a,2) = a$, $\pi_A(b,1) = \pi_A(b,2) = b$, $\pi_A(c,3) = c$; and $\pi_B(a,1) = \pi_B(b,1) = 1$, $\pi_B(a,2) = \pi_B(b,2) = 2$, $\pi_B(c,3) = 3$. Every element of A has a preimage in D, so π_A is onto; $4 \in B$ has none, so π_B is not onto.

Example 6.35 – Projections of curves

Let $A = B = \mathbb{R}$. Determine $\pi_A(D)$ and $\pi_B(D)$ for (i) $D = \{(x,y) \mid x = y^2,\ 0 \le y \le 2\}$ (ii) $D = \{(x,y) \mid y = \sin x,\ 0 \le x \le \pi\}$ (iii) $D = \{(x,y) \mid x^2 + y^2 = 1\}$.

Solution. $\pi_A(x,y) = x$ and $\pi_B(x,y) = y$; geometrically, $\pi_A(D)$ is the "shadow" of D on the x-axis and $\pi_B(D)$ its shadow on the y-axis.
(i) $\pi_A(D) = \{x \mid x = y^2,\ 0 \le y \le 2\} = [0,4]$ and $\pi_B(D) = \{y \mid y^2 = x,\ 0 \le x \le 4\} = [0,2]$.
(ii) $\pi_A(D) = \{x \mid x = \sin^{-1} y,\ 0 \le y \le 1\} = [0,\pi]$ and $\pi_B(D) = \{y \mid y = \sin x,\ 0 \le x \le \pi\} = [0,1]$.
(iii) $\pi_A(D) = \{x \mid x = \pm\sqrt{1-y^2},\ -1 \le y \le 1\} = [-1,1]$ and $\pi_B(D) = \{y \mid y = \pm\sqrt{1-x^2},\ -1 \le x \le 1\} = [-1,1]$.



(i) (ii) (iii)

Amber: $\pi_A(D)$ on the x-axis; blue: $\pi_B(D)$ on the y-axis.

Unary and binary operations

Definition: Unary and binary operations

For a non-empty set A, a function $f : A \to A$ is a **unary** (monary) operation on A; e.g. $x \mapsto x^n$, e^x, $\sin x$ on $\mathbb{R}$. A function $f : A \times A \to B$ is a **binary operation** on A; if $B = A$ it is **closed** on A. A binary operation f is

1. **commutative** if $f(a,b) = f(b,a)$ for all $(a,b) \in A \times A$;
2. **associative** (when $B \subseteq A$) if $f(a, f(b,c)) = f(f(a,b), c)$ for all $a,b,c \in A$;

and $x \in A$ is an **identity** (identity element) for f if $f(a,x) = f(x,a) = a$ for all $a \in A$.

Examples. $f(a,b) = a+b$ on $\mathbb{Z}$ is closed, commutative ($b+a = a+b$), associative ($a+(b+c) = (a+b)+c$), with identity 0 ($0 + a = a + 0 = a$). By contrast $g(a,b) = a - b$ on $\mathbb{Z}$ is closed but *not* commutative ($a - b \neq b - a$ in general), *not* associative ($a - (b - c) \neq (a - b) - c$ in general) and has *no identity*: $g(a,x) = a$ and $g(x,a) = a$ would need $a - x = a$ and $x - a = a$ for all a, which is impossible.

Example 6.36 – Two operations on $\mathbb{Z}$

Determine whether the following closed binary operations on $\mathbb{Z}$ are (i) commutative (ii) associative: (1) $f(a,b) = a + b - 3ab$ (2) $g(a,b) = a|b|$.

Solution. (1) $f(b,a) = b + a - 3ba = f(a,b)$: commutative. Also

$$f(a, f(b,c)) = a + (b + c - 3bc) - 3a(b + c - 3bc) = a + b + c - 3ab - 3bc - 3ac + 9abc,$$
$$f(f(a,b), c) = (a + b - 3ab) + c - 3(a + b - 3ab)c = a + b + c - 3ab - 3ac - 3bc + 9abc,$$

which are equal: associative.
(2) $g(a,b) = a|b|$ and $g(b,a) = b|a|$ differ in general: $g(2,-3) = 2 \times 3 = 6$ but $g(-3,2) = -3 \times 2 = -6$. Not commutative. But $g(a, g(b,c)) = a\,|b|c|| = a|b||c|$ and $g(g(a,b),c) = a|b|\,|c|$: associative.

Example 6.37 – Ceiling of a sum

Show that the closed binary operation $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $f(a,b) = \lceil a + b \rceil$, is commutative but not associative, and has no identity element.

Solution. $f(a,b) = \lceil a + b \rceil = \lceil b + a \rceil = f(b,a)$: commutative.
Take $a = 1.2$, $b = 2.4$, $c = 5.7$: $f(a,b) = \lceil 3.6 \rceil = 4$, $f(b,c) = \lceil 8.1 \rceil = 9$, so $f(f(a,b),c) = \lceil 9.7 \rceil = 10$ but $f(a, f(b,c)) = \lceil 10.2 \rceil = 11$: not associative.
An identity x would satisfy $\lceil a + x \rceil = a$ for all $a \in \mathbb{R}$; impossible, since $\lceil a + x \rceil$ is always an integer while a need not be.

Example 6.38 – The identity is unique

Let $f : A \times A \to B$ be a binary operation. If f has an identity, prove that it is unique.

Solution. Let x_1, x_2 be identities: $f(a, x_1) = f(x_1, a) = a$ and $f(a, x_2) = f(x_2, a) = a$ for all a. Putting $a = x_2$ in the first: $f(x_2, x_1) = f(x_1, x_2) = x_2$. Putting $a = x_1$ in the second: $f(x_1, x_2) = f(x_2, x_1) = x_1$. Hence $x_1 = x_2$. ■

Example 6.39 – Counting closed binary operations

Let $A = \{a, b, c, d\}$. Find the number of closed binary operations on A. How many are commutative?

Solution. A closed binary operation is a function $A \times A \to A$; there are $|A|^{|A \times A|} = 4^{16}$ of them.
Of the 16 ordered pairs in $A \times A$, 4 have the form (x,x) and 12 the form (x,y), $y \neq x$. For a commutative operation $f(x,y) = f(y,x)$, so only $12/2 = 6$ of the unequal pairs can be chosen freely. Each pair has 4 possible images, so the number of commutative operations is $4^4 \times 4^6 = 4^{10}$.

6.14 Exercise Set 6D — floor, ceiling, projections and operations

6D.1 Let $A = B = \mathbb{R}$ and $D = \{(x,y) \mid y = x^2\}$. Prove that the projection $\pi_A : D \to \mathbb{R}$ is one-to-one and onto, whereas $\pi_B : D \to \mathbb{R}$ is neither one-to-one nor onto.

6D.2 Compute $\lfloor 3.65 \rfloor$, $\lfloor -3.65 \rfloor$, $\lfloor -16 \rfloor$, $\lfloor 18.5 \rfloor$, $\lfloor -18.5 \rfloor$, $\lfloor \sqrt{5} \rfloor$, and $\lceil 3.65 \rceil$, $\lceil -3.65 \rceil$, $\lceil -16 \rceil$, $\lceil 18.5 \rceil$, $\lceil -18.5 \rceil$, $\lceil \sqrt{5} \rceil$.

6D.3 Evaluate (i) $\lfloor 2.3 - 1.6 \rfloor$ (ii) $\lfloor 2.3 \rfloor - \lfloor 1.6 \rfloor$ (iii) $\lceil 3.4 \rceil \lfloor 6.2 \rfloor$ (iv) $\lfloor 3.4 \rfloor \lceil 6.2 \rceil$ (v) $\lfloor 2\pi \rfloor$ (vi) $2\lceil \pi \rceil$ (take $\pi \approx 22/7$).

6D.4 If a is a specified integer, prove (i) $\lfloor x + a \rfloor = x + a$ for all $x \in \mathbb{Z}$ (ii) $\lfloor x + a \rfloor = \lfloor x \rfloor + a$ for all $x \in \mathbb{R}$.

6D.5 For $n, k \in \mathbb{Z}^+$, prove that $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.

6D.6 If $a \in \mathbb{R}^+$ and $0 < a < 1$, disprove that $\lfloor \lceil a \rceil / a \rfloor = 1$ and $\lceil \lfloor a \rfloor / a \rceil = 1$.

6D.7 Show that $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$, $f(a,b) = a - b$, is *not* closed.

6D.8 Determine whether the following binary operations f on $\mathbb{Z}$ are commutative and/or associative: (i) $f(x,y) = x + y - xy$ (ii) $f(x,y) = x + y - 1$ (iii) $f(x,y) =$ the larger of x, y (iv) $f(x,y) = x^y$.

6D.9 Let $A = \{1,2,3,4,5,6,7\}$ and $f : A \times A \to A$ be the closed binary operation $f(x,y) =$ the smaller of x, y. Show that f is commutative and associative and that 7 is the identity.

6D.10 Let $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$ be the closed binary operation $f(a,b) = \gcd(a,b)$. Show that f is commutative and associative but has no identity.

6D.11 Let A and B be sets contained in a universal set U. Show that $f : \mathcal{P}(U) \times \mathcal{P}(U) \to \mathcal{P}(U)$, $f(A,B) = A \cup B$, is a closed binary operation which is commutative and associative. Is there an identity for f?

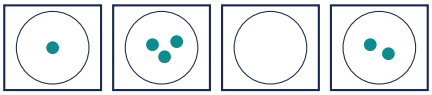
6D.12 Let $|A| = 5$. How many closed binary operations are there on A? How many of these are commutative?

6.15 The pigeonhole principle

Part (4) of the theorem on functions between finite sets gives the following principle (it is also used in the counting part of the syllabus).

Pigeonhole principle

If m pigeons occupy n pigeonholes and $m > n$, then two or more pigeons occupy the same pigeonhole. Equivalently: if $m > n$, at least one pigeonhole must contain two or more pigeons.



6 pigeons in 4 pigeonholes: some hole holds ≥ 2

A simple application: if 8 children are born in the same week, then two or more are born on the same day of the week.

Generalised pigeonhole principle

If m pigeons occupy n pigeonholes, then at least one pigeonhole contains $p + 1$ or more pigeons, where $p = \lfloor (m-1)/n \rfloor$.

Proof (by contradiction). Suppose every pigeonhole contains p or fewer pigeons. Then the total number of pigeons is at most

$$np = n\left\lfloor \frac{m-1}{n} \right\rfloor \leq n \cdot \frac{m-1}{n} = m - 1,$$

contradicting the fact that there are m pigeons. ■

Two further versions

(a) If $m = kn + 1$ pigeons ($k \geq 1$) occupy n pigeonholes, at least one pigeonhole contains $k + 1$ or more pigeons.
(b) If $m = p_1 + p_2 + \cdots + p_n - n + 1$ pigeons occupy n pigeonholes $H_1, \ldots, H_n$, then some pigeonhole H_j contains p_j or more pigeons.

These are proved in the worked examples below.

Examination tip

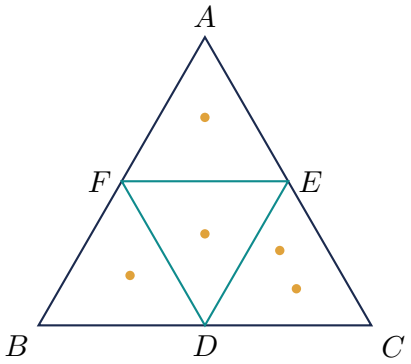
In every pigeonhole problem, write down explicitly (1) what the pigeons are, (2) what the pigeonholes are, and (3) why there are more pigeons than holes. Most marks are for setting up the holes correctly.

6.16 Worked examples on the pigeonhole principle

Example 6.40 – Five points in an equilateral triangle

ABC is an equilateral triangle whose sides are of length 1 cm. If we select 5 points inside the triangle, prove that at least two of them are at a distance less than $\frac{1}{2}$ cm.

Solution. Join the midpoints D, E, F of BC, CA, AB. This splits ABC into four equilateral triangles of side $\frac{1}{2}$ cm. Treating the four small triangles as pigeonholes and the five points as pigeons, some small triangle contains two or more points, and two points inside a triangle of side $\frac{1}{2}$ are less than $\frac{1}{2}$ cm apart.



Example 6.41 – Matching socks

A bag contains 12 pairs of socks, each pair in a different colour. If a person draws socks one by one at random, at most how many draws are needed to get at least one matched pair?

Solution. With $n \leq 12$ draws all socks could have different colours. With 13 draws (13 pigeons, 12 colours as pigeonholes) two socks must have the same colour. So at most **13** draws are required.

Example 6.42 – Strings on a tape

A magnetic tape contains 5 lakh strings, each made up of four or fewer English letters. Can all the strings be distinct?

Solution. An n-letter string can be formed in 26^n ways, so the number of distinct strings of four or fewer letters is $26^4 + 26^3 + 26^2 + 26 = 475254$. As $500000 > 475254$, treating the strings as pigeons and the 475254 possible strings as pigeonholes, at least one string is repeated. The strings cannot all be distinct.

Example 6.43 – Pages in dictionaries

Prove that if 30 dictionaries in a library contain a total of 61327 pages, then at least one of them has at least 2045 pages.

Solution. Pages are pigeons and dictionaries pigeonholes. By the generalised principle some dictionary has $p + 1$ or more pages, where $p = \lfloor (61327 - 1)/30 \rfloor = \lfloor 2044.2 \rfloor = 2044$. So some dictionary has at least 2045 pages.

Example 6.44 – Painting doors

If 5 colours are used to paint 26 doors, prove that at least 6 doors have the same colour.

Solution. Doors are pigeons, colours pigeonholes: some colour is used on at least $\lfloor (26 - 1)/5 \rfloor + 1 = 5 + 1 = 6$ doors.

Example 6.45 – Days of the week

Prove that in any set of 29 persons at least five must have been born on the same day of the week.

Solution. 7 days as pigeonholes, 29 persons as pigeons: some day has at least $\lfloor (29-1)/7 \rfloor + 1 = 4+1 = 5$ persons.

Example 6.46 – Birthdays in a month

How many persons must be chosen so that at least five of them have birthdays in the same calendar month?

Solution. With n persons and 12 months, the guaranteed number in some month is $\lfloor (n-1)/12 \rfloor + 1$. We need $\lfloor (n-1)/12 \rfloor + 1 = 5$, i.e. $\lfloor (n-1)/12 \rfloor = 4$; the least such n satisfies $n - 1 = 48$, so $n = \mathbf{49}$. (With 48 persons, four in each month is possible.)

Example 6.47 – Triples with the same sum

Find the least number of ways of choosing three different numbers from 1 to 10 so that all the choices have the same sum.

Solution. There are $\binom{10}{3} = 120$ choices. The smallest possible sum is $1 + 2 + 3 = 6$ and the largest $8 + 9 + 10 = 27$, so there are 22 possible sums. With 120 choices (pigeons) and 22 sums (pigeonholes), some sum is shared by at least $\lfloor (120-1)/22 \rfloor + 1 = 5 + 1 = 6$ choices. So at least 6 choices have the same sum.

Example 6.48 – Debating teams

Shirts numbered 1 to 20 are worn by 20 students. When any 3 of them form a debating team, the sum of their shirt numbers is the team's code number. Show that if any 8 of the 20 are selected, then from these 8 we can form at least two different teams with the same code number.

Solution. From 8 students, $\binom{8}{3} = 56$ teams can be formed. The smallest code is $1 + 2 + 3 = 6$ and the largest $18 + 19 + 20 = 57$, so only 52 codes (pigeonholes) are available for 56 teams (pigeons). Hence two different teams have the same code number.

Example 6.49 – Remainders modulo 36

Prove that every set of 37 positive integers contains at least two integers that leave the same remainder on division by 36.

Solution. The possible remainders are $0, 1, \ldots, 35$. Let A_r be the set of positive integers leaving remainder r; there are 36 such sets (pigeonholes). Any 37 positive integers (pigeons) must include two from the same A_r.

Example 6.50 – A pair summing to $2n+1$

Show that if any $n + 1$ numbers from 1 to $2n$ are chosen, then two of them have sum $2n + 1$.

Solution. Consider $A_1 = \{1, 2n\}$, $A_2 = \{2, 2n-1\}$, ..., $A_n = \{n, n+1\}$ — the only pairs from $1, \ldots, 2n$ with sum $2n + 1$. Every number from 1 to $2n$ lies in exactly one A_i. Since $n + 1$ numbers are chosen and there are only n sets, two chosen numbers lie in the same set; their sum is $2n + 1$.

Example 6.51 – Sum or difference divisible by 10

Show that every set of seven distinct integers includes two integers x and y such that at least one of $x + y$ or $x - y$ is divisible by 10.

Solution. Let $X = \{x_1, \dots, x_7\}$ and let r_i be the remainder of x_i on division by 10. Form the six subsets

$$A_1 = \{x_i \mid r_i = 0\}, \quad A_2 = \{x_i \mid r_i = 5\}, \quad A_3 = \{x_i \mid r_i = 1 \text{ or } 9\},$$
$$A_4 = \{x_i \mid r_i = 2 \text{ or } 8\}, \quad A_5 = \{x_i \mid r_i = 3 \text{ or } 7\}, \quad A_6 = \{x_i \mid r_i = 4 \text{ or } 6\}.$$

Seven integers (pigeons) in six subsets (pigeonholes): two, say x and y, lie in the same subset. In A_1, both are multiples of 10, so $x \pm y$ are too. In A_2, $x = 10k_1 + 5$, $y = 10k_2 + 5$, so $x + y = 10(k_1 + k_2 + 1)$ and $x - y = 10(k_1 - k_2)$. In any other subset, the remainders are equal (then $10 \mid x - y$) or add to 10 (then $10 \mid x + y$). ■

Example 6.52 – One divides the other

Prove that if 101 integers are selected from $S = \{1, 2, 3, \dots, 200\}$, then there are two such that one divides the other.

Solution. Let $X = \{1, 3, 5, \dots, 199\}$, the 100 odd numbers. Every $n \in S$ can be written as $n = 2^k x$ with $k \geq 0$ and $x \in X$ (its "odd part"). Selecting 101 numbers (pigeons) with only 100 possible odd parts (pigeonholes), two different selected numbers $a = 2^m x$ and $b = 2^n x$ have the same odd part. Then $a \mid b$ if $m \leq n$, and $b \mid a$ if $n < m$. ■

Example 6.53 – Exactly 14 pills

A patient is prescribed 45 pills, to take at least one pill a day for 30 days. Prove that there is a period of consecutive days during which the patient takes a total of exactly 14 pills.

Solution. Let a_i be the number of pills taken up to the end of day i. Since at least one pill is taken each day and 45 in all, $1 \leq a_1 < a_2 < \cdots < a_{30} \leq 45$, and hence $a_1 + 14 < a_2 + 14 < \cdots < a_{30} + 14 \leq 59$. The 60 numbers $a_1, \dots, a_{30}, a_1 + 14, \dots, a_{30} + 14$ all lie in $\{1, \dots, 59\}$: 60 pigeons in 59 pigeonholes, so two are equal. The a_i are distinct, and so are the $a_i + 14$; hence $a_i = a_j + 14$ for some $i > j$. Between days $j + 1$ and i the patient takes exactly $a_i - a_j = 14$ pills. ■

Example 6.54 – The $kn + 1$ form

Prove: if $m = kn + 1$ pigeons ($k \geq 1$) occupy n pigeonholes, then at least one pigeonhole contains $k + 1$ or more pigeons.

Solution. If every pigeonhole contained k or fewer pigeons, the total would be at most $kn < kn + 1 = m$ — a contradiction. ■

Example 6.55 – Marbles of three colours

A bag contains many red, white and blue marbles. What is the least number of marbles one should take out to be sure of getting at least six of the same colour?

Solution. Marbles are pigeons and the $n = 3$ colours pigeonholes. With $m = 3k + 1$ marbles some colour has $k + 1$ or more; we need $k + 1 = 6$, i.e. $k = 5$, so $m = 16$. With 15 marbles, five of each colour is possible. So **16** is the least number.

Example 6.56 – An extension of the principle

Suppose $m = p_1 + p_2 + \cdots + p_n - n + 1$ pigeons occupy n pigeonholes $H_1, \dots, H_n$. Prove that some pigeonhole H_j contains p_j or more pigeons.

Solution. If every H_j contained $p_j - 1$ or fewer pigeons, the total would be at most $(p_1 - 1) + \cdots + (p_n - 1) = p_1 + \cdots + p_n - n = m - 1$ — a contradiction. ■

6.17 Exercise Set 6E — pigeonhole principle

6E.1 Prove that in a set of 13 children at least two have birthdays during the same month.

6E.2 If three men and five women form a queue, show that at least two women will be next to each other.

6E.3 If seven cars carry 26 passengers, prove that at least one car must have 4 or more passengers.

6E.4 If six persons have a total of Rs. 2161 with them, show that one or more of them must have at least Rs. 361.

6E.5 Show that if 50 books in a library contain a total of 27551 pages, one of the books must have at least 552 pages.

6E.6 Suppose there are many red, white and blue socks in a box. What is the least number of socks one should take out to be sure of getting a matching pair?

6E.7 What is the minimum number of students needed so that at least two have last names beginning with the same English letter?

6E.8 Show that if five points are selected in a square whose sides have length 1 cm, then there are at least two points whose distance apart is at most $1/\sqrt{2}$ cm (less than $1/\sqrt{2}$ cm if the points are inside the square).

6E.9 If 10 points are selected from the interior of an equilateral triangle whose sides are 3 cm long, show that at least two points are within 1 cm of each other.

6E.10 Show that if any 6 numbers from 1 to 10 are chosen, then two of them have sum 11.

6E.11 Show that if any eight positive integers are chosen, two of them have the same remainder when divided by 7.

6E.12 Show that there are at least 90 ways to choose six numbers from 1 to 15 so that all the choices have the same sum.

6E.13 A set of children from primary Standards I to IV are assembled in a room. Find the minimum number of children one must choose from the set so that five of them belong to the same Standard.

6E.14 Show that if 11 numbers are chosen from $\{1, 2, 3, \dots, 20\}$, one of them is a multiple of another.

6E.15 Prove that if 151 integers are selected from $S = \{1, 2, 3, \dots, 300\}$, the selection must include two integers m, n such that $m \mid n$ or $n \mid m$.

6E.16 Let S be a set of 5 positive integers the maximum of which is at most 9. Prove that the sums of the elements in the non-empty subsets of S cannot all be distinct.

6E.17 If 11 integers are selected from $S = \{1, 2, 3, \dots, 100\}$, prove that there are at least two, say x and y, such that $0 < |\sqrt{x} - \sqrt{y}| < 1$.

6E.18 A student plays at least one set of tennis each day, but does not play more than 40 sets in a four-week period. Prove that, however he distributes his sets, there is a span of consecutive days during which he plays exactly 15 sets.

6E.19 Show that in any set of $n \geq 2$ persons there are at least two persons who know the same number of persons in the set.

6E.20 Consider a tournament in which each player plays against every other player and each player wins at least once. Show that at least two players have the same number of wins.

6.18 Solutions to Exercise Set 6E — pigeonhole principle

Every solution below follows the three-step plan of the examination tip: first name the pigeons, then name the pigeonholes, and finally apply the principle. The generalised form used throughout is

$$m \text{ pigeons in } n \text{ holes} \implies \text{some hole has at least } \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \text{ pigeons.}$$

6E.1 – 13 children and their birthday months

Pigeons: the 13 children ($m = 13$).
Pigeonholes: the 12 months of the year ($n = 12$); each child goes into the month of his or her birthday.

Apply. $m = 13 > n = 12$, so by the pigeonhole principle some month holds at least two children. (Generalised form: $\lfloor 12/12 \rfloor + 1 = 1 + 1 = 2$.)

Conclusion. At least two of the 13 children have birthdays in the same month. ■

Remark. 12 children could be born in 12 different months, so 13 is the smallest group size for which the conclusion is guaranteed.

6E.2 – Three men and five women in a queue

Make the holes. Look at the positions of the 3 men in the queue. They cut the queue into at most four *gaps*: G_1 before the first man, G_2 between the first and second men, G_3 between the second and third men, and G_4 after the third man.



5 women in 4 gaps: some gap (here G_1) holds two women, who stand side by side

Pigeons: the 5 women ($m = 5$).
Pigeonholes: the 4 gaps G_1, G_2, G_3, G_4 ($n = 4$); each woman goes into the gap she stands in.

Apply. $5 > 4$, so some gap contains at least $\lfloor 4/4 \rfloor + 1 = 2$ women. A gap contains no man, so two women in the same gap have nobody but women between them; in particular, two of them stand next to each other.

Conclusion. However the 8 people are arranged, at least two women are adjacent. ■

Remark. A direct check of all $\binom{8}{3} = 56$ arrangements of M,M,M,W,W,W,W,W confirms that every one contains two adjacent women.

6E.3 – Seven cars carrying 26 passengers

Pigeons: the 26 passengers ($m = 26$).
Pigeonholes: the 7 cars ($n = 7$).

Apply. $p = \left\lfloor \dfrac{m-1}{n} \right\rfloor = \left\lfloor \dfrac{25}{7} \right\rfloor = \lfloor 3.57 \rfloor = 3$, so some car carries at least $p + 1 = 4$ passengers.

Check by contradiction. If every car carried 3 or fewer passengers, the total would be at most $7 \times 3 = 21 < 26$, which is impossible.

Conclusion. At least one car carries 4 or more passengers. ■

6E.4 – Six persons with Rs. 2161

Pigeons: the 2161 rupees, thought of as 2161 one-rupee coins ($m = 2161$).
Pigeonholes: the 6 persons ($n = 6$); each coin goes to the person holding it.

Apply. $p = \left\lfloor \dfrac{2161-1}{6} \right\rfloor = \left\lfloor \dfrac{2160}{6} \right\rfloor = 360$, so some person holds at least $360 + 1 = 361$ coins.

Check by contradiction. If each person had at most Rs. 360, together they would have at most $6 \times 360 = 2160 < 2161$ rupees.

Conclusion. At least one person has Rs. 361 or more. ■

6E.5 – 50 books with 27551 pages

Pigeons: the 27551 pages ($m = 27551$).
Pigeonholes: the 50 books ($n = 50$).
Apply. $p = \left\lfloor \dfrac{27551 - 1}{50} \right\rfloor = \left\lfloor \dfrac{27550}{50} \right\rfloor = 551$, so some book has at least $551 + 1 = 552$ pages.
Check by contradiction. If every book had at most 551 pages, the total would be at most $50 \times 551 = 27550 < 27551$.
Conclusion. At least one book has 552 or more pages. ■

6E.6 – Least number of socks for a matching pair

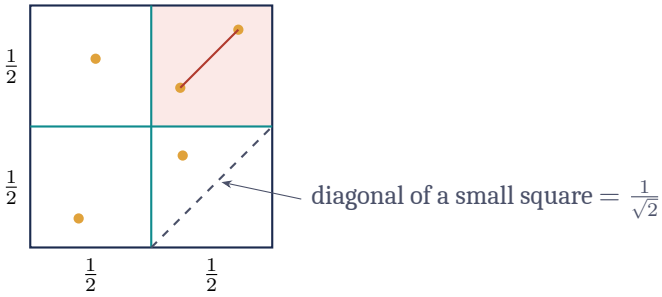
Pigeons: the socks drawn (m of them, to be found).
Pigeonholes: the 3 colours red, white, blue ($n = 3$).
Find m. We need some colour to receive at least 2 socks, i.e. $\lfloor (m-1)/3 \rfloor + 1 \geq 2$, i.e. $m - 1 \geq 3$, i.e. $m \geq 4$.
Show 3 is not enough. Three socks may be one red, one white and one blue — no matching pair.
Conclusion. The least number of socks to be sure of a matching pair is **4**.

6E.7 – Students with the same initial letter

Pigeons: the students (m of them, to be found).
Pigeonholes: the 26 letters A–Z ($n = 26$); each student goes to the first letter of his or her last name.
Find m. We need $\lfloor (m-1)/26 \rfloor + 1 \geq 2$, i.e. $m - 1 \geq 26$, i.e. $m \geq 27$.
Show 26 is not enough. 26 students could have last names starting with 26 different letters.
Conclusion. The minimum number of students is **27**.

6E.8 – Five points in a unit square

Make the holes. Join the mid-points of opposite sides of the square. This divides it into four small squares of side $\frac{1}{2}$ cm. (A point lying on a dividing line is placed in any one of the small squares containing it.)



Pigeons: the 5 points ($m = 5$).
Pigeonholes: the 4 small squares ($n = 4$).
Apply. $5 > 4$, so two of the points lie in the same small square.
Measure. The greatest distance between two points of a square of side $\frac{1}{2}$ is its diagonal:

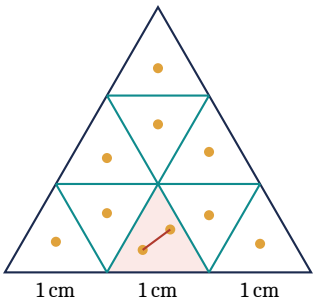
$$\sqrt{\left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2} = \sqrt{\tfrac{1}{2}} = \tfrac{1}{\sqrt{2}} \text{ cm } (\approx 0.707 \text{ cm}).$$

So those two points are at most $1/\sqrt{2}$ cm apart.
Interior points. The distance $1/\sqrt{2}$ is reached only by two opposite corners of a small square, and in every such pair at least one corner lies on the boundary of the big square. Hence, if all five points are inside the big square, the distance is strictly less than $1/\sqrt{2}$ cm.

Conclusion. Among any five points of the square, two are at distance at most $1/\sqrt{2}$ cm (less than $1/\sqrt{2}$ cm for interior points). ■

6E.9 – Ten points in an equilateral triangle of side 3 cm

Make the holes. Divide each side into three equal parts of 1 cm and, through the division points, draw lines parallel to the sides. The triangle splits into $3^2 = 9$ equilateral triangles of side 1 cm (6 pointing up, 3 pointing down).



Pigeons: the 10 points ($m = 10$).
Pigeonholes: the 9 small triangles ($n = 9$).

Apply. $10 > 9$, so two of the points lie in the same small triangle (shaded).
Measure. The greatest distance between two points of a triangle is its longest side; here every side is 1 cm. So these two points are at most 1 cm apart. (The distance 1 cm is reached only by two vertices of a small triangle; the only grid vertex inside the big triangle is its centre, so one of those two vertices would lie on the boundary. Hence two *interior* points are in fact less than 1 cm apart.)

Conclusion. At least two of the ten points are within 1 cm of each other. ■

6E.10 – Six numbers from 1 to 10 with a pair summing to 11

Make the holes. Split $\{1, 2, \dots, 10\}$ into five pairs, each with sum 11:

$$\{1, 10\}, \quad \{2, 9\}, \quad \{3, 8\}, \quad \{4, 7\}, \quad \{5, 6\}.$$

Pigeons: the 6 chosen numbers ($m = 6$).
Pigeonholes: the 5 pairs above ($n = 5$); each number goes into the pair containing it.

Apply. $6 > 5$, so some pair receives two of the chosen numbers. The chosen numbers are distinct, so they are *both* members of that pair, and their sum is 11.

Conclusion. Any 6 numbers chosen from 1 to 10 include two with sum 11. ■

Remark. Five numbers are not enough: $\{1, 2, 3, 4, 5\}$ contains no two numbers with sum 11.

6E.11 – Eight integers and remainders on division by 7

Pigeons: the 8 chosen positive integers ($m = 8$).
Pigeonholes: the possible remainders on division by 7, namely $0, 1, 2, 3, 4, 5, 6$ ($n = 7$).

Apply. $8 > 7$, so two of the integers, say a and b, leave the same remainder r: $a = 7q_1 + r$, $b = 7q_2 + r$.

Conclusion. Two of the integers have the same remainder on division by 7 (equivalently, $7 \mid a - b$). ■

Remark. Seven integers are not enough: $1, 2, \dots, 7$ leave the seven different remainders $1, 2, 3, 4, 5, 6, 0$.

6E.12 – At least 90 six-number choices from 1 to 15 with the same sum

Count the pigeons. The number of ways to choose six numbers from $\{1, \dots, 15\}$ is

$$\binom{15}{6} = \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10}{6!} = 5005.$$

Count the holes. The smallest possible sum is $1+2+3+4+5+6 = 21$ and the largest is $10+11+12+13+14+15 = 75$. So every sum is one of the integers $21, 22, \dots, 75$: there are $75 - 21 + 1 = 55$ possible sums.

Pigeons: the 5005 choices ($m = 5005$).

Pigeonholes: the 55 possible sums ($n = 55$); each choice goes to its sum.

Apply. Some sum is shared by at least

$$\left\lfloor \frac{5005 - 1}{55} \right\rfloor + 1 = \left\lfloor \frac{5004}{55} \right\rfloor + 1 = \lfloor 90.98 \rfloor + 1 = 90 + 1 = 91$$

choices.

Conclusion. There are at least 91 (≥ 90) ways to choose six numbers from 1 to 15 all having the same sum. ■

Remark. The bound 90 is far from tight: in fact the sums 47, 48 and 49 are each obtained in 227 ways.

6E.13 – Five children from the same Standard

Pigeons: the chosen children (m of them, to be found).

Pigeonholes: the 4 Standards I, II, III, IV ($n = 4$).

Find m. We want some Standard to receive $k+1 = 5$ children, so $k = 4$. By version (a) of the principle, $m = kn + 1 = 4 \times 4 + 1 = 17$ children suffice. Check: $\lfloor (17-1)/4 \rfloor + 1 = 4 + 1 = 5$.

Show 16 is not enough. 16 children could be 4 from each Standard, and then no five belong to the same Standard. (Indeed $\lfloor 15/4 \rfloor + 1 = 4$.)

Conclusion. The minimum number of children is **17**.

6E.14 – 11 numbers from 1 to 20: one is a multiple of another

Key fact. Every positive integer can be written uniquely as $n = 2^k x$ with $k \geq 0$ and x odd; x is called the *odd part* of n.

Make the holes. For $n \leq 20$ the odd part is one of $1, 3, 5, \dots, 19$ — ten values. Grouping $1, \dots, 20$ by odd part gives ten chains, and in each chain every number divides the next:

Odd part x	Numbers in $\{1, \dots, 20\}$ with odd part x
1	1, 2, 4, 8, 16
3	3, 6, 12
5	5, 10, 20
7	7, 14
9	9, 18
11, 13, 15, 17, 19	each alone: 11; 13; 15; 17; 19

Pigeons: the 11 chosen numbers ($m = 11$).

Pigeonholes: the 10 odd parts ($n = 10$).

Apply. $11 > 10$, so two chosen numbers $a \neq b$ have the same odd part x: $a = 2^i x$, $b = 2^j x$ with $i \neq j$. Say $i < j$; then $b = 2^{j-i}\, a$.

Conclusion. $a \mid b$: one of the chosen numbers is a multiple of another. ■

Remark. Ten numbers are not enough: in $\{11, 12, \dots, 20\}$ no number divides another (the smallest proper multiple of 11 is $22 > 20$).

6E.15 – 151 integers from 1 to 300: $m \mid n$ or $n \mid m$

Make the holes. As in 6E.14, write each selected integer as $2^k x$ with x odd. The odd part x is one of $1, 3, 5, \dots, 299$; there are $\frac{300}{2} = 150$ odd numbers in S.
Pigeons: the 151 selected integers ($m = 151$).
Pigeonholes: the 150 possible odd parts ($n = 150$).

Apply. $151 > 150$, so two selected integers have the same odd part: $m = 2^i x$ and $n = 2^j x$ with $i \neq j$. If $i < j$ then $n = 2^{j-i} m$, so $m \mid n$; if $j < i$ then $n \mid m$.

Conclusion. The selection contains two integers m, n with $m \mid n$ or $n \mid m$. ■

Remark. 150 integers are not enough: in $\{151, 152, \dots, 300\}$ no integer divides another, since $2 \times 151 = 302 > 300$.

6E.16 – Subset sums of a 5-element set with maximum at most 9

Idea. Using all $2^5 - 1 = 31$ non-empty subsets does not work directly: their sums can range from 1 up to $9 + 8 + 7 + 6 + 5 = 35$, giving 35 holes for only 31 pigeons. We therefore use only the *small* subsets.
Count the pigeons. The non-empty subsets of S with at most three elements number

$$\binom{5}{1} + \binom{5}{2} + \binom{5}{3} = 5 + 10 + 10 = 25.$$

Count the holes. The elements of S are distinct positive integers not exceeding 9, so a subset of at most three elements has sum at least 1 and at most $9+8+7 = 24$. There are only 24 possible sums $1, 2, \dots, 24$.
Pigeons: the 25 subsets of size 1, 2 or 3 ($m = 25$).
Pigeonholes: the 24 possible sums ($n = 24$).

Apply. $25 > 24$, so two different subsets have the same sum.

Conclusion. The sums of the non-empty subsets of S cannot all be distinct. ■

Remark. A computer check of all $\binom{9}{5} = 126$ possible sets S confirms the result.

6E.17 – 11 integers from 1 to 100 with $0 < |\sqrt{x} - \sqrt{y}| < 1$

Make the holes. Group the integers by the whole-number part of their square root: $C_k = \{x \mid k^2 \le x < (k+1)^2\}$ for $k = 1, \dots, 9$, together with $C_{10} = \{100\}$.

Class	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
Members	1–3	4–8	9–15	16–24	25–35	36–48	49–63	64–80	81–99	100
$\sqrt{x}$ lies in	$[1,2)$	$[2,3)$	$[3,4)$	$[4,5)$	$[5,6)$	$[6,7)$	$[7,8)$	$[8,9)$	$[9,10)$	10

These ten classes cover $\{1, \dots, 100\}$ exactly once.
Pigeons: the 11 selected integers ($m = 11$).
Pigeonholes: the 10 classes $C_1, \dots, C_{10}$ ($n = 10$).

Apply. $11 > 10$, so two selected integers $x < y$ lie in the same class. They cannot both be in $C_{10} = \{100\}$, so they lie in some C_k with $1 \le k \le 9$, and then $k \le \sqrt{x} < \sqrt{y} < k + 1$.

Conclusion. $0 < \sqrt{y} - \sqrt{x} < 1$, i.e. $0 < |\sqrt{x} - \sqrt{y}| < 1$. ■

Remark. Ten integers are not enough: the ten perfect squares $1, 4, 9, \dots, 100$ have square roots $1, 2, \dots, 10$, any two of which differ by at least 1.

6E.18 – Exactly 15 sets of tennis on consecutive days

Set up. A four-week period has 28 days. Let a_i be the total number of sets played on days $1, 2, \dots, i$ (so a_i is a running total). Since at least one set is played each day and at most 40 in all,

$$1 \le a_1 < a_2 < \cdots < a_{28} \le 40.$$

Adding 15 to each term,

$$16 \le a_1 + 15 < a_2 + 15 < \cdots < a_{28} + 15 \le 55.$$

Pigeons: the 56 numbers $a_1, \dots, a_{28},\ a_1 + 15, \dots, a_{28} + 15$ ($m = 56$).
Pigeonholes: the integers $1, 2, \dots, 55$ ($n = 55$); each of the 56 numbers lies between 1 and 55.

Apply. $56 > 55$, so two of these numbers are equal. The a_i are all different, and so are the $a_i + 15$; therefore the equal pair must be $a_i = a_j + 15$ for some i and j. Then $a_i > a_j$, so $i > j$.
Interpret. $a_i - a_j = 15$ is the number of sets played on days $j + 1, j + 2, \dots, i$.

Conclusion. There is a span of consecutive days (days $j + 1$ to i) during which he plays exactly 15 sets. ■

6E.19 – Two persons who know the same number of persons

Set up. "Knowing" is mutual, and nobody counts himself. So each of the n persons knows between 0 and $n - 1$ of the others. That is n possible values for n persons, which is not yet enough — we must remove one value.
Key observation. The values 0 and $n - 1$ cannot both occur. If person A knows all $n - 1$ others and person B knows nobody, then A knows B, so B knows A — a contradiction.
Two cases. If somebody knows nobody, the possible values are $0, 1, \dots, n - 2$. If nobody knows nobody, the possible values are $1, 2, \dots, n - 1$. In either case only $n - 1$ values are available.
Pigeons: the n persons ($m = n$).
Pigeonholes: the $n - 1$ available values ($n - 1$ holes).

Apply. $n > n - 1$, so two persons receive the same value.

Conclusion. In any set of $n \ge 2$ persons, two know the same number of persons in the set. ■

Remark. In graph language (persons as vertices, acquaintances as edges) this says: *every simple graph with at least two vertices has two vertices of the same degree.*

6E.20 – Two players with the same number of wins

Set up. Let there be n players (assume $n \ge 2$ and no drawn games). Each player plays the other $n - 1$ players once, so his number of wins is one of $0, 1, \dots, n - 1$. Since every player wins at least once, the value 0 is impossible, and the number of wins lies in $\{1, 2, \dots, n - 1\}$.
Pigeons: the n players ($m = n$).
Pigeonholes: the $n - 1$ possible numbers of wins $1, 2, \dots, n - 1$.

Apply. $n > n - 1$, so two players have the same number of wins.

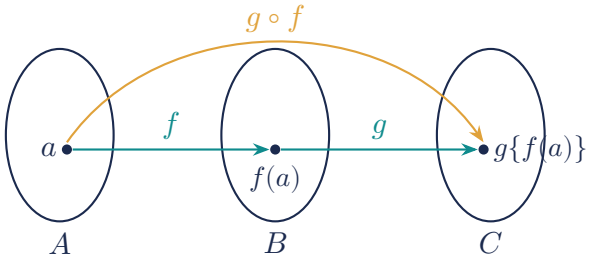
Conclusion. At least two players have the same number of wins. ■

Remark. The condition "each player wins at least once" cannot be dropped. In the tournament where player P_i beats every P_j with $j > i$, the five players $P_1, \dots, P_5$ have $4, 3, 2, 1, 0$ wins — all different.

6.19 Composition of functions

Definition: Composite function, powers

For non-empty sets A, B, C (not necessarily distinct) and functions $f : A \to B$, $g : B \to C$, the **composition** (or **product**) of f and g is the function $g \circ f : A \to C$ defined by $(g \circ f)(a) = g\{f(a)\}$ for all $a \in A$ — first f, then g. For $f : A \to A$, $f \circ f$ is written f^2, $f \circ f^2$ is f^3, and in general $f^1 = f$, $f^n = f \circ f^{n-1}$ for $n \geq 2$.



Example 6.57 – Composition on finite sets

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{w, x, y, z\}$, $f : A \to B$ and $g : B \to C$ with $f = \{(1, a), (2, a), (3, b), (4, c)\}$ and $g = \{(a, x), (b, y), (c, z)\}$. Find $g \circ f$.

Solution. $(g \circ f)(1) = g\{f(1)\} = g(a) = x$; $(g \circ f)(2) = g(a) = x$; $(g \circ f)(3) = g(b) = y$; $(g \circ f)(4) = g(c) = z$. Thus $g \circ f = \{(1, x), (2, x), (3, y), (4, z)\}$.

Example 6.58 – Composition is not commutative

Consider $f(x) = x^3$ and $g(x) = x^2 + 1$ on $\mathbb{R}$. Find $g \circ f$, $f \circ g$, f^2 and g^2.

Solution. All four are defined on $\mathbb{R}$:

$$(g \circ f)(x) = g(x^3) = x^6 + 1, \qquad (f \circ g)(x) = f(x^2 + 1) = (x^2 + 1)^3,$$
$$f^2(x) = f(x^3) = x^9, \qquad g^2(x) = g(x^2 + 1) = (x^2 + 1)^2 + 1.$$

Since $g \circ f \neq f \circ g$, composition of functions is **not commutative**, even when both composites exist.

Example 6.59 – Identity functions are neutral

Let I_A, I_B be the identity functions on A and B. For any $f : A \to B$, prove $f \circ I_A = f = I_B \circ f$.

Solution. For any $a \in A$: $(f \circ I_A)(a) = f\{I_A(a)\} = f(a)$ and $(I_B \circ f)(a) = I_B\{f(a)\} = f(a)$. ■

Example 6.60 – Finding coefficients from a composite

Let $f, g : \mathbb{R} \to \mathbb{R}$ be $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$, determine a and b.

Solution. $(g \circ f)(x) = g(ax + b) = 1 - (ax + b) + (ax + b)^2 = a^2x^2 + (2ab - a)x + (1 - b + b^2)$. Comparing coefficients with $9x^2 - 9x + 3$:

$$a^2 = 9, \qquad 2ab - a = -9, \qquad 1 - b + b^2 = 3.$$

The first gives $a = \pm 3$. For $a = 3$: $6b - 3 = -9$, so $b = -1$ (and $1 + 1 + 1 = 3$ ✓). For $a = -3$: $-6b + 3 = -9$, so $b = 2$ (and $1 - 2 + 4 = 3$ ✓). Hence $a = 3, b = -1$, or $a = -3, b = 2$.

Example 6.61 – When do two linear functions commute?

Let $f(x) = ax + b$ and $g(x) = cx + d$ on $\mathbb{R}$. What relationship between a, b, c, d makes $g \circ f = f \circ g$?

Solution. $(g\circ f)(x) = c(ax+b)+d = cax+(cb+d)$ and $(f\circ g)(x) = a(cx+d)+b = acx+(ad+b)$. These are equal for all x iff $cb + d = ad + b$.

Composition and one-to-one / onto

Let $f : A \to B$ and $g : B \to C$. Then (1) if f and g are one-to-one, so is $g \circ f$; (2) if $g \circ f$ is one-to-one, then f is one-to-one; (3) if f and g are onto, so is $g \circ f$; (4) if $g \circ f$ is onto, then g is onto.

Proof. (1) $(g \circ f)(a_1) = (g \circ f)(a_2) \Rightarrow g(f(a_1)) = g(f(a_2)) \Rightarrow f(a_1) = f(a_2)$ (g one-to-one) $\Rightarrow a_1 = a_2$ (f one-to-one).
(2) $f(a_1) = f(a_2) \Rightarrow g(f(a_1)) = g(f(a_2))$ (g is a function) $\Rightarrow (g \circ f)(a_1) = (g \circ f)(a_2) \Rightarrow a_1 = a_2$.
(3) Take any $c \in C$. As g is onto, $c = g(b)$ for some $b \in B$; as f is onto, $b = f(a)$ for some $a \in A$. Then $(g \circ f)(a) = g(b) = c$.
(4) For any $c \in C$ there is $a \in A$ with $g(f(a)) = c$; so $f(a) \in B$ is a preimage of c under g. ■

Composition is associative

For $f : A \to B, g : B \to C, h : C \to D$: $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. Both are functions $A \to D$. For $x \in A$, put $y = f(x)$ and $z = g(y)$. Then $[(h \circ g) \circ f](x) = (h \circ g)(y) = h(z)$ and $[h \circ (g \circ f)](x) = h[g(f(x))] = h(z)$. ■
Because of this we write $h \circ g \circ f$ without brackets.

Example 6.62 – Verifying associativity

Let $f, g, h : \mathbb{Z} \to \mathbb{Z}$ be $f(x) = x - 1$, $g(x) = 3x$, and $h(x) = 0$ if x is even, $h(x) = 1$ if x is odd. Determine $(f \circ (g \circ h))(x)$ and $((f \circ g) \circ h)(x)$ and verify that they are equal.

Solution. $(g\circ h)(x) = 3h(x)$, so $(f \circ (g \circ h))(x) = 3h(x) - 1$, which is -1 if x is even and 2 if x is odd. $(f \circ g)(x) = 3x - 1$, so $((f \circ g) \circ h)(x) = 3h(x) - 1$ — again -1 for even x and 2 for odd x. Hence $f \circ (g \circ h) = (f \circ g) \circ h$.

Example 6.63 – Composites of quadratics and linears

$f(x) = x^2 + 1, g(x) = 2x - 3$ on $\mathbb{R}$. Find $f \circ g, g \circ f$ and evaluate each at $x = 3$.

Solution. $(f \circ g)(x) = (2x - 3)^2 + 1 = 4x^2 - 12x + 10$; $(g \circ f)(x) = 2(x^2 + 1) - 3 = 2x^2 - 1$. At $x = 3$: $(f \circ g)(3) = 10$ and $(g \circ f)(3) = 17$.

6.20 Exercise Set 6F — composition

6F.1 Let $A = B = C = \mathbb{R}$, $f : A \to B$, $f(a) = a - 1$, and $g : B \to C$, $g(b) = b^2$. Find (i) $(g \circ f)(a)$ (ii) $(f \circ g)(b)$ (iii) $(f \circ f)(a)$ (iv) $(g \circ g)(b)$.

6F.2 Let $A = B = \mathbb{Z}$ and C the set of all even integers. Let $f : A \to B$, $f(a) = a + 1$, and $g : B \to C$, $g(b) = 2b$. Find $g \circ f$.

6F.3 Let $f, g : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$ and $g(x) = x + 5$. Prove that $g \circ f \neq f \circ g$.

6F.4 Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be $f(x) = x + 2$, $g(x) = x - 2$, $h(x) = 3x$. Find $g \circ f$, $f \circ g$, $f \circ f$, $g \circ g$, $f \circ h$, $h \circ g$, $h \circ f$ and $f \circ h \circ g$.

6F.5 Let $f : \mathbb{R}^+ \to \mathbb{R}$ be $f(x) = \log x$ and $g : \mathbb{R} \to \mathbb{R}^+$ be $g(x) = e^x$. Show that $f \circ g$ and $g \circ f$ are identity functions.

6F.6 Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2, g(x) = x+5$ and $h(x) = \sqrt{x^2 + 2}$. Verify that $(h\circ g)\circ f = h\circ(g\circ f)$.

6F.7 Let $g : \mathbb{N} \to \mathbb{N}$ be $g(n) = 2n$. If $A = \{1, 2, 3, 4\}$ and $f : A \to \mathbb{N}$ is $f = \{(1, 2), (2, 3), (3, 5), (4, 7)\}$, find $g \circ f$.

6F.8 Let $A = \{1, 2, 3\}$ and $f = \{(1, 2), (2, 3), (3, 1)\}$, $g = \{(1, 2), (2, 1), (3, 3)\}$, $h = \{(1, 1), (2, 2), (3, 1)\}$, $p = \{(1, 1), (2, 2), (3, 3)\}$ be functions on A. Find $f \circ g$, $g \circ f$, $p \circ g$, $g \circ p$, $f \circ p$ and $f \circ h \circ g$.

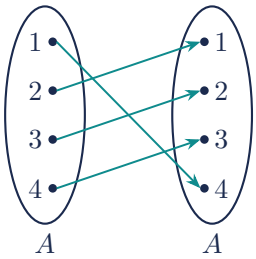
6F.9 With $A = \{1, 2, 3, 4\}$ and $f : A \to A$, $f = \{(1, 2), (2, 2), (3, 1), (4, 3)\}$, find f^2 and f^3.

6F.10 For the functions f, g, h of the associativity example above ($f(x) = x - 1$, $g(x) = 3x$, h = parity), determine $g \circ f$, $h \circ g$, f^2, g^2, h^2, f^3, g^3 and h^3.

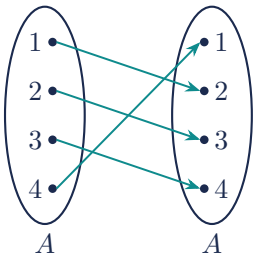
6.21 Invertible functions

Definition: Invertible function, inverse

A function $f : A \to B$ is **invertible** if there exists a function $g : B \to A$ such that $g \circ f = I_A$ and $f \circ g = I_B$. Then g is called an **inverse** of f, and we write $g = f^{-1}$.



f



$g = f^{-1}$ (arrows reversed)

Example 6.64 – A pair of inverse functions on a finite set

Let $A = \{1, 2, 3, 4\}$ and $f, g : A \to A$ be $f = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$ and $g = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Prove that f and g are inverses of each other.

Solution. $(g \circ f)(1) = g(4) = 1$, $(g \circ f)(2) = g(1) = 2$, $(g \circ f)(3) = g(2) = 3$, $(g \circ f)(4) = g(3) = 4$; and $(f \circ g)(1) = f(2) = 1$, $(f \circ g)(2) = f(3) = 2$, $(f \circ g)(3) = f(4) = 3$, $(f \circ g)(4) = f(1) = 4$. So $g \circ f = I_A = f \circ g$: g is an inverse of f, and f is an inverse of g.

Example 6.65 – Inverse of a linear function

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 5$, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \frac{1}{2}(x - 5)$. Prove that g is an inverse of f.

Solution. $(g \circ f)(x) = g(2x + 5) = \frac{1}{2}(2x + 5 - 5) = x = I_{\mathbb{R}}(x)$ and $(f \circ g)(x) = 2 \cdot \frac{1}{2}(x - 5) + 5 = x = I_{\mathbb{R}}(x)$. Hence g is an inverse of f (and f of g).

The functions in these examples are one-to-one and onto. The theorems below show that this is no accident.

Uniqueness of the inverse

If $f : A \to B$ is invertible, it has a unique inverse. Further, if $f(a) = b$, then $f^{-1}(b) = a$.

Proof. Suppose g and h are both inverses: $g \circ f = h \circ f = I_A$ and $f \circ g = f \circ h = I_B$. Then $h = h \circ I_B = h \circ (f \circ g) = (h \circ f) \circ g = I_A \circ g = g$. Next, if $f(a) = b$ and $g = f^{-1}$, then $a = I_A(a) = (g \circ f)(a) = g\{f(a)\} = g(b)$, i.e. $f^{-1}(b) = a$. ■

- For an invertible f, the statements $f(a) = b$ and $a = f^{-1}(b)$ are equivalent.
- If $f = \{(a, b) \mid a \in A,\ b \in B\}$ is invertible, then $f^{-1} = \{(b, a) \mid b \in B,\ a \in A\}$ — reverse every pair — and conversely.

- If f is invertible, then f^{-1} is invertible and $(f^{-1})^{-1} = f$.

Invertible $\iff$ bijective

A function $f : A \to B$ is invertible if and only if it is one-to-one and onto.

Proof. Suppose f is invertible with inverse g. If $f(a_1) = f(a_2)$, then $g\{f(a_1)\} = g\{f(a_2)\}$, i.e. $I_A(a_1) = I_A(a_2)$, so $a_1 = a_2$: f is one-to-one. For any $b \in B$, $g(b) \in A$ and $b = I_B(b) = f\{g(b)\}$, so b is the image of $g(b)$: f is onto.
Conversely, suppose f is one-to-one and onto. Then each $b \in B$ has a *unique* $a \in A$ with $f(a) = b$. Define $g : B \to A$ by $g(b) = a$. Then $(g \circ f)(a) = g(b) = a = I_A(a)$ and $(f \circ g)(b) = f(a) = b = I_B(b)$, so f is invertible with inverse g. ■

Finite sets of equal size

Let A, B be finite with $|A| = |B|$ and $f : A \to B$. The following are equivalent: (1) f is one-to-one (2) f is onto (3) f is invertible.

Proof. (1) $\iff$ (2) is the earlier theorem for finite sets of equal size; together they give a bijection, which is invertible by the previous theorem. Conversely an invertible f is one-to-one and onto. ■
Remark. So for $|A| = |B|$ finite, f is invertible iff it is one-to-one *or* onto.

Inverse of a composite

If $f : A \to B$ and $g : B \to C$ are invertible, then $g \circ f : A \to C$ is invertible and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof. f and g are one-to-one and onto, so $g \circ f$ is one-to-one and onto (composition theorem), hence invertible. Let $h = f^{-1} \circ g^{-1} : C \to A$. Then

$$(g \circ f) \circ h = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ I_B \circ g^{-1} = g \circ g^{-1} = I_C,$$
$$h \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ I_B \circ f = f^{-1} \circ f = I_A.$$

So h is the inverse of $g \circ f$: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ ("socks and shoes": undo the last step first). ■

Example 6.66 – Squaring is not invertible on $\mathbb{R}$

Is $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, invertible?

Solution. For any $a \in \mathbb{R}$, $f(a) = a^2 = f(-a)$, so a and $-a$ have the same image: f is not one-to-one, hence **not invertible**.

Example 6.67 – A square-root function and its inverse

Let $A = \{x \mid x \text{ real},\ x \geq -1\}$ and $B = \{x \mid x \text{ real},\ x \geq 0\}$. Show that $f : A \to B$, $f(a) = \sqrt{a+1}$, is invertible and determine f^{-1}.

Solution. *One-to-one:* $\sqrt{a_1+1} = \sqrt{a_2+1} \Rightarrow a_1 + 1 = a_2 + 1 \Rightarrow a_1 = a_2$.
Onto: for $b \in B$, $b = \sqrt{a+1}$ holds iff $b^2 = a + 1$, i.e. $a = b^2 - 1$; and $b^2 - 1 \geq -1$, so $a \in A$.
Hence f is invertible, with $f^{-1}(b) = b^2 - 1$ for all $b \in B$.

Example 6.68 – Exponential and logarithm

Find the inverse (if any) of $f(x) = e^x$ from $\mathbb{R}$ to $\mathbb{R}^+$.

Solution. $e^{x_1} = e^{x_2} \Rightarrow x_1 = x_2$, so f is one-to-one. For $y \in \mathbb{R}^+$, $x = \log_e y \in \mathbb{R}$ satisfies $e^x = y$, so f is onto. Hence f is invertible, with $f^{-1}(y) = \log_e y$ for $y \in \mathbb{R}^+$.

Example 6.69 – A cubic and a cube root

Let $A = B = \mathbb{R}$, $f(x) = 2x^3 - 1$ and $g(y) = \{\tfrac{1}{2}(y+1)\}^{1/3}$. Show that each of f and g is the inverse of the other.

Solution. $(g \circ f)(x) = \{\tfrac{1}{2}(2x^3 - 1 + 1)\}^{1/3} = (x^3)^{1/3} = x$, so $g \circ f = I_A$. And $(f \circ g)(y) = 2\left[\{\tfrac{1}{2}(y+1)\}^{1/3}\right]^3 - 1 = 2 \cdot \tfrac{1}{2}(y+1) - 1 = y$, so $f \circ g = I_B$. Hence each is invertible and is the inverse of the other.

Example 6.70 – Inverse of a composite

Let $A = B = C = \mathbb{R}$, $f(a) = 2a + 1$ and $g(b) = \tfrac{1}{3}b$. Compute $g \circ f$, show that it is invertible, and find $(g \circ f)^{-1}$.

Solution. $(g \circ f)(a) = g(2a+1) = \tfrac{1}{3}(2a+1)$. Now f is invertible with $f^{-1}(b) = \tfrac{1}{2}(b-1)$, and g with $g^{-1}(c) = 3c$. So $g \circ f$ is invertible and

$$(g \circ f)^{-1}(c) = (f^{-1} \circ g^{-1})(c) = f^{-1}(3c) = \tfrac{1}{2}(3c - 1).$$

Check: $\tfrac{1}{3}\left(2 \cdot \tfrac{1}{2}(3c-1) + 1\right) = c$. ✓

Example 6.71 – Inverse of a permutation

$A = \{1, 2, 3\}$; $f = \{(1,2), (2,3), (3,1)\}$, $g = \{(1,2), (2,1), (3,3)\}$. Find $g \circ f$, $f \circ g$ and f^{-1}.

Solution. $g \circ f = \{(1,1), (2,3), (3,2)\}$ and $f \circ g = \{(1,3), (2,2), (3,1)\}$. f is a bijection, so reverse its pairs: $f^{-1} = \{(2,1), (3,2), (1,3)\}$.

6.22 Exercise Set 6G — invertible functions

6G.1 Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$. Determine whether the following functions from A to B are invertible: (1) $f = \{(1,a), (2,a), (3,c), (4,d)\}$ (2) $g = \{(1,a), (2,c), (3,d), (4,d)\}$.

6G.2 Let $A = B = \mathbb{R}$. Show that $f : A \to B$, $f(a) = a + 1$, is invertible.

6G.3 Find the inverse of $f : A \to B$ in each case: (1) $A = B = \{1, 2, 3, 4, 5\}$, $f = \{(1,3), (2,2), (3,4), (4,5), (5,1)\}$ (2) $A = B = \mathbb{R}$, $f(x) = ax + b$, $a \neq 0$ and b constants (3) $f(a) = a^2 - 1$ from $\{x \mid x \text{ real},\ x \geq 0\}$ to $\{y \mid y \text{ real},\ y \geq -1\}$ (4) $A = \mathbb{R}$, $B = \mathbb{R}^+$, $f(x) = e^{2x+5}$.

6G.4 Let $f : \mathbb{Z} \to \mathbb{N}$ be $f(x) = 2x - 1$ if $x > 0$ and $f(x) = -2x$ if $x \leq 0$. Prove that f is a bijection and find f^{-1}.

6G.5 For the functions $f, g : \mathbb{R} \to \mathbb{R}$ below, verify that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$: (i) $f(x) = 2x$, $g(x) = 3x - 2$ (ii) $f(x) = \tfrac{1}{2}(x+1)$, $g(x) = \tfrac{1}{2}(x-1)$.

6G.6 Let $f : X \to Y$ be invertible and $C, D \subseteq Y$. Prove that if $C \subseteq D$ then $f^{-1}(C) \subseteq f^{-1}(D)$.

6G.7 For a function $f : X \to Y$ and $A \subset X$, disprove that $f^{-1}[f(A)] = A$ in general. (What happens if f is invertible?)

6.23 Solutions to Section 6 exercises

Set 6A

6A.1 f gives each of 1, 2, 3 exactly one image: f **is a function**. g gives no image to 2: not a function. h relates 1 to both 3 and 2: not a function.

6A.2 (i) **Yes**; range $\{a, b\}$. (ii) **No** — 1 is related to both a and b. (iii) **No** — 4 has no image. (iv) **Yes**; range

$\{a\}$ (a constant function).

6A.3 (i) Each integer x gives exactly one integer $y = 3x + 1$: a **function**, with range $\{3k + 1 \mid k \in \mathbb{Z}\} = \{\dots, -2, 1, 4, 7, \dots\}$.
(ii) A **function**, range $\{x^2 + 3 \mid x \in \mathbb{Z}\} = \{3, 4, 7, 12, 19, \dots\}$.
(iii) **Not a function**: $x = 1$ is related to $y = 1$ and $y = -1$ (and $x = -1$ to no y).
(iv) **Not a function**: $x = 0$ is related to $y = 1$ and $y = -1$ (and $x = 2$ to none).

6A.4 A function from A contains exactly one pair $(a, f(a))$ for each $a \in A$, so it has exactly $|A| = 5$ pairs. Since $|R| = 6 \neq 5$, some element of A appears in two pairs; hence R is not a function. ■

6A.5 $f(0) = 1$, $f(1) = 1$, $f(-1) = 3$, $f(2) = 3$, $f(-2) = 7$. Range $= \{1, 3, 7\}$.

6A.6 Range of x^2 on $\mathbb{R}$: $[0, \infty)$. $f(\mathbb{Z}) = \{0, 1, 4, 9, 16, \dots\}$, the perfect squares. $f([-2, 1]) = [0, 4]$ (the value 4 comes from $x = -2$, the value 0 from $x = 0$).

6A.7 $f(A_1) = \{w\}$, $f(A_2) = \{x\}$, $f(A_3) = \{w, x\}$, $f(A_4) = \{x, y\}$.

6A.8 $g(2) = 5$, $g(3) = 10$, $g(4) = 17$, $g(5) = 26$, $g(6) = 37$, and the same for h. (i) $A \subset \mathbb{Q} \subset \mathbb{R}$ and $f(x) = g(x) = h(x)$ on A: f is a restriction of g and of h. (ii) g agrees with f on A, so g extends f; and $g(x) = h(x)$ for all $x \in \mathbb{Q}$, so g is the restriction of h to $\mathbb{Q}$. (iii) h agrees with f on A and with g on $\mathbb{Q}$, so h extends both.

6A.9 $f^{-1}(3) = \{1, 2\}$, $f^{-1}(4) = \{3, 5\}$, $f^{-1}(B_1) = \{1, 2, 3, 5\}$, $f^{-1}(B_2) = \{3, 4, 5\}$.

6A.10 The domain is $\mathbb{Z}$. $x^2 + 5 = 6 \iff x = \pm 1$: $f^{-1}(\{6\}) = \{-1, 1\}$. $6 \le x^2 + 5 \le 7 \iff 1 \le x^2 \le 2$: $f^{-1}([6, 7]) = \{-1, 1\}$. $1 \le x^2 \le 5$: $f^{-1}([6, 10]) = \{-2, -1, 1, 2\}$. $x^2 + 5 \ge 5$ always, so $f^{-1}([-4, 5)) = \emptyset$ and $f^{-1}([-4, 5]) = \{0\}$. $f^{-1}([5, \infty)) = \mathbb{Z}$.

6A.11 Now the domain is $\mathbb{R}$. $1 \le x^2 \le 2$: $f^{-1}([6, 7]) = [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$. $1 \le x^2 \le 5$: $f^{-1}([6, 10]) = [-\sqrt{5}, -1] \cup [1, \sqrt{5}]$. $f^{-1}([5, \infty)) = \mathbb{R}$.

6A.12 The three pieces have ranges $(-\infty, 7]$ (for $x \le 0$), $(-1, 5)$ (for $0 < x < 3$) and $[2, \infty)$ (for $x \ge 3$). Solve on each piece and keep solutions lying in that piece:

Set	$x + 7, x \le 0$	$-2x + 5, 0 < x < 3$	$x - 1, x \ge 3$	Preimage
$\{-10\}$	-17	—	—	$\{-17\}$
$\{0\}$	-7	$5/2$	—	$\{-7, 5/2\}$
$\{4\}$	-3	$1/2$	5	$\{-3, 1/2, 5\}$
$\{6\}$	-1	—	7	$\{-1, 7\}$
$\{7\}$	0	—	8	$\{0, 8\}$
$\{8\}$	—	—	9	$\{9\}$
$[-5, -1]$	$[-12, -8]$	—	—	$[-12, -8]$
$[-5, 0]$	$[-12, -7]$	$[5/2, 3)$	—	$[-12, -7] \cup [5/2, 3)$
$[-2, 4]$	$[-9, -3]$	$[1/2, 3)$	$[3, 5]$	$[-9, -3] \cup [1/2, 5]$
$(5, 10)$	$(-2, 0]$	—	$(6, 11)$	$(-2, 0] \cup (6, 11)$
$[11, 17]$	—	—	$[12, 18]$	$[12, 18]$

For example, for $[-5, 0]$ on the middle piece: $-5 \le -2x + 5 \le 0 \iff 5/2 \le x \le 5$, intersected with $0 < x < 3$ gives $[5/2, 3)$. Watch the endpoints: $f(0) = 7$ belongs to $(5, 10)$, so 0 is included; $x = 3$ lies in the third piece, where $f(3) = 2$.

6A.13 $f^{-1}(\{6, 7, 8\}) = \{1, 2\}$ means: 1 and 2 map into $\{6, 7, 8\}$ (3 choices each), and 3, 4, 5 map into $B - \{6, 7, 8\} = \{9, 10, 11, 12\}$ (4 choices each). Number $= 3^2 \times 4^3 = 9 \times 64 = 576$.

6A.14 Each of the n elements has n possible images: n^n functions.

Set 6B

6B.1 (i) f_1: images b_2, b_3, b_1 distinct and all of B reached — **one-to-one and onto**. (ii) f_2: images d_2, d_1, d_4 distinct but d_3 missed — **one-to-one, not onto**. (iii) f_3: $b_1, b_2 \mapsto c_2$, and c_1, c_2 both reached — **onto, not one-to-one**. (iv) f_4: $d_1, d_3 \mapsto b_1$ and b_3 missed — **neither**.

6B.2 (i) The images w, x, y, z are distinct (one-to-one) and exhaust B (onto). (ii) $f(1) = f(2) = w$, so not one-to-one; w, x, y, z are all reached, so onto. (iii) f: images v, x, z, y are distinct, so one-to-one; w is missed, so not onto. g: $g(1) = g(2) = v$ (not one-to-one) and y, z are missed (not onto). ■

6B.3

Part	Answer	Reason
(i)	Both	images $1, 3, 4, 2$: a permutation of A
(ii)	Neither	$f(a) = f(b) = 1$; $2, 4$ missed
(iii)	One-to-one, not onto	images $1, 3, 4$ distinct; $2, 5$ missed
(iv)	Neither	$f(2) = f(3) = 3$; $2, 4, 5$ missed
(v)	Neither	$f(1) = f(2) = a$; b missed
(vi)	Onto, not one-to-one	$\lvert a\rvert = b$ has solutions $\pm b$
(vii)	Onto, not one-to-one	$f(c, 1) = c$; but $f(1, 2) = f(2, 1) = 2$
(viii)	Both	$(a + b, a - b) = (u, v) \iff a = \frac{u+v}{2},\ b = \frac{u-v}{2}$

6B.4

Part	One-to-one?	Onto?	Range / reason
(i) $2x - 3$	Yes	Yes	inverse $\frac{y+3}{2}$
(ii) x^3	Yes	Yes	inverse $y^{1/3}$
(iii) x^2	No	No	range $[0, \infty)$
(iv) $x^2 + x$	No	No	$x^2 + x = (x + \frac{1}{2})^2 - \frac{1}{4}$: range $[-\frac{1}{4}, \infty)$; $f(0) = f(-1)$
(v) e^{x^2}	No	No	$x^2 \geq 0$, so range $[1, \infty)$; $f(1) = f(-1)$
(vi) $\sin x$	No	No	range $[-1, 1]$; $\sin 0 = \sin \pi$

6B.5 *Not one-to-one:* $f(5, 0) = 20 = f(0, 4)$. *Onto:* for any $n \in \mathbb{Z}$, $f(-n, n) = -4n + 5n = n$. For $(n, 0) \in A$, $f(n, 0) = 4n$, so $f(A) = \{4n \mid n \in \mathbb{Z}^+\} = \{4, 8, 12, \ldots\}$.

6B.6 No. Take $X = \{1\}, Y = \{2\}, A = \{1, 2\}, B = \{b\}$ and $f(1) = f(2) = b$. Then f_1 and f_2 (each on a one-element set) are one-to-one, but f is not. (One-to-one on each piece does not prevent two pieces sharing an image.)

Set 6C

6C.1 $|A| = 3, |B| = 5$: $\dfrac{5!}{2!} = 5 \times 4 \times 3 = 60$.

6C.2 $\dfrac{n!}{(n-5)!} = n(n-1)(n-2)(n-3)(n-4) = 6720 = 8 \times 7 \times 6 \times 5 \times 4$, so $|B| = 8$.

6C.3 $p(4, 3) = 3^4 - 3 \times 2^4 + 3 \times 1^4 = 81 - 48 + 3 = 36$.

6C.4 $p(5, 3) = 3^5 - 3 \times 2^5 + 3 \times 1^5 = 243 - 96 + 3 = 150$.

6C.5 $|A| = 5, |B| = 8$.

(i) Functions $A \to B$: $8^5 = 32768$. One-to-one: $\dfrac{8!}{3!} = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6720$. Onto: 0 (as $5 < 8$).

(ii) Functions $B \to A$: $5^8 = 390625$. One-to-one: 0 (as $8 > 5$). Onto: $p(8, 5) = 5!\, S(8, 5) = 120 \times 1050 = 126000$.

6C.6 $n!$ (the permutations of A).

6C.7 There are $n!$ bijections in all; those with $f(1) = 1$ correspond to bijections of $\{2, \dots, n\}$, of which there are $(n-1)!$. Answer: $n! - (n-1)! = (n-1)(n-1)!$.

6C.8 $S(m,1) = \dfrac{1}{1!}\{1^m - \binom{1}{0}0^m\} = 1$ (for $m \geq 1$): all objects go into the single container. $S(m,m) = 1$: each of m objects must occupy its own container, and since the containers are identical there is only one way. (Formally $p(m,m) = m!$, so $S(m,m) = m!/m! = 1$.)

6C.9

$$S(5,3) = \tfrac{1}{3!}\{3^5 - 3\cdot 2^5 + 3\cdot 1^5\} = \tfrac{243-96+3}{6} = \tfrac{150}{6} = 25,$$
$$S(7,2) = \tfrac{1}{2!}\{2^7 - 2\cdot 1^7\} = \tfrac{126}{2} = 63,$$
$$S(8,5) = \tfrac{1}{5!}\{5^8 - 5\cdot 4^8 + 10\cdot 3^8 - 10\cdot 2^8 + 5\cdot 1^8\} = \tfrac{390625-327680+65610-2560+5}{120} = \tfrac{126000}{120} = 1050.$$

6C.10 Count all 6^8 functions from an 8-set A to a 6-set B by the size k of their range. Choose the range in $\binom{6}{k}$ ways; the number of functions from A *onto* a given k-set is $p(8,k) = k!\,S(8,k)$. Hence $\sum_{k=0}^{6}\binom{6}{k}k!\,S(8,k) = 6^8$. Numerically, with $S(8,0) = 0$:

$$6\cdot 1 + 15\cdot 2\cdot 127 + 20\cdot 6\cdot 966 + 15\cdot 24\cdot 1701 + 6\cdot 120\cdot 1050 + 1\cdot 720\cdot 266 = 1679616 = 6^8. \checkmark$$

Set 6D

6D.1 $D = \{(x, x^2) \mid x \in \mathbb{R}\}$. $\pi_A(x,x^2) = x$: distinct points of D have distinct first coordinates, so π_A is one-to-one; every real x is $\pi_A(x,x^2)$, so it is onto. $\pi_B(x,x^2) = x^2$: $\pi_B(1,1) = \pi_B(-1,1) = 1$, so not one-to-one; -1 is not a square, so not onto.

6D.2 $\lfloor 3.65\rfloor = 3$, $\lfloor -3.65\rfloor = -4$, $\lfloor -16\rfloor = -16$, $\lfloor 18.5\rfloor = 18$, $\lfloor -18.5\rfloor = -19$, $\lfloor\sqrt{5}\rfloor = 2$;
$\lceil 3.65\rceil = 4$, $\lceil -3.65\rceil = -3$, $\lceil -16\rceil = -16$, $\lceil 18.5\rceil = 19$, $\lceil -18.5\rceil = -18$, $\lceil\sqrt{5}\rceil = 3$.

6D.3 (i) $\lfloor 0.7\rfloor = 0$ (ii) $2 - 1 = 1$ (iii) $4\times 6 = 24$ (iv) $3\times 7 = 21$ (v) $2\pi \approx 44/7 \approx 6.29$, so $\lfloor 2\pi\rfloor = 6$ (vi) $\pi \approx 3.14$, so $2\lceil\pi\rceil = 2\times 4 = 8$.

6D.4 (i) If $x \in \mathbb{Z}$, then $x+a$ is an integer, and the floor of an integer is itself. (ii) Let $n = \lfloor x\rfloor$, so $n \leq x < n+1$. Then $n + a \leq x + a < n + a + 1$ with $n + a \in \mathbb{Z}$, so $\lfloor x + a\rfloor = n + a = \lfloor x\rfloor + a$. ■

6D.5 Divide: $n = qk + r$ with $0 \leq r < k$. If $r = 0$: $\lceil n/k\rceil = q$, and $\frac{n-1}{k} = q - \frac{1}{k}$ gives $\lfloor (n-1)/k\rfloor = q-1$; so RHS $= q$. If $1 \leq r < k$: $\lceil n/k\rceil = q+1$, and $\frac{n-1}{k} = q + \frac{r-1}{k}$ with $0 \leq \frac{r-1}{k} < 1$ gives $\lfloor (n-1)/k\rfloor = q$; so RHS $= q+1$. In both cases the sides agree. ■

6D.6 Take $a = 0.4$. Then $\lceil a\rceil = 1$ and $\lfloor 1/0.4\rfloor = \lfloor 2.5\rfloor = 2 \neq 1$. Also $\lfloor a\rfloor = 0$, so $\lceil 0/0.4\rceil = 0 \neq 1$. (In general for $0 < a < 1$: $\lceil\lfloor a\rfloor/a\rceil = 0$ always, and $\lfloor\lceil a\rceil/a\rfloor = \lfloor 1/a\rfloor \geq 1$, which exceeds 1 when $a \leq \frac{1}{2}$.)

6D.7 $f(1,2) = 1 - 2 = -1 \notin \mathbb{Z}^+$, so the operation is not closed on $\mathbb{Z}^+$.

6D.8 (i) $x+y-xy = y+x-yx$: commutative. $f(f(x,y),z) = x+y+z-xy-yz-zx+xyz = f(x,f(y,z))$ (symmetric in x,y,z): associative.
(ii) $x + y - 1$ is symmetric: commutative. $f(f(x,y),z) = x + y + z - 2 = f(x,f(y,z))$: associative.
(iii) $\max(x,y) = \max(y,x)$, and $\max(\max(x,y),z) = \max(x,y,z) = \max(x,\max(y,z))$: commutative and associative.
(iv) x^y: $2^3 = 8 \neq 9 = 3^2$, so not commutative; $2^{(3^2)} = 512 \neq 64 = (2^3)^2$, so not associative. (Strictly, x^y is not even closed on $\mathbb{Z}$, e.g. $2^{-1} = \frac{1}{2}$.)

6D.9 $\min(x,y) = \min(y,x)$; $\min(\min(x,y),z) = \min(x,y,z) = \min(x,\min(y,z))$; and for every $a \in A$, $\min(a,7) = \min(7,a) = a$ since $a \leq 7$. So f is commutative and associative with identity 7.

6D.10 $\gcd(a,b) = \gcd(b,a)$, and $\gcd(\gcd(a,b),c) = \gcd(a,b,c) = \gcd(a,\gcd(b,c))$. An identity e would need $\gcd(a,e) = a$, i.e. $a \mid e$, for *every* $a \in \mathbb{Z}^+$; no positive integer is divisible by every positive integer (e.g. $e + 1 \nmid e$). So there is no identity.

6D.11 $A \cup B \in \mathcal{P}(U)$ (closed); $A \cup B = B \cup A$; $(A \cup B) \cup C = A \cup (B \cup C)$. **The identity is** $\emptyset$, since $A \cup \emptyset = \emptyset \cup A = A$.

6D.12 $|A \times A| = 25$, so there are 5^{25} closed binary operations. For commutative ones, the 5 diagonal pairs and $\frac{25-5}{2} = 10$ unordered pairs can be assigned freely: $5^5 \times 5^{10} = 5^{15}$.

Set 6E

Complete step-by-step solutions to all twenty problems 6E.1–6E.20 (pigeons, pigeonholes, computation, conclusion and figures) are given in Section 6.18, directly after Exercise Set 6E, on pages 102–108.

Set 6F

6F.1 (i) $(g \circ f)(a) = (a-1)^2$ (ii) $(f \circ g)(b) = b^2 - 1$ (iii) $(f \circ f)(a) = a - 2$ (iv) $(g \circ g)(b) = b^4$.

6F.2 $(g \circ f)(a) = g(a+1) = 2(a+1), a \in \mathbb{Z}$.

6F.3 $(g \circ f)(x) = x^2 + 5$ and $(f \circ g)(x) = (x+5)^2$. At $x = 0$: $5 \neq 25$. Hence $g \circ f \neq f \circ g$.

6F.4 $g \circ f = f \circ g = I_{\mathbb{R}}$ (i.e. $x \mapsto x$); $f \circ f : x \mapsto x + 4$; $g \circ g : x \mapsto x - 4$; $f \circ h : x \mapsto 3x + 2$; $h \circ g : x \mapsto 3x - 6$; $h \circ f : x \mapsto 3x + 6$; $f \circ h \circ g : x \mapsto 3(x-2) + 2 = 3x - 4$.

6F.5 $(f \circ g)(x) = \log(e^x) = x$ for all $x \in \mathbb{R}$, so $f \circ g = I_{\mathbb{R}}$. $(g \circ f)(x) = e^{\log x} = x$ for all $x \in \mathbb{R}^+$, so $g \circ f = I_{\mathbb{R}^+}$.

6F.6 $(h \circ g)(x) = \sqrt{(x+5)^2 + 2}$, so $[(h \circ g) \circ f](x) = \sqrt{(x^2+5)^2 + 2}$. $(g \circ f)(x) = x^2 + 5$, so $[h \circ (g \circ f)](x) = \sqrt{(x^2+5)^2 + 2}$. Equal. ✓

6F.7 $g \circ f = \{(1,4), (2,6), (3,10), (4,14)\}$.

6F.8 $f \circ g = \{(1,3), (2,2), (3,1)\}$; $g \circ f = \{(1,1), (2,3), (3,2)\}$; $p \circ g = g = g \circ p$ (p is the identity); $f \circ p = f$; $f \circ h \circ g$: $1 \xrightarrow{g} 2 \xrightarrow{h} 2 \xrightarrow{f} 3$, $2 \to 1 \to 1 \to 2$, $3 \to 3 \to 1 \to 2$, so $f \circ h \circ g = \{(1,3), (2,2), (3,2)\}$.

6F.9 $f^2 = \{(1,2), (2,2), (3,2), (4,1)\}$ and $f^3 = f \circ f^2 = \{(1,2), (2,2), (3,2), (4,2)\}$.

6F.10 $(g \circ f)(x) = 3(x-1)$; $(h \circ g)(x) = h(3x) = h(x)$ (as $3x$ and x have the same parity); $f^2(x) = x - 2$, $f^3(x) = x - 3$; $g^2(x) = 9x$, $g^3(x) = 27x$; $h^2(x) = h(h(x)) = h(x)$ (as $h(0) = 0, h(1) = 1$), and likewise $h^3 = h$.

Set 6G

6G.1 (1) $f(1) = f(2) = a$: not one-to-one, so **not invertible**. (2) $g(3) = g(4) = d$ and b has no preimage, so g is also **not invertible**.

6G.2 $a_1 + 1 = a_2 + 1 \Rightarrow a_1 = a_2$ (one-to-one), and $f(b-1) = b$ (onto). So f is invertible with $f^{-1}(b) = b - 1$.

6G.3 (1) Reverse the pairs: $f^{-1} = \{(3,1), (2,2), (4,3), (5,4), (1,5)\}$.
(2) $y = ax + b \iff x = \frac{y-b}{a}$: $f^{-1}(y) = \frac{1}{a}(y-b), y \in \mathbb{R}$.
(3) $f(a) = a^2 - 1$ from $\{x \mid x \geq 0\}$ to $\{y \mid y \geq -1\}$ is a bijection: $y = a^2 - 1 \iff a = \sqrt{y+1}$ (taking $a \geq 0$). So $f^{-1}(y) = \sqrt{y+1}, y \geq -1$. (This is the reverse of the square-root example above.)
(4) $y = e^{2x+5} \iff \log y = 2x + 5 \iff x = \frac{1}{2}(\log y - 5)$: $f^{-1}(y) = \frac{1}{2}(\log y - 5), y \in \mathbb{R}^+$.

6G.4 (Here $\mathbb{N} = \{0, 1, 2, \ldots\}$.) f maps $x = 1, 2, 3, \ldots$ to $1, 3, 5, \ldots$ and $x = 0, -1, -2, \ldots$ to $0, 2, 4, \ldots$. Each piece is one-to-one, and the images (odd numbers versus even numbers) never overlap, so f is one-to-one; every natural number is odd or even, so f is onto. Inverse:

$$f^{-1}(y) = \begin{cases} \frac{1}{2}(y+1), & y = 1, 3, 5, 7, \ldots \\ -\frac{y}{2}, & y = 0, 2, 4, 6, \ldots \end{cases}$$

6G.5 (i) $(g \circ f)(x) = 3(2x) - 2 = 6x - 2$, so $(g \circ f)^{-1}(y) = \frac{y+2}{6}$. Also $f^{-1}(y) = \frac{y}{2}$, $g^{-1}(y) = \frac{y+2}{3}$, and $f^{-1}(g^{-1}(y)) = \frac{y+2}{6}$. ✓
(ii) $(g \circ f)(x) = \frac{1}{2}\left(\frac{x+1}{2} - 1\right) = \frac{x-1}{4}$, so $(g \circ f)^{-1}(y) = 4y + 1$. Also $f^{-1}(y) = 2y - 1$, $g^{-1}(y) = 2y + 1$, and $f^{-1}(g^{-1}(y)) = 2(2y+1) - 1 = 4y + 1$. ✓

6G.6 Let $x \in f^{-1}(C)$. Then $f(x) \in C \subseteq D$, so $x \in f^{-1}(D)$. Hence $f^{-1}(C) \subseteq f^{-1}(D)$. (This holds for

every function; invertibility is not needed.) ■

6G.7 In general only $A \subseteq f^{-1}[f(A)]$ holds. Counterexample: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $A = [-1, 0]$. Then $f(A) = [0, 1]$ and $f^{-1}([0, 1]) = [-1, 1] \neq A$.
If f is invertible (indeed, if f is merely one-to-one), equality *does* hold: $x \in f^{-1}[f(A)] \Rightarrow f(x) = f(a)$ for some $a \in A \Rightarrow x = a \in A$. So for an invertible f the equality always holds; the counterexample above works because x^2 is not invertible.

7 Size of a Set: Finite, Infinite, Countable, Uncountable

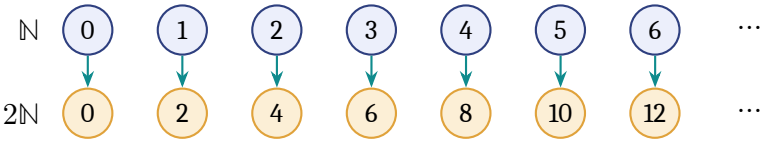
Definition: Cardinality

A and B have the **same cardinality**, $|A| = |B|$, if there is a bijection $A \to B$. We write $|A| \leq |B|$ if there is an injection $A \to B$, and $|A| < |B|$ if $|A| \leq |B|$ but $|A| \neq |B|$.

Definition: Finite, infinite, countable

A is **finite** if $A = \emptyset$ or $|A| = |\{1, \ldots, n\}|$ for some $n \in \mathbb{Z}^+$; otherwise it is **infinite**. A is **countably infinite** if $|A| = |\mathbb{N}|$ (written $\aleph_0$, "aleph-null"); **countable** if finite or countably infinite; **uncountable** otherwise.

Dedekind's characterisation: A is infinite iff it has the same cardinality as a proper subset of itself. A countably infinite set is exactly one whose elements can be listed as a sequence $a_0, a_1, a_2, \ldots$ with every element appearing.



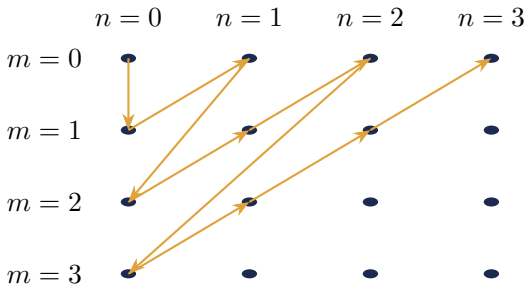
$f(n) = 2n$ is a bijection from $\mathbb{N}$ onto the even naturals, a proper subset — impossible for a finite set.

Basic facts on countable sets

1. Every subset of a countable set is countable; every infinite set contains a countably infinite subset.
2. If $f : A \to \mathbb{N}$ is injective, A is countable. If $f : \mathbb{N} \to A$ is surjective, A is countable.
3. $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$, $\mathbb{Q}$ and the set of all finite strings over a finite alphabet are countably infinite.
4. A finite or countable union of countable sets is countable.
5. $\mathbb{R}$, $(0, 1)$, the irrationals and $\mathcal{P}(\mathbb{N})$ are uncountable (Sections 8, 9).

Example 7.1 – $\mathbb{N} \times \mathbb{N}$ is countable — Cantor's pairing function

List the pairs diagonal by diagonal ($m + n = 0, 1, 2, \ldots$):



This gives the explicit bijection $\pi(m, n) = \dfrac{(m+n)(m+n+1)}{2} + n$: $\pi(0,0) = 0$, $\pi(1,0) = 1$,

$\pi(0,1)=2, \pi(2,0)=3, \pi(1,1)=4, \pi(0,2)=5, \pi(2,3)=18.$

Example 7.2 – $\mathbb{Q}$ is countable

Write each rational uniquely as p/q in lowest terms with $q>0$. The map $p/q \mapsto (p,q)$ is an injection $\mathbb{Q} \to \mathbb{Z}\times\mathbb{Z}^+$, which is countable (as $\mathbb{Z}\times\mathbb{Z}^+ \sim \mathbb{N}\times\mathbb{N}$). An infinite set with an injection into a countable set is countably infinite. Thus $\mathbb{Q}$, although dense in $\mathbb{R}$, has the same size as $\mathbb{N}$.

Example 7.3 – Intervals and $\mathbb{R}$ have the same size

(a) $(0,1) \to (a,b)$, $f(x)=a+(b-a)x$ is a bijection, so all open intervals are equinumerous. (b) $(0,1)\to\mathbb{R}$, $g(x)=\tan\left(\pi(x-\frac{1}{2})\right)$ is a bijection (tan maps $(-\frac{\pi}{2},\frac{\pi}{2})$ one-one onto $\mathbb{R}$). Hence $|(0,1)| = |\mathbb{R}|$, called $\mathfrak{c}$, the cardinality of the continuum.

7.1 Equivalent sets and countable sets

Definition: Equivalent sets

Non-empty sets A and B have the **same cardinality** (same size) if there is a one-to-one correspondence between their elements. They are then called **equivalent**, written $A \sim B$.

- $A=\{1,2,3\}$ and $B=\{a,b,c\}$: pairing $1\leftrightarrow a, 2\leftrightarrow b, 3\leftrightarrow c$ shows $A\sim B$, although $A\neq B$. **Equal sets are equivalent, but equivalent sets need not be equal.** Two finite sets are equivalent exactly when they have the same number of elements.
- $\mathbb{Z}^+ \sim E^+$ (positive even integers): $a\in\mathbb{Z}^+$ corresponds to $b=2a\in E^+$, and $b\in E^+$ to $a=b/2$. So **an infinite set can be equivalent to an infinite proper subset of itself**.

Definition: Countable (denumerable) set

A is **countable** if (i) A is finite, or (ii) $A\sim\mathbb{Z}^+$. Otherwise A is **uncountable**. (Since $\mathbb{Z}^+\sim\mathbb{N}$ via $n\mapsto n-1$, this agrees with the definition above.)

For example, $\{1,3,6,7\}$ is countable (finite), and E^+ is countable since $E^+\sim\mathbb{Z}^+$. An infinite countable set is one whose elements form a **sequence** $\{s_n\}=s_1,s_2,s_3,\ldots$ of distinct elements.

Three results

1. Every subset of a countable set is countable.
2. The union of two countable sets is countable.
3. The intersection of two uncountable sets need not be uncountable.

Example 7.4 – Reciprocals form a countable set

Show that $S=\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4},\ldots\}$ is countable.

Solution. The elements of S form the sequence $\{s_n\}$ with $s_n=\frac{1}{n}$, $n=1,2,3,\ldots$; that is, $n\mapsto\frac{1}{n}$ is a one-to-one correspondence $\mathbb{Z}^+\to S$. Hence S is countable.

Example 7.5 – A set given by a recurrence

Show that $A=\{1,2,4,7,11,16,\ldots\}$ is countable.

Solution. The elements form the sequence $\{a_n\}$ with $a_1=1$ and $a_n=a_{n-1}+(n-1)$ for $n\geq 2$ (so $a_2=2, a_3=4, a_4=7, a_5=11, a_6=16, a_7=22,\ldots$). Hence A is countable.

Aliter: $A \subseteq \mathbb{Z}^+$, which is countable, so A is countable by result 1.

Example 7.6 – The integers are countable

Show that $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$ is countable.

Solution. List the integers as $0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ 4,\ -4, \dots$: 0 is the first element, 1 the second, -1 the third, 2 the fourth, and so on. Every integer appears exactly once, so $\mathbb{Z}$ is the sequence $\{z_n\}$ with

$$z_n = -\frac{n-1}{2} \text{ if } n \text{ is odd}, \qquad z_n = \frac{n}{2} \text{ if } n \text{ is even} \qquad (n = 1, 2, 3, \dots).$$

Therefore $\mathbb{Z}$ is countable. (Indexing from $n = 0$ instead gives $f(n) = \frac{n}{2}$ for even n and $-\frac{n+1}{2}$ for odd n, listing $0, -1, 1, -2, 2, \dots$.)

Example 7.7 – The open interval $(0, 1)$ is not countable

Prove that $S = (0, 1) = \{x \mid x \text{ is a positive real number less than } 1\}$ is not countable.

Solution. Assume S is countable. Then its elements can be arranged as a sequence $S = \{s_1, s_2, s_3, \dots\} = \{s_n\}$. Write each in decimal form:

$$s_1 = 0.a_{11}a_{12}a_{13}a_{14}\dots, \quad s_2 = 0.a_{21}a_{22}a_{23}a_{24}\dots, \quad \dots, \quad s_n = 0.a_{n1}a_{n2}a_{n3}a_{n4}\dots, \ \dots$$

where each a_{ij} is an integer from 0 to 9. Now take $b = 0.b_1b_2b_3\dots$ with $b_1 \neq a_{11}, b_2 \neq a_{22}, b_3 \neq a_{33}, \dots$ (choose each b_i from 1 to 8). Then $b \in (0, 1)$, but b differs from every s_n in the n-th decimal place, so b is not a member of $\{s_n\}$. This contradicts $S = \{s_n\}$. Hence S is not countable. ■
(This is Cantor's diagonal argument; Section 8 explains why the digits 0 and 9 are avoided.)

Example 7.8 – Supersets of uncountable sets; intervals; $\mathbb{R}$

Prove: (1) If $A \subseteq B$ and A is uncountable, then B is uncountable. (2) The intervals $(0, 1]$, $[0, 1)$ and $[0, 1]$ are uncountable. (3) The set $\mathbb{R}$ of all real numbers is uncountable.

Solution. (1) Suppose B were countable. By result 1, every subset of B is countable, so A would be countable — contrary to what is given. Hence B is uncountable.
(2) $(0, 1] = \{x \mid 0 < x \leq 1\}$, $[0, 1) = \{x \mid 0 \leq x < 1\}$ and $[0, 1] = \{x \mid 0 \leq x \leq 1\}$ each contain the open interval $(0, 1)$, which is uncountable. By (1), they are uncountable.
(3) $(0, 1) \subset \mathbb{R}$ and $(0, 1)$ is uncountable, so $\mathbb{R}$ is uncountable by (1). ■

Example 7.9 – Union and differences of countable sets

Prove: (1) If A and B are countable, then $A \cup B$ is countable. (2) If A and B are countable, then $A - B$ and $B - A$ are countable.

Solution. (1) Write $A = \{s_1, s_2, s_3, \dots\}$ and $B = \{t_1, t_2, t_3, \dots\}$. Let $S = \{s_1, t_1, s_2, t_2, s_3, t_3, \dots\}$ (omitting any element already listed). Its elements form the sequence $a_1 = s_1$, $a_2 = t_1$, $a_3 = s_2$, $a_4 = t_2, \dots$, so S is countable. But $S = A \cup B$. (If A or B is finite, its list simply stops.)
(2) $A - B \subseteq A$ and $B - A \subseteq B$, so both are countable by result 1. ■

Example 7.10 – What can happen with uncountable sets

Illustrate: (1) a subset of an uncountable set can be countable; (2) the intersection of two uncountable sets can be countable; (3) if A and B are uncountable, $A - B$ can be countable.

Solution. (1) $\mathbb{R}$ is uncountable, $\mathbb{Z} \subset \mathbb{R}$, and $\mathbb{Z}$ is countable.
(2) Take $A = [0, 1]$ and $B = [1, 2]$, both uncountable (as in the previous example). Then $A \cap B = \{1\}$,

a finite and hence countable set.
(3) Take $A = [0,1]$ and $B = (0,1]$, both uncountable. Then $A - B = \{0\}$, which is countable.

Common pitfall

In (2) the two intervals must meet in a single point. For example, $[0,1] \cap (0,1] = (0,1]$ is uncountable, whereas $[0,1] \cap [1,2] = \{1\}$ is countable.

Cardinality	Examples
finite, n	$\{1, \dots, n\}$; any finite set
$\aleph_0$ (countably infinite)	$\mathbb{N}$, $\mathbb{Z}$, $2\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{N} \times \mathbb{N}$, finite binary strings, primes
$\mathfrak{c} = 2^{\aleph_0}$ (uncountable)	$\mathbb{R}$, $(0,1)$, $[0,1]$, irrationals, $\mathcal{P}(\mathbb{N})$, all infinite binary sequences
$2^{\mathfrak{c}}$	$\mathcal{P}(\mathbb{R})$, all functions $\mathbb{R} \to \mathbb{R}$

8 Cantor's Diagonal Argument

Cantor, 1891

The interval $(0,1)$ is uncountable. Consequently $\mathbb{R}$ is uncountable and $|\mathbb{N}| < |\mathbb{R}|$.

Proof (by contradiction).

Step 1: Suppose $(0,1)$ is countable. Then its elements can be listed as $r_1, r_2, r_3, \dots$, with every number appearing.

Step 2: Write each in decimal, $r_i = 0.d_{i1}d_{i2}d_{i3}\dots$ (choose the expansion not ending in all 9s, so it is unique).

Step 3: Build $x = 0.x_1x_2x_3\dots$ by changing the diagonal digits: $x_i = 5$ if $d_{ii} \neq 5$, and $x_i = 4$ if $d_{ii} = 5$.

Step 4: Then $x \in (0,1)$, and its digits are only 4s and 5s, so it has a unique expansion.

Step 5: For every i, x differs from r_i in the i-th digit, so $x \neq r_i$. Hence x is not in the list.

Step 6: This contradicts Step 1. So no list contains all of $(0,1)$: it is uncountable. Since $(0,1) \subseteq \mathbb{R}$, $\mathbb{R}$ is uncountable too. ■

r_i	1	2	3	4	5	⋯
$r_1 = 0.$	**3**	1	4	1	5	⋯
$r_2 = 0.$	2	**7**	1	8	2	⋯
$r_3 = 0.$	1	4	**1**	4	2	⋯
$r_4 = 0.$	1	2	3	**5**	0	⋯
$r_5 = 0.$	3	3	3	3	**3**	⋯
$x = 0.$	**5**	**5**	**5**	**4**	**5**	⋯

The diagonal is $3, 7, 1, 5, 3, \dots$; changing each digit gives $x = 0.55545\dots$, which differs from r_i in position i for every i.

Common pitfall

Why not flip digits to 0 or 9? Because $0.4999\dots = 0.5000\dots$: a number can have two decimal expansions, and the new x might coincide with a listed number via its other expansion. Using only 4s and 5s avoids

this.

Example 8.1 – Why the argument does not prove $\mathbb{N}$ or $\mathbb{Q}$ uncountable

For $\mathbb{N}$: natural numbers have only finitely many digits, so the "diagonal number" would have infinitely many non-zero digits and is not a natural number at all. For $\mathbb{Q} \cap (0, 1)$: the constructed x is generally irrational (its expansion need not repeat), so it does not belong to the set being listed — no contradiction arises.

Example 8.2 – Binary version

Show that the set S of all infinite binary sequences is uncountable.

Solution. If $s_1, s_2, \ldots$ listed all of S, define $d(i) = 1 - s_i(i)$. Then d differs from s_i in position i for each i, so d is missing — a contradiction. (Here no double-expansion problem arises.)

9 The Power Set Theorem

Cantor's Power Set Theorem

For every set A (finite or infinite), $|A| < |\mathcal{P}(A)|$. In particular there is no surjection $A \to \mathcal{P}(A)$.

Proof. *Part 1 ($|A| \le |\mathcal{P}(A)|$):* $a \mapsto \{a\}$ is an injection $A \to \mathcal{P}(A)$.
Part 2 (no surjection):

Step 1: Suppose $g : A \to \mathcal{P}(A)$ is onto.

Step 2: Define the diagonal set $D = \{a \in A : a \notin g(a)\}$. It is a subset of A, so $D \in \mathcal{P}(A)$.

Step 3: As g is onto, $D = g(a_0)$ for some $a_0 \in A$.

Step 4: If $a_0 \in D$, then by the definition of D, $a_0 \notin g(a_0) = D$ — contradiction. If $a_0 \notin D = g(a_0)$, then by definition $a_0 \in D$ — contradiction.

Step 5: Both cases are impossible, so g cannot be onto. Hence $|A| \neq |\mathcal{P}(A)|$, and with Part 1, $|A| < |\mathcal{P}(A)|$. ■

Consequence: $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \cdots$ — there is no largest infinity. Also $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = \mathfrak{c}$ (identify a subset $S \subseteq \mathbb{N}$ with its 0/1 indicator sequence).

Example 9.1 – The diagonal set for a concrete map

$A = \{1, 2, 3\}$, $g(1) = \{1, 2\}$, $g(2) = \{2\}$, $g(3) = \emptyset$. Find D and show g misses it.

Solution. $1 \in g(1)$, so $1 \notin D$; $2 \in g(2)$, so $2 \notin D$; $3 \notin g(3)$, so $3 \in D$. $D = \{3\}$, which equals none of $\{1, 2\}, \{2\}, \emptyset$. Of the 8 subsets of A only 3 can be hit by g; D is always one of those missed.

Examination tip

The Power Set proof and the diagonal argument are the same idea: D "disagrees with $g(a)$ at the point a", just as x disagrees with r_i at digit i. Say this in a long answer — it shows understanding.

10 Schröder–Bernstein Theorem

Cantor–Schröder–Bernstein

If there are injections $f : A \to B$ and $g : B \to A$, then there is a bijection $h : A \to B$. That is, $|A| \le |B|$ and $|B| \le |A|$ imply $|A| = |B|$.

Proof.

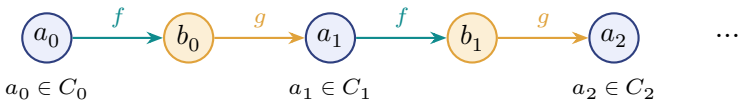
Step 1: Let $C_0 = A - g(B)$ (elements of A with no g-pre-image) and $C_{n+1} = g(f(C_n))$ for $n \ge 0$. Put $C = \bigcup_{n\ge 0} C_n$.

Step 2: Define $h(x) = f(x)$ if $x \in C$, and $h(x) = g^{-1}(x)$ if $x \notin C$. This is well defined: $x \notin C$ implies $x \notin C_0$, so $x \in g(B)$, and $g^{-1}(x)$ is unique because g is injective.

Step 3: *h is one-one.* Both points in C: f is injective. Both outside C: g^{-1} is injective. If $x \in C_n$, $y \notin C$ and $f(x) = g^{-1}(y)$, then $y = g(f(x)) \in C_{n+1} \subseteq C$ — contradiction.

Step 4: *h is onto.* Take $b \in B$. If $g(b) \notin C$, then $h(g(b)) = g^{-1}(g(b)) = b$. If $g(b) \in C_n$, then $n \ge 1$ (as $g(b) \in g(B)$), so $g(b) = g(f(x))$ for some $x \in C_{n-1}$; as g is injective, $b = f(x) = h(x)$.

Step 5: Hence h is a bijection $A \to B$. ■



A chain starting in C_0: on such chains h uses f; elsewhere it uses g^{-1}.

Why it matters: two easy injections replace one hard bijection.

Example 10.1 – $|[0,1]| = |(0,1)|$

$f : [0,1] \to (0,1)$, $f(x) = \frac{x+1}{3}$ is injective with values in $[\frac{1}{3}, \frac{2}{3}] \subset (0,1)$. $g : (0,1) \to [0,1]$, $g(x) = x$ (inclusion) is injective. By Schröder–Bernstein, $|[0,1]| = |(0,1)|$. (Writing down such a bijection directly is awkward, because the end-points 0 and 1 must be fitted in somewhere; the theorem avoids this.)

Example 10.2 – $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ without the pairing formula

$f(n) = (n, 0)$ is an injection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$; $g(m, n) = 2^m 3^n$ is an injection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by uniqueness of prime factorisation. Hence $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$.

Example 10.3 – $|[0,1]| = |\mathbb{R}|$

Inclusion $[0,1] \to \mathbb{R}$ is injective; $x \mapsto \frac{1}{2} + \frac{1}{\pi} \arctan x$ maps $\mathbb{R}$ injectively into $(0,1) \subset [0,1]$. Hence $|[0,1]| = |\mathbb{R}|$.

Example 10.4 – Finite binary strings

Let S be the set of finite binary strings (including the empty string ε). $f : \mathbb{N} \to S$, $n \mapsto$ its binary numeral, is injective. $g : S \to \mathbb{N}$, $w \mapsto$ the number whose binary numeral is "1" followed by w, minus 1 (so $\varepsilon \mapsto 0$, $0 \mapsto 1$, $1 \mapsto 2$, $00 \mapsto 3$, $101 \mapsto 12$), is injective because the leading 1 marks where w starts. Hence $|S| = \aleph_0$.

11 Solved University Examination Problems

These thirteen problems are typical of university examination questions. The marks shown are only a guide.

Problem 1: Set identity by laws — 7 marks

Q. Prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$ using the laws of set algebra.

Solution.

$$\begin{aligned}
(A \cap B) - (A \cap C) &= (A \cap B) \cap (A \cap C)^c && X - Y = X \cap Y^c \\
&= (A \cap B) \cap (A^c \cup C^c) && \textit{De Morgan} \\
&= (A \cap B \cap A^c) \cup (A \cap B \cap C^c) && \textit{Distributive} \\
&= \emptyset \cup (A \cap (B \cap C^c)) && \textit{Complement, Domination, Associative} \\
&= A \cap (B - C) && \textit{Identity}
\end{aligned}$$

Problem 2: Membership table — 5 marks

Q. Using a membership table, prove $A - (B - C) = (A - B) \cup (A \cap C)$.

Solution.

A	B	C	$B-C$	$A-(B-C)$	$A-B$	$A\cap C$	$(A-B)\cup(A\cap C)$
1	1	1	0	**1**	0	1	**1**
1	1	0	1	**0**	0	0	**0**
1	0	1	0	**1**	1	1	**1**
1	0	0	0	**1**	1	0	**1**
0	1	1	0	**0**	0	0	**0**
0	1	0	1	**0**	0	0	**0**
0	0	1	0	**0**	0	0	**0**
0	0	0	0	**0**	0	0	**0**

The bold columns agree in all 8 rows, so the identity holds.

Problem 3: Relation properties from a matrix — 7 marks

Q. On $A = \{1, 2, 3, 4\}$, $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$. Write M_R, test the four properties and find R^2.

Solution.

$$M_R = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

- Reflexive? No: $m_{33} = 0$, i.e. $(3,3) \notin R$.
- Symmetric? No: $(3,4) \in R$ but $(4,3) \notin R$.
- Antisymmetric? No: $(1,2)$ and $(2,1)$ both in R, $1 \neq 2$.
- Transitive? No: $(3,4), (4,1) \in R$ but $(3,1) \notin R$.

$R^2 = R \circ R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,4), (4,1), (4,2), (4,4)\}$. Since $R^2 \not\subseteq R$, this confirms R is not transitive.

Examination tip

A relation is transitive iff $R^2 \subseteq R$. Computing $M_R \odot M_R$ and comparing entry-wise is a fast, mechanical test for exam answers.

Problem 4: Equivalence relation and classes 7 marks

Q. On $\mathbb{Z}$, $a\,R\,b$ iff $3a+b$ is a multiple of 4. Prove R is an equivalence relation and find its classes.

Solution. Since $4a$ is a multiple of 4, $3a+b \equiv 0 \pmod 4 \iff b \equiv -3a \equiv a \pmod 4$. So $a\,R\,b \iff a \equiv b \pmod 4$.
Reflexive: $3a + a = 4a$. Symmetric: if $4 \mid 3a+b$ then $3b + a = 3(3a+b) - 8a$ is a multiple of 4. Transitive: $a \equiv b, b \equiv c \pmod 4$ give $a \equiv c \pmod 4$. Classes: $[0] = \{4k\}$, $[1] = \{4k+1\}$, $[2] = \{4k+2\}$, $[3] = \{4k+3\}$, $k \in \mathbb{Z}$.

Problem 5: Equivalence relation on pairs 7 marks

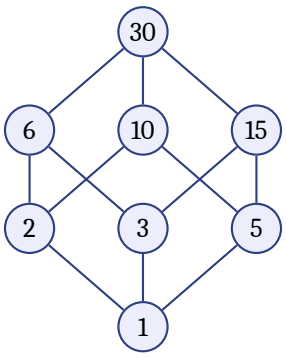
Q. On $A = \mathbb{Z}^+ \times \mathbb{Z}^+$ define $(a,b)\,R\,(c,d)$ iff $ad = bc$. Show R is an equivalence relation and describe $[(1,2)]$.

Solution. $ad = bc \iff \frac{a}{b} = \frac{c}{d}$ (all entries positive). Reflexive: $ab = ba$. Symmetric: $ad = bc \Rightarrow cb = da$. Transitive: $ad = bc$ and $cf = de$ give $adcf = bcde$; cancel $cd \neq 0$ to get $af = be$. $[(1,2)] = \{(k, 2k) : k \in \mathbb{Z}^+\}$ — all fractions equal to $\frac{1}{2}$. (This is how $\mathbb{Q}^+$ is constructed.)

Problem 6: Hasse diagram of D_{30} 7 marks

Q. Draw the Hasse diagram of $(D_{30}, |)$. Find the least and greatest elements, $\text{lub}\{6,10\}$, $\text{glb}\{6,10\}$, $\text{lub}\{2,15\}$.

Solution. $D_{30} = \{1,2,3,5,6,10,15,30\}$.

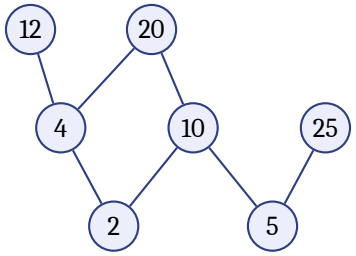


Least element 1; greatest 30. $\text{lub}\{6,10\} = \text{lcm} = 30$; $\text{glb}\{6,10\} = \text{gcd} = 2$; $\text{lub}\{2,15\} = 30$. The diagram is a cube, because $30 = 2 \cdot 3 \cdot 5$ has three distinct prime factors.

Problem 7: Extremal elements and bounds 10 marks

Q. For $A = \{2,4,5,10,12,20,25\}$ under divisibility, draw the Hasse diagram and find maximal, minimal, greatest, least elements; the upper bounds and LUB of $\{2,5\}$; the lower bounds and GLB of $\{12,20\}$.

Solution. Covering pairs: $2 \to 4$, $2 \to 10$, $5 \to 10$, $5 \to 25$, $4 \to 12$, $4 \to 20$, $10 \to 20$.



Item	Answer
Minimal elements	2, 5
Maximal elements	12, 20, 25
Least / greatest element	none (two minimal, three maximal)
Upper bounds of $\{2,5\}$; LUB	10, 20; LUB $= 10$
Lower bounds of $\{12,20\}$; GLB	2, 4; GLB $= 4$

Problem 8: Bijection and inverse — 7 marks

Q. Show that $f:\mathbb{R}-\{3\}\to\mathbb{R}-\{1\}$, $f(x)=\dfrac{x-2}{x-3}$, is a bijection and find f^{-1}.

Solution. *Well defined:* if $\frac{x-2}{x-3}=1$ then $x-2=x-3$, impossible; so $f(x)\neq 1$.
One-one: $\frac{x-2}{x-3}=\frac{y-2}{y-3}\Rightarrow(x-2)(y-3)=(y-2)(x-3)\Rightarrow -3x-2y=-3y-2x\Rightarrow x=y$.
Onto: for $y\neq 1$, solve $y=\frac{x-2}{x-3}$: $yx-3y=x-2\Rightarrow x(y-1)=3y-2\Rightarrow x=\frac{3y-2}{y-1}$. This $x\neq 3$ (else $3y-3=3y-2$). So every $y\neq 1$ has a pre-image.
Hence f is bijective and $f^{-1}(y)=\dfrac{3y-2}{y-1}$.

Common pitfall

Check the excluded points. Without proving $f(x)\neq 1$ and $x\neq 3$, the domain and codomain claims are unjustified and the answer loses marks.

Problem 9: Composition and the inverse of a composite — 7 marks

Q. $f(x)=2x+3$, $g(x)=3x-1$ on $\mathbb{R}$. Find $g\circ f$, $(g\circ f)^{-1}$ and verify $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$.

Solution. $(g\circ f)(x)=3(2x+3)-1=6x+8$, so $(g\circ f)^{-1}(y)=\frac{y-8}{6}$.
$f^{-1}(y)=\frac{y-3}{2}$, $g^{-1}(y)=\frac{y+1}{3}$, so $(f^{-1}\circ g^{-1})(y)=f^{-1}\left(\frac{y+1}{3}\right)=\frac{\frac{y+1}{3}-3}{2}=\frac{y-8}{6}$. ✓

Problem 10: Counting relations and functions — 5 marks

Q. $|A|=3$, $|B|=4$. Find the number of (a) relations from A to B; (b) functions $A\to B$; (c) one-one functions $A\to B$; (d) onto functions $A\to B$; (e) reflexive and (f) symmetric relations on A.

Solution. (a) $2^{3\cdot 4}=4096$. (b) $4^3=64$. (c) $4\cdot 3\cdot 2=24$. (d) 0, since $|A|<|B|$. (e) $2^{9-3}=64$. (f) $2^{3\cdot 4/2}=64$.

Problem 11: Countability — 10 marks

Q. (a) Prove that $\mathbb{N}\times\mathbb{N}$ is countable. (b) Deduce that $\mathbb{Q}$ is countable. (c) Prove that the union of two countable sets is countable.

Solution. (a) $\pi(m,n) = \frac{(m+n)(m+n+1)}{2} + n$ lists pairs diagonal by diagonal; each pair gets exactly one position, so π is a bijection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (Section 7). Alternatively, use $f(n) = (n, 0)$ and $g(m,n) = 2^m 3^n$ with Schröder–Bernstein.
(b) p/q in lowest terms with $q > 0 \mapsto (p, q)$ injects $\mathbb{Q}$ into $\mathbb{Z} \times \mathbb{Z}^+$, which is countable as in (a). So $\mathbb{Q}$ is countable; it is infinite, hence countably infinite.
(c) If $A = \{a_0, a_1, \dots\}$ and $B = \{b_0, b_1, \dots\}$, the list $a_0, b_0, a_1, b_1, \dots$ (deleting repeats) contains every element of $A \cup B$. If either set is finite, its list simply stops. So $A \cup B$ is countable. ■

Problem 12: Diagonal argument — 10 marks

Q. Prove by Cantor's diagonal argument that $\mathbb{R}$ is uncountable. Explain why the same argument fails for $\mathbb{Q}$.

Solution. Assume $(0,1)$ is listed as $r_1, r_2, \dots$ with $r_i = 0.d_{i1}d_{i2}\dots$. Define $x = 0.x_1x_2\dots$ with $x_i = 5$ if $d_{ii} \neq 5$, else $x_i = 4$. Then $x \in (0,1)$ and $x \neq r_i$ for every i (they differ in digit i), contradicting the completeness of the list. So $(0,1)$, and therefore $\mathbb{R}$, is uncountable.
For $\mathbb{Q}$: if we list $\mathbb{Q} \cap (0,1)$ and build x, there is no reason for x to be rational (its digits need not repeat). So x lies outside the set and no contradiction arises; indeed $\mathbb{Q}$ is countable. ■

Problem 13: Power Set and Schröder–Bernstein — 10 marks

Q. (a) State and prove the Power Set theorem. (b) Using the Schröder–Bernstein theorem, prove $|(0,1)| = |[0,1]|$.

Solution. (a) See Section 9: $a \mapsto \{a\}$ gives $|A| \leq |\mathcal{P}(A)|$; if $g : A \to \mathcal{P}(A)$ were onto, $D = \{a : a \notin g(a)\} = g(a_0)$ leads to $a_0 \in D \iff a_0 \notin D$. Hence $|A| < |\mathcal{P}(A)|$.
(b) $g : (0,1) \to [0,1]$, $g(x) = x$, is injective; $f : [0,1] \to (0,1)$, $f(x) = \frac{x+1}{3}$, is injective (linear with non-zero slope) with range $[\frac{1}{3}, \frac{2}{3}] \subset (0,1)$. By Schröder–Bernstein, $|(0,1)| = |[0,1]|$. ■

12 Exercises

Attempt these before consulting Section 13, which has complete solutions.

12.1 Part A — Short answer (2 marks each)

A1. Distinguish between $\emptyset$ and $\{\emptyset\}$. Write $\mathcal{P}(\{\emptyset\})$.

A2. If $A = \{1, 2\}$ and $B = \{2, 3\}$, find $A \times B$ and $|\mathcal{P}(A \times B)|$.

A3. State De Morgan's laws for sets.

A4. Find $A \triangle B$ for $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$.

A5. Is the empty relation on $A = \{1\}$ reflexive? Symmetric? Transitive?

A6. Give a relation on $\{1, 2\}$ that is both symmetric and antisymmetric.

A7. How many relations are there on a 4-element set? How many are reflexive?

A8. Is $(\mathbb{Z}, <)$ a poset? Justify.

A9. Write the equivalence classes of congruence modulo 2 on $\mathbb{Z}$.

A10. Find the range of $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + 1$. Is f onto?

A11. How many functions are there from a 3-element set to a 2-element set? How many are one-one?

A12. If $f(x) = 2x$ and $g(x) = x + 1$, find $f \circ g$ and $g \circ f$.

A13. Define a countable set and give a countably infinite set other than $\mathbb{N}$.

A14. State Cantor's Power Set theorem. If $|A| = 5$, find $|\mathcal{P}(A)|$.

A15. State the Schröder–Bernstein theorem.

12.2 Part B — Medium answer (5 / 7 marks)

B1. Prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ by the element method.

B2. Using the laws of sets, prove $[(A \cup B) \cap C] \cup (B^c \cap C) = C$.

B3. Prove $A \times (B - C) = (A \times B) - (A \times C)$.

B4. On $A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Find R^2 and test the properties reflexive, irreflexive, symmetric, antisymmetric, transitive.

B5. Show that $x \, R \, y$ iff $x - y \in \mathbb{Z}$ is an equivalence relation on $\mathbb{R}$, and describe $[\frac{1}{2}]$.

B6. Draw the Hasse diagram of $(\mathcal{P}(\{a, b, c\}), \subseteq)$. Find lub and glb of $\{\{a\}, \{b\}\}$ and of $\{\{a, b\}, \{b, c\}\}$.

B7. For $A = \{1, 2, 3, 4, 6, 8, 12\}$ under divisibility, draw the Hasse diagram; find maximal and minimal elements, lub$\{2, 3\}$, glb$\{8, 12\}$ and the upper bounds of $\{4, 6\}$.

B8. Show $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 4x - 7$, is bijective and find f^{-1}.

B9. For $f : A \to B$ and $C, D \subseteq B$, prove $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

B10. Prove that if $g \circ f$ is injective then f is injective. Give an example where $g \circ f$ is injective but g is not.

B11. Show that the set of all integer multiples of 5 is countably infinite.

B12. Show that $|(0, 1)| = |(2, 5)|$ and $|(0, 1)| = |\mathbb{R}|$.

12.3 Part C — Long answer (10 / 15 marks)

C1. (a) Prove $(A-B)-C = A-(B\cup C)$ using laws of sets. (b) Prove $A\cap(B\triangle C) = (A\cap B)\triangle(A\cap C)$ using a membership table.

C2. (a) On $A=\{1,2,\dots,12\}$, $a\,R\,b$ iff $a\equiv b \pmod 4$. List the classes and find $|R|$. (b) Write the equivalence relation induced by the partition $\{\{1,2\},\{3,4,5\},\{6\}\}$ and find its size.

C3. State and prove the theorem that the equivalence classes of an equivalence relation on A partition A.

C4. (a) Show that $f:\mathbb{R}-\{-1\}\to\mathbb{R}-\{1\}$, $f(x)=\frac{x-1}{x+1}$, is a bijection and find f^{-1}. (b) Prove that $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$ for bijections $f:A\to B, g:B\to C$.

C5. (a) Prove that $\mathbb{N}\times\mathbb{N}$ is countable. (b) Show that $\mathbb{Q}$ is countable. (c) Is the set of irrational numbers countable? Justify.

C6. (a) State and prove Cantor's Power Set theorem. (b) Prove $|[0,1]| = |\mathbb{R}|$ using the Schröder–Bernstein theorem.

12.4 Part D — Computation practice (5 marks each)

D1. For a 3-element set, count the relations that are (a) reflexive, (b) symmetric, (c) antisymmetric, (d) both reflexive and symmetric.

D2. $M_R=\begin{pmatrix}1&1&0\\0&1&1\\0&0&1\end{pmatrix}$ on $\{1,2,3\}$. Test the four properties and compute $M_R\odot M_R$.

D3. On $\{1,2,3\}$, $R=\{(1,2),(2,3)\}$, $S=\{(2,1),(3,3),(3,2)\}$. Find $R\circ S$, $S\circ R$ (first relation applied first) and R^{-1}.

D4. List all equivalence relations on $\{1,2,3\}$ (as partitions).

D5. From $A=\{a,b,c,d\}$ to $B=\{1,2,3\}$: how many functions, one-one functions and onto functions are there?

D6. $f:\mathbb{Z}\to\mathbb{Z}$, $f(x)=x^2-1$. Find $f(\{-2,0,3\})$ and $f^{-1}(\{0,3,5\})$.

D7. $f(x)=x+2$, $g(x)=x^2$ on $\mathbb{R}$. Find $f+g$, fg, $f\circ g$, $g\circ f$ and evaluate each at $x=-1$.

D8. On the set of lines in a plane, test the relations "is perpendicular to" and "is parallel to (or equal to)" for reflexivity, symmetry and transitivity.

13 Solutions to Exercises

13.1 Part A

A1. $\emptyset$ has no elements; $\{\emptyset\}$ has one element, namely $\emptyset$. $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

A2. $A \times B = \{(1,2),(1,3),(2,2),(2,3)\}$; $|\mathcal{P}(A \times B)| = 2^4 = 16$.

A3. $(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$.

A4. $A \triangle B = \{1,2\} \cup \{5\} = \{1,2,5\}$.

A5. Not reflexive ($(1,1) \notin \emptyset$); symmetric and transitive vacuously.

A6. $\{(1,1),(2,2)\}$ (or $\emptyset$): no pair (a,b) with $a \neq b$, so both conditions hold.

A7. $2^{16} = 65536$ relations; $2^{16-4} = 4096$ reflexive.

A8. No. $<$ is not reflexive ($1 \not< 1$), so it is not a partial order (it is a *strict* order).

A9. $[0] = \{\dots, -2, 0, 2, \dots\}$ (even), $[1] = \{\dots, -1, 1, 3, \dots\}$ (odd).

A10. Range $= [1, \infty)$; not onto $\mathbb{R}$, since e.g. 0 has no pre-image.

A11. $2^3 = 8$ functions; 0 one-one (3 elements cannot map injectively into 2).

A12. $(f \circ g)(x) = 2(x+1) = 2x + 2$; $(g \circ f)(x) = 2x + 1$.

A13. Countable: finite or in bijection with $\mathbb{N}$. Example: $\mathbb{Z}$, or the even numbers.

A14. $|A| < |\mathcal{P}(A)|$ for every set A. $|\mathcal{P}(A)| = 2^5 = 32$.

A15. If injections $A \to B$ and $B \to A$ exist, a bijection $A \to B$ exists.

13.2 Part B

B1. $x \in A \cup (B \cap C) \iff x \in A$ or ($x \in B$ and $x \in C$) $\iff$ ($x \in A$ or $x \in B$) and ($x \in A$ or $x \in C$) (distributivity of "or" over "and") $\iff x \in (A \cup B) \cap (A \cup C)$. ■

B2.

$$\begin{aligned}
[(A \cup B) \cap C] \cup (B^c \cap C) &= C \cap [(A \cup B) \cup B^c] && \textit{Commutative, Distributive} \\
&= C \cap [A \cup (B \cup B^c)] = C \cap (A \cup U) && \textit{Associative, Complement} \\
&= C \cap U = C && \textit{Domination, Identity}
\end{aligned}$$

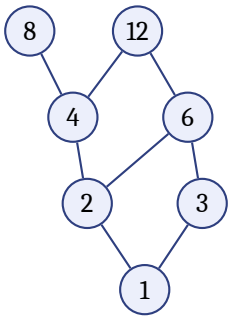
B3. $(x,y) \in A \times (B - C) \iff x \in A,\ y \in B,\ y \notin C \iff (x,y) \in A \times B$ and $(x,y) \notin A \times C$ (since $x \in A$, the pair fails to be in $A \times C$ exactly when $y \notin C$) $\iff (x,y) \in (A \times B) - (A \times C)$. ■

B4. $R^2 = \{(1,3),(2,4),(3,1),(4,2)\}$. Not reflexive, but irreflexive (no loops). Not symmetric ($(1,2) \in R$, $(2,1) \notin R$). Antisymmetric (no pair appears in both directions). Not transitive: $(1,2),(2,3) \in R$ but $(1,3) \notin R$.

B5. $x - x = 0 \in \mathbb{Z}$; $x - y \in \mathbb{Z} \Rightarrow y - x \in \mathbb{Z}$; $(x-y) + (y-z) = x - z$. So R is an equivalence relation. $[\frac{1}{2}] = \{\dots, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \dots\} = \{n + \frac{1}{2} : n \in \mathbb{Z}\}$: numbers with the same fractional part.

B6. The diagram is the cube in Section 4. lub$\{\{a\},\{b\}\} = \{a,b\}$, glb $= \emptyset$; lub$\{\{a,b\},\{b,c\}\} = \{a,b,c\}$, glb $= \{b\}$. In general lub $= \cup$ and glb $= \cap$.

B7.



Minimal: 1 (also least). Maximal: 8, 12 (no greatest). lub$\{2,3\} = 6$ (upper bounds 6, 12). glb$\{8,12\} = 4$ (lower bounds 1, 2, 4). Upper bounds of $\{4,6\}$: only 12.

B8. $4x_1 - 7 = 4x_2 - 7 \Rightarrow x_1 = x_2$; for any y, $x = \frac{y+7}{4}$ gives $f(x) = y$. Bijective, with $f^{-1}(y) = \frac{y+7}{4}$.

B9. $x \in f^{-1}(C \cap D) \iff f(x) \in C \cap D \iff f(x) \in C$ and $f(x) \in D \iff x \in f^{-1}(C) \cap f^{-1}(D)$. ■

B10. If $f(x_1) = f(x_2)$ then $g(f(x_1)) = g(f(x_2))$, so $x_1 = x_2$ as $g \circ f$ is injective. Example: $f : \{1\} \to \{a,b\}$, $f(1) = a$; $g : \{a,b\} \to \{0\}$ constant. $g \circ f$ has a one-element domain, so it is injective; g is not, as $g(a) = g(b)$.

B11. $5\mathbb{Z} = \{5k : k \in \mathbb{Z}\}$. $h : \mathbb{Z} \to 5\mathbb{Z}$, $h(k) = 5k$ is a bijection, and $\mathbb{Z}$ is countably infinite (Section 7), so composing with $\mathbb{N} \to \mathbb{Z}$ gives a bijection $\mathbb{N} \to 5\mathbb{Z}$.

B12. $f(x) = 2 + 3x$ is a bijection $(0,1) \to (2,5)$ (linear, inverse $\frac{y-2}{3}$). $g(x) = \tan(\pi(x - \frac{1}{2}))$ is a bijection $(0,1) \to \mathbb{R}$.

13.3 Part C

C1. (a) $(A - B) - C = (A \cap B^c) \cap C^c = A \cap (B^c \cap C^c) = A \cap (B \cup C)^c = A - (B \cup C)$ (difference law, associative, De Morgan).
(b)

A	B	C	$B \triangle C$	$A \cap (B \triangle C)$	$A \cap B$	$A \cap C$	$(A \cap B) \triangle (A \cap C)$
1	1	1	0	**0**	1	1	**0**
1	1	0	1	**1**	1	0	**1**
1	0	1	1	**1**	0	1	**1**
1	0	0	0	**0**	0	0	**0**
0	1	1	0	**0**	0	0	**0**
0	1	0	1	**0**	0	0	**0**
0	0	1	1	**0**	0	0	**0**
0	0	0	0	**0**	0	0	**0**

The bold columns agree, so the identity holds ($\cap$ distributes over $\triangle$).

C2. (a) $[1] = \{1,5,9\}$, $[2] = \{2,6,10\}$, $[3] = \{3,7,11\}$, $[4] = \{4,8,12\}$. Each block of 3 gives $3^2 = 9$ pairs: $|R| = 4 \times 9 = 36$.
(b) $R = \{(1,1),(1,2),(2,1),(2,2)\} \cup \{(a,b) : a,b \in \{3,4,5\}\} \cup \{(6,6)\}$; $|R| = 4 + 9 + 1 = 14$.

C3. Statement and proof as in Section 5 (Fundamental theorem): show $a \in [a]$, $a \,R\, b \iff [a] = [b]$, and that two classes are equal or disjoint; conclude the classes are non-empty, pairwise disjoint and cover A.

C4. (a) $f(x) \neq 1$, since $x - 1 = x + 1$ is impossible. One-one: $\frac{x-1}{x+1} = \frac{y-1}{y+1} \Rightarrow (x-1)(y+1) = (y-1)(x+1) \Rightarrow x - y = y - x \Rightarrow x = y$. Onto: $y(x+1) = x - 1 \Rightarrow x(y-1) = -1 - y \Rightarrow x = \frac{1+y}{1-y}$, defined for $y \neq 1$, and $x \neq -1$ (else $1 + y = -1 + y$). So $f^{-1}(y) = \dfrac{1+y}{1-y}$.
(b) $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ I_B \circ g^{-1} = I_C$; likewise $(f^{-1} \circ g^{-1}) \circ (g \circ f) = I_A$. Since inverses are unique, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. ■

C5. (a), (b) as in Problem 11. (c) No. $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} - \mathbb{Q})$. If the irrationals were countable, $\mathbb{R}$ would be a union of two countable sets, hence countable — contradicting Cantor's theorem.

C6. (a) As in Section 9. (b) Inclusion $[0,1] \hookrightarrow \mathbb{R}$ is injective; $h(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$ is a strictly increasing map $\mathbb{R} \to (0,1) \subset [0,1]$, hence injective. By Schröder–Bernstein, $|[0,1]| = |\mathbb{R}|$.

13.4 Part D

D1. (a) $2^6 = 64$ (b) $2^6 = 64$ (c) $2^3 \cdot 3^3 = 216$ (d) $2^3 = 8$.

D2. Reflexive (diagonal all 1). Not symmetric ($m_{12} = 1$, $m_{21} = 0$). Antisymmetric (no $m_{ij} = m_{ji} = 1$ with $i \neq j$). $M_R \odot M_R = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, which has a 1 at $(1,3)$ where M_R has 0: **not transitive** ($(1,2)$, $(2,3) \in R$, $(1,3) \notin R$).

D3. $R \circ S$ (first R): $1 \to 2 \to 1$; $2 \to 3 \to 3$; $2 \to 3 \to 2$: $\{(1,1),(2,3),(2,2)\}$. $S \circ R$ (first S): $2 \to 1 \to 2$; $3 \to 2 \to 3$: $\{(2,2),(3,3)\}$. $R^{-1} = \{(2,1),(3,2)\}$.

D4. Five: $\{\{1\},\{2\},\{3\}\}$; $\{\{1,2\},\{3\}\}$; $\{\{1,3\},\{2\}\}$; $\{\{2,3\},\{1\}\}$; $\{\{1,2,3\}\}$. (The number of equivalence relations on an n-set is the Bell number B_n; $B_3 = 5$.)

D5. Functions $3^4 = 81$; one-one 0 (as $4 > 3$); onto $3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 = 81 - 48 + 3 = 36$.

D6. $f(-2) = 3$, $f(0) = -1$, $f(3) = 8$: $f(S) = \{-1, 3, 8\}$. $x^2 - 1 = 0 \Rightarrow x = \pm 1$; $x^2 - 1 = 3 \Rightarrow x = \pm 2$; $x^2 = 6$ has no integer root. $f^{-1}(T) = \{-2, -1, 1, 2\}$.

D7. $(f+g)(x) = x^2 + x + 2$ [2 at $x = -1$]; $(fg)(x) = x^3 + 2x^2$ [1]; $(f \circ g)(x) = x^2 + 2$ [3]; $(g \circ f)(x) = (x+2)^2$ [1].

D8. Perpendicular: not reflexive (no line is perpendicular to itself), symmetric, not transitive ($l_1 \perp l_2, l_2 \perp l_3$ gives $l_1 \parallel l_3$). Parallel or equal: reflexive, symmetric, transitive — an equivalence relation whose classes are the "directions".

14 One-Page Revision Sheet

Sets

$A - B = A \cap B^c$; $A \triangle B = (A \cup B) - (A \cap B)$
$(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$
$A \cup (A \cap B) = A$; $A \cap (A \cup B) = A$
$A \subseteq B \iff A \cup B = B \iff A \cap B = A$
$|\mathcal{P}(A)| = 2^{|A|}$; $|A \cup B| = |A| + |B| - |A \cap B|$

Cartesian product

$|A \times B| = |A||B|$; $A \times B \neq B \times A$
$\times$ distributes over $\cup, \cap, -$

Relations ($|A| = n$)

All: 2^{n^2}; reflexive: 2^{n^2-n}; symmetric: $2^{n(n+1)/2}$
Antisymmetric: $2^n 3^{n(n-1)/2}$
Reflexive: diagonal 1s. Symmetric: $M = M^T$.
Transitive $\iff R^2 \subseteq R$ ($M \odot M \leq M$).
$R \circ S$: first R then S; $M_{R \circ S} = M_R \odot M_S$
$M(R^c) = M_R^T$; out-degree = row sum, in-degree = column sum

Orders

Partial order = R + AS + T. Equivalence = R + S + T.
Hasse: drop loops, transitive edges, arrows.
Greatest/least unique; maximal/minimal may be many.
In $(D_n, |)$: lub = lcm, glb = gcd.
In $(\mathcal{P}(S), \subseteq)$: lub $= \cup$, glb $= \cap$.
Lattice: every pair has a lub and a glb.

Equivalence classes

$[a] = [b] \iff aRb$; classes are equal or disjoint.
Equivalence relations $\leftrightarrow$ partitions.
Block of size k gives k^2 pairs.
No. of equivalence relations ($n = 3, 4, 5, 6$): 5, 15, 52, 203.

Functions ($|A| = m$, $|B| = n$)

Functions n^m; one-one $\frac{n!}{(n-m)!}$; bijections $n!$ ($m = n$)
Onto: $\sum_k (-1)^k \binom{n}{k} (n-k)^m$
$f(S_1 \cup S_2) = f(S_1) \cup f(S_2)$; $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$
f^{-1} of a set preserves $\cup, \cap$, complement.
$(f+g)(x) = f(x) + g(x)$; $(fg)(x) = f(x)g(x)$
$(g \circ f)(x) = g(f(x))$; associative, not commutative
Inverse exists $\iff$ bijective; $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
$g \circ f$ 1-1 $\Rightarrow f$ 1-1; $g \circ f$ onto $\Rightarrow g$ onto
Stirling: $S(m,n) = p(m,n)/n!$; $S(m+1,n) = S(m,n-1) + nS(m,n)$
Pigeonhole: some hole has $\geq \lfloor \frac{m-1}{n} \rfloor + 1$
$\lceil -x \rceil = -\lfloor x \rfloor$; $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ ($n \in \mathbb{Z}$)

Cardinality

$|A| = |B|$: bijection; $|A| \leq |B|$: injection
Countable: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{N} \times \mathbb{N}$, finite strings
$\pi(m,n) = \frac{(m+n)(m+n+1)}{2} + n$
Uncountable: $\mathbb{R}$, $(0,1)$, irrationals, $\mathcal{P}(\mathbb{N})$
$(0,1) \to \mathbb{R}$: $\tan(\pi(x - \frac{1}{2}))$

Big theorems

Cantor diagonal: $x_i = 5$ if $d_{ii} \neq 5$, else 4.
Power set: $D = \{a : a \notin g(a)\}$ is never hit; $|A| < |\mathcal{P}(A)|$.
Schröder–Bernstein: injections both ways $\Rightarrow$ bijection.
$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| = |\mathbb{R}| < |\mathcal{P}(\mathbb{R})|$

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